TEJAS200

BITSAT

Physics

Chapter-wise Question Book

50 MCQs per chapter / One answer-key page after every chapter

30 chapter units

1,500 questions

Revised practice edition V2 / October 2026

Using this book

Each chapter contains 50 single-correct, four-option questions. Select one option, A–D, for each question. The next page after the chapter contains its complete answer key. Solutions are not included. Questions 1–10 retain the strongest ten questions selected from the previous middle block, 11–40 are newly written middle-to-hard replacements, and 41–50 preserve the previous hard block. All previous easy questions and ten weaker middle questions have been removed. These are intended practice levels, not official difficulty grades or empirical calibration. Work through the chapter before opening its answer key. Use the questions to check understanding, selection of the first step, calculation, and the stated conditions. Constants and idealisations given in a question take priority for that question. Chapter units follow the updated Tejas200 chapter engine. They are study groupings, rather than a claim that the official syllabus uses the same chapter headings. This is original practice material, not an official BITS question paper. **Reference conventions.** Use SI units unless specified. Where a question does not state a value, use $g = 10\,\mathrm{m\,s^{-2}}$, $c = 3.0 \times 10^{8}\,\mathrm{m\,s^{-1}}$, $k = 9.0 \times 10^{9}\,\mathrm{N\,m^{2}\,C^{-2}}$, $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$, $e = 1.6 \times 10^{-19}\,\mathrm{C}$, $h = 6.626 \times 10^{-34}\,\mathrm{J\,s}$, $hc = 1240\,\mathrm{eV\,nm}$, and hydrogen $E_n = -13.6/n^{2}\,\mathrm{eV}$. Numerical approximations explicitly supplied in a question take priority. Keep π exact unless specified. A logarithm without a stated base denotes the natural logarithm, unless a question specifies otherwise.

Contents

1 Units & Measurements 1

2 Kinematics 9

3 Laws of Motion 17

4 Circular Motion 25

5 Work Energy Power 33

6 Centre of Mass & Collisions 41

7 Rotational Motion 49

8 Gravitation 57

9 Elasticity 65

10 Fluid Mechanics 73

11 Thermal Properties 81

12 Thermodynamics 89

13 Kinetic Theory 97

14 Oscillations 105

15 Waves 114

16 Electrostatics 123

17 Capacitance 131

18 Current Electricity 140

19 Magnetic Effects of Current 148

20 Electromagnetic Induction **157**

21 Alternating Current **166**

22 Electromagnetic Waves **174**

23 Ray Optics **183**

24 Wave Optics **192**

25 Dual Nature **200**

26 Atoms **209**

27 Nuclei **218**

28 Semiconductors **227**

29 Magnetism & Matter **236**

30 Communication Systems **245**

Chapter 1

Units & Measurements

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. For $z = x - y$, $x = (5.0 \pm 0.1)$ and $y = (4.0 \pm 0.1)$ in the same unit. The maximum percentage error in z is

(A) 40 (B) 19

(C) 21 (D) 20

02. A screw gauge has pitch 0.5 mm and 50 divisions. A wire gives pitch-scale reading 2.5 mm, circular reading 18 divisions and positive zero error 3 divisions. Diameter in mm is

(A) 2.65 (B) 1.65

(C) 5.3 (D) 3.65

03. A vernier with least count 0.02 mm has zero error -0.06 mm. During measurement the main-scale reading is 8 mm and division 13 coincides. The true reading in mm is

(A) 8.32 (B) 16.64

(C) 9.32 (D) 7.32

04. The pendulum formula $T = 2\pi\sqrt{L/g}$ is used to determine g. If maximum percentage errors in L, T are $1, 0.5$ respectively, the first-order error in g is

(A) 1 (B) 4

(C) 2 (D) 3

05. If the unit of length is doubled and the unit of time is halved while mass is unchanged, the numerical value of a fixed power becomes what fraction of its old value?

(A) -0.96875 (B) 0.0625

(C) 0.03125 (D) 1.03125

06. A length is inferred by subtracting two positions: (24.60 ± 0.02) cm and (18.20 ± 0.02) cm. The first-order maximum percentage error in the length is

(A) 1.625 (B) 0.625

(C) 1.25 (D) -0.375

07. The force law is $F = kv^a r^b$, with $[k] = MT^{-1}$. Dimensional consistency gives (a, b) equal to

(A) $(2, -1)$ (B) $(1, 1)$

(C) $(1, 0)$ (D) $(0, 1)$

08. For an instrument, 25 vernier divisions equal 24 main divisions, each 0.5 mm. If division 12 coincides at zero with the vernier zero to the right, zero error in mm is

(A) -0.76 (B) 1.24

(C) 0.24 (D) 0.48

09. A wire diameter changes from 1.00 to 1.01 mm after correcting an instrument error. The exact fractional change in the inferred cross-sectional area is

(A) -0.9799 (B) 0.0402

(C) 1.0201 (D) 0.0201

10. A physical quantity is $Q = (a + b)/(a - b)$ for $a = 10, b = 6$. Each has maximum absolute error 0.1. Using first-order partial derivatives, maximum absolute error in Q is

(A) 1.2 (B) 0.2

(C) 0.4 (D) -0.8

11–40 / New middle-to-hard practice

11. A length scale reads 2.14 cm and 10.30 cm for true lengths 2.00 cm and 10.00 cm. A specimen reads 6.73 cm. Assuming an affine calibration law, its true length in cm is

(A) 6.75 (B) 6.5

(C) 6.63 (D) 6.59

12. A gauge obeys $r = a\ell + b$. Standards $\ell = 2.0, 8.0$ mm give $r = 2.3, 8.9$ mm. Two readings on a hollow part are 5.6 mm and 3.4 mm. The true difference of the two lengths, in mm, is

(A) 1.9 (B) 2.2

(C) 2.3 (D) 2

13. A screw gauge has pitch 0.50 mm and 100 divisions. Closed jaws give circular reading 96, with the zero lying below the reference line by 4 divisions. A wire reads main scale 1.50 mm and circular division 40. Its corrected cross-sectional area, in mm^2, is

(A) $\frac{1849\pi}{2500}$ (B) $\frac{1681\pi}{2500}$

(C) $\frac{441\pi}{625}$ (D) $\frac{9\pi}{16}$

14. A vernier has 50 divisions equal to 49 main divisions, each 1 mm. Its positive zero error is 3 least counts. A solid cylinder gives diameter MSR 12 mm, coincidence 18, and length MSR 50 mm, coincidence 3. The corrected ratio L/d is

(A) $\frac{2503}{618}$ (B) $\frac{500}{123}$

(C) 4 (D) $\frac{2500}{121}$

15. A shared additive radius offset is bounded by 0.02 cm when an annulus is measured as outer radius 5.00 cm and inner radius 3.00 cm. Additional independent reading errors are bounded by 0.01 cm for each radius. The first-order maximum percentage error in the annulus area is

(A) 2 (B) 1

(C) 0.5 (D) 1.5

16. Two positions are read with the same arbitrary additive zero error, a common scale uncertainty of 0.1%, and independent reading uncertainties 0.01 cm each. To keep the first-order maximum fractional error in their difference at most 0.5%, their nominal separation must be at least, in cm,

(A) 10 (B) 5

(C) 6 (D) 4

17. A sphere has nominal radius 1 cm and a symmetric radius uncertainty δ. The difference between its largest and smallest allowed volumes is $301\pi/375\,\mathrm{cm}^3$. The unique positive value of δ, in cm, is

(A) 0.05 (B) 0.2

(C) 0.1 (D) 0.3

18. Two positive quantities are $a = 10 \pm 1$ and $b = 4 \pm 1$, with errors independently bounded. For $Q = (a + b)/(a - b)$, the exact largest possible value is

(A) 4 (B) 3

(C) $\frac{8}{3}$ (D) 3.5

19. For a wire, $Y = 4FL/(\pi d^2 \Delta L)$. The same calibration factor multiplies both L and ΔL and is uncertain by 2%. Independent errors in F and d are at most 1% and 0.5%. The first-order maximum percentage error in Y is

(A) 2 (B) 3

(C) 6 (D) 4

20. A density is obtained as $\rho = m/(\pi r^2 L)$. A common length scale reads every length 1% too high, while a mass balance reads 2% too high. The ratio of reported density to true density is

(A) $\frac{102}{101}$ (B) $\frac{1020000}{1030301}$

(C) $\frac{102}{103}$ (D) $\frac{101}{102}$

21. A cylindrical wire has diameter 2.000 mm and length 20.0 cm. The length uncertainty is at most 0.1 cm. A diameter gauge has reading uncertainty equal to half its least count plus an independent zero-correction uncertainty 0.001 mm. To keep the first-order maximum volume error at 2%, the largest allowed gauge least count, in mm, is

(A) 0.02 (B) 0.014

(C) 0.028 (D) 0.03

22. In a new unit system, the numerical values of a fixed velocity, acceleration and force are respectively 1/2, 1/4 and 1/12 of their SI values. The numerical value of a fixed action is multiplied relative to SI by

(A) $\frac{1}{6}$ (B) $\frac{1}{12}$

(C) 6 (D) $\frac{1}{3}$

23. An experiment fits $F = av + bv^2$. In a new unit system the mass, length and time units are $2, 3, 4$ times the old units. The factor multiplying the old numerical value of a/b to give its new numerical value is

(A) 0.75 (B) $\frac{4}{3}$

(C) $\frac{16}{9}$ (D) $\frac{8}{3}$

24. In a new unit system the numerical values of a fixed acceleration, density and pressure become 2, 8 and 4 times their SI values. The numerical value of a fixed speed becomes what factor of its SI value?

(A) 0.5 (B) $\frac{\sqrt{2}}{2}$

(C) 2 (D) $\sqrt{2}$

25. A quantity $X = G^a h^b c^d$ has the dimensions of length; G, h, c are the gravitational constant, Planck constant and light speed. The value of $a + b + d$ is

(A) -1.5 (B) 1.5

(C) 0.5 (D) -0.5

26. A time scale is formed as $\tau = \varepsilon_0^a m^b q^c \ell^d$, where q, m, ℓ are charge, mass and length. Dimensional consistency gives $c + d$ equal to

(A) 1 (B) 0.5

(C) -0.5 (D) 2

27. A wave phase is $\phi = Bt^2 - Cx + \phi_0$. At $x = 0$, consecutive crests pass at $t = 2$ s and $t = 6$ s. At one instant, adjacent crests are 0.50 m apart. The instantaneous rightward phase speed at $t = 8$ s, in $\mathrm{m\,s^{-1}}$, is

(A) 0.125 (B) 1

(C) 0.5 (D) 0.25

28. A drag model is $F = \rho^a \nu^b v^c r^d$, with a dimensionless coefficient suppressed. Here ν is kinematic viscosity, of dimensions L^2T^{-1}. Doubling both r and v, with ρ, ν fixed, multiplies F by 8. Dimensional consistency then gives d/b equal to

(A) 3 (B) 2

(C) 1 (D) 4

29. Vernier A has n divisions equal to $n - 1$ main divisions, each 1 mm. Vernier B has $n + 10$ divisions equal to $n + 9$ main divisions, each 0.5 mm. Their least-count difference is 1/30 mm. A has positive zero error 3 least counts. It reads main scale 4 mm and coincidence 7. The corrected reading in mm is

(A) 4.15 (B) 4.35

(C) 4.4 (D) 4.2

30. Two screw gauges have N circular divisions each, pitches 1 mm and 0.5 mm, and the same zero-correction uncertainty z mm. Each reading uncertainty is half a circular division plus z. Measuring diameters 1 mm and 0.5 mm gives maximum first-order area errors 1.2% and 1.6% respectively. The value of N is

(A) 150 (B) 250

(C) 125 (D) 100

31. A caliper adds the same unknown zero offset z to every radius. It reports outer and inner radii of an annulus as 5 cm and 3 cm. The true annulus area is $12\pi\,\mathrm{cm}^2$. The same caliper reports a solid sphere radius as 4 cm. The true sphere volume divided by π, in cm^3, is

(A) 36 (B) 108

(C) $\frac{256}{3}$ (D) $\frac{32}{3}$

32. A stopwatch obeys $r = at + b$. Intervals truly lasting 20 s and 100 s read 20.6 s and 102.2 s. It reads 81.8 s for 40 oscillations of a 1 m pendulum. The inferred g/π^2, in $\mathrm{m\,s^{-2}}$, is

(A) 4 (B) 1.02

(C) 0.25 (D) 1

33. Two positive lengths rounded to the nearest 0.1 cm read $x = 2.0$ cm and $y = 3.0$ cm. An independent rounded reading of their sum is 5.0 cm. The least upper bound on xy, in cm^2, consistent with all three readings is

(A) 6.2525 (B) 6.15

(C) 6.05 (D) 6

34. Air and submerged force readings for a solid are 10.0 N and 6.0 N. Both have a common additive zero error z, $|z| \le 0.2$ N, and additional independent errors e_1, e_2, each bounded by 0.1 N, with reading=true+$z + e_i$. Neglect buoyancy in air. The largest possible density ratio to water is

(A) $\frac{101}{38}$ (B) $\frac{103}{38}$

(C) 2.5 (D) $\frac{33}{14}$

35. A drag law is modeled as $F = av + bv^2$. Steady-speed measurements give $F = 12$ N at $v = 2\,\mathrm{m\,s^{-1}}$ and $F = 40$ N at $v = 4\,\mathrm{m\,s^{-1}}$. With the same drag law, a constant thrust of 84 N produces terminal speed, in $\mathrm{m\,s^{-1}}$, equal to

(A) 4 (B) 6

(C) 5 (D) 7

36. Drag on a sphere is modeled as $F = \eta^a \rho^b v^c r^d$. Dimensional consistency alone leaves one free exponent. If a separate experiment finds $F \propto v^2$, the value of $a + d$ is

(A) 1 (B) 0

(C) 2 (D) 3

37. A spring constant is inferred from a straight-line fit $T^2 = Am + B$. At $m = 0.10, 0.30$ kg the values of T^2 are $0.60, 1.40$ s^2. The fitted effective extra oscillating mass, in kg, is

(A) 0.025 (B) 0.05

(C) 0.1 (D) 0.15

38. A resistance thermometer obeys $R = A + B\theta + C\theta^2$, with θ in °C. It reads 100, 130 and 180 Ω at 0, 50 and 100°C. A reading 152.5 Ω corresponds to a temperature between 0 and 100°C equal to

(A) 62.5 (B) 65.625

(C) 75 (D) 80

39. An experiment infers $q = xy/(x + y)$. At $x = y = 10$, a common multiplicative scale error is at most 1%, and additional independent absolute reading errors are at most 0.1 in each variable. The first-order maximum percentage error in q is

(A) 2 (B) 3

(C) 1 (D) 0.5

40. An angle is inferred from $\tan\theta = y/x$, with $x = 3$, $y = 4$. Both readings have a common scale error of at most 3%, and additional independent absolute errors at most 0.1 each. The first-order maximum absolute error in θ, in radians, is

(A) 0.02 (B) 0.03

(C) 0.056 (D) 0.028

41–50 / Preserved hard practice

41. A uniform wire is tested at lengths (1.000 ± 0.005) m and (2.000 ± 0.005) m, with unchanged diameter and resistivity. Measured resistances are (10.0 ± 0.1) and $(18.0 \pm 0.1)\,\Omega$, each including the same unknown contact resistance. Diameter has 1% error. All stated errors are independent bounded errors. Find first-order maximum percentage error in inferred resistivity.

(A) 4.5 (B) 11

(C) 5.5 (D) 6.5

42. A freely falling body has measured downward positions $(1.00 \pm 0.01), (4.00 \pm 0.01), (9.00 \pm 0.01)$ m at three equally spaced times. Time spacing is (0.500 ± 0.001) s. All reading errors are independently bounded; an additional common unknown position zero offset is identical for all readings. Find first-order maximum percentage error in inferred g.

(A) 4.8 (B) 1.4

(C) 3.4 (D) 2.4

43. A cell is measured with two exactly known loads R and $2R$. The corresponding terminal voltages are (6.00 ± 0.06) and (8.00 ± 0.08) V, with independent bounded errors. Cell emf and internal resistance are constant but unknown. Find first-order maximum percentage error in inferred emf.

(A) 5 (B) 10

(C) 6 (D) 4

44. A sphere terminal speed is measured in a viscous fluid. Fluid density and viscosity are exact, while sphere mass m and radius r each have independent 1% bounded errors. At nominal values the sphere density is twice fluid density. Using Stokes drag and buoyancy, first-order maximum percentage error in predicted terminal speed is

(A) 5 (B) 12

(C) 7 (D) 6

45. A rectangular loop has perimeter $P = (30.0 \pm 0.3)$ cm and side ratio $r = a/b = (2.00 \pm 0.02)$. A perpendicular magnetic field has 1% independent error. Find the first-order maximum percentage error in magnetic flux.

(A) $\frac{10}{3}$ (B) $\frac{13}{3}$

(C) $\frac{7}{3}$ (D) $\frac{20}{3}$

46. A sphere falls at terminal speed through a viscous liquid. Its measured mass is (10.0 ± 0.1) g; independently measured displaced-liquid mass is (8.0 ± 0.1) g. Fall time, radius and fall distance each have independent 1% errors. Take g exact. Find the first-order maximum percentage error in inferred viscosity.

(A) 14 (B) 26

(C) 13 (D) 12

47. A spring has an unknown fixed effective moving mass. Attached masses (1.00 ± 0.01) and (3.00 ± 0.01) kg produce measured periods (2.00 ± 0.01) and (3.00 ± 0.01) s. All four errors are independent bounded errors. Find the first-order maximum percentage error in inferred spring constant.

(A) 3 (B) 2

(C) 4 (D) 6

48. A wire has resistances $(100 \pm 1)\,\Omega$ and $(104 \pm 1)\,\Omega$ at temperatures separated by an exactly known 20 K. The two resistance errors are independent and bounded. Using a linear resistance-temperature law referenced to the first temperature, find the first-order maximum percentage error in the temperature coefficient.

(A) 102 (B) 50

(C) 51 (D) 52

49. A test mass m on a spring is displaced by a source mass M through an equilibrium shift x. Its spring period before the source is brought near is T, and source-test separation at the shifted equilibrium is r. Infer G using the spring calibration. Independent errors in m, M, x, T, r are $5, 2, 1, 1, 1\%$, respectively. Find the first-order maximum percentage error in G.

(A) 7 (B) 14

(C) 6 (D) 8

50. A bullet of mass m embeds in a stationary block of mass M. All collision energy loss heats both bodies, of the same unknown specific heat c. Nominally $m = M$. Independent bounded errors in m, M, bullet speed u and common temperature rise are $4, 5, 2, 3\%$. Find first-order maximum percentage error in inferred c.

(A) 8 (B) 7

(C) 14 (D) 6

Answer key

Units & Measurements

P-UNME / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**B**	21	**C**	31	**A**	41	**C**
02	**A**	12	**D**	22	**A**	32	**D**	42	**D**
03	**A**	13	**A**	23	**B**	33	**B**	43	**A**
04	**C**	14	**B**	24	**B**	34	**A**	44	**D**
05	**C**	15	**D**	25	**D**	35	**B**	45	**A**
06	**B**	16	**B**	26	**B**	36	**C**	46	**C**
07	**C**	17	**C**	27	**D**	37	**B**	47	**A**
08	**C**	18	**D**	28	**A**	38	**C**	48	**C**
09	**D**	19	**A**	29	**D**	39	**A**	49	**A**
10	**B**	20	**B**	30	**C**	40	**D**	50	**B**

Chapter 2

Kinematics

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A particle starts with speed $4\,\mathrm{m\,s^{-1}}$ and acceleration $2\,\mathrm{m\,s^{-2}}$. Time to cover $32\,\mathrm{m}$, in seconds, is

(A) 3 (B) 4

(C) 5 (D) 8

02. A stone is dropped from rest. It travels $45\,\mathrm{m}$ in the last second before impact. With $g = 10$, total fall time in seconds is

(A) 5 (B) 6

(C) 10 (D) 4

03. A ball is projected at $25\,\mathrm{m\,s^{-1}}$ with horizontal component $15\,\mathrm{m\,s^{-1}}$. With $g = 10$, its maximum rise in metres is

(A) 21 (B) 19

(C) 20 (D) 40

04. A boat speed in still water is $5\,\mathrm{m\,s^{-1}}$, current speed $3\,\mathrm{m\,s^{-1}}$. For zero drift, its effective cross-river speed in m/s is

(A) 4 (B) 8

(C) 3 (D) 5

05. A river is $120\,\mathrm{m}$ wide. A boat points straight across at $4\,\mathrm{m\,s^{-1}}$ relative to water; current is $3\,\mathrm{m\,s^{-1}}$. Downstream drift in metres is

(A) 89 (B) 90

(C) 91 (D) 180

06. A car goes at $10\,\mathrm{m\,s^{-1}}$ for 5 s and then accelerates uniformly to $20\,\mathrm{m\,s^{-1}}$ in 5 s. Total distance in metres is

(A) 126 (B) 250

(C) 124 (D) 125

07. A ball thrown upward passes a point at $t = 1$ and $t = 3$ s. With $g = 10$, launch speed in m/s is

(A) 40 (B) 19

(C) 20 (D) 21

08. A ball thrown upward passes a point at $t = 1$ and 3 s. With $g = 10$, the height of that point in metres is

(A) 15 (B) 16

(C) 14 (D) 30

09. A particle has constant acceleration. It covers 10 m in first 2 s and 18 m in next 2 s. Its acceleration in m/s squared is

(A) 4 (B) 2

(C) 1 (D) 3

10. Two vectors of equal magnitude 10 have resultant magnitude $10\sqrt{3}$. Angle between them in degrees is

(A) 120 (B) 61

(C) 60 (D) 59

11–40 / New middle-to-hard practice

11. A particle has $v = 3t^2 - 12t + 9$ in SI units. Its total distance from $t = 0$ to 4 s, in metres, is

(A) 16 (B) 8

(C) 4 (D) 12

12. A particle has $x = t^3 - 3t^2 - 9t$ in SI units. The ratio of distance travelled to displacement magnitude over $0 \le t \le 4$ s is

(A) 1.7 (B) 1.4

(C) $\frac{10}{17}$ (D) 1

13. A particle starts from rest, accelerates at a for time τ and then at $-2a$ until it first returns to its initial position. The total elapsed time divided by τ is

(A) 2 (B) 3

(C) $1 + \sqrt{2}$ (D) $\frac{\sqrt{3}}{2} + \frac{3}{2}$

14. A car starts from rest, accelerates uniformly for 4 s, and then brakes at twice that acceleration to rest. Its total distance is 72 m. Its peak speed, in $\mathrm{m\,s^{-1}}$, is

(A) 12 (B) 18

(C) 36 (D) 24

15. A ball passes a height h at $t = 1$ s and $t = 5$ s after an upward launch. Taking $g = 10$, its speed at height h, in $\mathrm{m\,s^{-1}}$, is

(A) 30 (B) 25

(C) 20 (D) 10

16. A projectile crosses a horizontal level at 2 s and 6 s after launch. With $g = 10$, the maximum height above that level, in metres, is

(A) 80 (B) 60

(C) 40 (D) 20

17. A projectile launched from the origin has trajectory $y = 2x - x^2/20$, with coordinates in metres. A wall at $x = 10$ m has its top exactly on the trajectory. The difference between the trajectory maximum height and this wall-top height, in m, is

(A) 20 (B) 5

(C) 10 (D) 15

18. A projectile passes through $(10, 15)$ m and $(30, 15)$ m after launch from the origin. Taking $g = 10$, its horizontal launch component, in $\mathrm{m\,s^{-1}}$, is

(A) 15 (B) 5

(C) 20 (D) 10

19. A projectile is launched with components $(20, 30)\,\mathrm{m\,s^{-1}}$. At the same instant a target at $(40, 60)$ m is dropped from rest. With $g = 10$, the collision height above the launch level, in metres, is

(A) 20 (B) 50

(C) 40 (D) 30

20. A projectile is aimed directly at a target initially 60 m horizontally and 80 m vertically above launch. The target is released when the projectile launches at $50\,\mathrm{m\,s^{-1}}$. With $g = 10$, the ratio of projectile kinetic energy at collision to its launch kinetic energy is

(A) 0.36 (B) 0.52

(C) 0.64 (D) 0.8

21. A boat moves at $5\,\mathrm{m\,s^{-1}}$ relative to water in a river flowing at $3\,\mathrm{m\,s^{-1}}$. It crosses a width 80 m directly opposite, then returns along the same ground line. The round-trip time, in seconds, is

(A) 40 (B) 50

(C) 32 (D) 64

22. A boat speed relative to water is $5\,\mathrm{m\,s^{-1}}$ and a current is $3\,\mathrm{m\,s^{-1}}$. To reach a point 60 m downstream on the opposite bank, 80 m away, in the least possible time, the crossing time, in seconds, is

(A) 25 (B) 20

(C) 16 (D) $-\frac{45}{4}+\frac{5\sqrt{769}}{4}$

23. Rain is vertical relative to the ground. A runner sees it at 45° to the vertical while running east at $6\,\mathrm{m\,s^{-1}}$. When he runs north at $8\,\mathrm{m\,s^{-1}}$, the apparent rain speed in $\mathrm{m\,s^{-1}}$ is

(A) 14 (B) 10

(C) 12 (D) 8

24. A person walking east at $3\,\mathrm{m\,s^{-1}}$ sees rain fall vertically. Walking west at the same speed, he sees rain at 45° to the vertical. The actual rain speed in $\mathrm{m\,s^{-1}}$ is

(A) 6 (B) $3\sqrt{5}$

(C) $3\sqrt{2}$ (D) $3\sqrt{3}$

25. Two cars start 100 m apart on one straight road, moving in the same direction. Rear car starts at $20\,\mathrm{m\,s^{-1}}$ and brakes at $2\,\mathrm{m\,s^{-2}}$; front car maintains $10\,\mathrm{m\,s^{-1}}$. The smallest separation, in metres, before the rear car stops is

(A) 100 (B) 50

(C) 75 (D) 25

26. Two particles start simultaneously from $(0,0)$ and $(30,40)$ m, with velocities $(4,0)$ and $(0,-3)\,\mathrm{m\,s^{-1}}$. Their minimum separation, in metres, is

(A) 20 (B) 10

(C) 14 (D) 18

27. A rod of fixed length 5 m has ends sliding on perpendicular axes. When its coordinates are $(x,y)=(3,4)$ m, the horizontal end moves outward at $2\,\mathrm{m\,s^{-1}}$ with zero acceleration. The magnitude of the vertical end acceleration, in $\mathrm{m\,s^{-2}}$, is

(A) 1.5625 (B) 0.5625

(C) 1 (D) 1.5

28. A rod of length 10 m slides between floor and wall. Its lower end is 6 m from the wall and moves away at $4\,\mathrm{m\,s^{-1}}$. The instantaneous rate of decrease of the rod midpoint height, in $\mathrm{m\,s^{-1}}$, is

(A) 0.75 (B) 2

(C) 1.5 (D) 3

29. Two projectiles launched from the same point with the same speed have equal nonzero ranges, at angles $\alpha < 45^\circ < \beta$. If their maximum heights are in ratio $1:3$, the smaller angle in degrees is

(A) 30 (B) 60

(C) 15 (D) 22.5

30. Two projectiles launched from the same point at the same speed u have equal nonzero range R on level ground and flight times in ratio $1:2$. The value of $u^2/(gR)$ is

(A) 0.8 (B) 1

(C) 1.25 (D) 1.5

31. A trolley moves horizontally at $5\,\mathrm{m\,s^{-1}}$. A ball launches with velocity $(5\hat{i}+20\hat{j})\,\mathrm{m\,s^{-1}}$ relative to it. Immediately afterward the trolley accelerates rightward at $2\,\mathrm{m\,s^{-2}}$. With $g=10$, when the ball returns to launch height its horizontal position relative to the launch point on the trolley, in m rightward, is

(A) 20 (B) 4

(C) 16 (D) -4

32. A train moving at $12\,\mathrm{m\,s^{-1}}$ launches a ball vertically relative to itself at $15\,\mathrm{m\,s^{-1}}$. It then accelerates horizontally at $2\,\mathrm{m\,s^{-2}}$. With $g=10$, the ball lands how far behind its launch point on the train, in metres?

(A) 6 (B) 3

(C) 9 (D) 18

33. Two stones are dropped from the same point 1 s apart. With $g=10$, when their separation first becomes 15 m, the lower stone speed, in $\mathrm{m\,s^{-1}}$, is

(A) 20 (B) 25

(C) 15 (D) 10

34. A stone is dropped from a height 80 m. After 1 s a second is thrown downward from the same point so they land together. With $g=10$, the required initial speed of the second stone, in $\mathrm{m\,s^{-1}}$, is

(A) $\frac{35}{3}$ (B) 15

(C) 10 (D) $\frac{25}{3}$

35. A driver at $20\,\mathrm{m\,s^{-1}}$ reacts for 0.5 s, then brakes uniformly at $5\,\mathrm{m\,s^{-2}}$. If a barrier is 45 m away initially, the impact speed, in $\mathrm{m\,s^{-1}}$, is

(A) $5\sqrt{2}$ (B) 10

(C) 0 (D) 5

36. Two cars each initially travel at $20\,\mathrm{m\,s^{-1}}$ with separation 15 m. Front car brakes at $5\,\mathrm{m\,s^{-2}}$; rear driver reacts for 0.5 s then brakes at the same rate. Ignoring car lengths, the minimum separation over the entire motion, in metres, is

(A) 0 (B) 5

(C) 10 (D) 7.5

37. A particle moving right obeys $a=-kv$, with constant $k>0$. Its speeds at $x=2$ m and $x=5$ m are 6 and $3\,\mathrm{m\,s^{-1}}$. Starting the clock when it crosses $x=0$, its displacement in the first $\ln 2$ s, in m, is

(A) 4 (B) 3

(C) 6 (D) 5

38. A particle obeys $a = kx$, with constant $k > 0$. At $x = 2$ m it moves toward the origin at $1\,\mathrm{m\,s^{-1}}$; later at $x = 4$ m its speed is $\sqrt{13}\,\mathrm{m\,s^{-1}}$. The smallest coordinate it reaches in m is

(A) $\sqrt{2}$ (B) 1

(C) 0 (D) $\sqrt{3}$

39. A particle has $\mathbf{r} = 3t\hat{i} + (4t - 5t^2)\hat{j}$ in SI units. When its velocity is perpendicular to its initial velocity, its height in metres is

(A) 0.546875 (B) 0.625

(C) 0.9375 (D) 0.390625

40. A projectile has horizontal speed $6\,\mathrm{m\,s^{-1}}$. Two seconds after an instant when its velocity makes an angle greater than 45° above horizontal, its velocity is perpendicular to that earlier velocity. With $g = 10$, the vertical displacement between the two instants, in m, is

(A) -16 (B) 16

(C) 36 (D) 20

41–50 / Preserved hard practice

41. A block released from rest slides 5.00 m down an incline with $\sin\theta = 3/5$, $\cos\theta = 4/5$ and unknown kinetic coefficient μ. It then stops after 2.50 m on a horizontal surface whose kinetic coefficient is 2μ. Take $g = 10$ and neglect the short junction. Total travel time, in seconds, is

(A) $\frac{3\sqrt{3}}{2}$ (B) $2\sqrt{3}$

(C) 3 (D) $\sqrt{3}$

42. A projectile is launched at 20 m/s at 45° on level ground. At its highest point it splits into two equal masses; one fragment is instantaneously at rest. With $g = 10$ and no air resistance, horizontal separation of their ground impact points in metres is

(A) 41 (B) 80

(C) 39 (D) 40

43. A 2 kg particle initially at rest receives horizontal force $F(t) = 6t\,\mathrm{N}$ for $0 \le t \le 2\,\mathrm{s}$. It then receives constant force $-6\,\mathrm{N}$. Distance travelled up to its first stop, in metres, is

(A) 9 (B) 10

(C) 11 (D) 20

44. A 1 kg particle carrying $+1\,\mathrm{C}$ is projected vertically upward at 20 m/s in an upward electric field $5\,\mathrm{N/C}$. Take $g = 10$. The field is switched off at the highest point. Neglect air resistance. Ground-impact speed is

(A) $10\sqrt{2}\,\mathrm{m/s}$ (B) 40 m/s

(C) $20\sqrt{2}\,\mathrm{m/s}$ (D) 20 m/s

45. A projectile is launched at 20 m/s and 45° on level ground. An additional horizontal force provides acceleration 5 m/s^2 along its horizontal motion only until the highest point; it is then switched off. Take $g = 10$. Find the range, in metres.

(A) 54 (B) 110

(C) 56 (D) 55

46. A 1 kg particle starts from rest at $x = 0$ and is acted on by $F_x = 6t - 6$ N for $0 \le t \le 3$ s. Find its total distance travelled, in metres.

(A) 7 (B) 9

(C) 16 (D) 8

47. A 1 kg block is launched horizontally by a 200 N/m spring from a 20 m high table. At launch, a ground target 16 m from the table edge moves toward the edge at 2 m/s. Take $g = 10$ and neglect losses. Find the spring compression needed to hit the target, in metres.

(A) $1 - \frac{3\sqrt{2}}{10}$ (B) $\frac{3\sqrt{2}}{20}$

(C) $\frac{3\sqrt{2}}{10}$ (D) $\frac{3\sqrt{2}}{20} + \frac{1}{2}$

48. A 1 kg block at 6 m/s sticks to a stationary equal block. The pair climbs a 30° incline with kinetic coefficient $1/(2\sqrt{3})$, then slides back. Take $g = 10$ and sufficient gravity to overcome static friction at the turn. Find time from collision until first return to the incline foot, in seconds.

(A) $\frac{2\sqrt{3}}{5} + \frac{7}{5}$ (B) $\frac{4}{5} + \frac{4\sqrt{3}}{5}$

(C) $-\frac{3}{5} + \frac{2\sqrt{3}}{5}$ (D) $\frac{2}{5} + \frac{2\sqrt{3}}{5}$

49. A particle starts from rest on a circle of radius 2 m. Its tangential force divided by mass is $4t$ m/s^2. At the first positive time when its total force makes 45° with the velocity, find its angular displacement in radians.

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{5}{6}$ (D) $\frac{2}{9}$

50. An airless planet has radius 2×10^7 m and surface escape speed 2×10^4 m/s. A ball drops from rest 40 m above the initial position of a massive platform that moves upward at 10 m/s. It rebounds perfectly elastically in the platform frame. Treat gravity as constant. Find its greatest subsequent height above the initial platform position, in metres.

(A) 101 (B) 100

(C) 99 (D) 200

Answer key

Kinematics

P-KINE / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**D**	21	**A**	31	**B**	41	**A**
02	**A**	12	**A**	22	**D**	32	**C**	42	**D**
03	**C**	13	**D**	23	**B**	33	**A**	43	**B**
04	**A**	14	**D**	24	**B**	34	**A**	44	**C**
05	**B**	15	**C**	25	**C**	35	**A**	45	**D**
06	**D**	16	**D**	26	**C**	36	**B**	46	**D**
07	**C**	17	**B**	27	**A**	37	**A**	47	**C**
08	**A**	18	**D**	28	**C**	38	**D**	48	**D**
09	**B**	19	**C**	29	**A**	39	**A**	49	**B**
10	**C**	20	**B**	30	**C**	40	**B**	50	**B**

Chapter 3

Laws of Motion

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A $2\,\text{kg}$ block on a rough horizontal table has $\mu_s = 0.5, \mu_k = 0.3$. A horizontal $12\,\text{N}$ pull is applied. With $g = 10$, acceleration in m/s squared is

(A) 6 (B) 4

(C) 3 (D) 2

02. A $5\,\text{kg}$ block is pulled with a $25\,\text{N}$ force at 37° above horizontal. Take $\sin 37 = 3/5, \cos 37 = 4/5, g = 10$ and $\mu_k = 0.2$. Acceleration in m/s squared is

(A) 5.2 (B) 3.6

(C) 1.6 (D) 2.6

03. A $5\,\text{kg}$ block is pushed with $25\,\text{N}$ at 37° below horizontal. Use $\sin 37 = 3/5, \cos 37 = 4/5, g = 10, \mu_k = 0.2$. Acceleration in m/s squared is

(A) 1.4 (B) 2.8

(C) 2.4 (D) 0.4

04. Two blocks $2, 3\,\text{kg}$ touch on a smooth table. A $20\,\text{N}$ horizontal force is applied to the $2\,\text{kg}$ block toward the other. Contact force in newtons is

(A) 24 (B) 12

(C) 13 (D) 11

05. A $2, 3\,\text{kg}$ Atwood pair uses a light string and pulley. With $g = 10$, string tension in newtons is

(A) 25 (B) 24

(C) 23 (D) 48

06. A 4 kg table block is connected to a hanging 1 kg mass. Table is smooth and $g = 10$. String tension in newtons is

(A) 9 (B) 8

(C) 16 (D) 7

07. A block slides down an incline with $\sin\theta = 3/5, \cos\theta = 4/5$ and $\mu_k = 0.25$. With $g = 10$, its acceleration in m/s squared is

(A) 8 (B) 4

(C) 3 (D) 5

08. A horizontal force holds a 2 kg block against a vertical wall. With $\mu_s = 0.5, g = 10$, minimum horizontal force in newtons is

(A) 41 (B) 80

(C) 40 (D) 39

09. A mass hangs from two strings. One is horizontal and the other makes 30° with vertical. If the weight is 30 N, inclined-string tension is

(A) $30\sqrt{3}$ N (B) 60 N

(C) $20\sqrt{3}$ N (D) $10\sqrt{3}$ N

10. On a smooth incline $\sin\theta = 3/5$, a force F parallel upward makes a 2 kg block accelerate upward at $2\,\mathrm{m\,s^{-2}}$. With $g = 10$, F in newtons is

(A) 16 (B) 17

(C) 32 (D) 15

11–40 / New middle-to-hard practice

11. Two contacting blocks of masses 2 and 3 kg lie on a horizontal floor with kinetic friction coefficient 0.2 for both. A 30 N force pushes the 2 kg block toward the other. With $g = 10$, the contact force in N is

(A) 12 (B) 20

(C) 18 (D) 24

12. Two blocks of masses 2 and 3 kg contact on a smooth line. A 20 N force pushes the 2 kg block rightward and a 5 N force pushes the 3 kg block leftward. The contact force in N is

(A) 14 (B) 9

(C) 15 (D) 11

13. A 2 kg block rests on a horizontal floor with $\mu_s = 0.5$. It is pulled by a force at 37° above horizontal. Take $g = 10$, $\sin 37^\circ = 3/5$, $\cos 37^\circ = 4/5$. The minimum pull needed to start sliding, in N, is

(A) $\frac{100}{11}$ (B) 10

(C) $\frac{100}{7}$ (D) 20

14. A 2 kg block rests on a horizontal floor with $\mu_s = 0.5$. A force pushes it at 37° below horizontal. Take $g = 10$, $\sin 37^\circ = 3/5$, $\cos 37^\circ = 4/5$. The minimum force needed to start sliding, in N, is

(A) $\frac{100}{11}$ (B) 20

(C) 25 (D) 10

15. A 3 kg block on a horizontal table is joined over a light fixed pulley to a hanging 2 kg block. The table has $\mu_s = \mu_k = 0.2$. Taking $g = 10$, the string tension after release, in N, is

(A) 16 (B) 12

(C) 14.4 (D) 20

16. A 3 kg table block is connected over a light fixed pulley to a hanging 2 kg block. The hanging block accelerates downward at $2\,\mathrm{m\,s^{-2}}$. With $g = 10$, the table kinetic friction coefficient is

(A) 0.2 (B) 0.5

(C) 0.4 (D) $\frac{1}{3}$

17. A light rope is fixed to a ceiling, passes under a light movable pulley carrying 4 kg, then over a fixed pulley and ends in a hanging 1 kg mass. With $g = 10$, acceleration of the movable pulley downward, in $\mathrm{m\,s^{-2}}$, is

(A) 2 (B) 2.5

(C) 5 (D) 4

18. A rope fixed at one ceiling point passes under a light movable pulley carrying 6 kg, then over a fixed pulley to a hanging 1 kg mass. With $g = 10$, the rope tension in N after release is

(A) 20 (B) 18

(C) 15 (D) 12

19. A 2 kg block rests on a 3 kg block; the lower block is on a smooth floor. Friction coefficient between blocks is 0.4. A horizontal force acts on the lower block. Taking $g = 10$, the largest force for which they move together, in N, is

(A) 12 (B) 30

(C) 8 (D) 20

20. A 2 kg block rests on a 3 kg block on a smooth floor. Between them $\mu_s = \mu_k = 0.3$. A 20 N horizontal force is applied to the upper block. With $g = 10$, the lower block acceleration, in $\mathrm{m\,s^{-2}}$, is

(A) 2 (B) 7

(C) 4 (D) $\frac{20}{3}$

21. A 2 kg block is pressed horizontally against a vertically accelerating wall. With $g = 10$, horizontal force P just prevents slipping when wall acceleration is $2\,\mathrm{m\,s^{-2}}$ upward. Force $P - 20$ N just prevents slipping when acceleration is $2\,\mathrm{m\,s^{-2}}$ downward. The common static friction coefficient is

(A) 0.8 (B) 0.48

(C) 0.5 (D) 0.4

22. A 2 kg block moves upward along a fixed vertical wall. A horizontal force of 40 N presses it against the wall; $\mu_k = 0.25$. It has initial upward speed $6\,\mathrm{m\,s^{-1}}$. With $g = 10$, the time until it stops, in s, is

(A) 0.4 (B) 0.5

(C) 0.6 (D) 1.2

23. A smooth incline rises rightward at an unknown angle. A block stays at rest relative to it when the incline accelerates horizontally leftward at $6\,\mathrm{m\,s^{-2}}$. The same incline is placed on a platform accelerating vertically upward at $2\,\mathrm{m\,s^{-2}}$. With $g = 10$, horizontal leftward acceleration needed for the same relative equilibrium, in $\mathrm{m\,s^{-2}}$, is

(A) 7.2 (B) 6

(C) 8 (D) 12

24. A smooth 45° incline rises rightward on a platform accelerating upward at $2\,\mathrm{m\,s^{-2}}$ and leftward at unknown A. A 2 kg block has relative acceleration $2\sqrt{2}\,\mathrm{m\,s^{-2}}$ down the plane. With $g = 10$, normal reaction in N is

(A) $12\sqrt{2}$ (B) $20\sqrt{2}$

(C) $24\sqrt{2}$ (D) $16\sqrt{2}$

25. A 2 kg block slides down a rough 37° incline with $\mu_k = 0.25$. A 10 N horizontal force acts toward the uphill horizontal direction. Take $g = 10$, $\sin 37 = 3/5$, $\cos 37 = 4/5$. Its signed acceleration down the incline, in $\mathrm{m\,s^{-2}}$, is

(A) 1.25 (B) -0.75

(C) 2 (D) 1.75

26. A 2 kg block slides upward on an incline with $\sin\theta = 3/5$, $\cos\theta = 4/5$ and $\mu_k = 0.25$. It is pulled upward along the plane by 25 N. With $g = 10$, its acceleration up the plane, in $\mathrm{m\,s^{-2}}$, is

(A) 6.5 (B) 3.5

(C) 2.5 (D) 4.5

27. Two masses 2 kg and 3 kg hang in sequence from light vertical strings in a lift. The upper string can sustain 80 N, the lower 60 N. With $g = 10$, the largest upward lift acceleration allowed, in $\mathrm{m\,s^{-2}}$, is

(A) 6 (B) 10

(C) 12 (D) 16

28. A 1 kg mass hangs below a 2 kg mass using two light strings in a lift. Their maximum tensions are 45 N for the upper string and 20 N for the lower. With $g = 10$, the maximum upward acceleration in $\mathrm{m\,s^{-2}}$ is

(A) 15 (B) 12.5

(C) 10 (D) 5

29. A 2 kg block on a horizontal floor has $\mu_s = 0.5$, $\mu_k = 0.3$. A horizontal force rises slowly from 0 to 12 N; inertia of the forcing apparatus is negligible. Immediately after the block starts moving, its acceleration in $\mathrm{m\,s^{-2}}$ approaches

(A) 3 (B) 5

(C) 0 (D) 2

30. A 1 kg block has $\mu_s = 0.6$, $\mu_k = 0.4$ on a horizontal floor, $g = 10$. A horizontal force is held at 5 N until $t = 2$ s and then raised instantly to 8 N. Starting from rest, the displacement by $t = 3$ s, in m, is

(A) 4 (B) 2

(C) 1.5 (D) 0.5

31. A block lands with zero horizontal speed on a long belt moving right at $4\,\mathrm{m\,s^{-1}}$. With $\mu_k = 0.2$, $g = 10$, the distance it slips relative to the belt before moving with it, in m, is

(A) 4 (B) 8

(C) 2 (D) 16

32. A 1 kg block initially moves right at $6\,\mathrm{m\,s^{-1}}$ on a long belt moving right at $2\,\mathrm{m\,s^{-1}}$. With $\mu_k = 0.2$, $g = 10$, the block ground displacement before its speed equals the belt speed, in m, is

(A) 8 (B) 4

(C) 6 (D) 12

33. A 2 kg block starts from rest on a horizontal floor with $\mu_s = \mu_k = 0.5$, $g = 10$. Two strings pull upward at 30° to horizontal: 4 N leftward and 12 N rightward. The block acceleration rightward, in $\mathrm{m\,s^{-2}}$, is

(A) $-3 + 2\sqrt{3}$ (B) $-1 + \sqrt{3}$

(C) $2\sqrt{3}$ (D) $-1 + 2\sqrt{3}$

34. A 2 kg block slides right on a horizontal floor with $\mu_k = 0.5$. An unknown pull makes 37° above horizontal and gives acceleration $2\,\mathrm{m\,s^{-2}}$. Take $g = 10$, $\sin 37 = 3/5$, $\cos 37 = 4/5$. The normal reaction in N is

(A) 20 (B) $\frac{136}{11}$

(C) 14 (D) 12

35. A uniform chain of length 2 m and mass 4 kg has 0.5 m hanging over a smooth table edge; the table has $\mu_s = \mu_k = 0.2$. It is released from rest. With $g = 10$, the initial chain tension at the edge, in N, is

(A) 10 (B) 9

(C) 6 (D) 8

36. A uniform vertical rope has mass 3 kg and length 3 m, with an unknown point mass attached at its bottom. Everything has the same vertical acceleration. Tension 2 m below the top is 12 N, and tension 1 m below the top is 20 N. The attached mass in kg is

(A) 0.25 (B) 1

(C) 1.5 (D) 0.5

37. A 2 kg block rests on a horizontal platform carriage with $\mu_s = 0.5$. The carriage accelerates horizontally at $3\,\mathrm{m\,s^{-2}}$ while also accelerating vertically upward at $2\,\mathrm{m\,s^{-2}}$. With $g = 10$, the friction force as a fraction of its limiting value is

(A) 0.75 (B) 0.5

(C) 1 (D) 0.6

38. A horizontal platform has unknown horizontal acceleration A and unknown upward acceleration. A scale gives a passenger apparent weight 1.2 times true weight. A pendulum at rest relative to the platform has string angle with $\tan\theta = 1/2$. With $g = 10$, A in $\mathrm{m\,s^{-2}}$ is

(A) 10 (B) 5

(C) 6 (D) 12

39. A 1 kg block slides on a horizontal floor with $\mu_k = 0.2$. The applied horizontal force is $F = 2 + x$ N to the right, where x is in metres. At $x = 0$ it has speed $2\,\mathrm{m\,s^{-1}}$ rightward. With $g = 10$, its speed at $x = 2$ m, in $\mathrm{m\,s^{-1}}$, is

(A) 4 (B) 2

(C) 3 (D) $2\sqrt{2}$

40. A 2 kg body initially moves right at $4\,\mathrm{m\,s^{-1}}$. For $0 \le t \le 4$ s a force $F = 4t$ N acts leftward. Its total distance over this interval, in m, is

(A) $\frac{16}{3}$ (B) $\frac{64}{3}$

(C) 16 (D) $\frac{32}{3}$

41–50 / Preserved hard practice

41. A 2.00 kg block starts from rest on a level floor with unknown constant kinetic friction. A constant horizontal force acts for 2.00 s, then is removed. The block stops 4.00 s later; total travel is 30.0 m. Friction opposes motion throughout. Work done by the applied force, in joules, is

(A) 100 (B) 300

(C) 150 (D) 200

42. A 1 kg table block with kinetic coefficient 0.20 is connected to a hanging 1 kg mass over a fixed solid-disc pulley of mass 2 kg. The string does not slip. From rest the hanging mass falls 1 m and hits the floor; the string then becomes slack. Take $g = 10$. Find the further distance travelled by the table block, in metres.

(A) $\frac{8}{3}$ (B) $\frac{7}{3}$

(C) $\frac{1}{3}$ (D) $\frac{4}{3}$

43. A 2 kg block on a rough horizontal table is connected over a light smooth pulley to a hanging 1 kg mass. They start from rest; the hanging mass falls 1 m and stops on the floor. The table block then moves a further 1 m before stopping. Take $g = 10$ and a slack string after floor contact. Find the kinetic friction coefficient.

(A) 0.8 (B) 0.6

(C) 0.1 (D) 0.2

44. A 1 kg block on a horizontal spring of stiffness 100 N/m is released from rest at $x = +0.30$ m, measured from the unstretched position. It slides with kinetic coefficient 0.20; take $g = 10$. Find its coordinate, in metres, at its first subsequent instant of rest.

(A) -0.52 (B) 0.74

(C) -0.26 (D) -1.26

45. A 1 kg point block rests on a 3 kg plank on a smooth floor, 1 m from its left edge. A 12 N horizontal force pulls the plank to the right. Both static and kinetic coefficients between them are 0.20, and $g = 10$. Find the work done by friction on the block before it leaves the plank, in joules.

(A) 3 (B) 4

(C) 6 (D) 2

46. A block is projected up an incline with $\sin\theta = 3/5$, $\cos\theta = 4/5$. It returns to the launch point at speed $u/\sqrt{2}$ when launched at speed u. Kinetic friction is constant in both directions, and it does slide back. Find its coefficient.

(A) 0.25 (B) 0.125

(C) 0.75 (D) 0.625

47. Two blocks of 1 and 3 kg are connected on a level floor, with kinetic coefficients 0.40 and 0.20. A 16 N force pulls the 3 kg block for 2 s from rest, when the connection breaks. Both then eventually stop. Take $g = 10$. Find total frictional heat generated under the 1 kg block during the entire motion, in joules.

(A) 33 (B) 17.5

(C) 15.5 (D) 16.5

48. A 2 kg block initially rests on a 30° incline. Both friction coefficients are $1/\sqrt{3}$. A force $F = 6t$ N acts up the incline from $t = 0$. With $g = 10$, find the block momentum up the incline at $t = 4$ s, in kg m/s.

(A) $\frac{1}{3}$ (B) $\frac{7}{3}$

(C) $\frac{4}{3}$ (D) $\frac{8}{3}$

49. Two spheres of equal radius have material densities $2\rho_0$ and $3\rho_0$. In the same viscous liquid, their downward terminal speeds are in ratio 1 : 3. Neglect fluid inertia. Immediately after the first sphere is released from rest, find its downward acceleration divided by g.

(A) 0.75 (B) 0.125

(C) 0.25 (D) 0.625

50. A 1 kg block on a rough horizontal floor is attached to a spring of stiffness 20 N/m. Both static and kinetic friction coefficients are 0.20; take $g = 10\,\text{m/s}^2$. The spring is initially compressed 0.50 m and the block is released from rest. The floor extends indefinitely. Total distance travelled before permanent rest, in metres, is

(A) 1 (B) 1.6

(C) 1.2 (D) 0.8

Answer key

Laws of Motion

P-LAMO / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**C**	21	**D**	31	**A**	41	**C**
02	**D**	12	**A**	22	**A**	32	**A**	42	**D**
03	**A**	13	**A**	23	**A**	33	**A**	43	**D**
04	**B**	14	**B**	24	**B**	34	**B**	44	**C**
05	**B**	15	**C**	25	**B**	35	**B**	45	**A**
06	**B**	16	**D**	26	**D**	36	**D**	46	**A**
07	**B**	17	**B**	27	**A**	37	**B**	47	**D**
08	**C**	18	**B**	28	**D**	38	**C**	48	**C**
09	**C**	19	**D**	29	**D**	39	**D**	49	**C**
10	**A**	20	**A**	30	**B**	40	**C**	50	**C**

Chapter 4

Circular Motion

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A 1000 kg car rounds a level circle of radius 50 m. With $\mu_s = 0.2, g = 10$, limiting speed in m/s is

(A) 10 (B) 20

(C) 11 (D) 9

02. A conical pendulum has string tension 20 N and bob weight 10 N. The string angle to vertical in degrees is

(A) 61 (B) 59

(C) 120 (D) 60

03. A conical pendulum has radius 1 m and $\tan\theta = 0.4$. With $g = 10$, bob speed in m/s is

(A) 3 (B) 4

(C) 2 (D) 1

04. A 2 kg car model crosses the top of a convex circular bridge of radius 5 m at $5\,\mathrm{m\,s^{-1}}$. With $g = 10$, normal reaction in newtons is

(A) 9 (B) 11

(C) 20 (D) 10

05. A particle circles with radial acceleration $12\,\mathrm{m\,s^{-2}}$ and tangential acceleration $5\,\mathrm{m\,s^{-2}}$. Magnitude of acceleration in m/s squared is

(A) 26 (B) 12

(C) 13 (D) 14

06. A bob circles in a vertical plane. At the side point its weight is 10 N and tension 24 N. Magnitude of net force in newtons is

(A) 27 (B) 25

(C) 52 (D) 26

07. A conical pendulum string makes 60° to vertical and has length 0.8 m. With $g = 10$, angular speed squared in rad squared/s squared is

(A) 26 (B) 24

(C) 50 (D) 25

08. A turntable has $\mu_s = 0.5$ and angular speed $5\,\mathrm{rad\,s^{-1}}$. With $g = 10$, largest no-slip radius in metres is

(A) -0.8 (B) 1.2

(C) 0.2 (D) 0.4

09. A car on a frictionless bank of 45° has design speed $20\,\mathrm{m\,s^{-1}}$. With $g = 10$, curve radius in metres is

(A) 80 (B) 40

(C) 39 (D) 41

10. For uniform circular motion, velocity vectors separated by one quarter-period have equal magnitude v. Magnitude of their difference divided by v is

(A) 3 (B) $\sqrt{2}$

(C) 2 (D) $\sqrt{3}$

11–40 / New middle-to-hard practice

11. A road is banked at an angle with $\tan\theta = 1/2$ and has tyre-road coefficient $\mu = 1/3$. If r is the curve radius, the maximum allowed value of $v^2/(rg)$ without sideways slip is

(A) 1 (B) $\frac{5}{6}$

(C) $\frac{7}{6}$ (D) 0.5

12. A road is banked with $\tan\theta = 1/2$ and has coefficient $\mu = 1/3$. The minimum allowed value of $v^2/(rg)$ without sliding down the bank is

(A) 0.2 (B) $\frac{1}{7}$

(C) $\frac{1}{3}$ (D) 0

13. A conical pendulum has an unknown string length. Its orbit radius is $\sqrt{3}$ m and its period is $2\pi/\sqrt{5}$ s. With $g = 10$, the string length in m is

(A) $\sqrt{13}$ (B) $\sqrt{7}$

(C) 2 (D) $\sqrt{5}$

14. A 2 kg conical-pendulum bob has string tension 30 N and string length 0.9 m. Taking $g = 10$, its speed squared, in $\mathrm{m^2\,s^{-2}}$, is

(A) 7.5 (B) 5

(C) 10 (D) 13.5

15. A bob on a light string completes a smooth vertical circle of radius r. Its bottom tension is six times its top tension. The ratio $v_{\text{bottom}}^2/(gr)$ is

(A) 5 (B) 6.2

(C) 7 (D) 6

16. A bob tied to a string has bottom speed $\sqrt{6gr}$ in a vertical circle of radius r. Ignoring air resistance, the ratio of bottom to top string tension is

(A) 6 (B) 7

(C) 5 (D) 8

17. A bob has speed $\sqrt{6gr}$ at the bottom of its vertical circular path of radius r. At 60° from the downward vertical, the tension divided by mg is

(A) 6 (B) 5.5

(C) 4.5 (D) 5

18. A bob has bottom speed $\sqrt{6gr}$ in a vertical circle of radius r. At the side point, the magnitude of its resultant force divided by its weight is

(A) 4 (B) $\sqrt{15}$

(C) $\sqrt{17}$ (D) 5

19. A small bead slides from rest at height h above the bottom of a smooth vertical loop of radius r. A rough entry section before the loop removes energy $mgr/2$. The minimum h/r for completing the loop without losing contact is

(A) 4 (B) 3.5

(C) 3 (D) 2.5

20. A bead enters the bottom of a smooth vertical loop of radius r with kinetic energy $3mgr$. The top normal reaction divided by mg is

(A) 3 (B) 0

(C) 2 (D) 1

21. A small block rests on a horizontal rotating disc at radius 0.5 m. The disc has angular speed $4\,\mathrm{rad\,s^{-1}}$ and angular acceleration $6\,\mathrm{rad\,s^{-2}}$. With $g = 10$, the minimum friction coefficient needed at that instant is

(A) $\frac{\sqrt{73}}{10}$ (B) 1.1

(C) 0.8 (D) 0.3

22. A block on a horizontal disc at radius r has tangential acceleration $3g/10$ while the maximum available friction acceleration is $g/2$. Its largest allowed radial acceleration divided by g is

(A) 0.4 (B) 0.5

(C) 0.3 (D) 0.2

23. A 1 kg body moves on a horizontal circle of radius 2 m, pulled radially by a light string through a smooth hole to a hanging 2 kg mass. The hanging mass remains at rest. With $g = 10$, the period of the circling body, in s, is

(A) $\frac{\sqrt{10}\pi}{10}$ (B) $\sqrt{10}\pi$

(C) $\frac{2\sqrt{5}\pi}{5}$ (D) $\frac{\sqrt{10}\pi}{5}$

24. A mass m circles on a smooth table with a central string passing through a hole to a stationary hanging mass M. A central adjustment doubles the hanging load, with no external torque on the circling mass, and a new steady circle is established. The final period divided by the initial period is

(A) $\frac{\sqrt{2}}{2}$ (B) $\frac{\sqrt[3]{2}}{2}$

(C) $\frac{2^{\frac{2}{3}}}{2}$ (D) 0.5

25. A 1 kg bead rotates with a horizontal disc and is attached to its centre by a spring of natural length 0.5 m and constant 100 N/m. At angular speed $5\,\mathrm{rad\,s^{-1}}$, its equilibrium radius in m is

(A) $\frac{2}{3}$ (B) 0.625

(C) 0.5 (D) 1

26. A 1 kg bead on a smooth horizontal rotating disc is joined to the centre by a spring of unknown stiffness and natural length. Its steady radius is 0.50 m at angular speed $2\,\mathrm{rad\,s^{-1}}$, and 1.00 m at $4\,\mathrm{rad\,s^{-1}}$. The spring natural length in m is

(A) 0.5 (B) 0.25

(C) $\frac{3}{7}$ (D) $\frac{2}{7}$

27. A car negotiates a level circular curve of radius 50 m at $10\,\mathrm{m\,s^{-1}}$. Its tyres have $\mu_s = 0.3$ and $g = 10$. The maximum simultaneous tangential braking acceleration, in $\mathrm{m\,s^{-2}}$, is

(A) $\sqrt{5}$ (B) 1

(C) 3 (D) 2

28. A car on a level circle has maximum friction acceleration $5\,\mathrm{m\,s^{-2}}$. It is braking tangentially at $3\,\mathrm{m\,s^{-2}}$ on a radius 25 m. The greatest speed allowed at that instant, in $\mathrm{m\,s^{-1}}$, is

(A) 10 (B) 5

(C) $5\sqrt{3}$ (D) $5\sqrt{5}$

29. A car moves at constant speed on a convex circular bridge. At the top its normal reaction is half its weight. At a point where the radius makes 60° with the upward vertical, the normal reaction divided by weight would be

(A) 1.5 (B) 0

(C) 0.5 (D) 1

30. A bead starts tangentially at the top of a smooth fixed sphere of radius r with unknown speed u. It loses contact where the radius makes an angle θ to the upward vertical with $\cos\theta = 3/4$. The value of $u^2/(gr)$ is

(A) 0.75 (B) 0.25

(C) 0 (D) 0.5

31. A particle executes uniform circular motion with speed v and radius r. The magnitude of its average acceleration over one-sixth of a revolution, divided by v^2/r, is

(A) $\frac{6}{\pi}$ (B) $\frac{3}{\pi}$

(C) 0.5 (D) 1

32. A particle moves uniformly in a circle. The magnitude of its average velocity over one-third of a revolution, divided by its speed, is

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{3}}{\pi}$

(C) $\frac{3\sqrt{3}}{2\pi}$ (D) $\frac{3}{2\pi}$

33. A particle moves around a radius 2 m circle with speed satisfying $v^2 = 4 + 6s$, where s is arc length in m. At $s = 2$ m, the angle of acceleration relative to the inward radial direction has tangent

(A) 0.25 (B) $\frac{8}{3}$

(C) 0.75 (D) 0.375

34. A particle moves on a radius 4 m circle with speed $v = 2t$ in SI units. At the first instant when radial and tangential accelerations are equal, its speed in $\mathrm{m\,s^{-1}}$ is

(A) $2\sqrt{2}$ (B) $\sqrt{2}$

(C) 2 (D) 4

35. A pilot crosses the bottom of a vertical circular path of radius 100 m at $40\,\mathrm{m\,s^{-1}}$. The maximum allowed apparent weight is 3 times true weight. With $g = 10$, the speed can be increased from its current value by at most, in $\mathrm{m\,s^{-1}}$,

(A) 20 (B) 10

(C) $-40 + 20\sqrt{5}$ (D) $10\sqrt{5}$

36. A rider is strapped into a seat on the inside of a vertical circular loop and moves at constant speed. The inward support force at the bottom is $3mg$. The inward support force at the top, divided by mg, is

(A) 0 (B) 1

(C) 2 (D) 3

37. A mass m circles below a smooth tube, supported by a string whose other end carries a stationary mass $2m$. The exposed slant string length is free to adjust. The circular angular speed is $\sqrt{5}\,\mathrm{rad\,s^{-1}}$. With $g = 10$, the orbit radius in m is

(A) 4 (B) 2

(C) $\sqrt{3}$ (D) $2\sqrt{3}$

38. A conical pendulum string is lengthened from l to $2l$. Its angular speed is kept unchanged and its orbit radius becomes $\sqrt{7}$ times the original. The original string angle to the vertical, in degrees, is

(A) 60 (B) 30

(C) 45 (D) 75

39. A bead of mass m moves on a smooth horizontal circular track of radius r. A spring to the centre has natural length $r/2$ and constant k, with $kr = 2F$. A constant tangential force F acts from rest. After a half-circle, the inward track reaction divided by F is

(A) 2π (B) $1 + 2\pi$

(C) $-1 + \pi$ (D) $-1 + 2\pi$

40. A particle on a smooth horizontal circular track of radius r is acted on by a constant tangential force F. Starting from rest, when it has travelled a quarter circle, the normal reaction divided by F is

(A) π (B) 2π

(C) $\frac{\pi}{2}$ (D) 1

41–50 / Preserved hard practice

41. A bob on a string of length R is launched from the lowest point at the minimum speed needed for a complete vertical circle. While rising, the string is cut at 60° from the downward vertical. Find its greatest height above the initial lowest point, divided by R.

(A) 3 (B) 1

(C) 2 (D) 4

42. A 1 kg bob on a 1 m string is launched from the lowest point with speed $\sqrt{40}\,\mathrm{m\,s^{-1}}$. With $g = 10$, the cosine of the angle measured from downward vertical where string first goes slack is

(A) $\frac{1}{3}$ (B) $\frac{-5}{3}$

(C) $\frac{-4}{3}$ (D) $\frac{-2}{3}$

43. A smooth track ends at a point 60° around a vertical circular arc of radius R, measured from the downward vertical. A block released from rest at height h above the arc bottom exits tangentially there while rising. It lands at bottom level, a horizontal distance $\sqrt{3}R$ beyond the exit. Find h/R.

(A) $\frac{19}{14}$ (B) $\frac{33}{14}$

(C) $\frac{5}{14}$ (D) $\frac{19}{7}$

44. A bead of mass m is released from rest an infinitesimal distance from the top of a smooth fixed sphere of radius R, and slides on the outside. In the limit of vanishing initial displacement, its height above the sphere centre at loss of contact, divided by R, is

(A) $\frac{2}{3}$ (B) $\frac{-1}{3}$

(C) $\frac{5}{3}$ (D) $\frac{4}{3}$

45. A pendulum of length l is released from rest at 60° to the downward vertical. At its lowest point, the string catches a smooth peg $3l/4$ below the original pivot. Let ϕ be the angle from the downward vertical about the peg. At the instant the string first becomes slack, $\cos\phi$ is

(A) -0.5 (B) -0.75

(C) $-\frac{2}{3}$ (D) $-\frac{1}{3}$

46. A mass on a horizontal frictionless table circles at radius r under a string through a central hole. It is slowly pulled to radius $r/2$ with no external torque about the hole. The ratio of final to initial string tension is

(A) 9 (B) 8

(C) 7 (D) 16

47. A 1 kg bead circles inside a vertical ring of radius 1 m. Speed at the lowest point is $\sqrt{60}\,\mathrm{m\,s^{-1}}$ and $g = 10$. Ratio of normal-force magnitudes at bottom and top is

(A) 6 (B) 8

(C) 14 (D) 7

48. A bob on a 1 m string has speed $\sqrt{70}$ m/s at its lowest point, relative to a support accelerating uniformly upward at a. Take $g = 10$. Find the largest a in $\mathrm{m/s^2}$ for which the string remains taut throughout the vertical circle.

(A) 8 (B) 3

(C) 4 (D) 5

49. A uniform rod of mass M, length L pivots freely at one end and is released from horizontal. At the lowest vertical position, find the hinge force magnitude divided by Mg.

(A) 3.5 (B) 5

(C) 1.5 (D) 2.5

50. A car negotiates a banked curve with $\tan\theta = 1/2$ and $\mu_s = 1/4$. Taking limiting upward friction for the lowest possible speed, the ratio $v_{\min}^2/(rg)$ is

(A) $\frac{2}{9}$ (B) $\frac{-7}{9}$

(C) $\frac{11}{9}$ (D) $\frac{4}{9}$

Answer key

Circular Motion

P-CIMO / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**A**	21	**A**	31	**B**	41	**C**
02	**D**	12	**B**	22	**A**	32	**C**	42	**D**
03	**C**	13	**B**	23	**D**	33	**D**	43	**A**
04	**D**	14	**A**	24	**B**	34	**A**	44	**A**
05	**C**	15	**B**	25	**A**	35	**C**	45	**C**
06	**D**	16	**B**	26	**C**	36	**B**	46	**B**
07	**D**	17	**B**	27	**A**	37	**D**	47	**D**
08	**C**	18	**C**	28	**A**	38	**C**	48	**C**
09	**B**	19	**C**	29	**B**	39	**D**	49	**D**
10	**B**	20	**D**	30	**B**	40	**A**	50	**A**

Chapter 5

Work Energy Power

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A force $F = 6 - 2x$ newtons acts along x from 0 to 3 m. Work in joules is

(A) 9 (B) 10

(C) 8 (D) 18

02. Potential is $U = x^3 - 3x$ joules. Force at $x = 2$ m in newtons is

(A) -18 (B) -10

(C) -8 (D) -9

03. A 2 kg block is pulled 5 m by 10 N on a rough horizontal surface with friction 4 N. Starting from rest, final kinetic energy in joules is

(A) 31 (B) 30

(C) 29 (D) 60

04. A block falls from height 8 m and compresses an ideal spring. Neglect other effects. If spring energy at maximum compression is 100 J and total gravitational drop is 10 m, with $g = 10$, block mass in kg is

(A) 1 (B) 0.5

(C) 0 (D) 2

05. A 2 kg body moves under constant power 20 W from rest. Its speed after 5 s in m/s is

(A) 20 (B) 9

(C) 10 (D) 11

06. Two springs of equal constant k are connected in series and stretched a total x. Ratio of total stored energy to kx^2 is

(A) 0.5 (B) -0.75

(C) 1.25 (D) 0.25

07. A 4 kg body at rest explodes into two 2 kg pieces moving oppositely, each at $3\,\mathrm{m\,s^{-1}}$. Released kinetic energy in joules is

(A) 36 (B) 18

(C) 17 (D) 19

08. A spring stores 18 J at extension 0.3 m. Its force magnitude at extension 0.1 m in newtons is

(A) 39 (B) 40

(C) 41 (D) 80

09. A 2 kg block rises vertically at constant speed $3\,\mathrm{m\,s^{-1}}$. With $g = 10$, work done by lifting force in 4 s in joules is

(A) 240 (B) 480

(C) 241 (D) 239

10. A 1 kg body has speed $2\,\mathrm{m\,s^{-1}}$ at $x = 0$ in potential $U = 2x^2$ joules. Its turning point at positive x in metres is

(A) 0.5 (B) 0

(C) 2 (D) 1

11–40 / New middle-to-hard practice

11. A 2 kg particle at $x = 0$ moves right at $2\,\mathrm{m\,s^{-1}}$. The only force is $F = 6 - 2x$ N. Its speed at $x = 4$ m, in $\mathrm{m\,s^{-1}}$, is

(A) $2\sqrt{5}$ (B) 4

(C) $2\sqrt{3}$ (D) 2

12. A 1 kg body starts at $x = 0$ with speed $2\,\mathrm{m\,s^{-1}}$ to the right under $F = 2 - 2x$ N. Its first positive turning position in m is

(A) 3 (B) $1 + \sqrt{2}$

(C) $1 + \sqrt{3}$ (D) 2

13. A force $F = 6t$ N acts on a 3 kg particle moving initially at $2\,\mathrm{m\,s^{-1}}$ along the force. Work done in the first 2 s, in J, is

(A) 36 (B) 48

(C) 54 (D) 24

14. A 2 kg body starts from rest under force $F = 4t$ N. The instantaneous power at $t = 2$ s, in W, is

(A) 16 (B) 32

(C) 24 (D) 64

15. A 2 kg body moves along a smooth straight line under constant positive power 12 W. It initially moves at $2\,\mathrm{m\,s^{-1}}$. The distance covered before its speed reaches $4\,\mathrm{m\,s^{-1}}$, in m, is

(A) 2 (B) $\frac{28}{9}$

(C) $\frac{14}{3}$ (D) $\frac{8}{3}$

16. An engine delivers 10 kW to a car opposed by resistance $R = 100 + v^2$ N, where v is in $\mathrm{m\,s^{-1}}$. The terminal speed in $\mathrm{m\,s^{-1}}$ is

(A) 30 (B) 15

(C) 25 (D) 20

17. A spring of constant k is cut into pieces of natural lengths in ratio 1 : 2. The pieces are joined in parallel. The energy required to extend the pair by x, divided by the energy to extend the original spring by x, is

(A) 1.5 (B) 3

(C) 4.5 (D) 4

18. Two springs of constants k and $3k$ are connected in series. When the total stored energy is E, the energy in the softer spring divided by E is

(A) 0.25 (B) 0.5

(C) $\frac{1}{3}$ (D) 0.75

19. A 1 kg block is released from rest 0.2 m above an uncompressed vertical spring of constant 200 N/m. With $g = 10$, its maximum spring compression, in m, is

(A) 0.1 (B) 0.2

(C) 0.3 (D) 0.25

20. A mass m rests at equilibrium on a vertical spring, compressing it by x_0. It is lifted until the spring is uncompressed and released from rest without losing contact. Its maximum compression divided by x_0 is

(A) $\sqrt{2}$ (B) 2

(C) 3 (D) 1

21. A 1 kg block moving at $4\,\mathrm{m\,s^{-1}}$ hits an uncompressed horizontal spring $k = 100$ N/m. The surface has $\mu_k = 0.2$ and $g = 10$. The compression at first stop, in m, is

(A) 0.2 (B) $-\frac{1}{50} + \frac{\sqrt{401}}{50}$

(C) 0.3 (D) 0.5

22. A 1 kg block slides at $3\,\mathrm{m\,s^{-1}}$ into an uncompressed horizontal spring of unknown constant. On the rough floor $\mu_s = \mu_k = 0.5$, $g = 10$. Its first maximum compression is 0.30 m. It then rebounds. Its speed at its first return to the spring natural length, in $\mathrm{m\,s^{-1}}$, is

(A) $\sqrt{3}$ (B) $\sqrt{6}$

(C) 3 (D) $\frac{\sqrt{6}}{2}$

23. A particle follows $O(0,0) \to A(2,0) \to B(2,3)$ in a force field $\mathbf{F} = (2y\hat{i} + x\hat{j})$ N, coordinates in m. The difference between work along this path and along the straight line OB, in J, is

(A) -6 (B) 3

(C) 6 (D) -3

24. A particle moves around the square $(0,0) \to (1,0) \to (1,1) \to (0,1) \to (0,0)$ in force $\mathbf{F} = (y\hat{i} + 2x\hat{j})$ N. The net work in J is

(A) 2 (B) 1

(C) -1 (D) 0

25. A 1 kg pendulum of length 1 m is released from rest at 60° to the downward vertical. With $g = 10$, the power of gravity at 30° while descending, in W, is

(A) $5\sqrt{-10 + 10\sqrt{3}}$ (B) $10\sqrt{-10 + 10\sqrt{3}}$

(C) 0 (D) $5\sqrt{5}$

26. A pendulum bob descends from rest at 90° to the downward vertical. At 60°, its gravitational power is P. At 30°, it is Q. The ratio P/Q is

(A) 3 (B) $\sqrt{3}$

(C) 1 (D) $\sqrt[4]{3}$

27. A uniform chain of length L and mass M has half its length hanging over a smooth table edge. It is pulled slowly until only $L/4$ hangs. Work done divided by MgL is

(A) 0.125 (B) 0.1875

(C) 0.09375 (D) 0.03125

28. A uniform rope of length L and mass M hangs vertically. It is folded by lifting its lower end slowly until that end reaches the upper support. Work done divided by MgL is

(A) 0.25 (B) 0.375

(C) 0.125 (D) 0.5

29. Water of density $1000\,\mathrm{kg\,m^{-3}}$ is pumped from a tank at ground level to a spout 10 m higher at $0.01\,\mathrm{m^3\,s^{-1}}$, leaving at $4\,\mathrm{m\,s^{-1}}$. With $g = 10$ and pump efficiency 0.5, required input power in kW is

(A) 2.16 (B) 1.08

(C) 2 (D) 1

30. A pump lifts water at mass flow 5 kg/s by 8 m and increases its speed from 2 to $6\,\mathrm{m\,s^{-1}}$. With $g = 10$, the useful pump power in W is

(A) 400 (B) 490

(C) 480 (D) 450

31. A gun continuously fires bullets at speed $600\,\mathrm{m\,s^{-1}}$ and experiences mean recoil force 120 N. Bullet kinetic energy is 50% of the supplied energy. The required input power in kW is

(A) 72 (B) 144

(C) 36 (D) 24

32. A machine launches 2 kg parcels vertically at $10\,\mathrm{m\,s^{-1}}$, one every 0.5 s, from a platform 5 m above the incoming parcel level. With $g = 10$ and 80% efficiency, input power in W is

(A) 500 (B) 400

(C) 250 (D) 200

33. A particle has potential $U = x^4 - 4x^2$ J. The separation in m between its two stable equilibrium positions is

(A) 2 (B) 4

(C) $\sqrt{2}$ (D) $2\sqrt{2}$

34. A particle in $U = 3x^2 - 2x^3$ J is initially at $x = 0$ with kinetic energy K. The barrier height that K must exceed to cross toward positive x, in J, is

(A) 3 (B) 0

(C) 1 (D) 2

35. A particle starts at $x = 0$ with kinetic energy 3 J in potential $U = x^4 - 4x^2$ J. Its positive turning point in m is

(A) $\sqrt{2 - \sqrt{3}}$ (B) $\sqrt{\sqrt{3} + 2}$

(C) $\sqrt{2 + \sqrt{7}}$ (D) $\sqrt{3}$

36. A particle of mass 1 kg starts at $x = 1$ m moving right at $2\,\mathrm{m\,s^{-1}}$ in $U = (x - 2)^2$ J. Its right turning point in m is

(A) $\sqrt{2} + 2$ (B) 4

(C) 3 (D) $\sqrt{3} + 2$

37. A 2 kg block moves 5 m upward on an incline with $\sin\theta = 3/5$, $\cos\theta = 4/5$, $\mu_k = 0.25$. A horizontal 20 N force pushes it uphill. With $g = 10$, the change in kinetic energy, in J, is

(A) 30 (B) -30

(C) 15 (D) -15

38. A 1 kg block is pulled 2 m horizontally by a 10 N force at 30° upward. With $g = 10$ and $\mu_k = 0.2$, net work in J is

(A) $8\sqrt{3}$ (B) $10\sqrt{3}$

(C) $-2 + 10\sqrt{3}$ (D) $-4 + 10\sqrt{3}$

39. A motor lifts a 100 kg load steadily at $1\,\mathrm{m\,s^{-1}}$ using 2.0 kW, and at $2\,\mathrm{m\,s^{-1}}$ using 3.5 kW. Its model is $P_{\text{in}} = P_{\text{mechanical}}/\eta + P_0$ with constants η, P_0. With $g = 10$, input power to lift 200 kg at $1.5\,\mathrm{m\,s^{-1}}$, in kW, is

(A) 4.5 (B) 6

(C) 3.5 (D) 5

40. A lift of mass 600 kg raises a 200 kg passenger load through 10 m, starting and ending at rest. A 500 kg counterweight descends by 10 m. Friction dissipates 5000 J. With $g = 10$ and 70% motor efficiency, required electrical energy in kJ is

(A) 50 (B) 35

(C) 70 (D) 30

41–50 / Preserved hard practice

41. A 1 kg block first contacts a right-hand spring at speed $\sqrt{30}$ m/s. Two compression-only springs, each of stiffness 100 N/m, have uncompressed inner ends 1 m apart on a level floor. Kinetic friction is 5 N throughout. The block compresses the right spring, returns, and reaches the left spring. Find its first maximum left-spring compression, in metres.

(A) $-\frac{1}{20} + \frac{\sqrt{41}}{20}$ (B) $\frac{21}{20} - \frac{\sqrt{41}}{20}$

(C) $-\frac{1}{40} + \frac{\sqrt{41}}{40}$ (D) $\frac{\sqrt{41}}{40} + \frac{19}{40}$

42. A 1 kg block falls from rest 0.75 m above a vertical spring unloaded tip. During compression the spring upward force is $100x$ N; during unloading it is $50x$ N, where x is compression in metres. The block is not attached. Take $g = 10$. Find its greatest rebound height above the unloaded tip, in metres.

(A) 0.0625 (B) 0.125

(C) 0.5625 (D) 0.875

43. A 0.01 kg bullet embeds in a 0.99 kg bob on a 1 m string. While rising the string is cut at 60° from downward vertical. The bob then reaches a maximum height 0.80 m above its original lowest point. Take $g = 10$. Find bullet speed in m/s.

(A) $1 + 300\sqrt{2}$ (B) $300\sqrt{2}$

(C) $600\sqrt{2}$ (D) $-1 + 300\sqrt{2}$

44. A block of mass m slides from rest down a smooth track from height $4R$. At the bottom it sticks to a stationary block of mass qm. The pair then enters a smooth inner vertical loop of radius R. Find the largest q for which they maintain contact throughout the loop.

(A) $\frac{\sqrt{10}}{5}$ (B) $-\frac{1}{2} + \frac{\sqrt{10}}{5}$

(C) $-1 + \frac{2\sqrt{10}}{5}$ (D) $2 - \frac{2\sqrt{10}}{5}$

45. A uniform disc of mass M rolls without slipping from rest down a vertical height h. At the bottom it picks up and retains a stationary point mass M at its centre in a brief inelastic collision. The floor supplies the impulse needed for rolling without slipping immediately after the collision; neglect all other impulsive forces. Final kinetic energy divided by Mgh is

(A) 0.6 (B) 0.75

(C) 0.4 (D) 0.36

46. A 1 kg block at 6 m/s strikes a stationary block of unknown mass attached to a 12 N/m spring, initially unstretched. Restitution is 1/2, and 9 J of kinetic energy is lost during the short collision. Ignore spring impulse during impact. Find first maximum spring compression in metres.

(A) $1+\frac{\sqrt{6}}{2}$ (B) $\sqrt{6}$

(C) $\frac{\sqrt{6}}{2}$ (D) $-1+\frac{\sqrt{6}}{2}$

47. A 1 kg block at rest on a smooth table is driven by constant power 8 W. Its distance travelled in first 2 s, in metres, is

(A) $\frac{16\sqrt{2}}{3}$ (B) $8\sqrt{2}$

(C) 4 (D) $\frac{8\sqrt{2}}{3}$

48. Two equal masses 1 kg hang on a light string over a pulley with fixed axle and moment of inertia $I = 0.02\,\mathrm{kg\,m^2}$, radius 0.1 m. One mass is replaced by 3 kg. Starting from rest, after the heavier mass descends 1 m, its speed squared in SI units is

(A) $\frac{17}{3}$ (B) $\frac{23}{3}$

(C) $\frac{40}{3}$ (D) $\frac{20}{3}$

49. A 1 kg particle is at $x = 1$ m with velocity +2 m/s, under force $F_x = 6x - 3x^2$ N. Find its first subsequent turning-point coordinate, in metres.

(A) 6 (B) 2

(C) 3 (D) 4

50. A 2 kg block on a level floor is attached to a 100 N/m spring. Both friction coefficients are 0.25, and $g = 10$. Released at coordinate $x = -0.32$ m from rest, where $x = 0$ is the natural spring length, find its coordinate of permanent rest, in metres.

(A) 0.51 (B) 0.02

(C) 0.01 (D) 0.98

Answer key

Work Energy Power

P-WEP / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**C**	21	**B**	31	**A**	41	**A**
02	**D**	12	**C**	22	**A**	32	**A**	42	**B**
03	**B**	13	**B**	23	**D**	33	**D**	43	**B**
04	**A**	14	**B**	24	**B**	34	**C**	44	**C**
05	**C**	15	**B**	25	**A**	35	**C**	45	**A**
06	**D**	16	**D**	26	**D**	36	**D**	46	**C**
07	**B**	17	**C**	27	**C**	37	**D**	47	**A**
08	**B**	18	**D**	28	**A**	38	**C**	48	**D**
09	**A**	19	**B**	29	**A**	39	**D**	49	**C**
10	**D**	20	**B**	30	**C**	40	**A**	50	**B**

Chapter 6

Centre of Mass & Collisions

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A $1\,\text{kg}$ ball moving at $6\,\text{m}\,\text{s}^{-1}$ collides head-on with a stationary $2\,\text{kg}$ ball, with $e = 1/2$. Final velocity of the first ball in m/s is

(A) 0 (B) -1

(C) 2 (D) 1

02. A $1\,\text{kg}$ ball at $6\,\text{m}\,\text{s}^{-1}$ collides with a stationary $2\,\text{kg}$ ball, $e = 1/2$. Final speed of second ball in m/s is

(A) 2 (B) 6

(C) 4 (D) 3

03. Two $1\,\text{kg}$ masses at velocities $+3, -1\,\text{m}\,\text{s}^{-1}$ stick. Kinetic energy lost in joules is

(A) 8 (B) 4

(C) 5 (D) 3

04. A $2\,\text{kg}$ body at $3\,\text{m}\,\text{s}^{-1}$ sticks to a $1\,\text{kg}$ body at $-3\,\text{m}\,\text{s}^{-1}$. Common velocity in m/s is

(A) 0 (B) 1

(C) 0.5 (D) 2

05. A particle of mass m strikes a stationary particle of mass $3m$ elastically head-on. Ratio of final projectile velocity to its initial velocity is

(A) -0.5 (B) 0.5

(C) -1 (D) -1.5

06. A particle of mass m strikes stationary mass $3m$ elastically head-on. Target final speed divided by projectile initial speed is

(A) 1.5 (B) 0.5

(C) -0.5 (D) 1

07. A 1 kg particle moves east at $3\,\mathrm{m\,s^{-1}}$ and a 1 kg particle north at $4\,\mathrm{m\,s^{-1}}$. They stick. Speed in m/s is

(A) 1.5 (B) 5

(C) 2.5 (D) 3.5

08. A 50 kg person jumps at $3\,\mathrm{m\,s^{-1}}$ relative to a 100 kg boat initially at rest. Boat recoil speed in m/s is

(A) 0.5 (B) 0

(C) 1 (D) 2

09. Masses $1, 2$ kg have velocities $3, 0\,\mathrm{m\,s^{-1}}$. Total kinetic energy in centre-of-mass frame in joules is

(A) 2 (B) 3

(C) 4 (D) 6

10. A bullet of mass 0.02 kg moving at $100\,\mathrm{m\,s^{-1}}$ embeds in 0.98 kg wood at rest. Energy lost in joules is

(A) 99 (B) 196

(C) 98 (D) 97

11–40 / New middle-to-hard practice

11. A particle of mass m moving at speed u hits a stationary particle of mass M elastically along one line. The projectile rebounds at $u/3$. The fraction of initial kinetic energy transferred to M is

(A) $\frac{8}{9}$ (B) $\frac{2}{3}$

(C) $\frac{4}{9}$ (D) $\frac{1}{3}$

12. A ball of mass m hits a stationary ball of mass $3m$ head-on. After impact the first ball is at rest. The coefficient of restitution is

(A) $\frac{2}{3}$ (B) 1

(C) 0.5 (D) $\frac{1}{3}$

13. Three collinear balls of masses m, $2m$, $4m$ are initially separated. The first moves right at u; the other two rest. Collisions are elastic and instantaneous. Immediately after the $2m$ ball first hits the $4m$ ball, the $4m$ ball speed divided by u is

(A) $\frac{4}{9}$ (B) $\frac{2}{3}$

(C) $\frac{8}{9}$ (D) $\frac{1}{3}$

14. A 1 kg ball moving right at $6\,\mathrm{m\,s^{-1}}$ hits a stationary 2 kg ball with $e = 1/2$. That ball then hits a stationary 3 kg ball elastically. Immediately after the second collision, the 3 kg ball speed in $\mathrm{m\,s^{-1}}$ is

(A) 3.6 (B) 3

(C) 2.4 (D) 1.2

15. A ball moving right at $8\,\mathrm{m\,s^{-1}}$ hits a massive wall moving left at $2\,\mathrm{m\,s^{-1}}$. With restitution $e = 0.5$, the final ball velocity, taking right positive, in $\mathrm{m\,s^{-1}}$, is

(A) -7 (B) -3

(C) -6 (D) -5

16. A ball falls vertically at $10\,\mathrm{m\,s^{-1}}$ onto a massive platform moving upward at $2\,\mathrm{m\,s^{-1}}$. With restitution $e = 1/2$ and $g = 10$, the maximum rise of the ball above its collision point, in m, is

(A) 1.8 (B) 0.8

(C) 5 (D) 3.2

17. A ball is dropped from 20 m onto a fixed floor with $e = 1/2$. Taking $g = 10$, the total time from release to the third floor impact, in s, is

(A) 4 (B) 6

(C) 3 (D) 5

18. A ball dropped from height h rebounds from a fixed floor with constant restitution $e = 1/2$. The total path length up to and including its third impact, divided by h, is

(A) 1.625 (B) 1.5

(C) 1.75 (D) 1.3125

19. A smooth fixed vertical wall is struck by a 1 kg ball whose incoming velocity is $(6\hat{i} + 8\hat{j})\,\mathrm{m\,s^{-1}}$, with $\hat{i}$ normal toward the wall. If $e = 1/2$, the fraction of kinetic energy lost is

(A) 0.25 (B) 0.75

(C) 0.36 (D) 0.27

20. A ball strikes a smooth plane with incoming speed $10\,\mathrm{m\,s^{-1}}$. Its velocity makes 30° with the plane; after collision it makes an angle β with the plane. With $e = 1/2$, the value of $\tan\beta$ is

(A) $\frac{\sqrt{3}}{6}$ (B) $\frac{\sqrt{3}}{3}$

(C) $\frac{\sqrt{3}}{2}$ (D) $\frac{2\sqrt{3}}{3}$

21. A 3 kg body moving horizontally at $4\,\mathrm{m\,s^{-1}}$ explodes into masses 1 and 2 kg. The 1 kg piece moves vertically upward at $6\,\mathrm{m\,s^{-1}}$. The kinetic energy released in J is

(A) 33 (B) 39

(C) 45 (D) 24

22. At its highest point, a projectile of mass $2m$ has horizontal speed u. It splits into equal fragments; one moves vertically downward with speed u. The ratio of total kinetic energy after to before explosion is

(A) 3 (B) 4

(C) 2.5 (D) 2

23. A 0.01 kg bullet embeds in a suspended 0.99 kg block. The combined body rises 0.20 m. With $g = 10$, bullet speed in $\mathrm{m\,s^{-1}}$ is

(A) 200 (B) 100

(C) 400 (D) 20

24. A bullet of mass m embeds in a pendulum block of mass $9m$. The pendulum length is 1 m and the combined body rises until the string makes 60° to vertical. With $g = 10$, the fraction of initial bullet kinetic energy remaining immediately after collision is

(A) 0.01 (B) 0.1

(C) 0.2 (D) 0.9

25. A 0.02 kg bullet enters a stationary 1 kg block at $100\,\mathrm{m\,s^{-1}}$ and exits in the same direction at $50\,\mathrm{m\,s^{-1}}$. The block is attached to a horizontal spring of constant 100 N/m on a smooth surface. Its maximum displacement in m is

(A) 0.05 (B) 0.2

(C) 0.3 (D) 0.1

26. A bullet passes through a block resting on a smooth floor. Its speed falls from u to $u/2$, and the block mass is 10 times the bullet mass. The fraction of the initial bullet kinetic energy lost as internal energy is

(A) 0.25 (B) 0.7

(C) 0.725 (D) 0.75

27. A 50 kg person walks 3 m relative to a 100 kg boat floating without horizontal resistance. Starting from rest, the boat displacement relative to shore, in m and opposite the walk, is

(A) 3 (B) 2

(C) 1 (D) 1.5

28. A 60 kg person on a 90 kg raft throws a 10 kg parcel horizontally at $8\,\mathrm{m\,s^{-1}}$ relative to the person. Initially all are at rest. The raft recoil speed relative to shore, in $\mathrm{m\,s^{-1}}$, is

(A) $\frac{8}{15}$ (B) 0.5

(C) 0.8 (D) $\frac{8}{17}$

29. A uniform square lamina of side 2 m has a square of side 1 m removed from its upper right corner. With origin at the lower left corner, the remaining centre of mass has x coordinate in m

(A) 1 (B) 0.75

(C) $\frac{5}{6}$ (D) $\frac{2}{3}$

30. A uniform circular disc of radius R has a circular hole of radius $R/2$ removed, internally tangent at the right rim. The remaining centre of mass is displaced left of the original centre by distance divided by R equal to

(A) 0.25 (B) 0.125

(C) $\frac{1}{3}$ (D) $\frac{1}{6}$

31. A rod lies on $0 \le x \le L$ with linear density $\lambda = \lambda_0(1 + x/L)$. Its centre of mass coordinate divided by L is

(A) 0.5 (B) 0.6

(C) $\frac{5}{9}$ (D) $\frac{2}{3}$

32. A rod of length L has linear density proportional to x measured from its left end. Its left half is removed. The centre of mass of what remains has x/L equal to

(A) $\frac{7}{9}$ (B) 0.75

(C) $\frac{5}{6}$ (D) $\frac{2}{3}$

33. Sand falls vertically at 2 kg/s into a cart moving horizontally at constant $3\,\mathrm{m\,s^{-1}}$. A horizontal force keeps its speed constant. The rate of mechanical energy converted to internal energy by capture, in W, is

(A) 0 (B) 6

(C) 18 (D) 9

34. A smooth cart of initial mass 2 kg moves at $6\,\mathrm{m\,s^{-1}}$. It captures 1 kg of sand falling with zero horizontal speed. The horizontal kinetic energy lost, in J, is

(A) 18 (B) 12

(C) 6 (D) 24

35. A 2 kg body initially moves at $3\,\mathrm{m\,s^{-1}}$. A force $F = 4 - 2t$ N acts along the positive direction for $0 \le t \le 4$ s. Its displacement over these 4 s, in m, is

(A) $\frac{52}{3}$ (B) 20

(C) $\frac{40}{3}$ (D) 12

36. A 1 kg ball initially moves right at $4\,\mathrm{m\,s^{-1}}$. A leftward triangular force pulse rises from 0 to 12 N in 1 s and falls to 0 in the next 1 s. Its final speed in $\mathrm{m\,s^{-1}}$ is

(A) 16 (B) 4

(C) 8 (D) 12

37. Two particles of masses m and $3m$ are joined by a light spring on a smooth line, with their centre of mass initially at rest. When the spring releases energy E, the kinetic energy of the $3m$ particle divided by E is

(A) $\frac{1}{3}$ (B) 0.25

(C) 0.5 (D) 0.75

38. Two masses 1 and 3 kg joined by a spring move with centre speed $2\,\mathrm{m\,s^{-1}}$. At an instant their total kinetic energy is 14 J. The relative speed between the masses in $\mathrm{m\,s^{-1}}$ is

(A) 3 (B) 4

(C) 2 (D) 6

39. A stationary cart of mass M is struck by a mass m moving at u and they stick. A second identical mass m also moving at u later sticks to them. If the final speed is $u/2$, the ratio M/m is

(A) 2 (B) 3

(C) 1 (D) 4

40. A cart of mass $2m$ moving at speed u captures a stationary mass m, then a second stationary mass m. The fraction of its original kinetic energy lost after both captures is

(A) 0.25 (B) 0.75

(C) 0.5 (D) $\frac{1}{3}$

41–50 / Preserved hard practice

41. A 2 kg block on a 12 N/m horizontal spring is at extension 0.50 m and moving toward equilibrium at 1 m/s. A 1 kg block moving away from equilibrium at 6 m/s strikes and sticks to it at that instant. Find the subsequent amplitude in metres.

(A) $\frac{1}{6}$ (B) $\frac{5}{6}$

(C) $\frac{5}{12}$ (D) $\frac{11}{12}$

42. A 1 kg ball moving at 5 m/s strikes a stationary 2 kg ball. Thereafter both slide independently on a level floor with kinetic coefficient 0.20. They stop after travelling 0.25 m and 2.25 m respectively. Take $g = 10$. Find the coefficient of restitution.

(A) 0.4 (B) 0.2

(C) 0.8 (D) 0.9

43. A bullet of mass 0.01 kg embeds in a 0.99 kg bob on a string. For the combined body to just complete a vertical circle of radius 0.2 m, with $g = 10$, the bullet speed must be at least

(A) $100\sqrt{2}\,\mathrm{m\,s^{-1}}$ (B) $10\sqrt{10}\,\mathrm{m\,s^{-1}}$

(C) $200\,\mathrm{m\,s^{-1}}$ (D) $100\sqrt{10}\,\mathrm{m\,s^{-1}}$

44. A 3 kg body moves at 2 m/s along $+x$ and explodes into masses 1 kg and 2 kg. The first fragment moves perpendicular to x. The explosion increases total kinetic energy by 12 J. Find the angle between the fragment velocities, in degrees.

(A) 119 (B) 240

(C) 121 (D) 120

45. At the highest point of its flight, a 2 kg projectile is 20 m above horizontal ground and moves horizontally at 10 m/s. It explodes into equal fragments. One fragment moves vertically downward at 10 m/s immediately after the explosion. Take $g = 10\,\text{m/s}^2$. The other fragment's horizontal distance from the explosion point when it lands, in metres, is

(A) $20\sqrt{5}$ (B) $20 + 20\sqrt{5}$

(C) $-20 + 20\sqrt{5}$ (D) 40

46. A 1 kg block moving uphill at 5 m/s along a smooth 30° incline sticks to a stationary 4 kg cart at its foot. The pair immediately moves uphill, compressing an initially unstretched spring of 100 N/m along the incline. Take $g = 10$. Find maximum compression in metres.

(A) $\frac{5}{4} - \frac{3\sqrt{5}}{20}$ (B) $\frac{3\sqrt{5}}{40} + \frac{3}{8}$

(C) $-\frac{1}{8} + \frac{3\sqrt{5}}{40}$ (D) $-\frac{1}{4} + \frac{3\sqrt{5}}{20}$

47. A 50 kg person walks 3 m relative to a 100 kg boat on still water and comes to rest relative to it. He then jumps horizontally at 6 m/s relative to the boat, from a height 1.25 m above water. Ignore horizontal water forces, take $g = 10$, and treat both as point masses horizontally. Find his horizontal landing displacement from his initial position, in metres.

(A) 8 (B) 3

(C) 5 (D) 4

48. A uniform disc originally has mass M and radius R. A hole of radius $R/2$, centred $R/2$ from the original centre, is cut out. Find the remaining disc moment of inertia about the axis normal to the disc through its new centre of mass, divided by MR^2.

(A) $\frac{37}{192}$ (B) $\frac{37}{96}$

(C) $\frac{59}{96}$ (D) $\frac{133}{192}$

49. A 2 kg ball moving at +3 m/s collides with a 1 kg ball moving at −3 m/s. Restitution between balls is 1/2. The lighter ball then rebounds elastically from a massive wall moving at −1 m/s and collides with the heavier ball again. Find the heavier ball velocity just after this second ball-ball collision, in m/s.

(A) −1.5 (B) −3.5

(C) −2.5 (D) −5

50. A uniform rod of mass M and length L pivots at one end. A point mass M at speed u strikes and sticks to its free end perpendicular to the rod. Find the magnitude of the hinge impulse during impact divided by Mu.

(A) 0.125 (B) 0.875

(C) 0.5625 (D) 0.0625

Answer key

Centre of Mass & Collisions

P-CMC / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**A**	21	**B**	31	**C**	41	**B**
02	**D**	12	**D**	22	**A**	32	**A**	42	**C**
03	**B**	13	**A**	23	**A**	33	**D**	43	**D**
04	**B**	14	**C**	24	**B**	34	**B**	44	**D**
05	**A**	15	**A**	25	**D**	35	**A**	45	**B**
06	**B**	16	**D**	26	**C**	36	**C**	46	**D**
07	**C**	17	**D**	27	**C**	37	**B**	47	**D**
08	**C**	18	**A**	28	**B**	38	**B**	48	**B**
09	**B**	19	**D**	29	**C**	39	**A**	49	**C**
10	**C**	20	**A**	30	**D**	40	**C**	50	**A**

Chapter 7

Rotational Motion

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A $2\,\text{kg}$ disc of radius $1\,\text{m}$ is acted on by tangential force $3\,\text{N}$. Angular acceleration in rad/s squared is

(A) 2 (B) 3

(C) 4 (D) 6

02. A body with $I_{cm} = 2\,\text{kg}\,\text{m}^2$ and mass $3\,\text{kg}$ rotates about a parallel axis $2\,\text{m}$ away. Its moment of inertia in SI units is

(A) 13 (B) 14

(C) 28 (D) 15

03. A uniform disc rolls without slipping. Fraction of total kinetic energy that is rotational is

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$

(C) $\frac{-2}{3}$ (D) $\frac{4}{3}$

04. A constant torque increases angular speed from 2 to $6\,\text{rad}\,\text{s}^{-1}$ in $2\,\text{s}$. With $I = 3\,\text{kg}\,\text{m}^2$, torque in N m is

(A) 6 (B) 7

(C) 5 (D) 12

05. A wheel at $10\,\text{rad}\,\text{s}^{-1}$ stops with uniform angular retardation $2\,\text{rad}\,\text{s}^{-2}$. Angle travelled in radians is

(A) 24 (B) 25

(C) 26 (D) 50

06. A particle of mass 2 kg moves at $3\,\mathrm{m\,s^{-1}}$ along a straight line 4 m from a fixed origin. Angular momentum magnitude about the origin in SI units is

(A) 23 (B) 24

(C) 48 (D) 25

07. A uniform disc and ring have equal mass and radius, and equal rotational kinetic energy about symmetry axes. The ratio $\omega_{disc}^2/\omega_{ring}^2$ is

(A) 3 (B) 4

(C) 1 (D) 2

08. A solid cylinder rolls down a rough incline without slipping. For $\sin\theta = 0.3, g = 10$, centre acceleration in m/s squared is

(A) 4 (B) 1

(C) 2 (D) 3

09. A thin ring rolls down an incline without slipping. With $g = 10, \sin\theta = 0.4$, acceleration in m/s squared is

(A) 2 (B) 3

(C) 4 (D) 1

10. A 2 kg uniform disc and 1 kg point mass at its rim have radius 1 m. Combined central moment of inertia in SI units is

(A) 2 (B) 4

(C) 3 (D) 1

11–40 / New middle-to-hard practice

11. A rigid body rolls without slipping down an incline from rest. Its centre covers 3 m in 2 s on a plane with $\sin\theta = 1/5$, $g = 10$. The ratio $I/(mR^2)$ is

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) 1 (D) 0.5

12. A rolling body loses gravitational energy 12 J descending a smooth-in-energy rough incline without slipping. At the bottom its translational kinetic energy is 8 J. The minimum friction coefficient required on a 45° incline is

(A) $\frac{2}{3}$ (B) 0.5

(C) $\frac{1}{3}$ (D) 1

13. A uniform solid cylinder rolls without slipping on a horizontal floor. A horizontal force F acts through its centre. The friction force divided by F, taking the applied-force direction positive, is

(A) -0.5 (B) $\frac{1}{3}$

(C) $-\frac{1}{3}$ (D) 0

14. A uniform solid cylinder rolls on a horizontal floor under a horizontal force F applied at its uppermost point. With no slip, the friction force divided by F, positive in the force direction, is

(A) 0.5 (B) $-\frac{1}{3}$

(C) 0 (D) $\frac{1}{3}$

15. A wheel has $I = \frac{1}{2}mR^2$ and rolls without slipping under a horizontal pull applied at height h above its centre. The floor friction vanishes when h/R equals

(A) 1.5 (B) 0.5

(C) 0 (D) 1

16. A body with $I = \frac{2}{5}mR^2$ rolls without slipping under horizontal force F acting at height $R/5$ above its centre. The signed friction force divided by F is

(A) $\frac{3}{7}$ (B) $-\frac{2}{7}$

(C) $\frac{1}{7}$ (D) $-\frac{1}{7}$

17. A uniform solid cylinder unwinds downward from a light string wound round its circumference, with the free end fixed. With $g = 10$, its downward acceleration in $\mathrm{m\,s^{-2}}$ is

(A) $\frac{10}{3}$ (B) 10

(C) $\frac{20}{3}$ (D) 5

18. A yo-yo has outer radius R, axle radius $R/2$ and moment of inertia $mR^2/2$. It descends while its axle string unwinds without slip. Its acceleration divided by g is

(A) $\frac{1}{3}$ (B) 0.2

(C) 0.5 (D) $\frac{2}{3}$

19. Two masses 1 kg and 3 kg hang over a pulley of radius R and moment of inertia $2R^2\,\mathrm{kg\,m^2}$. String does not slip. With $g = 10$, the ratio of tensions on the heavy and light sides is

(A) 1.5 (B) $\frac{4}{3}$

(C) 2 (D) 3

20. An Atwood pair has masses 2 and 3 kg and accelerates at $1\,\mathrm{m\,s^{-2}}$. With $g = 10$, the pulley moment of inertia divided by radius squared, in kg, is

(A) 10 (B) 1

(C) 2 (D) 5

21. A uniform rod of length L is hinged at one end and released from rest horizontally. Immediately after release, the vertical hinge reaction divided by rod weight is

(A) 0.75 (B) 0.5

(C) 0 (D) 0.25

22. A uniform rod hinged at one end falls from rest horizontally. When it first reaches vertical downward, the hinge reaction divided by its weight is

(A) 2.5 (B) 1.5

(C) 1 (D) 3

23. A particle of mass m moving at u perpendicular to a uniform rod of mass $3m$ and length L hits and sticks at its free end. The other end is hinged. The final angular speed divided by u/L is

(A) 1 (B) $\frac{1}{3}$

(C) 0.5 (D) $\frac{2}{3}$

24. A bullet of mass m moving perpendicular to a uniform rod of mass $6m$ and length L embeds at its free end; the rod is hinged at its other end. The fraction of initial bullet energy remaining after impact is

(A) $\frac{1}{7}$ (B) $\frac{2}{3}$

(C) 0.5 (D) $\frac{1}{3}$

25. A turntable has moment of inertia I. Two particles each of mass m sit opposite at radius R, with $2mR^2 = I$. They move to radius $R/2$ without external torque. The ratio of final to initial rotational energy is

(A) $\frac{4}{3}$ (B) 2

(C) 0.625 (D) 1.6

26. A disc has moment of inertia $3\,\mathrm{kg\,m^2}$ and rotates at $2\,\mathrm{rad\,s^{-1}}$. A person standing at its rim contributes another $6\,\mathrm{kg\,m^2}$. The person walks to the axis with no external torque. The work done by the person in J is

(A) 18 (B) 36

(C) 54 (D) 24

27. Two coaxial discs with moments of inertia I and $2I$ rotate at angular speeds 3ω and 0. They are coupled by a friction clutch with no external torque. The fraction of initial rotational energy lost is

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) 0.5 (D) 0.75

28. Two identical coaxial discs rotate in opposite directions at speeds 3ω and ω. They are pressed together until they rotate together. The fraction of their initial rotational energy remaining is

(A) 0.8 (B) 0.5

(C) 0.4 (D) 0.2

29. A uniform disc of mass M and radius R has a circular hole of radius $R/2$ removed, internally tangent to the rim. The remaining moment of inertia about the original central axis divided by MR^2 is

(A) 0.46875 (B) 0.375

(C) 0.4375 (D) 0.40625

30. From a uniform disc of mass M and radius R, a concentric disc of radius $R/2$ is removed. The radius of gyration of the remaining annulus, divided by R, is

(A) $\frac{\sqrt{2}}{2}$ (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{\sqrt{10}}{4}$ (D) $\frac{\sqrt{5}}{4}$

31. A rod of length L has linear density proportional to x from one end. Its moment of inertia about that end divided by ML^2 is

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) 0.5 (D) 0.25

32. A rod on $0 \le x \le L$ has density proportional to x. Its moment of inertia about an axis through $x = L$, perpendicular to the rod, divided by ML^2 is

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$

(C) $\frac{1}{12}$ (D) 0.5

33. A solid sphere is projected on a rough horizontal floor with speed u and no spin. It eventually rolls without slipping. The fraction of initial kinetic energy lost is

(A) $\frac{5}{7}$ (B) $\frac{3}{7}$

(C) $\frac{2}{7}$ (D) $\frac{1}{7}$

34. A solid cylinder is projected at speed u on a rough floor with angular velocity zero. With kinetic coefficient μ, the distance travelled before pure rolling starts, divided by $u^2/(\mu g)$, is

(A) $\frac{5}{18}$ (B) $\frac{1}{6}$

(C) $\frac{1}{3}$ (D) $\frac{2}{9}$

35. A solid sphere initially moves right at speed u and spins so its bottom point moves right relative to its centre at speed u. After friction establishes rolling, the signed centre speed divided by u is

(A) $\frac{1}{7}$ (B) $\frac{5}{7}$

(C) $-\frac{1}{7}$ (D) $\frac{3}{7}$

36. A uniform disc has initial centre speed u rightward and clockwise spin with $\omega R = 3u$. On a rough floor it settles into pure rolling. Its final centre speed divided by u is

(A) $\frac{2}{3}$ (B) $\frac{5}{3}$

(C) $\frac{4}{3}$ (D) 2

37. A body of moment of inertia $2\,\mathrm{kg\,m^2}$ initially rotates at $1\,\mathrm{rad\,s^{-1}}$. Torque $\tau = 4t$ N m acts in the same direction. The work done by $t = 2$ s, in J, is

(A) 8 (B) 16

(C) 24 (D) 32

38. A rotor with $I = 1\,\mathrm{kg\,m^2}$ receives constant mechanical power 6 W. Its angular speed increases from 2 to $4\,\mathrm{rad\,s^{-1}}$. The angle turned during this increase, in rad, is

(A) 2 (B) $\frac{14}{3}$

(C) $\frac{28}{9}$ (D) 4

39. A particle of mass 2 kg has position $\mathbf{r} = (t\hat{i} + t^2\hat{j})$ m. At $t = 2$ s, the magnitude of angular momentum about the origin in SI units is

(A) 4 (B) 8

(C) 16 (D) 12

40. A particle of mass 2/3 kg has position $\mathbf{r} = (2t\hat{i} + 3t^2\hat{j})$ m. The torque of its net force about the origin at $t = 2$ s, in N m, is

(A) 0 (B) 16

(C) 12 (D) 8

41–50 / Preserved hard practice

41. A uniform rod pivots freely at one end and falls from horizontal. When it is 45° below horizontal, find the upward vertical hinge-force component divided by its weight.

(A) 2.75 (B) 2.375

(C) 1.375 (D) 0.375

42. A light string unwinds without slip from a fixed solid cylindrical pulley of mass 2 kg and radius 0.2 m. A 1 kg hanging mass is also attached to a vertical spring of stiffness 20 N/m fixed above it, initially unstretched. Released from rest, find its speed squared after descending 0.50 m, using $g = 10$.

(A) 2.5 (B) 5

(C) 1.5 (D) 3.5

43. At the instant considered, a solid cylinder of mass M and radius R rolls on a level floor under a horizontal force $F = 3Mg$ applied at height $R/4$ above its centre. Find the minimum static friction coefficient allowing rolling without slip.

(A) 0.5 (B) 0.25

(C) 0.75 (D) $\frac{1}{6}$

44. A solid cylinder on a level floor is pulled by a string tangent to its rim and directed 30° above the horizontal to the right, producing clockwise torque. The static coefficient is 1/3. Find the largest $F/(Mg)$ that permits rolling without slip.

(A) $\frac{6}{11} - \frac{\sqrt{3}}{11}$ (B) $\frac{\sqrt{3}}{22} + \frac{8}{11}$

(C) $\frac{\sqrt{3}}{11} + \frac{5}{11}$ (D) $\frac{\sqrt{3}}{22} + \frac{5}{22}$

45. A thin ring has centre velocity u to the right and clockwise angular speed $2u/R$ when placed on a rough floor. After slipping ceases, find the fraction of initial kinetic energy converted to heat.

(A) 0.9 (B) 0.55

(C) 0.05 (D) 0.1

46. A uniform disc of radius R is placed on a rough horizontal floor with centre initially at rest and angular speed ω_0. Kinetic coefficient is μ, and no other horizontal force acts. Before pure rolling begins, find its centre displacement divided by $R^2\omega_0^2/(\mu g)$.

(A) $\frac{17}{18}$ (B) $\frac{1}{18}$

(C) $\frac{1}{36}$ (D) $\frac{19}{36}$

47. A uniform rod of mass M, length L hangs vertically from a frictionless end pivot. A particle of mass M moving horizontally at u strikes and sticks to its free end, where $u^2 = 2gL$. Find the greatest rod angle from the downward vertical, in degrees.

(A) 61 (B) 120

(C) 59 (D) 60

48. A uniform disc of mass M, radius R rotates freely at ω. A point mass M strikes the rim tangentially at speed $R\omega$ opposite the local rim velocity and sticks. All lost mechanical energy heats both bodies, each of specific heat c. Find $c\Delta T/(R^2\omega^2)$.

(A) $\frac{1}{3}$ (B) $\frac{1}{9}$

(C) $\frac{1}{6}$ (D) $\frac{2}{3}$

49. A fixed pulley has radius r, unknown moment of inertia I, and constant opposing axle-friction torque. A light string does not slip. Hanging masses 3 and 1 kg produce downward acceleration $2\,\mathrm{m/s^2}$ of the heavier mass. Replacing 3 kg by 5 kg gives downward acceleration $4\,\mathrm{m/s^2}$. Take $g = 10$. In the first setup, axle-friction heat after the heavier mass descends 1 m, in joules, is

(A) 8 (B) 16

(C) 12 (D) 4

50. A uniform solid sphere rolls without slipping toward a smooth vertical wall at speed v. The wall collision reverses its centre velocity with coefficient of restitution $1/2$ and leaves its spin unchanged. Neglect floor impulse during the brief wall collision. It then slips on the rough horizontal floor until rolling resumes. Final signed centre velocity divided by v, positive toward the wall, is

(A) -0.5 (B) $-\frac{1}{14}$

(C) $\frac{3}{7}$ (D) $\frac{1}{14}$

Answer key

Rotational Motion

P-ROMO / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**A**	21	**D**	31	**C**	41	**C**
02	**B**	12	**C**	22	**A**	32	**A**	42	**A**
03	**B**	13	**C**	23	**C**	33	**C**	43	**A**
04	**A**	14	**D**	24	**D**	34	**A**	44	**C**
05	**B**	15	**B**	25	**D**	35	**D**	45	**D**
06	**B**	16	**D**	26	**B**	36	**B**	46	**B**
07	**D**	17	**C**	27	**B**	37	**C**	47	**D**
08	**C**	18	**A**	28	**D**	38	**C**	48	**A**
09	**A**	19	**A**	29	**D**	39	**B**	49	**A**
10	**A**	20	**D**	30	**C**	40	**B**	50	**B**

Chapter 8

Gravitation

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Two planets have equal density and radii in ratio $2:1$. Escape speeds are in ratio

(A) 4 (B) 2

(C) 1 (D) 3

02. At altitude $h = R$, weight is 25 N. Surface weight in newtons is

(A) 200 (B) 100

(C) 101 (D) 99

03. Circular satellites at radii r and $9r$ have periods T_1, T_2. The ratio T_2/T_1 is

(A) 28 (B) 54

(C) 26 (D) 27

04. A test mass at distance R from a point mass has potential energy -12 J. Work needed to move it slowly to $3R$ in joules is

(A) 9 (B) 8

(C) 7 (D) 16

05. Two masses M and $4M$ are $3d$ apart. The zero-field point between them is at distance x from mass M. The ratio x/d is

(A) 0.5 (B) 0

(C) 2 (D) 1

06. A planet has mass $8M$ and radius $2R$, compared with reference M, R. Escape speed ratio is

(A) 1 (B) 2

(C) 4 (D) 3

07. A body is raised from planet surface to altitude R. Work done against gravity divided by GMm/R is

(A) 1 (B) -0.5

(C) 0.5 (D) 1.5

08. A geostationary satellite has orbital period 24 hours. A satellite at one quarter its orbit radius has period in hours equal to

(A) 3 (B) 4

(C) 2 (D) 6

09. A field obeys $g = kr$ inside a uniform planet. The potential difference between its centre and surface, $V(R) - V(0)$, divided by kR^2 is

(A) 0.5 (B) 1

(C) -0.5 (D) 1.5

10. A satellite moves in an ellipse around a point mass. At periapsis and apoapsis distances r_p, r_a, the ratio v_p/v_a equals

(A) 1 (B) r_p/r_a

(C) $\sqrt{r_p/r_a}$ (D) r_a/r_p

11–40 / New middle-to-hard practice

11. For a uniform spherical planet of radius R, gravity at depth d equals gravity at altitude R. The ratio d/R is

(A) $\frac{2}{3}$ (B) 0.25

(C) 0.75 (D) 0.5

12. For a uniform spherical planet, gravity at altitude h equals gravity at depth $R/2$. The ratio h/R is

(A) 0.25 (B) 0.5

(C) $-1 + \sqrt{2}$ (D) $\sqrt{2}$

13. A straight narrow tunnel passes through the centre of a uniform nonrotating planet of radius R and surface gravity g. A body is released from rest at the surface. Its speed at distance $R/2$ from the centre, divided by $\sqrt{gR}$, is

(A) 1 (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{\sqrt{2}}{2}$ (D) 0.5

14. A uniform nonrotating planet has radius R and surface escape speed v_e. The speed needed at its centre to escape through a narrow tunnel to infinity, divided by v_e, is

(A) $\sqrt{2}$ (B) $\frac{\sqrt{6}}{3}$

(C) $\sqrt{3}$ (D) $\frac{\sqrt{6}}{2}$

15. A spherical shell of radius R and mass $3M/4$ surrounds a central point mass $M/4$. The escape speed at radius $R/2$, divided by $\sqrt{GM/R}$, is

(A) $\sqrt{3}$ (B) $\sqrt{2}$

(C) $\frac{\sqrt{6}}{2}$ (D) $\frac{\sqrt{10}}{2}$

16. A shell of radius R and mass $3M/4$ surrounds a point mass $M/4$. A test body released from rest just inside the shell falls to $r = R/2$. Its speed divided by $\sqrt{GM/R}$ is

(A) $\frac{\sqrt{2}}{2}$ (B) $\sqrt{2}$

(C) 0.5 (D) 1

17. A small spherical cavity of radius $R/2$ is made inside a uniform sphere of radius R and density ρ, with cavity centre displaced by $R/2$. The gravitational field magnitude anywhere in the cavity divided by $4\pi G\rho R/3$ is

(A) 0 (B) 0.5

(C) 1 (D) 0.25

18. A uniform sphere of mass M and radius R has a spherical cavity of radius $R/2$ removed, its centre at distance $R/2$ from the original centre. At the original centre, the potential due to what remains, divided by GM/R, is

(A) -1.125 (B) -1.375

(C) -1.25 (D) -1.5

19. A body is launched radially from the surface of a nonrotating planet at $v_e/2$, where v_e is surface escape speed. Its maximum altitude divided by planet radius is

(A) 0.5 (B) $\frac{1}{3}$

(C) 1 (D) 0.25

20. A body projected vertically from the surface of a nonrotating planet reaches altitude $3R$. The launch speed divided by surface escape speed is

(A) $\frac{\sqrt{2}}{2}$ (B) 0.5

(C) 0.75 (D) $\frac{\sqrt{3}}{2}$

21. A body is released from rest at radius $3R$ above a nonrotating planet centre. When it reaches the surface R, the ratio of its speed to surface escape speed is

(A) $\frac{\sqrt{6}}{3}$ (B) $\frac{\sqrt{6}}{2}$

(C) $\frac{\sqrt{3}}{3}$ (D) $\frac{2}{3}$

22. A body is dropped from rest at altitude R and another from altitude $3R$ above a planet of radius R. The ratio of their impact kinetic energies, first to second, is

(A) $\frac{1}{3}$ (B) 0.75

(C) 0.5 (D) $\frac{2}{3}$

23. A satellite shifts between two circular orbits. Its orbital speed becomes half the original. The required increase in total energy divided by its initial kinetic energy is

(A) 0.75 (B) 3

(C) 0.25 (D) 0.5

24. A satellite initially in a circular orbit is given additional total mechanical energy equal to 3/4 of its initial kinetic energy, and settles into another circular orbit. The increase in angular momentum divided by the initial angular momentum is

(A) 1 (B) 2

(C) 3 (D) 0.5

25. A satellite on an elliptical orbit has apoapsis three times periapsis. If v_c is circular speed at periapsis radius r_p, then v_p^2/v_c^2 is

(A) 2 (B) 1.5

(C) 3 (D) 0.5

26. An elliptical satellite orbit has apoapsis radius $3r_p$. Its period divided by that of a circular orbit of radius r_p is

(A) $\sqrt{2}$ (B) 3

(C) $3\sqrt{3}$ (D) $2\sqrt{2}$

27. Two point masses m and $3m$ orbit their common centre of mass at constant separation d. The speed of mass m divided by $\sqrt{Gm/d}$ is

(A) 0.5 (B) 2

(C) 1.5 (D) $\sqrt{3}$

28. Two equal stars of mass M circle their common centre at radius R, with period T. A negligible test mass orbits a single star of mass $2M$ at radius $2R$. Its period divided by T is

(A) 0.5 (B) 1

(C) $\sqrt{2}$ (D) 2

29. Planet A has twice the surface gravity and twice the escape speed of planet B. The ratio of their mean densities ρ_A/ρ_B is

(A) 0.5 (B) 2

(C) 1 (D) 4

30. Two spherical planets have surface gravities in ratio $3:1$ and escape speeds in ratio $\sqrt{6}:1$. Their masses are in ratio

(A) 6 (B) 24

(C) 12 (D) 3

31. A 10 kg test body weighs 80 N at a spherical planet pole and 60 N at its equator. The planet radius is 2×10^6 m. Ignoring deformation, its rotation period divided by π, in s, is

(A) 4000 (B) 1000

(C) $200\sqrt{10}$ (D) 2000

32. A planet rotates so that a body at its equator has apparent weight half its polar weight. The ratio of its rotation period to the period of a circular orbit just above its surface is

(A) $\sqrt{2}$ (B) $\frac{\sqrt{2}}{2}$

(C) 2 (D) 0.5

33. Two point masses M and $4M$ are held $3a$ apart. At the zero-field point between them, the potential divided by GM/a is

(A) -3 (B) -2.5

(C) -2 (D) -4

34. Two point masses with total mass M are separated by d. Their zero-field point lies $d/3$ from the smaller mass. At that point, the potential divided by GM/d is

(A) -3 (B) -1.8

(C) -1.5 (D) $-\frac{5}{3}$

35. A uniform ring of mass M and radius R attracts a test mass on its axis. The positive axial distance of maximum field divided by R is

(A) 1 (B) $\sqrt{2}$

(C) 0.5 (D) $\frac{\sqrt{2}}{2}$

36. A test particle is released from rest on the axis of a uniform ring of mass M and radius R, at distance $\sqrt{3}R$. Its speed when crossing the ring centre divided by $\sqrt{GM/R}$ is

(A) $\frac{\sqrt{2}}{2}$ (B) 0.5

(C) 1 (D) $\sqrt{2}$

37. Two homogeneous solid spheres of the same density are tangent. Their radii are R and $2R$. The ratio of their mutual force to that between two tangent spheres each of radius R and that density is

(A) $\frac{16}{9}$ (B) 4

(C) $\frac{32}{9}$ (D) 8

38. A homogeneous sphere is divided into two equal-density spheres with equal masses. If the two new spheres just touch, their mutual attraction divided by GM^2/R^2, where M, R describe the original sphere, is

(A) $\frac{\sqrt[3]{2}}{8}$ (B) 0.0625

(C) 0.125 (D) $\frac{2^{\frac{2}{3}}}{16}$

39. A total mass M is split into two point masses held at fixed separation. If their mutual attraction is $3/4$ of its largest possible value, the ratio of larger to smaller mass is

(A) 4 (B) 2

(C) 9 (D) 3

40. A total mass M is split into two point masses separated by d. Their mutual attraction is $2GM^2/(9d^2)$. The smaller mass divided by M is

(A) $\frac{1}{3}$ (B) $\frac{2}{9}$

(C) 0.25 (D) 0.5

41–50 / Preserved hard practice

41. A body is launched from a spherical planet surface at half escape speed, directed 45° above the horizontal. Ignore atmosphere and rotation. Find its greatest altitude divided by planet radius R.

(A) $\frac{4}{3} - \frac{\sqrt{10}}{6}$ (B) $-\frac{1}{3} + \frac{\sqrt{10}}{6}$

(C) $-\frac{1}{6} + \frac{\sqrt{10}}{12}$ (D) $\frac{\sqrt{10}}{12} + \frac{1}{3}$

42. A straight frictionless tunnel through a uniform spherical planet passes at distance $R/2$ from its centre. A particle is released from rest at one tunnel entrance. If the surface gravity is g, find its greatest speed squared divided by gR.

(A) 0.75 (B) 0.25

(C) 0.375 (D) 0.875

43. A satellite in a circular orbit of radius r receives a tangential impulse raising its speed to $\sqrt{3/2}$ times circular speed. Find its greatest subsequent distance from the central point mass, divided by r.

(A) 4 (B) 6

(C) 3 (D) 2

44. A uniform sphere of radius R has a spherical cavity of radius $R/2$, whose centre is $R/2$ from the original sphere centre. The remaining mass is M. A very short pendulum of length ℓ hangs inside the cavity. Find its period squared divided by $4\pi^2\ell R^2/(GM)$.

(A) 1.75 (B) 0.75

(C) 2.75 (D) 3.5

45. Two point masses M and $2M$ are initially $2a$ apart. The lighter moves toward the stationary heavier one at speed u, where $u^2 = GM/a$. Under only mutual gravity, find the lighter velocity component along its initial direction, divided by u, when separation first becomes a.

(A) $\frac{8}{3}$ (B) $\frac{10}{3}$

(C) $\frac{5}{3}$ (D) $\frac{2}{3}$

46. A particle falls radially from rest at distance $2R$ toward a thin uniform spherical shell of mass M and radius R, through a negligible hole. Find the time it spends crossing the shell interior to the opposite surface, divided by the period of a circular orbit just outside radius R.

(A) $\frac{1}{\pi}$
(B) $\frac{1}{2\pi} + \frac{1}{2}$
(C) $\frac{1}{2\pi}$
(D) $1 - \frac{1}{\pi}$

47. A particle released from rest at one end of a diameter tunnel through a uniform spherical planet crosses the central half of the tunnel, from $x = R/2$ to $x = -R/2$. Find its mean speed during this segment divided by the planet surface escape speed.

(A) $\frac{3\sqrt{2}}{4\pi} + \frac{1}{2}$
(B) $-\frac{3\sqrt{2}}{2\pi} + 1$
(C) $\frac{3\sqrt{2}}{2\pi}$
(D) $\frac{3\sqrt{2}}{4\pi}$

48. A satellite of mass m orbits a rigid central body of gravitational mass $M \gg m$ in a circle of radius r. The body has moment of inertia mr^2 and initial spin $\sqrt{GM/r^3}$. The body spin axis is parallel to the orbital angular momentum and its spin is prograde. An internal mechanism transfers angular momentum until the satellite circular-orbit angular momentum doubles. With no external work or torque, find heat generated divided by GMm/r.

(A) 0.125
(B) 0.5625
(C) 0.0625
(D) 0.875

49. A uniform spherical planet contracts to half its radius without losing mass and remains uniform. A very short simple pendulum is at a fixed point $R/4$ from the centre, in a negligible tunnel. Find its new small-angle period divided by its old period; R is the original radius.

(A) $1 - \frac{\sqrt{2}}{4}$
(B) $\frac{\sqrt{2}}{4}$
(C) $\frac{\sqrt{2}}{8}$
(D) $\frac{\sqrt{2}}{8} + \frac{1}{2}$

50. A satellite circles a fixed point mass at radius R. A brief tangential braking impulse reduces its speed to half the original circular speed without changing its position. Neglect all further non-gravitational forces and assume the central body is smaller than the orbit's closest radius. The subsequent minimum distance divided by R is

(A) 0.5
(B) $\frac{1}{7}$
(C) $\frac{2}{7}$
(D) 0.25

Answer key

Gravitation

P-GRAV / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**C**	21	**A**	31	**D**	41	**B**
02	**B**	12	**C**	22	**D**	32	**A**	42	**A**
03	**D**	13	**B**	23	**A**	33	**A**	43	**C**
04	**B**	14	**D**	24	**A**	34	**B**	44	**A**
05	**D**	15	**D**	25	**B**	35	**D**	45	**C**
06	**B**	16	**A**	26	**D**	36	**C**	46	**A**
07	**C**	17	**B**	27	**C**	37	**C**	47	**C**
08	**A**	18	**C**	28	**B**	38	**D**	48	**A**
09	**A**	19	**B**	29	**C**	39	**D**	49	**B**
10	**D**	20	**D**	30	**C**	40	**A**	50	**B**

Chapter 9

Elasticity

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A wire is cut into four equal pieces, then all four are connected in parallel. Equivalent spring constant divided by original is

(A) 32 (B) 15

(C) 16 (D) 17

02. A wire radius is halved and length doubled at unchanged material. New spring constant divided by old is

(A) 0.25 (B) -0.875

(C) 1.125 (D) 0.125

03. Two series wires of equal L, A have moduli $Y, 2Y$. Equivalent modulus for their total length and common area divided by Y is

(A) $\frac{4}{3}$ (B) $\frac{8}{3}$

(C) $\frac{7}{3}$ (D) $\frac{1}{3}$

04. A material has Young modulus 200 GPa and tensile strain 0.001. Elastic energy density in kJ per cubic metre is

(A) 99 (B) 101

(C) 100 (D) 200

05. A wire has Poisson ratio 0.25 and longitudinal strain 0.002. First-order volume strain is

(A) 0.002 (B) -0.999

(C) 1.001 (D) 0.001

06. Identical wires in parallel carry total force 100 N. One wire has twice the Young modulus of the other. Force in the stiffer wire, in newtons, is

(A) $\frac{200}{3}$ (B) $\frac{197}{3}$

(C) $\frac{400}{3}$ (D) $\frac{203}{3}$

07. Two wires have equal Young modulus and equal lengths, but radii in ratio 1 : 2. Under equal tension, their extension ratio $\Delta L_1/\Delta L_2$ is

(A) 8 (B) 5

(C) 3 (D) 4

08. Two wires have equal material and volume, with lengths L and $2L$. Their spring constants ratio k_L/k_{2L} is

(A) 3 (B) 8

(C) 4 (D) 5

09. A wire is stretched by x under force F. Work needed to stretch it further from x to $3x$ divided by Fx is

(A) 8 (B) 4

(C) 3 (D) 5

10. A rod has $Y = 100$ GPa and thermal expansion coefficient $10^{-5}\,\mathrm{K}^{-1}$. Rigid ends prevent expansion on heating by 20 K. Compressive stress in MPa is

(A) 40 (B) 21

(C) 20 (D) 19

11–40 / New middle-to-hard practice

11. Two elastic wires in series extend in ratio 3 : 1 under a common load. They are reconnected in parallel and given the same total extension as the original pair. The ratio of new stored energy to original stored energy is

(A) $\frac{4}{3}$ (B) $\frac{16}{3}$

(C) 2.25 (D) 3

12. Two wires of the same natural length carry loads in ratio 1 : 4 when connected in parallel. They are reconnected in series and loaded within the elastic range. The fraction of total stored elastic energy in the first wire is

(A) 0.75 (B) 0.2

(C) 0.8 (D) 0.25

13. Two parallel wires of equal natural length support a rigid horizontal plate without rotation. Their areas are $A, 2A$ and Young moduli $2Y, Y$. A total load 120 N is applied. The load in the first wire, in N, is

(A) 40 (B) 90

(C) 60 (D) 80

14. Two parallel wires of equal length have radii $r, 2r$ and Young moduli $3Y, Y$. If the thinner wire carries 30 N, total supported load in N is

(A) 40 (B) 70

(C) 90 (D) 120

15. Two vertical wires of constants 100 and 200 N/m hang from one support to a load plate. The second wire initially has 0.10 m slack; the first is just taut. A 50 N load is applied slowly. The plate displacement in m is

(A) $\frac{7}{30}$ (B) 0.25

(C) 0.5 (D) $\frac{1}{6}$

16. Two parallel wires have stiffnesses k and $2k$. The second is slack until displacement a, while the first is taut initially. A slowly applied load $4ka$ produces final displacement x. The fraction of load carried by the second wire is

(A) 0.75 (B) 0.5

(C) $\frac{2}{3}$ (D) $\frac{1}{3}$

17. A uniform elastic wire is cut into lengths in ratio $1:2$. The two pieces are joined in parallel. The ratio of force needed for an extension x to that for the uncut wire at the same extension is

(A) 4.5 (B) 6

(C) 1.5 (D) 3

18. A uniform wire is melted and redrawn at the same volume and material into a wire of half its original length. This new wire is then cut into two equal pieces and put in series. The final effective stiffness divided by the original is

(A) 8 (B) 4

(C) 16 (D) 2

19. A wire is made from two equal-length segments of the same material with radii r and $2r$. Compared with a uniform wire of the same total length and radius r, its extension under the same force is multiplied by

(A) 0.5 (B) 0.75

(C) 0.625 (D) 1.25

20. A rod has two equal-length parts in series. One has area A and modulus Y, the other area $2A$ and modulus $2Y$. Treating the whole rod as if its area were A, its effective Young modulus divided by Y is

(A) 1.6 (B) $\frac{4}{3}$

(C) 2 (D) 1.25

21. A uniform vertical wire of length L, density ρ, and modulus Y hangs under its own weight. A point mass equal to the wire mass is added at its lower end. The total extension divided by $\rho g L^2/Y$ is

(A) 2 (B) 1.5

(C) 1 (D) 0.5

22. A uniform wire hangs vertically. Its extension under self-weight is x. A point load at the bottom is then added, making total extension $5x$. The added mass divided by the wire mass is

(A) 2.5 (B) 2

(C) 1 (D) 4

23. A wire has Poisson ratio 1/4 and resistivity that stays constant under strain. Its electrical resistance increases by 0.6% under a small tensile load. The longitudinal strain is

(A) 0.006 (B) $\frac{1}{150}$

(C) 0.002 (D) 0.004

24. For a stretched isotropic wire, first-order volume strain is 0.1% while first-order resistance increase is 0.3%; resistivity stays constant. Its Poisson ratio is

(A) 0.5 (B) 0.25

(C) 0.2 (D) $\frac{1}{3}$

25. A wire carries tensile stress 40 MPa at 20°C between rigid fixed supports. Its Young modulus is 100 GPa and $\alpha = 10^{-5}\,\mathrm{K}^{-1}$. The temperature at which its stress first becomes zero, in °C, is

(A) 80 (B) 20

(C) 40 (D) 60

26. A rod between rigid walls initially has compressive stress 20 MPa. Its $Y = 100$ GPa, $\alpha = 2\times10^{-5}\,\mathrm{K}^{-1}$. The temperature must fall by how many kelvin before its stress becomes tensile 30 MPa?

(A) 15 (B) 50

(C) 10 (D) 25

27. Two equal-length rods in series have equal area, moduli $Y, 2Y$ and expansion coefficients $\alpha, 2\alpha$. Rigid supports fix their total length. On heating by ΔT, the compressive stress divided by $Y\alpha\Delta T$ is

(A) 2 (B) 1

(C) 3 (D) 1.5

28. Two equal-length rods with equal area and modulus Y have coefficients α and 3α. Their ends together span rigid walls. When both are heated by ΔT, the strain of the first rod divided by $\alpha\Delta T$ is

(A) -1 (B) 1

(C) 2 (D) 0

29. A body is scaled geometrically by an unknown factor λ, so its mass becomes λ^3 times the original. Its load-bearing legs have negligible mass, unchanged Young modulus, and radii enlarged by 3. If their compressive strain becomes twice the original, λ is

(A) $3^{\frac{2}{3}}$ (B) 2

(C) 3 (D) $\sqrt[3]{18}$

30. An isotropic material has bulk modulus $K = 2Y/3$. Under uniaxial tensile strain 0.004, its first-order volume strain is

(A) 0.002 (B) 0

(C) 0.004 (D) 0.001

31. A liquid occupies volume V in a rigid vessel and has bulk modulus K. A piston of area A is pushed inward by x, where $Ax \ll V$. The work required for this quasistatic compression divided by KA^2x^2/V is

(A) 0.5 (B) 0.25

(C) 2 (D) 1

32. A liquid of bulk modulus K fills volume V behind a piston of area A. The piston is attached to a spring of constant $k = KA^2/V$. For small inward displacement x, the fraction of total stored energy in the compressed liquid is

(A) $\frac{1}{3}$ (B) 0.25

(C) 0.5 (D) $\frac{2}{3}$

33. A wire of length 2 m, area $10^{-6}\,\mathrm{m}^2$ and density $8000\,\mathrm{kg\,m^{-3}}$ transmits a longitudinal pulse in 0.4 ms. Under a 100 N load, its extension in mm is

(A) 4 (B) 1

(C) 0.5 (D) 2

34. Two wires of the same density have equal longitudinal wave transit times. Their lengths are L and $2L$, areas equal. Under equal loads, the ratio of their extensions, first to second, is

(A) 0.5 (B) 1

(C) 2 (D) 4

35. A load stretches a wire by x at equilibrium. The same mass then oscillates vertically on the wire, treated as an ideal spring of negligible mass. Its period is doubled by changing only the wire length at fixed area and material. The new equilibrium extension divided by x is

(A) 0.5 (B) 0.25

(C) 4 (D) 2

36. A wire under a 10 N end force stores 0.05 J. If a 1 kg body oscillates on the same wire as a spring, its angular frequency squared in s^{-2} is

(A) 200 (B) 1000

(C) 100 (D) 2000

37. An elastic support has constant stiffness k and an effective moving mass m_0. Its period satisfies $T^2 = 4\pi^2(m + m_0)/k$. Loads $m = 1$ kg and $m = 3$ kg give $T^2 = 2\pi^2/5$ and $4\pi^2/5\,\mathrm{s}^2$. The value of m_0, in kg, is

(A) 3 (B) 2

(C) 0.5 (D) 1

38. A model and a geometrically similar giant are made of the same material. The giant is 3 times as tall and carries only its own weight. To keep the same compressive leg strain as the model, the giant leg radius, relative to the model radius, must be at least

(A) $3\sqrt{3}$ (B) $\sqrt{3}$

(C) 9 (D) 3

39. An elastic connector obeys $F = kx + bx^3$, with x in m and F in N. Measurements give $F = 3$ N at $x = 1$ m and $F = 18$ N at $x = 2$ m. The work to stretch it from $x = 1$ m to $x = 2$ m, in J, is

(A) 9 (B) 6

(C) 10 (D) 12

40. A wire stores energy U under tension F. A second wire has the same density, twice the length, twice the radius, and twice the longitudinal sound speed. If the second wire stores $2U$, its tension divided by F is

(A) 4 (B) $2\sqrt{2}$

(C) 8 (D) 2

41–50 / Preserved hard practice

41. A uniform wire of length L, density ρ and Young modulus Y hangs vertically under its own weight. The extension divided by $\rho g L^2/Y$ is

(A) 1.5 (B) -0.5

(C) 0.5 (D) 1

42. A mass m hangs at rest from two identical parallel light elastic wires, each of stiffness k. One wire is suddenly cut without giving the mass an impulse. The other wire stays elastic and damping is negligible. Find its greatest tension divided by mg.

(A) 3 (B) 0.5

(C) 2.5 (D) 1.5

43. A 1 kg mass hangs at rest from a light elastic wire of stiffness 100 N/m. Its support is suddenly lowered by 0.20 m while the mass position is unchanged. The wire can pull but cannot push. Take $g = 10$. Find the greatest subsequent tension divided by the weight.

(A) $1 + \sqrt{3}$ (B) $\sqrt{3} + 2$

(C) $2 + 2\sqrt{3}$ (D) $\sqrt{3}$

44. A 1 kg mass hangs from a light elastic wire of stiffness 100 N/m in a lift initially accelerating upward at 2 m/s^2. At relative equilibrium the lift acceleration abruptly changes to 6 m/s^2 downward. Take $g = 10$; wire cannot push. Find the relative speed, in m/s, when the wire first becomes slack.

(A) $\frac{\sqrt{3}}{5} + \frac{1}{2}$ (B) $\frac{\sqrt{3}}{5}$

(C) $1 - \frac{2\sqrt{3}}{5}$ (D) $\frac{2\sqrt{3}}{5}$

45. A 1 kg mass falls 0.10 m before the first of two parallel light wires becomes taut. Both have stiffness 100 N/m; the second remains slack until the first has stretched by another 0.10 m. Take $g = 10$. Find maximum extension of the first wire, in metres.

(A) $\frac{9}{10} - \frac{\sqrt{6}}{20}$ (B) $\frac{1}{20} + \frac{\sqrt{6}}{40}$

(C) $\frac{1}{10} + \frac{\sqrt{6}}{20}$ (D) $\frac{\sqrt{6}}{40} + \frac{11}{20}$

46. A uniform rod of length L, area A, modulus Y rests on a smooth horizontal surface. Opposite outward forces F at left end and $3F$ at right end accelerate it. Its extension divided by $FL/(AY)$ is

(A) 3 (B) 4

(C) 1 (D) 2

47. Two identical vertical wires of stiffness k support a rigid horizontal light bar at its ends, each initially sharing a central load W equally. If the load is moved to a point one quarter of bar length from the left end, the sum of elastic energies divided by W^2/k is

(A) 1.3125 (B) 0.3125

(C) 0.625 (D) -0.6875

48. A rigid light bar is supported at its ends by two initially equal-length vertical wires. Their stiffnesses are $2k$ on the left and k on the right. The left wire is heated so that its free elongation is $W/(2k)$; the right is unheated. A load W is placed to keep the bar horizontal. Find the load position measured from the left end, as a fraction of bar length.

(A) $\frac{2}{9}$ (B) $\frac{2}{3}$

(C) $\frac{1}{3}$ (D) $\frac{5}{6}$

49. A uniform wire of length L and mass M hangs vertically with an additional mass M at its lower end. Its extension divided by $MgL/(AY)$ is

(A) 3 (B) 0.5

(C) 2.5 (D) 1.5

50. A light linear elastic vertical wire supports a mass m at rest with extension x from its natural length. A second mass m falls from rest through height $3x$ and sticks to the first. The wire impulse during impact is negligible and the wire stays elastic. Maximum wire extension divided by x is

(A) 5 (B) 4

(C) 3 (D) 2

Answer key

Elasticity

P-ELAS / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**B**	21	**B**	31	**A**	41	**C**
02	**D**	12	**C**	22	**B**	32	**C**	42	**D**
03	**A**	13	**C**	23	**D**	33	**B**	43	**A**
04	**C**	14	**B**	24	**B**	34	**C**	44	**D**
05	**D**	15	**A**	25	**D**	35	**C**	45	**C**
06	**A**	16	**B**	26	**D**	36	**B**	46	**D**
07	**D**	17	**A**	27	**A**	37	**D**	47	**B**
08	**C**	18	**B**	28	**A**	38	**A**	48	**B**
09	**B**	19	**C**	29	**D**	39	**A**	49	**D**
10	**C**	20	**A**	30	**A**	40	**A**	50	**B**

Chapter 10

Fluid Mechanics

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Water flows horizontally, density $1000\,\mathrm{kg\,m^{-3}}$, speeds 2 and $4\,\mathrm{m\,s^{-1}}$. Pressure drop from slower to faster point in kPa is

(A) 6 (B) 12

(C) 7 (D) 5

02. A body density is $800\,\mathrm{kg\,m^{-3}}$ and volume $0.01\,\mathrm{m^3}$. Held fully submerged in water ($\rho = 1000, g = 10$), required downward force in newtons is

(A) 21 (B) 19

(C) 20 (D) 40

03. For complete wetting, $T = 0.075\,\mathrm{N\,m^{-1}}$, capillary radius $0.5\,\mathrm{mm}$, $\rho = 1000, g = 10$. Rise in centimetres is

(A) 2 (B) 6

(C) 3 (D) 4

04. A drop divides into eight equal-volume drops. Total new surface area divided by original area is

(A) 1 (B) 4

(C) 3 (D) 2

05. A sphere of density 3ρ falls in a liquid of density ρ; the same sphere falls in a second liquid of density 2ρ and equal viscosity. Terminal-speed ratio first to second is

(A) 4 (B) 3

(C) 1 (D) 2

06. A sphere weighs 12 N in air and 8 N in water. Its density divided by water density is

(A) 6 (B) 2

(C) 3 (D) 4

07. A closed pipe area decreases by factor 3. In steady incompressible flow, kinetic energy per unit mass increases by factor

(A) 18 (B) 9

(C) 8 (D) 10

08. A water jet leaves a hole horizontally at $4\,\mathrm{m\,s^{-1}}$. Hole is 1.25 m above ground, $g = 10$. Horizontal range in metres is

(A) 2 (B) 4

(C) 3 (D) 1

09. In laminar pipe flow, volume flow rate is proportional to r^4 at fixed length, pressure drop and viscosity. Radius increases by 50%. Flow-rate multiplier is

(A) 10.125 (B) 5.0625

(C) 6.0625 (D) 4.0625

10. The Reynolds number is dimensionless. For $\rho = 1000\,\mathrm{kg\,m^{-3}}$, speed $0.1\,\mathrm{m\,s^{-1}}$, tube diameter 0.01 m, viscosity 0.001 Pa s, its value is

(A) 2000 (B) 1000

(C) 1001 (D) 999

11–40 / New middle-to-hard practice

11. Water flows steadily through a pipe rising by 1 m. Area changes from $4A$ at the lower point to A at the upper. Lower speed is $1\,\mathrm{m\,s^{-1}}$. With $\rho = 1000$ and $g = 10$, pressure drop from lower to upper point, in kPa, is

(A) 7.5 (B) 10

(C) 17.5 (D) 25

12. In a horizontal water pipe, area falls from $2A$ to A and static pressure falls by 6 kPa. Taking $\rho = 1000\,\mathrm{kg\,m^{-3}}$, the speed in the wider part, in $\mathrm{m\,s^{-1}}$, is

(A) 4 (B) 1

(C) 2 (D) $2\sqrt{2}$

13. A large water tank has free surface 5 m above the ground. A small hole is 1 m above ground. The hole is moved to another height giving the same horizontal jet range. The new height in m is

(A) 2.5 (B) 2

(C) 3 (D) 4

14. A large tank has free surface 8 m above ground. A hole is chosen to maximize the horizontal jet range. With $g = 10$, the exit speed squared, in $\mathrm{m^2\,s^{-2}}$, at that hole is

(A) 80 (B) 160

(C) 64 (D) 40

15. A homogeneous block floats at an interface with one fourth of its volume in liquid of density 1200 and the rest in liquid of density $800\,\mathrm{kg\,m^{-3}}$. It is then placed in a single liquid of density $1000\,\mathrm{kg\,m^{-3}}$. The fraction of its volume above that liquid is

(A) 0.1 (B) 0.25

(C) 0.2 (D) 0.9

16. A block of density $900\,\mathrm{kg\,m^{-3}}$ floats at the interface of liquids of densities 600 and $1200\,\mathrm{kg\,m^{-3}}$. If the upper liquid is replaced by one of density 750, the new fraction of block volume in the lower liquid is

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$

(C) 0.25 (D) 0.5

17. A block of volume $0.002\,\mathrm{m^3}$ and density $800\,\mathrm{kg\,m^{-3}}$ is held fully submerged in water by a vertical spring attached below. The spring constant is 100 N/m, $g = 10$. Its extension in cm is

(A) 16 (B) 8

(C) 2 (D) 4

18. A body of density $\rho/2$ is fully immersed in a liquid of density ρ and tethered downward by a spring. The equilibrium spring extension is x. The liquid density is increased to $3\rho/2$ while the body remains immersed. The new extension divided by x is

(A) 2 (B) 3

(C) 1.5 (D) 0.5

19. A capillary gives 4 cm rise in a stationary lift. The lift is accelerated upward at $g/2$ and the tube is replaced by one of half the radius. With all wetting properties unchanged, the new rise in cm is

(A) 6 (B) 8

(C) $\frac{8}{3}$ (D) $\frac{16}{3}$

20. A capillary of radius r gives rise h for a perfectly wetting liquid. A tube of radius $2r$ is placed in a lift accelerating downward. It gives the same rise h. The downward acceleration divided by g is

(A) 0.25 (B) 0.5

(C) 1 (D) 0.75

21. Eight identical liquid drops each of radius r coalesce without evaporation. The fraction of their initial total surface energy released is

(A) 0.5 (B) 0.25

(C) 0.75 (D) 0.875

22. Two equal drops coalesce in an ideal device anchored to a fixed support. If all released surface energy is transferred to the kinetic energy of a particle of mass m, what is $mv^2/(8\pi Tr^2)$, where r is each initial radius?

(A) $-1+2^{\frac{2}{3}}$ (B) $1-\frac{2^{\frac{2}{3}}}{2}$

(C) $2-2^{\frac{2}{3}}$ (D) $\sqrt[3]{2}$

23. A soap bubble expands quasistatically from radius r to $2r$. Neglect external atmospheric work by taking the outside pressure as zero. The work required divided by its initial surface energy is

(A) 3 (B) 1

(C) 2 (D) 4

24. A soap bubble of radius r expands to $2r$ in atmosphere of pressure p_0. If $p_0 r = 3T$, the total quasistatic work done by the gas divided by $8\pi Tr^2$ is

(A) 6.5 (B) 10

(C) 3 (D) 3.5

25. Two capillary tubes of equal length and radii $r, 2r$ carry a liquid in laminar flow. They are joined in series under pressure difference Δp. Compared with the flow through the narrower tube alone at the same pressure difference, the rate is multiplied by

(A) $\frac{16}{17}$ (B) 1.0625

(C) 17 (D) $\frac{1}{17}$

26. Two capillaries of lengths $L, 2L$ and radii $r, 2r$ are connected in parallel across the same pressure difference. The wider tube carries what fraction of the total volume rate?

(A) $\frac{16}{17}$ (B) $\frac{2}{3}$

(C) $\frac{8}{9}$ (D) $\frac{1}{9}$

27. A capillary carries laminar volume rate Q under pressure difference Δp. Its radius is doubled and length tripled. To keep hydraulic power ΔpQ unchanged, the new pressure difference divided by the old is

(A) 0.75 (B) 0.1875

(C) 0.25 (D) $\frac{\sqrt{3}}{4}$

28. Two laminar-flow capillaries contain the same liquid. The second has twice the pressure gradient, while its radius is chosen so that the Reynolds numbers based on mean speed and diameter are equal. The ratio of second to first volume flow rate is

(A) 0.5 (B) $\frac{2^{\frac{2}{3}}}{4}$

(C) $\frac{2^{\frac{2}{3}}}{2}$ (D) $\sqrt[3]{2}$

29. A sphere settles at speed v in liquid of density ρ and at $v/3$ in liquid of density 2ρ; both viscosities are η. A sphere of the same material must settle at v in a third liquid of density $\rho/2$ and viscosity 2η. Its radius divided by the first sphere radius is

(A) 2 (B) $\frac{\sqrt{6}}{2}$

(C) $\frac{\sqrt{6}}{3}$ (D) $\sqrt{2}$

30. A small sphere of density 3ρ is released in liquid of density ρ and has terminal speed $2\,\mathrm{m\,s^{-1}}$. Its speed follows $v = v_T(1 - e^{-t/\tau})$ under Stokes drag. With $g = 10$, its displacement before reaching half terminal speed, in m, is

(A) $\frac{3\log(2)}{5}$ (B) $\frac{3\log(2)}{10}$

(C) 0.3 (D) $-\frac{3}{10} + \frac{3\log(2)}{5}$

31. A liquid has $v_x(y) = k(2y^2/a^2 - y^3/a^3)$ and viscosity 2η. At $y = a$, shear stress is measured as 6 Pa. If $k/a = 2\,\mathrm{s^{-1}}$, the value of η in Pa s is

(A) 6 (B) 3

(C) 1.5 (D) 0.75

32. A fluid velocity profile is $v(y) = ay - by^2$. The speed is maximal at $y = 2$ mm and the wall shear stress at $y = 0$ is 4 Pa. If viscosity is 0.01 Pa s, the maximum speed in $\mathrm{m\,s^{-1}}$ is

(A) 0.4 (B) 0.8

(C) 0.1 (D) 0.2

33. A soap film with two surfaces spans a movable wire of length 0.10 m. Moving the wire quasistatically by 2 mm requires $40\,\mu$J of work. A mass hung from that wire could be balanced by the film at $g = 10$; that mass in g is

(A) 10 (B) 4

(C) 2 (D) 1

34. A soap film pulls a wire of length l against a spring of constant k. The equilibrium extension is x. A second film of the same surface tension uses wire length $2l$ and spring constant $3k$. Its stored spring energy divided by the first is

(A) $\frac{2}{3}$ (B) $\frac{4}{9}$

(C) 2 (D) $\frac{4}{3}$

35. A rectangular tank of length 2 m is accelerated horizontally at $3\,\mathrm{m\,s^{-2}}$. Water depth was initially 0.5 m and no water spills. With $g = 10$, the maximum water depth in m after equilibrium is

(A) 1.1 (B) 0.8

(C) 0.6 (D) 1

36. A tank 4 m long contains water of uniform depth 0.5 m. It accelerates horizontally until the free surface just touches the bottom at one end, without spilling. With $g = 10$, the acceleration in $\mathrm{m\,s^{-2}}$ is

(A) 5 (B) 2.5

(C) 10 (D) 1.25

37. A U-tube of uniform area contains water. A 10 cm column of oil of density $800\,\mathrm{kg\,m^{-3}}$ is poured into one arm. The water level in the other arm rises by how many cm from its initial level?

(A) 8 (B) 4

(C) 2 (D) 5

38. A U-tube has arm areas A and $2A$. It contains water, initially level. A 12 cm column of oil of density $750\,\mathrm{kg\,m^{-3}}$ is added to the narrower arm. The water level in the wider arm rises by how many cm?

(A) 6 (B) 4

(C) 9 (D) 3

39. A thin vertical tube of length 1 m is closed at its upper end. Initially it contains air at 10 kPa and is open to that atmosphere below. Its upper end is then held at a water surface while its lower end is submerged. Assume isothermal air and neglect capillarity. With $\rho = 1000$, $g = 10$, the length of water that enters the tube, in m, is

(A) $\frac{3}{2} - \frac{\sqrt{5}}{2}$ (B) $-\frac{1}{2} + \frac{\sqrt{5}}{2}$

(C) 0.5 (D) $2 - \sqrt{3}$

40. A 1 m vertical tube, closed at the top, initially contains air at atmospheric pressure 10^5 Pa. It is submerged open end downward. The trapped air contracts isothermally to length 0.5 m. With water density 1000 and $g = 10$, the depth of the closed top below the water surface, in m, is

(A) 10.5 (B) 9.5

(C) 9 (D) 10

41–50 / Preserved hard practice

41. A tank free surface is at height H above its wall base. Ground beyond the wall rises at 45° from that base. A small hole at depth h sends a horizontal jet. Ignoring losses, find h/H for greatest horizontal range before the jet meets the inclined ground.

(A) 0.625 (B) 0.75

(C) 0.25 (D) 0.125

42. A cube of side a and density half a liquid density is released from rest with its centre at depth a below the liquid surface. Its faces stay horizontal. Neglect drag and fluid inertia. Find the greatest height of its bottom above the liquid surface, divided by a.

(A) $\frac{1}{6}$ (B) 0.75

(C) 0.25 (D) 0.5

43. A cube of side a and density $3\rho/2$ rests at an interface between deep liquids of densities ρ above and 2ρ below. It is fully covered by liquid and has faces horizontal. Ignoring drag and added-fluid inertia, find small vertical angular frequency squared divided by g/a.

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{5}{6}$ (D) $\frac{2}{9}$

44. A conducting liquid drop of radius R, charge Q and surface tension γ splits into eight identical drops with equal charges, which separate to infinity. Ignore other energy channels. Its electrostatic self-energy is $Q^2/(8\pi\varepsilon_0 R)$. Find the minimum $Q^2/(\pi^2\varepsilon_0\gamma R^3)$ for splitting to be energetically allowed.

(A) $\frac{131}{3}$ (B) $\frac{125}{3}$

(C) $\frac{256}{3}$ (D) $\frac{128}{3}$

45. Eight identical liquid drops of radius r, density ρ and surface tension γ coalesce into one spherical drop. All released surface energy heats the liquid of specific heat c. Find $\rho cr\,\Delta T/\gamma$.

(A) 2.5 (B) 1.5

(C) 0.5 (D) 3

46. A sphere of mass 0.50 kg falls in a viscous liquid at terminal speed 3 m/s. Pulling it upward at steady 2 m/s requires a vertical upward force of 5 N. Drag is linear and direction-opposing, with the same coefficient both ways. Take $g = 10$. Find sphere density divided by liquid density.

(A) 2.5 (B) 3.5

(C) 5 (D) 1.5

47. A container of liquid density ρ accelerates vertically upward at g. Its pressure field has settled relative to the container. A small body of density $3\rho/2$ is released from rest relative to the liquid. Neglect drag and added-fluid inertia. Find its initial laboratory acceleration divided by g, taking upward as positive.

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{1}{9}$ (D) $\frac{1}{6}$

48. Water of density 1000 kg/m^3 flows steadily from area $2A$ to area A, 0.20 m higher. The pressure taps connect to a differential U-tube containing a liquid of density 14000 kg/m^3. Its interface under the lower tap is 0.030 m lower than the other interface. Take $g = 10$, ignore losses. Find the speed squared in the wider pipe, in m^2/s^2.

(A) 5.2 (B) 1.6

(C) 3.6 (D) 2.6

49. A wooden block of density $3\rho/4$ floats in a liquid of density ρ. A metal load of density 3ρ hangs below it, fully immersed. When the block is just fully submerged, find load mass divided by block mass.

(A) $\frac{1}{6}$ (B) 0.5

(C) 0.25 (D) 0.75

50. A coherent water jet of speed v, cross-section A, and density ρ strikes a freely moving plate of mass M, initially at rest. The plate moves on frictionless horizontal rails along the jet; water leaves sideways with the plate's instantaneous axial speed, without remaining on it. When plate speed first becomes $v/2$, its displacement divided by $M/(\rho A)$ is

(A) 0.5 (B) $\log(2)$

(C) $1 - \log(2)$ (D) $-1 + 2\log(2)$

Answer key

Fluid Mechanics

P-FLME / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**C**	21	**A**	31	**C**	41	**C**
02	**C**	12	**C**	22	**C**	32	**A**	42	**D**
03	**C**	13	**D**	23	**A**	33	**C**	43	**B**
04	**D**	14	**A**	24	**A**	34	**D**	44	**D**
05	**D**	15	**A**	25	**A**	35	**B**	45	**B**
06	**C**	16	**B**	26	**C**	36	**B**	46	**A**
07	**B**	17	**D**	27	**D**	37	**B**	47	**A**
08	**A**	18	**A**	28	**C**	38	**D**	48	**D**
09	**B**	19	**D**	29	**B**	39	**A**	49	**B**
10	**B**	20	**B**	30	**D**	40	**B**	50	**C**

Chapter 11

Thermal Properties

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Water masses 1 and 3 kg at 80 and 20°C are mixed adiabatically. Final Celsius temperature is

(A) 36 (B) 70

(C) 34 (D) 35

02. A 100 W heater warms 0.5 kg water, $c = 4000\,\mathrm{J\,kg^{-1}\,K^{-1}}$, with no losses. Time for 10 K rise in seconds is

(A) 200 (B) 199

(C) 201 (D) 400

03. Two equal-thickness slabs of equal area with conductivities k and $3k$ are in series. Equivalent conductivity divided by k is

(A) 2.5 (B) 1.5

(C) 0.5 (D) 3

04. Two equal-thickness slabs with equal areas and conductivities $k, 3k$ conduct side by side. Effective conductivity for combined area divided by k is

(A) 1 (B) 3

(C) 2 (D) 4

05. A 1 m rod lengthens by 0.4 mm for 20 K rise. Expansion coefficient in units of $10^{-5}\,\mathrm{K^{-1}}$ is

(A) 4 (B) 1

(C) 2 (D) 3

06. A body cools from 80 to 60°C in a room at 20°C. In Newton cooling, ratio of final to initial instantaneous cooling-rate magnitudes is

(A) $\frac{-1}{3}$ (B) $\frac{4}{3}$

(C) $\frac{5}{3}$ (D) $\frac{2}{3}$

07. For Newton cooling, temperature excess drops to half in 10 min. Time to drop to one eighth of initial excess in minutes is

(A) 30 (B) 60

(C) 31 (D) 29

08. A vessel contains 1 kg water at 20°C with $c = 4000\,\mathrm{J\,kg^{-1}\,K^{-1}}$. Vessel heat capacity is 1000 J/K. Heat for 10 K rise in kJ is

(A) 51 (B) 100

(C) 49 (D) 50

09. A black surface at temperature T faces surroundings at $T/2$. Net radiated power divided by σAT^4 is

(A) 0.9375 (B) 1.875

(C) −0.0625 (D) 1.9375

10. A steel ruler expands uniformly by 0.1%. A fixed true length measured using its new scale has a numerical reading that

(A) Increases (B) Decreases

(C) Remains unchanged (D) Must double

11–40 / New middle-to-hard practice

11. A vessel has unknown heat capacity C and holds a liquid of unknown specific heat c. Raising 0.5 kg of liquid and the vessel by 10 K takes 25 kJ; using 1.0 kg takes 45 kJ. Raising 0.25 kg and the same vessel by 20 K takes how many kJ?

(A) 30 (B) 35

(C) 25 (D) 20

12. In a calorimeter initially at 20°C, mixing 1 kg water at 80°C with 1 kg water at 20°C gives 45°C. Take $c_w = 4000$ J/kg K. Heat needed to warm this calorimeter and 0.5 kg water by 10 K, in kJ, is

(A) 20 (B) 36

(C) 40 (D) 28

13. In an insulated vessel, 0.20 kg water at 40°C meets 0.15 kg ice at 0°C. Use $c_w = 4000$ J/kg K and $L = 320$ kJ/kg. The mass of ice remaining after equilibrium, in g, is

(A) 100 (B) 50

(C) 0 (D) 150

14. 0.20 kg water at 30°C is mixed adiabatically with 0.10 kg ice at −20°C. Use $c_w = 4200$, $c_i = 2100$ J/kg K and $L = 336$ kJ/kg. Ice remaining at equilibrium, in g, is

(A) 50 (B) 25

(C) 37.5 (D) 0

15. In an insulated vessel, 0.10 kg steam at 100°C is mixed with 1.0 kg water at 20°C. Take $L = 2.4$ MJ/kg and $c_w = 4000$ J/kg K. The final temperature in °C is

(A) 80 (B) 100

(C) 90 (D) $\frac{900}{11}$

16. An insulated container at fixed atmospheric pressure receives 0.20 kg steam at 100°C into 1.0 kg water at 20°C. With $L = 2.4$ MJ/kg and $c_w = 4000$ J/kg K, what fraction of the added steam remains uncondensed at equilibrium? All material remains inside a freely moving piston.

(A) 0 (B) 0.5

(C) $\frac{1}{3}$ (D) $\frac{2}{3}$

17. Three insulated-sided rods join at a negligible-capacity junction. Their conductances are k, k, $2k$ W/K, with other ends fixed at 100, 0, 20°C respectively. The steady junction temperature, in °C, is

(A) 25 (B) 40

(C) 35 (D) 50

18. A junction receives 20 W from a heater. It connects to reservoirs at 0°C and 20°C through thermal resistances 2 and 4 K/W. Its steady temperature, in °C, is

(A) 20 (B) $\frac{100}{3}$

(C) 40 (D) $\frac{80}{3}$

19. A cylinder of radius R and conductivity k is surrounded by a shell of outer radius $2R$ and unknown conductivity k_s. Their common length is L. Axial effective conductivity is $3k/2$. The core and shell are then separated and joined in series, retaining lengths and areas. Heat rate relative to the core alone at the same end temperatures is

(A) $\frac{2}{3}$ (B) $\frac{5}{6}$

(C) 0.6 (D) 0.5

20. Two rods initially carry equal heat rates at the same end temperatures. The second rod is replaced by the same material at half its original length and twice its original radius. It is joined in series with the unchanged first rod. The series heat rate divided by the original first-rod rate is

(A) 0.8 (B) $\frac{1}{9}$

(C) 0.5 (D) $\frac{8}{9}$

21. A fixed black body radiates net power P at 400 K and $16P/3$ at 600 K in the same surroundings. The surroundings temperature in K is

(A) 300 (B) 0

(C) 400 (D) 200

22. A black sphere has temperature T and surroundings at $T/2$. It is replaced by a sphere of half its radius whose emission peak wavelength is half that of the first. The ratio of new to old net radiation power is

(A) 16 (B) 4.25

(C) 3.75 (D) 4

23. A body in a 20°C room cools from 80 to 50°C in 10 min under Newton cooling. A fixed heater would maintain it at 80°C. When that heater is turned on while the body is at 50°C, its initial heating rate, in K/min, is

(A) $\log(8)$ (B) $\frac{3\log(2)}{2}$

(C) $\log(64)$ (D) $\log(2)$

24. A body cools from 60 to 40°C in 5 min in a 20°C room. A 100 W heater maintains it at 40°C. In the Newton cooling model, its heat capacity in J/K is

(A) $\frac{300}{\log(2)}$ (B) $\frac{3000}{\log(2)}$

(C) $\frac{1500}{\log(2)}$ (D) 1500

25. A body has heat capacity $C(T) = aT$, with T absolute temperature and $a > 0$. Heat Q raises it from T_0 to $2T_0$. The same additional heat raises it from $2T_0$ to final temperature T_f. The ratio T_f/T_0 is

(A) $\sqrt{7}$ (B) $\sqrt{6}$

(C) $2\sqrt{2}$ (D) 3

26. A body has heat capacity $C(\theta) = a + b\theta$ J/K, where θ is in °C. Heating from 0 to 10°C takes 150 J; from 10 to 20°C takes 250 J. Adding 150 J while initially at 20°C gives a final temperature in °C of

(A) $-10 + 20\sqrt{3}$ (B) $10\sqrt{3}$

(C) $-10 + 20\sqrt{2}$ (D) 25

27. A rod is measured using a ruler whose linear expansion coefficient is 2×10^{-5} K^{-1}. Both are initially calibrated at 20°C, reading 50.00 cm. After both warm by 100 K, the reading is 50.02 cm. To first order, the rod expansion coefficient in units 10^{-5} K^{-1} is

(A) 1.6 (B) 2.4

(C) 4 (D) 2

28. Two rulers have expansion coefficients α and 2α. When they and a rod are heated by 100 K, their reported rod-length changes are +0.2% and −0.1% respectively. To first order, rod coefficient in units 10^{-5} K^{-1} is

(A) 2 (B) 5

(C) 6 (D) 3

29. At one temperature, a rod radius is 20.02 mm and a ring-hole radius 20.00 mm. Their linear coefficients are 10^{-5} and 4×10^{-5} K^{-1}. Neglect products of the initial fractional gap and thermal strain. The common temperature rise needed for first fit, in K, is

(A) 50 (B) $\frac{200}{3}$

(C) $\frac{100}{3}$ (D) 25

30. Two rods of lengths L and $2L$ point toward each other across a gap a. Their coefficients are α and 2α. They are heated together until the gap first closes. The first rod extension divided by a is

(A) $\frac{1}{3}$ (B) 0.4

(C) 0.2 (D) 0.5

31. A vessel initially full of a liquid overflows by 2% of its initial volume after heating by 100 K. Its volume expansion coefficient is 10^{-4} K^{-1}. The same liquid in a second initially full vessel of coefficient 2×10^{-4} K^{-1} is heated by 100 K. To first order, overflow as a percentage of initial volume is

(A) 1 (B) 3

(C) 2 (D) 0

32. A vessel capacity rises by 1% on heating, while its contained liquid volume would rise by 4%. After heating, overflow equals 1% of the initial vessel capacity. The initial filled fraction of the vessel is

(A) $\frac{25}{26}$ (B) $\frac{51}{52}$

(C) 0.99 (D) 1

33. An electrical heater at fixed voltage has $R = R_0(1 + \alpha\Delta T)$ and heats a body of constant heat capacity without loss. Time to rise by $2D$ is $5/2$ times the time to rise by D. The time for a rise $3D$ divided by the time for rise D is

(A) 5 (B) 3

(C) 4.5 (D) 4

34. A heater at fixed current has $R = R_0(1 + \alpha\Delta T)$ and heats a constant-capacity body with no loss. The time to rise by $3D$ is twice the time to rise by D. With $\alpha > 0$, the resistance after rise D divided by R_0 is

(A) 2 (B) 4

(C) 3 (D) 1.5

35. A solid sphere of radius r and a hollow sphere of outer radius $2r$, inner radius r, have thermally insulated inner cavities where present and the same material and surface heat-transfer coefficient. Under Newton cooling from the same initial excess temperature, the ratio of their excess half-times, hollow to solid, is

(A) 1.75 (B) 2

(C) $\frac{4}{7}$ (D) 1.5

36. A solid sphere of radius r and a hollow sphere of outer radius $2r$ have equal Newton-cooling time constants. Both use the same material and external surface heat-transfer coefficient; the inner cavity is insulated. The hollow inner radius divided by r is

(A) $2^{\frac{2}{3}}$ (B) 2

(C) $\sqrt[3]{2}$ (D) $\sqrt{2}$

37. A heater warms a solid to its melting point through 30 K in 60 s, then completely melts it in 100 s. Its power and heat-loss rate stay constant in both stages. The ratio of latent heat per kg to specific heat, in K, is

(A) 100 (B) 30

(C) 60 (D) 50

38. A 0.20 kg solid of specific heat 1000 J/kg K is heated by 100 W in a 20°C room. At its 60°C melting point its temperature is rising at 0.30 K/s just before melting. Complete melting takes 100 s with unchanged losses. The latent heat in kJ/kg is

(A) 20 (B) 40

(C) 50 (D) 30

39. A grey radiator of area A has steady temperature $2T$ in surroundings at T under a fixed heater. Its area is doubled while heater power and emissivity remain fixed. The new steady absolute temperature divided by T is

(A) $\frac{\sqrt[4]{17}\cdot 2^{\frac{3}{4}}}{2}$ (B) $\sqrt[4]{17}$

(C) $\sqrt{2}$ (D) 2

40. Two spherical planets have uniform surface temperature and negligible internal heat. The second is twice as far from the same star, absorbs 3/4 of incident short-wave radiation and has long-wave emissivity 1/2. The first absorbs and emits as a black body. The ratio of their temperatures is

(A) $\frac{\sqrt{2}\sqrt[4]{3}}{2}$ (B) $\frac{\sqrt{2}}{2}$

(C) $\frac{2^{\frac{3}{4}}\sqrt[4]{3}}{2}$ (D) $\frac{\sqrt[4]{6}}{2}$

41–50 / Preserved hard practice

41. Two rods of conductance 1 and 2 W/K connect a junction to 100°C and 0°C respectively. A resistor R at the junction is supplied by a 20 V emf with internal resistance 5 Ω. All resistor heat enters the junction, which reaches 40°C. Find R in ohms.

(A) 10 (B) 5

(C) 6 (D) 4

42. A wire supports a fixed-mass elastic oscillator. On heating by small ΔT, its isotropic linear expansion coefficient is α, and its Young modulus decreases by fraction $3\alpha\Delta T$. Find the first-order fractional period increase divided by $\alpha\Delta T$.

(A) 2 (B) 0

(C) 0.5 (D) 1

43. A thin conducting ring initially has radius R and linear expansion coefficient α. A concentric disc of radius $0.999R$ has coefficient 3α. Ring material resistivity has temperature coefficient 3α; expansion is isotropic. To first order, find the fractional increase of ring electrical resistance when the disc first touches it.

(A) $\frac{1999}{3994}$ (B) $\frac{1995}{1997}$

(C) $\frac{2}{1997}$ (D) $\frac{1}{1997}$

44. A heater has resistance $R = 4(1 + 0.10\theta)\,\Omega$, where θ is its excess temperature in kelvin. It is connected to an ideal 20 V supply and loses heat at 5θ W. Find its positive steady excess temperature, in kelvin.

(A) 10 (B) 9

(C) 11 (D) 20

45. A heater has $R = 4(1+0.10\theta)\,\Omega$ and is supplied at 20 V. Its heat capacity C and heat-loss coefficient G in $G\theta$ W are unknown constants. Warming rates are 1 K/s at $\theta = 0$ and 1/4 K/s at $\theta = 10$ K. Find its eventual steady excess temperature, in kelvin.

(A) $-4 + 5\sqrt{17}$ (B) $-6 + 5\sqrt{17}$

(C) $-5 + 5\sqrt{17}$ (D) $-10 + 10\sqrt{17}$

46. A 2 kg block is projected up an incline at 10 m/s. The incline has $\sin\theta = 3/5$, $\cos\theta = 4/5$, and kinetic coefficient 1/4. All frictional heat enters the block, of specific heat 100 J/(kg K). Take $g = 10$. Find its temperature rise by the first stop, in kelvin.

(A) 0.5625 (B) 0.0625

(C) 0.875 (D) 0.125

47. A 1 kg copper block at 100°C is placed with 0.10 kg of ice at −10°C in an insulated container. Take $c_{Cu} = 400$, $c_{ice} = 2000$, $c_{water} = 4000$ J/(kg K), and $L_f = 3.2 \times 10^5$ J/kg. Find the final temperature in Celsius.

(A) 8.5 (B) 7.5

(C) 15 (D) 6.5

48. Two rigid spherical black containers, with negligible wall heat capacity, contain monatomic ideal gas of N and $8N$ molecules. Their radii are R and $2R$ and initial temperatures T and $2T$. Each cools only by radiation into negligible-temperature surroundings, with rapid gas-wall equilibration. Find the cooling time for the larger container divided by that for the smaller container, each cooling to half its own initial temperature.

(A) 0.625 (B) 0.75

(C) 0.125 (D) 0.25

49. A $4\,\Omega$ junction heater is connected to a 12 V emf with internal resistance $2\,\Omega$. Its heat escapes by two paths: a rod of conductance 1 W/K to 0°C, and two rods each 4 W/K in series to 100°C. All heater power enters the junction. Find steady junction temperature in Celsius.

(A) 144 (B) 71

(C) 72 (D) 73

50. A uniform wire is held at fixed length. It has Young's modulus Y, thermal expansion coefficient α, constant heat capacity C, and electrical resistance $R_0(1 + \beta\Delta T)$. An ideal source maintains voltage V. Ignore heat loss and assume all stated linear laws remain valid. Time to reach compressive stress $Y\alpha/\beta$, divided by $CR_0/(\beta V^2)$, is

(A) 0.5 (B) 1.5

(C) 2 (D) 1

Answer key

Thermal Properties

P-THPR / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**A**	21	**D**	31	**A**	41	**B**
02	**A**	12	**B**	22	**B**	32	**B**	42	**D**
03	**B**	13	**B**	23	**A**	33	**C**	43	**C**
04	**C**	14	**C**	24	**C**	34	**A**	44	**A**
05	**C**	15	**D**	25	**A**	35	**A**	45	**C**
06	**D**	16	**C**	26	**A**	36	**A**	46	**D**
07	**A**	17	**C**	27	**B**	37	**D**	47	**B**
08	**D**	18	**B**	28	**B**	38	**D**	48	**D**
09	**A**	19	**B**	29	**C**	39	**A**	49	**C**
10	**B**	20	**D**	30	**C**	40	**D**	50	**B**

Chapter 12

Thermodynamics

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Two moles of an ideal gas expand isothermally at 300 K from V to $2V$. Taking $R = 8$ and $\ln 2 = 0.7$, work in joules is

(A) 3361 (B) 3359

(C) 3360 (D) 6720

02. A Carnot engine operates between 500 and 300 K. To double efficiency while keeping cold reservoir fixed, new hot temperature in kelvin is

(A) 1501 (B) 1500

(C) 1499 (D) 3000

03. A gas follows a straight line from (V, P) to $(3V, 3P)$ on a pressure-volume graph. Work divided by PV is

(A) 3 (B) 5

(C) 8 (D) 4

04. A gas goes from $(V, 3P)$ to $(3V, P)$ on a straight line in a P-V graph. Work divided by PV is

(A) 5 (B) 8

(C) 4 (D) 3

05. A gas expands at pressure $3P$ from V to $2V$ and returns at pressure P. Clockwise rectangular-cycle net work divided by PV is

(A) 3 (B) 2

(C) 1 (D) 4

06. For an ideal gas with $\gamma = 3/2$, reversible adiabatic expansion quadruples volume. Temperature becomes what fraction of initial?

(A) -0.5 (B) 0.5

(C) 1.5 (D) 1

07. A monatomic ideal gas undergoes an isobaric process. Fraction of absorbed heat converted to work is

(A) 1.4 (B) -0.6

(C) 0.8 (D) 0.4

08. An ideal diatomic gas with active rotation is heated at constant pressure. Fraction of absorbed heat increasing internal energy is

(A) $\frac{5}{7}$ (B) $\frac{12}{7}$

(C) $\frac{10}{7}$ (D) $\frac{-2}{7}$

09. A heat engine has efficiency 25% and rejects 600 J each cycle. Heat absorbed per cycle in joules is

(A) 799 (B) 1600

(C) 801 (D) 800

10. A gas absorbs 50 J along a path where its internal energy decreases 20 J. Work done by gas in joules is

(A) 140 (B) 69

(C) 70 (D) 71

11–40 / New middle-to-hard practice

11. A Carnot engine initially has efficiency 1/5. Lowering its cold reservoir by 80 K doubles the efficiency, with hot reservoir fixed. At the original temperatures it rejects 480 J per cycle. Its original work per cycle, in J, is

(A) 120 (B) 160

(C) 240 (D) 80

12. A Carnot engine has initial efficiency 1/4. Raising its source by 100 K while lowering its sink by 50 K changes the efficiency to 1/2. The initial source-sink temperature difference, in K, is

(A) 300 (B) 200

(C) 50 (D) 100

13. Two Carnot engines operate in cascade between 600 K, an intermediate reservoir at T, and 150 K. Heat rejected by the first is all absorbed by the second. If they deliver equal work, T in K is

(A) 300 (B) 225

(C) 375 (D) 450

14. Two Carnot engines operate in cascade from 900 K to an intermediate reservoir and then to 100 K. Their efficiencies are equal and all first-stage rejected heat enters the second stage. The ratio of first-stage work to second-stage work is

(A) 9 (B) 3

(C) 2 (D) 1

15. A Carnot engine between 600 and 300 K absorbs 1200 J per cycle. All its work drives a Carnot refrigerator between 250 and the same 300 K reservoir. Total heat delivered to the 300 K reservoir per engine cycle, in J, is

(A) 4800 (B) 3600

(C) 4200 (D) 3000

16. A Carnot engine between 800 and 400 K absorbs 1000 J. Its work drives a reversible heat pump between 300 and that same 400 K reservoir. The total heat received by the 400 K reservoir, in J, is

(A) 1500 (B) 3000

(C) 2000 (D) 2500

17. Two bodies have equal constant heat capacities C and temperatures 900 K and 100 K. A reversible engine extracts the maximum possible work while they approach a common temperature. The maximum work divided by $100C$, where C is in J/K, is

(A) 4 (B) 2

(C) 3 (D) 6

18. Two bodies have constant heat capacities C and $2C$ and temperatures 800 K and 100 K respectively. A reversible engine runs until they reach a common temperature. Maximum work divided by $100C$ is

(A) 2 (B) 4

(C) 5 (D) 3

19. A monatomic ideal gas runs the clockwise rectangle $A(V,P) \to B(V,3P) \to C(2V,3P) \to D(2V,P) \to A$. Its thermal efficiency is

(A) $\frac{2}{9}$ (B) $\frac{1}{3}$

(C) $\frac{4}{21}$ (D) $\frac{4}{15}$

20. A monatomic ideal gas runs $A(V,P) \to B(V,3P) \to C(3V,P) \to A$, with BC a straight segment and CA at pressure P. Heat absorbed means only positive heat, including subintervals of BC. The efficiency is

(A) $\frac{2}{7}$ (B) $\frac{1}{3}$

(C) $\frac{2}{9}$ (D) $\frac{4}{15}$

21. A monatomic ideal gas expands quasistatically along $P \propto V$. The fraction of absorbed heat converted to work is

(A) 0.4 (B) $\frac{1}{3}$

(C) 0.25 (D) 0.5

22. An ideal gas follows $p \propto T^4$ during a reversible adiabatic change. Two moles of the same gas are then heated at fixed volume from 300 to 400 K. With $R = 8$ J/mol K, required heat in J is

(A) 4800 (B) 2400

(C) 6400 (D) 3200

23. A monatomic ideal gas at temperature T_0 expands reversibly adiabatically from V_0 to $8V_0$. It is then heated at its new constant pressure until temperature returns to T_0. Total work done divided by nRT_0 is

(A) 1.25 (B) 1.875

(C) 0.75 (D) 1.125

24. An ideal gas has $\gamma = 3/2$. It expands reversibly adiabatically to 4 times initial volume, then is heated at fixed volume until pressure returns to its initial value. Heat in the fixed-volume step divided by initial internal energy is

(A) 4 (B) 2

(C) 3.5 (D) 1.5

25. A monatomic ideal gas initially at (P_0, V_0, T_0) expands adiabatically against a suddenly fixed external pressure $P_0/2$. After motion damps, final pressure equals external pressure and no energy escapes except pressure work. Final T/T_0 is

(A) 0.8 (B) 0.5

(C) $\frac{2^{\frac{3}{5}}}{2}$ (D) 0.75

26. An ideal gas has $C_V = 2R$ per mole. It starts at (P_0, V_0) and expands adiabatically against constant external pressure $P_0/4$ until final equilibrium, with damping retained internally. Final volume divided by V_0 is

(A) 3 (B) $2\sqrt[3]{2}$

(C) 2 (D) 4

27. A monatomic gas supports a weightless piston against a spring whose force is proportional to piston height from the cylinder bottom. Volume is proportional to that height. It is cooled quasistatically to half its initial volume. Heat released divided by initial P_0V_0 is

(A) 1.5 (B) 0.75

(C) 1.125 (D) 1.875

28. A monatomic gas under a piston has initial pressure P_0, volume V_0. A constant external load gives P_0 and an initially unstretched spring adds pressure $k_V(V - V_0)$ with $k_V V_0 = P_0$. It expands slowly to $2V_0$. Heat absorbed divided by P_0V_0 is

(A) 3 (B) 7.5

(C) 4.5 (D) 6

29. Two equal rigid chambers contain equal moles of ideal gases, one monatomic at 400 K and one diatomic at 200 K. Rotation is active and vibration frozen. They are thermally connected inside an insulated enclosure. Final pressure in the first chamber divided by its initial pressure is

(A) 0.625 (B) 0.6875

(C) 0.875 (D) 0.75

30. One mole of monatomic gas at 600 K and one mole of diatomic gas at 200 K occupy fixed-volume chambers connected thermally inside an insulated enclosure. Rotation is active and vibration frozen. Heat transferred from the monatomic gas divided by R, in K, is

(A) 500 (B) 375

(C) 300 (D) 450

31. A monatomic gas goes from (P, V) to $(2P, 2V)$. Path 1 first heats at volume V, then expands at $2P$. Path 2 first expands at P, then heats at volume $2V$. The ratio of absorbed heats Q_1/Q_2 is

(A) 2 (B) 1.5

(C) $\frac{11}{13}$ (D) $\frac{13}{11}$

32. Two quasistatic paths join (V_0, P_0) and $(2V_0, 4P_0)$. Path 1 follows $P = P_0(V/V_0)^2$; path 2 is straight between the endpoints. Heat absorbed along path 2 minus that along path 1, divided by P_0V_0, is

(A) $\frac{1}{3}$ (B) $\frac{1}{6}$

(C) $\frac{7}{3}$ (D) $-\frac{1}{6}$

33. A monatomic ideal gas fills volume V_0 beside an evacuated chamber of volume $2V_0$, both inside an insulated rigid container. The partition is removed. The gas is then heated at fixed total volume until its original pressure returns. Heat added divided by initial internal energy is

(A) 1.5 (B) 3

(C) 2 (D) 1

34. An ideal gas initially at (P_0, V_0, T_0) expands freely into an insulated evacuated chamber of volume $3V_0$. It is then compressed reversibly isothermally back to V_0. Net heat absorbed over both steps divided by nRT_0 is

(A) $\log(4)$ (B) 0

(C) $-\log(2)$ (D) $-\log(4)$

35. A monatomic ideal gas in a rigid vessel initially has $PV = 200$ J. A $10\,\Omega$ heater carries 2 A for 10 s without heat loss. Final pressure divided by initial pressure is

(A) $\frac{7}{3}$ (B) 2

(C) $\frac{5}{3}$ (D) 3

36. An ideal gas absorbs 600 J while expanding at constant pressure and doing 200 J work. It then expands reversibly adiabatically to four times its volume. The temperature ratio in this adiabatic step is

(A) 0.25 (B) 0.125

(C) 0.5 (D) $\frac{\sqrt{2}}{2}$

37. An ideal gas follows $P = P_0(3 - V/V_0)$ for $V_0 \le V \le 2V_0$. Its maximum temperature divided by its temperature at $V = V_0$ is

(A) $\frac{4}{3}$ (B) 1

(C) 1.5 (D) 1.125

38. A monatomic ideal gas follows $P = P_0(5 - V/V_0)$ as it expands from V_0 to $4V_0$. At what V/V_0 does heat flow change from absorption to rejection?

(A) 3 (B) 3.5

(C) 2.5 (D) 3.125

39. An ideal gas has adiabatic sound-speed squared $5.0 \times 10^4\,\mathrm{m^2\,s^{-2}}$ at pressure 10^5 Pa and density $3\,\mathrm{kg\,m^{-3}}$. Two moles at 300 K have internal energy, taking $R = 8$ J/mol K, in J equal to

(A) 7200 (B) 9600

(C) 14400 (D) 12000

40. For an ideal gas at one state, adiabatic sound speed is $\sqrt{3/2}$ times isothermal sound speed. It is compressed reversibly adiabatically to one fourth its volume. Work done on the gas divided by its initial internal energy is

(A) 1 (B) 0.5

(C) 2 (D) 3

41–50 / Preserved hard practice

41. A monatomic ideal gas expands quasistatically from (P_0, V_0) to volume $2V_0$, following $P = KV^n$ with constant K and unknown $n \neq -1$. During this expansion, heat absorbed is four times the work done by the gas. It is then cooled at fixed volume to its initial temperature. Net heat absorbed over both steps divided by P_0V_0 is

(A) 6 (B) 3

(C) 1.5 (D) 4.5

42. A monatomic ideal gas expands quasistatically from volume V_0 to $2V_0$ along $P = KV^{-n}$, where K, n are constant and $n \neq 1$. It releases heat equal to half the positive work it does. The gas is then reversibly compressed adiabatically until its temperature regains its initial value. Final volume divided by V_0 is

(A) 0.5 (B) $\frac{\sqrt{2}}{2}$

(C) $\sqrt{2}$ (D) 1

43. A monatomic ideal gas supports a frictionless piston of area $0.10\ \mathrm{m^2}$ and mass 100 kg, with vacuum above. Initial gas volume is $0.010\ \mathrm{m^3}$. A spring of stiffness 10000 N/m, initially unstretched, opposes further rise. A $10\,\Omega$ heater at 20 V expands the gas slowly to twice its initial volume. Take $g = 10$, no heat loss. Find heating time in seconds.

(A) 15 (B) 16

(C) 14 (D) 30

44. A monatomic ideal gas starts at (P_0, V_0) with $P_0V_0 = 100$ J in an insulated piston cylinder. External pressure is suddenly held at $P_0/2$. During expansion, a falling 1 kg weight supplies 25 J through a paddle. After internal motion has damped, the final piston is in equilibrium; no energy escapes except external-pressure work. Find V_f/V_0.

(A) 1.8 (B) 3.6

(C) 0.8 (D) 2.8

45. A monatomic ideal gas initially supports a piston of area A and weight P_0A, with vacuum above, at volume V_0. The piston is clamped while heat Q is added, then released. During its motion the gas remains internally quasistatic and adiabatic. Its first turning point is $8V_0$. Neglect damping. Find $Q/(P_0V_0)$.

(A) $\frac{53}{6}$ (B) $\frac{47}{6}$

(C) $\frac{41}{6}$ (D) $\frac{47}{3}$

46. Two equal-volume rigid chambers contain 2 moles of monatomic gas at 300 K and 1 mole of diatomic gas at 600 K. Removing the partition in an insulated outer vessel mixes them without reaction. Diatomic rotation is active and vibration frozen. Find final pressure divided by the initial pressure in the monatomic chamber.

(A) $\frac{12}{11}$ (B) $\frac{23}{11}$

(C) $\frac{1}{11}$ (D) $\frac{24}{11}$

47. A reversible engine takes 300 J at 600 K and rejects heat to a 300 K reservoir. It first lifts 5 kg by 1 m with $g = 10$; its remaining work drives a reversible refrigerator between 200 K and that same 300 K reservoir. Find the total heat delivered to the 300 K reservoir, in joules.

(A) 451 (B) 449

(C) 450 (D) 900

48. A reversible engine operates between 600 K and 300 K. All its work slowly lifts a 1 kg mass from radius R to $2R$ above an airless spherical planet whose surface escape speed is 20 m/s. Find the engine heat input, in joules.

(A) 199 (B) 200

(C) 201 (D) 400

49. A body of constant heat capacity 1 J/K cools reversibly from 600 K to 300 K by supplying a reversible engine that rejects heat to a 300 K reservoir. All work is stored in a horizontal 600 N/m spring. Find final spring compression squared, in m^2.

(A) $1 - \frac{\log(2)}{2}$ (B) $\log(2)$

(C) $\frac{1}{2} - \frac{\log(2)}{2}$ (D) $1 - \log(2)$

50. A monatomic ideal gas lies under a weightless frictionless piston of area A, with vacuum above. A spring of stiffness k opposes piston height x; gas volume is Ax and spring force is kx. Initially $x = x_0$ and the gas is in equilibrium at T_0. The assembly is insulated. A paddle supplies energy $6kx_0^2$ quasistatically, all retained by gas and spring. Final gas temperature divided by T_0 is

(A) 4 (B) 2

(C) 7 (D) 3

Answer key

Thermodynamics

P-THER / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**A**	21	**C**	31	**D**	41	**C**
02	**B**	12	**D**	22	**A**	32	**B**	42	**B**
03	**D**	13	**C**	23	**B**	33	**C**	43	**A**
04	**C**	14	**B**	24	**C**	34	**D**	44	**A**
05	**B**	15	**C**	25	**A**	35	**A**	45	**B**
06	**B**	16	**D**	26	**A**	36	**C**	46	**A**
07	**D**	17	**A**	27	**A**	37	**D**	47	**C**
08	**A**	18	**B**	28	**D**	38	**D**	48	**B**
09	**D**	19	**C**	29	**B**	39	**B**	49	**D**
10	**C**	20	**D**	30	**B**	40	**A**	50	**A**

Chapter 13

Kinetic Theory

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Gas RMS speed doubles on heating from 27°C. Taking Celsius offset 273, final Celsius temperature is

(A) 1854 (B) 927

(C) 928 (D) 926

02. At the same temperature, helium and oxygen have molar masses $4,32\,\mathrm{g/mol}$. Ratio of their mean-square molecular speeds is

(A) 7 (B) 16

(C) 9 (D) 8

03. Molecular collision diameter doubles at fixed number density. Mean free path becomes what fraction?

(A) -0.75 (B) 0.5

(C) 0.25 (D) 1.25

04. At fixed temperature for one species, pressure halves. Mean free path is multiplied by

(A) 3 (B) 2

(C) 1 (D) 4

05. A gas with $f = 6$ active quadratic degrees has molar C_P/R equal to

(A) 3 (B) 5

(C) 4 (D) 8

06. For a gas with $f = 5$ degrees, heat-capacity ratio is

(A) 1.4 (B) 0.4

(C) 2.8 (D) 2.4

07. A mole of ideal monatomic gas gains 120 J internal energy. With $R = 8$, temperature rise in kelvin is

(A) 11 (B) 20

(C) 10 (D) 9

08. Two ideal gases of molar masses $M, 4M$ have equal RMS speeds. Required temperature ratio T_{4M}/T_M is

(A) 5 (B) 3

(C) 8 (D) 4

09. A gas has pressure P, volume V, active degrees $f = 3$. Internal energy divided by PV is

(A) 0.5 (B) 3

(C) 1.5 (D) 2.5

10. For an ideal diatomic gas with frozen vibrations, total rotational energy per molecule divided by total translational energy is

(A) $\frac{2}{3}$ (B) $\frac{5}{3}$

(C) $\frac{-1}{3}$ (D) $\frac{4}{3}$

11–40 / New middle-to-hard practice

11. A mixture of monatomic helium ($M = 4$ g/mol) and diatomic nitrogen ($M = 28$ g/mol) has $\gamma = 3/2$. Rotation is active and vibration frozen. Its mean molar mass in g/mol is

(A) 24 (B) 12

(C) 16 (D) 18

12. Equal masses of helium (4 g/mol, monatomic) and oxygen (32 g/mol, diatomic with active rotation and frozen vibration) are mixed. The heat-capacity ratio is

(A) 1.5 (B) $\frac{5}{3}$

(C) 1.4 (D) $\frac{47}{29}$

13. A gas has molar $C_V = 5R/2$ below T_* and $7R/2$ above T_*. Treat this as a continuous-energy piecewise heat-capacity model with no latent heat. Heating one mole at fixed volume from $T_*/2$ to $2T_*$ requires heat divided by RT_* equal to

(A) 5.25 (B) 4.75

(C) 3.75 (D) 5

14. One mole has $C_V = 3R/2$ below an unknown activation temperature T_* and $5R/2$ above it, with continuous energy and no latent heat. Heating it at fixed volume from 100 to 400 K requires $650R$ J. Given $100 < T_* < 400$, T_* in K is

(A) 150 (B) 200

(C) 300 (D) 250

15. A dilute gas has one sixth of its molecules in each velocity beam $\pm v\hat{i}$, $\pm 2v\hat{j}$, $\pm 3v\hat{k}$. Collisions are negligible. Pressure normal to a plane whose unit normal is $(\hat{i}+\hat{j})/\sqrt{2}$, divided by pressure on a plane normal to $\hat{i}$, is

(A) $\frac{5}{3}$ (B) 1.5

(C) 2.5 (D) 5

16. Half the molecules in a gas move along $\pm\hat{i}$ at speed v, one fourth along $\pm\hat{j}$ at speed $2v$, and one fourth along $\pm\hat{k}$ at speed $2v$. At fixed volume they collide and become isotropic without energy exchange. Final pressure divided by initial pressure on a wall normal to $\hat{i}$ is

(A) 2 (B) 1.5

(C) 2.5 (D) $\frac{5}{3}$

17. A rigid insulated vessel contains a diatomic gas with active rotation and frozen vibration at T. Its bulk speed equals the gas initial molecular RMS speed. The vessel stops, converting all bulk kinetic energy to internal energy. Final molecular RMS speed divided by initial RMS speed is

(A) 1.6 (B) $\sqrt{2}$

(C) $\frac{2\sqrt{10}}{5}$ (D) $\frac{\sqrt{15}}{3}$

18. Two rigid insulated vessels contain equal moles of helium ($M = 4$, monatomic) and nitrogen ($M = 28$, active rotation, frozen vibration), initially at equal T. Both move at the initial helium molecular RMS speed and stop, retaining all bulk kinetic energy. The final nitrogen to helium temperature ratio is

(A) 2.6 (B) 2.5

(C) 1.4 (D) $\frac{7}{3}$

19. For one ideal gas species, temperature is quadrupled and pressure doubled. Collision diameter stays fixed. The characteristic collision frequency $\nu = \bar{v}/\lambda$ is multiplied by

(A) 2 (B) 1

(C) 0.5 (D) 4

20. Two dilute gases have the same molar mass. Compared with the first, the second has temperature $9T$, pressure $3p$, and twice the collision diameter. In the dilute hard-sphere model, the collision frequency ratio is

(A) 4 (B) 3

(C) 12 (D) 0.75

21. Gas A and gas B have equal pressure and temperature. Their mean free paths satisfy $\lambda_A = 2\lambda_B$ and their molar masses satisfy $M_B = 4M_A$. The ratio of collision frequencies ν_B/ν_A, using $\nu = \bar{v}/\lambda$, is

(A) 2 (B) 4

(C) 1 (D) 0.5

22. Two dilute gases have equal collision diameter. Gas B has four times A temperature and twice its pressure. Their collision frequencies satisfy $\nu_B = \nu_A/2$, where $\nu = \bar{v}/\lambda$. The molar-mass ratio M_B/M_A is

(A) 4 (B) 2

(C) 16 (D) 1

23. A gas mixture contains equal masses of species of molar masses M and $4M$ at the same temperature. Number effusion rate through a small hole is proportional to molecule number times inverse square root of molar mass. The lighter fraction of effusing molecules is

(A) 0.5 (B) $\frac{2}{3}$

(C) 0.8 (D) $\frac{8}{9}$

24. Across identical small apertures, number effusion rate is proportional to $p/\sqrt{MT}$. Gas A has (p, T, M) and gas B has $(p/2, 4T, 9M)$. The ratio of escaping mass rates, A to B, is

(A) 6 (B) $\frac{2}{3}$

(C) 12 (D) $\frac{4}{3}$

25. Equal-pressure, equal-temperature tanks of volumes V and $2V$ contain helium (4 g/mol) and oxygen (32 g/mol). Their contents mix at the same temperature. The RMS speed of a randomly selected molecule divided by the RMS speed of pure oxygen at that temperature is

(A) $\frac{\sqrt{51}}{3}$ (B) $\sqrt{3}$

(C) $\frac{\sqrt{30}}{3}$ (D) $\sqrt{2}$

26. Equal masses of helium (4 g/mol) and oxygen (32 g/mol) form an equilibrium mixture. The mean square speed of a randomly selected molecule divided by that of pure oxygen at the same temperature is

(A) 8 (B) $\frac{65}{9}$

(C) 4.5 (D) $\frac{17}{3}$

27. A helium tank at pressure $2p$, volume V, temperature $2T$ is connected to an oxygen tank at pressure p, volume $2V$, temperature T. They reach temperature T in total rigid volume $3V$. Oxygen partial pressure divided by initial p is

(A) $\frac{1}{3}$ (B) $\frac{4}{3}$

(C) 1 (D) $\frac{2}{3}$

28. A rigid tank at fixed temperature contains helium (4 g/mol) and oxygen (32 g/mol). Selectively removing all helium reduces pressure to 3/4 of its original value. The removed mass divided by original total mass is

(A) 0.04 (B) $\frac{1}{9}$

(C) 0.25 (D) $\frac{1}{7}$

29. An ideal gas has total internal energy $U = 4PV$. The ratio of adiabatic sound speed to molecular RMS speed is

(A) $\frac{2\sqrt{3}}{3}$ (B) $\frac{\sqrt{15}}{5}$

(C) $\frac{\sqrt{5}}{2}$ (D) $\frac{\sqrt{15}}{6}$

30. A gas has adiabatic sound speed divided by molecular RMS speed equal to $\sqrt{7/15}$. Only translational and rotational modes are active. The fraction of its internal energy in rotation is

(A) 0.6 (B) 0.4

(C) $\frac{2}{3}$ (D) $\frac{1}{3}$

31. A gas has half its molecules at speed v, one third at $2v$, and one sixth at $3v$. Its RMS speed divided by its mean speed is

(A) $\frac{\sqrt{10}}{3}$ (B) $\frac{\sqrt{6}}{2}$

(C) $\sqrt{3}$ (D) $\frac{\sqrt{30}}{5}$

32. An isotropic dilute sample contains molecules only at speeds v and $3v$. Its RMS speed is $\sqrt{5}v$. If n is number density and m molecular mass, the ratio $P/[nm(\bar{v})^2]$, where $\bar{v}$ is mean speed, is

(A) 0.6 (B) $\frac{5}{12}$

(C) $\frac{5}{3}$ (D) $\frac{1}{3}$

33. A molecular beam carries kinetic power 60 W at speed $20\,\mathrm{m\,s^{-1}}$ normally toward a target. Some molecules are absorbed; the rest reflect elastically. Measured mean force is 9 N. The absorbed fraction is

(A) 0.5 (B) 0.75

(C) 0.25 (D) $\frac{1}{3}$

34. A molecular beam has kinetic power P and speed v, incident at 45° to a smooth target normal. One fourth of molecules are absorbed and the rest reflect elastically. Normal force F satisfies $Fv/P =$

(A) 1.75 (B) $\sqrt{2}$

(C) $\frac{7\sqrt{2}}{2}$ (D) $\frac{7\sqrt{2}}{4}$

35. An isothermal atmosphere obeys $p(z) = p_0 e^{-Mgz/(RT)}$, with constant g. At height H, pressure is $p_0/4$. The molecular RMS speed squared divided by gH is

(A) $\frac{3}{2\log(2)}$ (B) $\frac{3}{\log(2)}$

(C) $\frac{1}{2\log(2)}$ (D) $\log(4)$

36. An isothermal gas rotating rigidly obeys $p(r) = p(0)e^{M\omega^2 r^2/(2RT)}$. At radius R, its pressure is $e^2 p(0)$. Thermal molecular RMS speed relative to the local gas flow, divided by ωR is

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{6}}{2}$

(C) $\sqrt{3}$ (D) 0.5

37. A monatomic ideal gas in an open rigid vessel is heated at atmospheric pressure until only one third its initial mass remains. It is then sealed and compressed reversibly adiabatically until its density equals its initial density. Final temperature divided by initial temperature is

(A) 9 (B) $3 \cdot 3^{\frac{2}{3}}$

(C) 3 (D) $3^{\frac{2}{3}}$

38. A diatomic ideal gas with active rotation and frozen vibration is heated in an open rigid vessel until half its mass remains. After sealing, it is compressed reversibly adiabatically until density is twice its initial value. Final molecular RMS speed divided by initial RMS speed is

(A) 2 (B) $2^{\frac{9}{10}}$

(C) $2^{\frac{7}{10}}$ (D) $2^{\frac{2}{5}}$

39. An initially isotropic ideal gas is compressed to half its volume by an operation doubling each molecular x velocity component while leaving y, z components unchanged. It then becomes isotropic at fixed volume with no further energy exchange. Final equilibrium pressure divided by initial pressure is

(A) 2 (B) 8

(C) 4 (D) 6

40. An initially isotropic ideal gas is manipulated to double molecular x components and halve y components, leaving z unchanged, while its volume is halved. It then thermalizes without energy exchange. Final pressure divided by initial pressure is

(A) 1.75 (B) 4

(C) 2.5 (D) 3.5

41–50 / Preserved hard practice

41. A rigid insulated vessel contains N monatomic ideal-gas molecules at temperature T. Exactly $N/4$ molecules are removed, carrying mean translational kinetic energy $3k_BT$ per removed molecule. No other energy is transferred. The remainder re-equilibrates. Find final temperature divided by T.

(A) $\frac{2}{3}$ (B) $\frac{2}{9}$

(C) $\frac{1}{3}$ (D) $\frac{5}{6}$

42. One mole of monatomic gas at 600 K mixes with one mole of diatomic gas at 300 K in a thermally insulated cylinder under constant external pressure. Rotation is active and vibration frozen. Both samples are initially at that same pressure. The piston can move, and gases do not react. Find final equilibrium temperature, in kelvin.

(A) 425 (B) 426

(C) 850 (D) 424

43. A mixture contains monatomic molecules of molar mass M and diatomic molecules of molar mass $2M$, with rotation active and vibration frozen. Its heat-capacity ratio is 19/13. Find its number-averaged mean squared molecular speed divided by that of the pure monatomic gas at the same temperature.

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$

(C) $\frac{2}{9}$ (D) $\frac{5}{6}$

44. An ideal mixture of monatomic and diatomic gases is reversibly compressed adiabatically to one eighth of its volume. Rotations are active and vibrations frozen. Each species rms speed increases by factor $2^{3/4}$. Find the monatomic mole fraction.

(A) 0.25 (B) 0.75

(C) 0.5 (D) $\frac{1}{6}$

45. One mole of monatomic ideal gas, molar mass 0.024 kg/mol, is heated at constant pressure by a $10\,\Omega$ resistor carrying 1 A for 10 s. Take $R = 8$ J/(mol K), neglect heat losses. Find the increase in mean squared molecular speed, in m^2/s^2.

(A) 10000 (B) 5000

(C) 4999 (D) 5001

46. A freely rotating flywheel of moment of inertia 0.20 kg m^2 and angular speed 20 rad/s is stopped by a paddle inside an insulated rigid vessel. All lost energy heats 0.50 mole of monatomic ideal gas. Take $R = 8$. Find its temperature increase, in kelvin.

(A) $\frac{23}{3}$ (B) $\frac{20}{3}$

(C) $\frac{40}{3}$ (D) $\frac{17}{3}$

47. Two moles of monatomic gas of molar mass M and one mole of diatomic gas of molar mass $2M$ have equal initial rms molecular speeds. They mix in a rigid insulated vessel without reaction. Diatomic rotations are active and vibrations frozen. Find the diatomic gas final mean squared molecular speed divided by its initial value.

(A) $\frac{3}{11}$ (B) $\frac{19}{22}$

(C) $\frac{8}{11}$ (D) $\frac{4}{11}$

48. In an idealised diatomic-gas model, molar $C_V = 5R/2$ below 600 K and $7R/2$ above 600 K, with internal energy continuous at the threshold. A reversible adiabatic compression raises temperature from 300 K to 1200 K. Find final volume divided by initial volume.

(A) 0.984375 (B) 0.015625

(C) 0.5078125 (D) 0.0078125

49. A rigid insulated vessel contains one mole of monatomic and one mole of diatomic ideal gas at a common temperature. Rotations are active and vibrations frozen. A paddle supplies work W, and the mixture re-equilibrates. Find the increase in total translational kinetic energy divided by W.

(A) 0.375 (B) 0.875

(C) 0.25 (D) 0.75

50. A mixture of monatomic and rigid diatomic ideal gases is heated at constant pressure. Vibrations are frozen and its internal-energy rise is 2/3 of the heat supplied. In a separate sample of the same composition at common temperature and fixed number densities, all monatomic molecular speeds are instantaneously doubled, while diatomic speeds remain unchanged. Immediately, before energy redistribution between the species, the additional force on a piston divided by its initial pressure force is

(A) 1.5 (B) 2

(C) 1 (D) 0.5

Answer key

Kinetic Theory

P-KITH / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**C**	21	**C**	31	**D**	41	**A**
02	**D**	12	**D**	22	**A**	32	**B**	42	**A**
03	**C**	13	**B**	23	**D**	33	**A**	43	**A**
04	**B**	14	**B**	24	**D**	34	**D**	44	**C**
05	**C**	15	**C**	25	**C**	35	**A**	45	**B**
06	**A**	16	**D**	26	**B**	36	**A**	46	**B**
07	**C**	17	**C**	27	**D**	37	**B**	47	**C**
08	**D**	18	**A**	28	**A**	38	**B**	48	**B**
09	**C**	19	**B**	29	**D**	39	**C**	49	**D**
10	**A**	20	**A**	30	**B**	40	**D**	50	**A**

Chapter 14

Oscillations

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. An SHM has maximum speed $4\,\text{m/s}$ and maximum acceleration $8\,\text{m/s}^2$. Angular frequency in rad/s is

(A) 3 (B) 2

(C) 4 (D) 1

02. An SHM has maximum speed $4\,\text{m/s}$ and maximum acceleration $8\,\text{m/s}^2$. Amplitude in metres is

(A) 2 (B) 3

(C) 1 (D) 4

03. A pendulum of length $1\,\text{m}$ is in a lift accelerating upward at $2\,\text{m/s}^2$, with $g = 10$. Its angular frequency squared in SI units is

(A) 24 (B) 13

(C) 12 (D) 11

04. A $1\,\text{kg}$ vertical spring oscillator stretches the spring $0.1\,\text{m}$ at equilibrium, with $g = 10$. Angular frequency in rad/s is

(A) 20 (B) 10

(C) 11 (D) 9

05. Two equal springs k are joined in series to mass m. Angular frequency squared divided by k/m is

(A) 0.5 (B) 1.5

(C) 1 (D) -0.5

06. A spring is cut into two equal halves. One half holds the original mass. New period divided by original period is

(A) 2 (B) $1/2$

(C) $1/\sqrt{2}$ (D) $\sqrt{2}$

07. For SHM starting at equilibrium moving positively, time to first reach $x = A/2$ divided by period is

(A) $\frac{-11}{12}$ (B) $\frac{1}{6}$

(C) $\frac{1}{12}$ (D) $\frac{13}{12}$

08. A compound pendulum has moment of inertia I about pivot and centre-of-mass distance d. Small-angle angular frequency squared is

(A) mgd/I (B) $mg/(Id)$

(C) gd/I (D) $I/(mgd)$

09. For a uniform rod length L pivoted at one end, small-angle angular frequency squared divided by g/L is

(A) 1.5 (B) 3

(C) 0.5 (D) 2.5

10. A spring-mass SHM has frequency f. The kinetic energy repeats at frequency divided by f equal to

(A) 1 (B) 2

(C) 4 (D) 3

11–40 / New middle-to-hard practice

11. An SHM has speeds $\sqrt{21}$ and $3\,\mathrm{m\,s^{-1}}$ at displacements 1 and 2 m respectively. Its amplitude in m is

(A) 2 (B) $\sqrt{6}$

(C) 3 (D) 2.5

12. In an SHM, speeds at $x = 1$ m and $x = 3$ m are 4 and $2\,\mathrm{m\,s^{-1}}$. Its maximum acceleration in $\mathrm{m\,s^{-2}}$ is

(A) $\sqrt{105}$ (B) 1.5

(C) $\frac{\sqrt{105}}{2}$ (D) $\sqrt{21}$

13. An SHM has angular frequency $2\,\mathrm{rad\,s^{-1}}$. At one instant its displacement is 3 m and velocity is $-8\,\mathrm{m\,s^{-1}}$. The first time after this instant when it reaches the positive extreme, in s, is

(A) $\pi - \frac{\operatorname{atan}\left(\frac{4}{3}\right)}{2}$ (B) $\frac{\operatorname{atan}\left(\frac{4}{3}\right)}{2}$

(C) $\frac{\operatorname{atan}\left(\frac{4}{3}\right)}{2} + \pi$ (D) $\frac{\pi}{2}$

14. An SHM starts at equilibrium moving positively. Between the first instants at which its kinetic energy is $3E/4$ and $E/4$, respectively, the distance travelled divided by amplitude is

(A) $\frac{1}{2}+\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{3}}{2}$

(C) 0.5 (D) $-\frac{1}{2}+\frac{\sqrt{3}}{2}$

15. A 1 kg mass oscillates on a horizontal spring of constant 100 N/m, amplitude 0.20 m. At equilibrium it sticks to an initially stationary 3 kg mass. The new amplitude in m is

(A) 0.05 (B) 0.1

(C) 0.2 (D) 0.4

16. A mass m rests on a vertical spring at extension x_0. A second mass m, initially at rest, is attached gently at the same position and both are released without an impulse. The maximum spring extension divided by x_0 is

(A) 3 (B) 2.5

(C) 4 (D) 2

17. A horizontal spring oscillator has amplitude A. When its displacement is $A/2$, the spring constant is doubled instantaneously without changing position or velocity. The new amplitude divided by A is

(A) $\frac{\sqrt{5}}{2}$ (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{\sqrt{2}}{2}$ (D) $\frac{\sqrt{10}}{4}$

18. An oscillator stiffness is reduced to one fourth at a displacement x without changing velocity. Its new amplitude is $\sqrt{13}A/2$, where A was the old amplitude. The magnitude $|x|/A$ at switching was

(A) 0.75 (B) $\frac{\sqrt{3}}{2}$

(C) 0.25 (D) 0.5

19. A mass at rest on a horizontal spring is suddenly subjected to constant force F along its axis. The force is removed after one half of the resulting SHM period. Its later oscillation amplitude about the original equilibrium divided by F/k is

(A) 3 (B) $\sqrt{2}$

(C) 2 (D) 1

20. A resting horizontal spring oscillator receives a constant force F for one third of its natural period, then force is removed. Its final mechanical energy divided by $F^2/(2k)$ is

(A) 4 (B) 1

(C) 2 (D) 3

21. A solid cylinder of mass m rolls without slipping on a horizontal floor while a horizontal spring k is attached to its centre. A point mass $m/2$ is fixed at the cylinder axis. The angular oscillation frequency squared divided by k/m is

(A) 0.5 (B) $\frac{2}{3}$

(C) $\frac{1}{3}$ (D) 1

22. A uniform solid sphere rolls without slipping and is attached at its centre to a spring. Its oscillation period is T. The same spring is attached to a sliding mass equal to the sphere mass on a smooth floor. The period ratio T_{slide}/T is

(A) $\frac{5}{7}$ (B) $\frac{\sqrt{35}}{5}$

(C) $\frac{\sqrt{35}}{7}$ (D) $\frac{\sqrt{10}}{5}$

23. A mass m is connected between two horizontal fixed supports by springs k and $3k$, both along its line of motion. A second identical mass is then rigidly attached to it. The new angular frequency squared divided by k/m is

(A) 1 (B) 2

(C) 3 (D) 4

24. A mass on a smooth incline is supported by a spring parallel to the incline. It is initially at rest with extension x_0. A constant extra downslope force equal to its downslope weight is suddenly added. Maximum extension divided by x_0 is

(A) 3 (B) 4

(C) $\sqrt{3}$ (D) 2

25. A pendulum has small-oscillation period T on a platform accelerating horizontally at $3g/4$. The same pendulum on a stationary platform has period T_0. The ratio T/T_0 is

(A) $\frac{2\sqrt{3}}{3}$ (B) $\frac{2\sqrt{5}}{5}$

(C) 0.8 (D) $\frac{\sqrt{5}}{2}$

26. A pendulum has period T_u in a lift accelerating upward at a and T_d when it accelerates downward at the same a. If $T_d/T_u = 2$, the horizontal platform acceleration needed to give this pendulum period T_u, divided by g, is

(A) 0.8 (B) 0.6

(C) $\frac{\sqrt{39}}{5}$ (D) 1.6

27. A uniform rod of length L swings as a compound pendulum about a point $L/4$ from its centre. Another pivot on the rod gives the same small-angle period. Its distance from the centre divided by L is

(A) $\frac{1}{6}$ (B) 0.5

(C) 0.25 (D) $\frac{1}{3}$

28. A uniform rod of length L is used as a compound pendulum. The pivot distance from its centre that minimizes its small-angle period, divided by L, is

(A) $\frac{\sqrt{6}}{6}$ (B) $\frac{1}{3}$

(C) 0.5 (D) $\frac{\sqrt{3}}{6}$

29. A floating vertical cylinder has unknown mass and cross-section. Adding 1 kg increases its draft by 1 cm in water. Its original small vertical oscillation period was $2\pi/\sqrt{50}$ s. With $\rho_w = 1000$, $g = 10$, its original mass in kg is

(A) 25 (B) 20

(C) 10 (D) 50

30. A uniform cylinder of height 0.40 m and density one fourth that of its liquid floats vertically. It oscillates vertically while remaining partly immersed; instantaneous boundary touching is allowed. With $g = 10$, the greatest speed compatible with this condition, in $\mathrm{m\,s^{-1}}$, is

(A) $\sqrt{2}$ (B) 0.5

(C) 1 (D) 2

31. A spring-piston system of mass 10 kg has angular frequency squared $10\,\mathrm{s^{-2}}$ when its gas chamber is vented. Sealing the chamber gives $25\,\mathrm{s^{-2}}$ for rapid adiabatic oscillations. Equilibrium gas pressure is 10^5 Pa, volume $0.10\,\mathrm{m^3}$, piston area $0.010\,\mathrm{m^2}$. The gas γ is

(A) 2 (B) $\frac{5}{3}$

(C) 1.4 (D) 1.5

32. A piston oscillator has angular frequency squared 10 when vented, 20 with sealed isothermal gas, and $25\,\mathrm{s^{-2}}$ with sealed adiabatic gas at the same equilibrium. The gas molar C_V/R is

(A) 3 (B) 2

(C) 1.5 (D) 2.5

33. A lightly damped 1 kg oscillator has spring constant 100 N/m and amplitude 0.10 m. Its amplitude halves in 10 cycles. Use the undamped period and $E \simeq kA^2/2$. The initial cycle-averaged power loss in W is

(A) $\frac{\log(2)}{4\pi}$ (B) $\frac{\log(2)}{\pi}$

(C) $\frac{1}{2\pi}$ (D) $\frac{\log(2)}{2\pi}$

34. A lightly damped oscillator has mass 1 kg, angular frequency $10\,\mathrm{rad\,s^{-1}}$, and amplitude halves in 10 cycles. At resonance a force maintains amplitude 0.10 m. Using damping force $-2m\beta\dot{x}$ and the undamped period, the needed force amplitude in N is

(A) $\frac{\log(2)}{2\pi}$ (B) $\frac{\log(2)}{\pi}$

(C) $\frac{2\log(2)}{\pi}$ (D) $\frac{\log(2)}{10\pi}$

35. A driven oscillator of mass 1 kg has force amplitude F_0, damping constant $b = 2$ kg/s, and displacement amplitude $F_0/\sqrt{(k-\omega^2)^2 + 4\omega^2}$. Its amplitudes at angular frequencies 1 and 3 are equal. The undamped natural frequency is

(A) $\sqrt{10}$ (B) 3

(C) $\sqrt{5}$ (D) $\sqrt{7}$

36. A driven oscillator of mass 1 kg and stiffness 10 N/m has amplitude $F_0/\sqrt{(10-\omega^2)^2+b^2\omega^2}$. Amplitudes at $\omega = 1, 3$ rad/s are equal. The angular frequency of maximum amplitude in rad/s is

(A) 3 (B) $\sqrt{5}$

(C) $\sqrt{10}$ (D) $\sqrt{3}$

37. A positively charged bead moves along the line joining two fixed equal positive charges at distances a on either side of equilibrium. A mechanical spring also restores it to the midpoint. Its frequency squared is $9\,\mathrm{s}^{-2}$; doubling both fixed-charge distances makes it $2\,\mathrm{s}^{-2}$. Mechanical spring frequency alone in rad/s is

(A) $\sqrt{2}$ (B) 2

(C) 1 (D) 3

38. A negatively charged bead is held at the midpoint between two equal fixed positive charges and attached to a stabilizing spring. Switching the charges on doubles its period compared with the spring alone. Both fixed charges are then doubled. The fixed-charge distance multiplier at the stability boundary, where the squared frequency is zero, is

(A) $\sqrt[3]{2}$ (B) 1.5

(C) $\frac{2^{\frac{2}{3}}\sqrt[3]{3}}{2}$ (D) $\frac{6^{\frac{2}{3}}}{3}$

39. Two collinear SHMs have the same angular frequency. One has amplitude 3 cm; the other has unknown amplitude. Their sum has amplitude 5 cm and their difference amplitude $\sqrt{13}$ cm. The cosine of their phase difference is

(A) $-\frac{\sqrt{10}}{10}$ (B) 0.5

(C) $\frac{1}{3}$ (D) $\frac{\sqrt{10}}{10}$

40. Two SHMs are $x_1 = 3\cos\omega t$ and $x_2 = 4\cos(\omega t + \phi)$, in cm. Their sum amplitude is 5 cm. At the first instant when x_1 passes zero moving negatively, x_2 is positive. For $0 \le \phi < 2\pi$, ϕ is

(A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{2}$

(C) 0 (D) π

41–50 / Preserved hard practice

41. A mass m on a spring k oscillates with amplitude A. At $x = A/2$ moving outward, it collides and sticks to an equal free mass moving in the same direction. The new amplitude is again A. Find the free mass speed divided by $A\sqrt{k/m}$.

(A) $-\frac{\sqrt{3}}{2}+\frac{\sqrt{6}}{2}$ (B) $-\frac{\sqrt{6}}{2}+\frac{\sqrt{3}}{2}+1$

(C) $-\frac{\sqrt{3}}{4}+\frac{\sqrt{6}}{4}$ (D) $-\frac{\sqrt{3}}{4}+\frac{1}{2}+\frac{\sqrt{6}}{4}$

42. A mass m on a horizontal spring oscillates with amplitude A. At $x = A/2$ moving outward an equal mass with zero horizontal velocity is attached in a brief perfectly inelastic collision, with negligible spring impulse. At the next turning point this added mass is removed without impulse. Find the remaining mass speed at its next equilibrium crossing divided by $A\sqrt{k/m}$.

(A) $1-\frac{\sqrt{10}}{4}$ (B) $\frac{\sqrt{10}}{4}$

(C) $\frac{\sqrt{10}}{8}+\frac{1}{2}$ (D) $\frac{\sqrt{10}}{8}$

43. A mass m oscillates vertically on a light spring of stiffness k with amplitude $mg/(2k)$ about equilibrium. At its upper turning point an equal mass is attached at rest without changing its position. Find new amplitude divided by the old amplitude.

(A) 3 (B) 2

(C) 4 (D) 6

44. A mass m is attached to a light spring of stiffness k and natural length a, anchored at the origin. It moves without friction along $x > 0$. Without charge its period is T_0. A repulsive fixed charge at the origin gives Coulomb force magnitude K/x^2, where $K > 0$. The new small-oscillation period is $T_0\sqrt{2/5}$. Find $K/(ka^3)$.

(A) 48 (B) 49

(C) 96 (D) 47

45. A driven horizontal platform executes prescribed vertical motion $y = -A\cos\omega t$, with upward y positive, and $A\omega^2 = 2g$. A small body initially rests on it at $t = 0$ at the lowest position. It follows the platform until contact is lost, then moves freely under gravity. During the first free flight, maximum body height above the platform's mean level, divided by A, is

(A) 1.25 (B) 1.5

(C) 0.5 (D) 1

46. A pendulum initially hangs at rest in a stationary train. The train begins constant horizontal acceleration $\sqrt{3}g$ without a velocity jump. Neglect resistance. At the first passage through its new equilibrium direction, find the string tension divided by bob weight.

(A) 3 (B) 5

(C) 4 (D) 8

47. A solid uniform sphere radius R rolls without slipping at the bottom of a fixed smooth-shape concave circular track of radius $5R$, with sufficient static friction and small oscillations. Centre follows radius $4R$. Angular frequency squared divided by g/R is

(A) $\frac{-23}{28}$ (B) $\frac{5}{14}$

(C) $\frac{5}{28}$ (D) $\frac{33}{28}$

48. A uniform rod length L pivots at a point $L/4$ from one end. Small-angle period squared divided by $4\pi^2 L/g$ is

(A) $\frac{7}{12}$ (B) $\frac{7}{6}$

(C) $\frac{19}{12}$ (D) $\frac{-5}{12}$

49. A mass m hangs at rest on a vertical spring k in a stationary lift. The lift begins accelerating upward at $3g$ with no instantaneous velocity change. At the first instant the spring extension reaches $2mg/k$, find the relative speed squared divided by g^2m/k.

(A) 6 (B) 4

(C) 10 (D) 5

50. A cylinder of area A and mass m floats at depth $d = m/(\rho A)$. A point mass m falling vertically at speed u sticks to its top, where $u^2 = 2gd$. Neglect drag and added-fluid inertia, and let cylinder length exceed $4d$. Find the greatest immersion depth divided by d.

(A) $1 + \sqrt{2}$

(B) $\sqrt{2} + 2$

(C) $\sqrt{2} + 3$

(D) $2\sqrt{2} + 4$

Answer key

Oscillations

P-OSCI / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**D**	21	**A**	31	**D**	41	**A**
02	**A**	12	**C**	22	**C**	32	**B**	42	**B**
03	**C**	13	**A**	23	**B**	33	**D**	43	**A**
04	**B**	14	**D**	24	**A**	34	**B**	44	**A**
05	**A**	15	**B**	25	**B**	35	**D**	45	**A**
06	**C**	16	**A**	26	**C**	36	**B**	46	**C**
07	**C**	17	**D**	27	**D**	37	**C**	47	**C**
08	**A**	18	**D**	28	**D**	38	**C**	48	**A**
09	**A**	19	**C**	29	**B**	39	**D**	49	**D**
10	**B**	20	**D**	30	**C**	40	**B**	50	**B**

Chapter 15

Waves

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A transverse wave is $y = 0.01\sin(20x - 100t)$ in SI units. Wave speed in m/s is

(A) 4 (B) 6

(C) 10 (D) 5

02. A transverse wave is $y = 0.02\sin(10x - 50t)$ in SI units. Maximum particle speed in m/s is

(A) 0 (B) 2

(C) 0.5 (D) 1

03. Two equal-amplitude waves interfere with phase difference 120°. Resultant amplitude divided by either amplitude is

(A) 0.5 (B) 1

(C) 2 (D) 0

04. A string fixed at both ends has length 2 m and wave speed 120 m/s. Third harmonic frequency in hertz is

(A) 180 (B) 90

(C) 91 (D) 89

05. An open pipe and a one-end-closed pipe have equal lengths and same sound speed. Ratio of their fundamental frequencies is

(A) 1 (B) 2

(C) 3 (D) 4

06. A stationary source emits 300 Hz. An observer approaches it at 30 m/s, sound speed 300 m/s. Observed frequency in hertz is

(A) 660 (B) 331

(C) 329 (D) 330

07. A source emits 270 Hz while approaching a stationary observer at 30 m/s. Sound speed is 300 m/s. Observed frequency in hertz is

(A) 299 (B) 300

(C) 301 (D) 600

08. A source emits 330 Hz while receding from stationary observer at 30 m/s. Sound speed 300 m/s. Observed frequency in hertz is

(A) 299 (B) 301

(C) 600 (D) 300

09. A stretched string fundamental frequency is 100 Hz. Tension increases by 44%. New fundamental frequency in hertz is

(A) 240 (B) 119

(C) 121 (D) 120

10. A sound intensity level rises by 20 dB. Intensity is multiplied by

(A) 99 (B) 101

(C) 100 (D) 200

11–40 / New middle-to-hard practice

11. A sinusoidal wave travels at $1\,\mathrm{m\,s^{-1}}$ with wavelength 0.20 m. Its maximum particle speed is $\pi/10\,\mathrm{m\,s^{-1}}$. Displacement amplitude in mm is

(A) 5 (B) 20

(C) 50 (D) 10

12. A sinusoidal transverse wave travels at $4\,\mathrm{m\,s^{-1}}$. Its maximum particle speed is $2\,\mathrm{m\,s^{-1}}$ and maximum particle acceleration $20\,\mathrm{m\,s^{-2}}$. Wavelength in m is

(A) $\frac{2\pi}{5}$ (B) $\frac{\pi}{5}$

(C) $\frac{4\pi}{5}$ (D) 4π

13. For a standing wave, the nearest two points on opposite sides of a node with amplitude half the antinodal amplitude are 0.10 m apart. Frequency is 100 Hz. Wave speed in $\mathrm{m\,s^{-1}}$ is

(A) 40 (B) 120

(C) 60 (D) 20

14. A standing wave has adjacent nodes 0.30 m apart. At 0.05 m from a node maximum particle speed is $2\,\mathrm{m\,s^{-1}}$; at 0.10 m from that node maximum particle acceleration is $20\sqrt{3}\,\mathrm{m\,s^{-2}}$. Wave speed in $\mathrm{m\,s^{-1}}$ is

(A) $\frac{3}{\pi}$ (B) $\frac{6}{\pi}$

(C) 3 (D) 6

15. Two oppositely travelling sinusoidal waves have the same frequency on one string. Their interference envelope has maximum amplitude 5 mm and minimum 1 mm. Net transmitted power divided by power carried by the stronger wave alone is

(A) $\frac{5}{9}$ (B) 0.2

(C) $\frac{4}{9}$ (D) $\frac{8}{9}$

16. On a string let P_0 be power of a progressive wave of amplitude A_0 at the relevant frequency. Two opposite waves have net power $8P_0$ and maximum interference-envelope amplitude $4A_0$. Minimum envelope amplitude divided by A_0 is

(A) $2\sqrt{2}$ (B) 3

(C) 2 (D) 1

17. Two steel wires A and B have diameters in ratio 2 : 1, tensions in ratio 1 : 2, and lengths in ratio $1 : \sqrt{2}$. The ratio of pulse transit times t_A/t_B is

(A) 2 (B) 4

(C) $\frac{\sqrt{2}}{2}$ (D) $\sqrt{2}$

18. Two strings of equal length L and equal tension have linear densities μ and 4μ. They are joined and both outer ends fixed. Among modes having a node at the join, the lowest frequency mode has how many antinodes in total?

(A) 2 (B) 4

(C) 3 (D) 5

19. A uniform string is divided into three pieces whose fundamental frequencies at the original tension are 300, 400, 600 Hz. Rejoining the original string and raising its tension by a factor 9/4 gives fundamental frequency in Hz

(A) 200 (B) 400

(C) 150 (D) 300

20. A uniform string is cut into pieces of lengths in ratio 2 : 3. Their fundamentals at the same tension beat at 50 Hz. The rejoined original string must have fundamental 120 Hz. Required tension divided by original tension is

(A) 2 (B) 4

(C) 9 (D) 2.25

21. At 300 Hz, a resonance tube has first and second closed-end resonance lengths 24 and 74 cm. End correction is constant. At 500 Hz, its first resonance length in cm is

(A) 16 (B) 24

(C) 15 (D) 14

22. A closed-end resonance tube at 500 Hz has first and third resonance lengths 16 and 86 cm. The same end correction applies at every open end. A 37 cm pipe open at both ends has fundamental frequency in Hz equal to

(A) 500 (B) 437.5

(C) 400 (D) 350

23. An open pipe and a closed pipe have lengths in ratio 2 : 3. Their lowest common resonance frequency is 300 Hz. With sound speed $360\,\mathrm{m\,s^{-1}}$ and no end corrections, closed-pipe length in m is

(A) 0.3 (B) 0.6

(C) 0.9 (D) 1.2

24. The second overtones of an open and a closed pipe coincide. Their first overtones differ by 20 Hz, the open-pipe value being larger. With sound speed $300\,\mathrm{m\,s^{-1}}$ and no end correction, open-pipe length in m is

(A) 2.5 (B) 0.75

(C) 1.25 (D) 1.5

25. Two fixed-length organ pipes have resonance wavelengths 0.40 and 0.50 m in a gas at 300 K, where sound speed is $300\,\mathrm{m\,s^{-1}}$. Ignoring pipe expansion, at 363 K their beat frequency in Hz is

(A) 180 (B) 150

(C) 165 (D) 135

26. An unknown tuning fork gives 4 beats/s with a 256 Hz standard. Adding wax lowers the unknown frequency by 2 Hz, after which it gives 6 beats/s with the standard. The original unknown frequency in Hz is

(A) 260 (B) 254

(C) 258 (D) 252

27. A source moving at one constant speed produces 800 Hz while approaching a stationary observer and 600 Hz while receding. Source rest frequency in Hz is

(A) 700 (B) 750

(C) $\frac{4800}{7}$ (D) 650

28. An observer moving toward a stationary source hears 480 Hz; moving away at the same speed, the observer hears 320 Hz. With sound speed $340\,\mathrm{m\,s^{-1}}$, observer speed in $\mathrm{m\,s^{-1}}$ is

(A) 34 (B) 68

(C) 85 (D) 136

29. A sonar vehicle moves at $20\,\mathrm{m\,s^{-1}}$ behind a reflector moving in the same direction at $10\,\mathrm{m\,s^{-1}}$. It emits 280 Hz and receives its echo. Sound speed is $300\,\mathrm{m\,s^{-1}}$. Echo frequency in Hz is

(A) 290 (B) $\frac{9280}{31}$

(C) 300 (D) $\frac{8680}{29}$

30. A stationary sonar emits 300 Hz and receives 340 Hz from a reflector approaching along the line of sight. Sound speed is $340\,\mathrm{m\,s^{-1}}$. Reflector speed in $\mathrm{m\,s^{-1}}$ is

(A) 21.25 (B) 20

(C) 40 (D) 42.5

31. Two coherent sound sources produce maximum and minimum intensities $25I_0$ and I_0 at a point when their relative phase is varied. At the actual phase, intensity is $7I_0$. For phase in $[0, \pi]$, its value in degrees is

(A) 90 (B) 120

(C) 150 (D) 60

32. Two coherent sound sources individually give intensities $4I_0$ and $9I_0$; together they give $7I_0$. Keeping phase unchanged, the displacement amplitude of the first source doubles. New combined intensity divided by I_0 is

(A) 13 (B) 25

(C) 19 (D) 7

33. At a point, background sound level is 50 dB. Switching on one machine makes total level 60 dB. Four identical mutually incoherent machines then operate there, each contributing the same intensity. New total level in dB is

(A) $\log\left(3700000^{\frac{10}{\log(10)}}\right)$ (B) $\log\left(3000000^{\frac{10}{\log(10)}}\right)$

(C) $\log\left(3600000^{\frac{10}{\log(10)}}\right)$ (D) $\log\left(2000^{\frac{20}{\log(10)}}\right)$

34. A point source in fixed background gives total sound levels $10\log_{10}30$ dB at distance r and $10\log_{10}9$ dB at distance $2r$, relative to the same reference intensity. At distance $3r$, total level in dB is

(A) $\frac{10\log(6)}{\log(10)}$

(B) $\log\left(\left(\frac{46}{9}\right)^{\frac{10}{\log(10)}}\right)$

(C) $\log\left(\left(\frac{46}{3}\right)^{\frac{10}{\log(10)}}\right)$

(D) $\log\left(\left(\frac{10}{3}\right)^{\frac{10}{\log(10)}}\right)$

35. A string has tension 100 N and linear density 0.010 kg/m. It carries a sinusoidal wave of frequency 50 Hz and amplitude 1.0 mm. Mean wave power in W is

(A) $\frac{\pi^2}{200}$ (B) $\frac{\pi^2}{400}$

(C) $\frac{\pi^2}{20}$ (D) $\frac{\pi^2}{100}$

36. A wave is sent along a wire. A second wire of the same material has twice the radius, is stretched at nine times the tension, and is driven at twice the frequency. To transmit equal mean power, its amplitude divided by the first amplitude must be

(A) $\frac{\sqrt{6}}{12}$ (B) $\frac{1}{6}$

(C) $\frac{\sqrt{6}}{6}$ (D) $\frac{\sqrt{2}}{4}$

37. A sinusoidal wave has amplitude 2 cm. At one point and instant its displacement is 1 cm and its slope is $-\sqrt{3}/5$. Its wavelength in m is

(A) $\frac{\pi}{5}$ (B) $\frac{\pi}{20}$

(C) $\frac{\pi}{10}$ (D) $\frac{2\pi}{5}$

38. A standing wave has a node at $x = 0$. At $x = \lambda/12$, simultaneous particle data are displacement 4 cm, velocity 6 cm/s and acceleration -16 cm/s^2. Antinodal amplitude in cm is

(A) 12 (B) 8

(C) 10 (D) 5

39. An open pipe and its gas are heated from 300 to 363 K. Its fundamental increases from 1001 to 1100 Hz. Sound speed is proportional to $\sqrt{T}$ and pipe length obeys $L = L_0(1 + \alpha\Delta T)$. The linear expansion coefficient in K^{-1} is

(A) $\frac{1}{6300}$ (B) 0.00001

(C) $\frac{1}{630000}$ (D) $\frac{1}{63000}$

40. A stretched wire has fixed endpoints. Cooling it by 20 K doubles its fundamental frequency. Neglect changes in linear density and take the tension change as linear in temperature. Starting from the original state, the warming needed to make it first go slack, in K, is

(A) $\frac{40}{3}$ (B) 20

(C) 10 (D) $\frac{20}{3}$

41–50 / Preserved hard practice

41. A 4 kg solid cube of density 2000 kg/m^3 hangs fully immersed in a liquid of density 1000 kg/m^3. Its apparent weight tensions a sonometer string of vibrating length 1 m and linear density 0.010 kg/m. Take $g = 10$, ignore pulley losses. Find the fundamental frequency in hertz.

(A) $10\sqrt{5}$ (B) $-1 + 10\sqrt{5}$

(C) $1 + 10\sqrt{5}$ (D) $20\sqrt{5}$

42. A uniform elastic sonometer wire has fixed mass. At tension 10 N its length is 1 m. Its longitudinal stiffness dF/dL is a constant 100 N/m. A hanging load is increased until its fundamental frequency becomes 3/2 of the initial value. Take $g = 10$. Find the added mass in kg; include changes of both wire length and linear density.

(A) $\frac{100}{31}$ (B) $\frac{19}{31}$

(C) $\frac{50}{31}$ (D) $\frac{81}{31}$

43. A stationary source emits 500 Hz in air with uniform wind. Two observers, one downwind and one upwind, each move toward the source at 30 m/s relative to the ground. They hear 1625/3 Hz and 550 Hz respectively. Find wind speed in m/s.

(A) 31 (B) 30

(C) 60 (D) 29

44. A stationary source emits 330 Hz toward a wall approaching it at 30 m/s. With sound speed 330 m/s, the source hears reflected frequency in hertz equal to

(A) 396 (B) 395

(C) 792 (D) 397

45. A car siren is heard by a stationary observer at 550 Hz while approaching and 450 Hz while receding at the same constant speed. Sound speed is 330 m/s. During approach, the car also hears its sound reflected from a stationary wall ahead. Find that reflected frequency heard in the car, in hertz.

(A) 606 (B) 604

(C) 605 (D) 1210

46. An open pipe has length L_0 at 300 K. Over the model range $T \geq 300$ K its length is $L_0[1 + 10^{-3}(T - 300)]$, and sound speed is proportional to $\sqrt{T}$. Find the absolute temperature in kelvin at which its fundamental frequency is greatest.

(A) 700 (B) 699

(C) 1400 (D) 701

47. A vertical air column, open at its top and closed by a liquid surface, has fundamental frequency 300 Hz. A large reservoir supplying the liquid is pressurised by 1000 Pa; the liquid rises in the tube and fundamental becomes 600 Hz. Air above the liquid remains at atmospheric pressure. Take sound speed 340 m/s, $g = 10$, and unchanged end correction. Find liquid density in kg/m^3.

(A) $\frac{11983}{17}$ (B) $\frac{12000}{17}$

(C) $\frac{12017}{17}$ (D) $\frac{24000}{17}$

48. Two strings joined between fixed supports have lengths 1 m and 1.5 m and linear densities 0.010 and 0.040 kg/m. Their common tension is supplied by an ideal 100 N/m spring extended 0.40 m. Consider normal modes with a node at the junction and frequency at most 100 Hz. Find the greatest number of internal nodes, including the junction but excluding both supports.

(A) 10 (B) 11

(C) 12 (D) 22

49. Three coherent harmonic waves of the same frequency and medium separately have intensities $I, 4I, 9I$ and phases $0, \theta, 2\theta$, where $0 \leq \theta \leq \pi$. Their resultant intensity is $4I$. Find the sum of all possible values of $\cos\theta$.

(A) $-\frac{8}{3}$ (B) $-\frac{1}{3}$

(C) $-\frac{7}{3}$ (D) $-\frac{4}{3}$

50. Three coherent waves on one ideal string are $y = A\cos(kx-\omega t)+A\cos(kx-\omega t+\phi)+A\cos(kx+\omega t)$. At $x = 0$, oscillation amplitude is $\sqrt{7}A$. Let P_0 be average power of one travelling wave of amplitude A at this k, ω. Net average power toward $+x$, divided by P_0, is

(A) 2 (B) 4

(C) 3 (D) 1

Answer key

Waves

P-WAVE / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**D**	21	**D**	31	**B**	41	**A**
02	**D**	12	**C**	22	**B**	32	**A**	42	**C**
03	**B**	13	**C**	23	**C**	33	**A**	43	**B**
04	**B**	14	**A**	24	**D**	34	**B**	44	**A**
05	**B**	15	**A**	25	**C**	35	**A**	45	**C**
06	**D**	16	**C**	26	**D**	36	**A**	46	**A**
07	**B**	17	**A**	27	**C**	37	**C**	47	**B**
08	**D**	18	**C**	28	**B**	38	**C**	48	**B**
09	**D**	19	**A**	29	**B**	39	**D**	49	**D**
10	**C**	20	**B**	30	**A**	40	**D**	50	**A**

Chapter 16

Electrostatics

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A uniformly charged nonconducting sphere has radius R and total Q. Field at $r = R/2$ divided by surface field is

(A) -0.5 (B) 1

(C) 0.5 (D) 1.5

02. A uniformly charged ring has radius R. At axial distance $x = R$, potential divided by centre potential is

(A) 1 (B) $1/2$

(C) $1/\sqrt{2}$ (D) $\sqrt{2}$

03. Two charges q and $4q$ are $3a$ apart. Their zero-field point between them is distance a from which charge?

(A) Neither; no zero point exists (B) The charge $4q$

(C) The charge q (D) Neither; exactly the midpoint

04. An electric dipole far-field axial field is compared with its equatorial field magnitude at equal distance. Ratio is

(A) 3 (B) 4

(C) 2 (D) 1

05. A point charge q moves from infinity to distance r from fixed charge Q. External quasistatic work is

(A) 0 (B) $-kQq/r$

(C) kQq/r^2 (D) kQq/r

06. Three equal charges q occupy vertices of an equilateral triangle side a. Total interaction energy divided by kq^2/a is

(A) 6 (B) 3

(C) 2 (D) 4

07. Four equal charges q occupy corners of a square side a. Potential at its centre divided by kq/a is

(A) 8 (B) $2\sqrt{2}$

(C) 4 (D) $4\sqrt{2}$

08. A potential is $V = 3x^2$ volts in SI units. Field x component at $x = 2\,\text{m}$, in V/m, is

(A) -24 (B) -13

(C) -11 (D) -12

09. A potential is $V = 5x - 2y$ in volts with coordinates in metres. Electric field magnitude in V/m is

(A) 3 (B) $\sqrt{29}$

(C) 7 (D) $\sqrt{21}$

10. A dipole rotates quasistatically from parallel to perpendicular to field. External work divided by pE is

(A) 0 (B) 0.5

(C) 2 (D) 1

11–40 / New middle-to-hard practice

11. Two identical small conducting spheres have opposite charges with total $+4\,\mu$C. Their attraction at a large fixed separation is three times the repulsion after contact and reseparation to that distance. The original positive charge in μC is

(A) 8 (B) 6

(C) 3 (D) 4

12. Two conducting spheres of radii R and $2R$, far apart, initially carry $5Q$ and $-2Q$. A thin wire joins them, then is removed. Their separation is doubled. Final force magnitude divided by initial force magnitude is

(A) 0.025 (B) 0.05

(C) 0.1 (D) 0.2

13. Eight identical charged conducting liquid drops coalesce without charge or volume loss. Initial total electrostatic self-energy is U. Final electrostatic self-energy minus U, divided by U, is

(A) 7 (B) 3

(C) 1 (D) 2

14. Identical charged conducting drops coalesce with no loss, producing a drop whose potential is nine times each original potential. Surface tension stays constant. The fraction of original total surface energy lost is

(A) 0.5 (B) $\frac{2}{3}$

(C) $\frac{8}{9}$ (D) $\frac{1}{3}$

15. Four equal charges $+Q$ occupy the corners of a square of side a and charge $-q$ is at its centre. The value of q is such that every corner charge has zero force. Total interaction energy, with zero at infinite separation, divided by k_eQ^2/a, is

(A) 0 (B) $-4-\sqrt{2}$

(C) $\sqrt{2}+4$ (D) 2

16. Three charges $+Q$ at an equilateral triangle corners are balanced by a central negative charge. The central charge sign is then reversed without changing magnitude or positions. Net force on a corner divided by k_eQ^2/a^2, with side a, is

(A) $2\sqrt{3}$ (B) 2

(C) $\sqrt{3}$ (D) 3

17. A dipole at 60° to a uniform field experiences torque magnitude $6\sqrt{3}$ N m. External quasistatic work to rotate it from that angle to unstable equilibrium, in J, is

(A) 18 (B) 24

(C) 6 (D) 12

18. Rotating a dipole quasistatically from parallel to 60° in a uniform field requires 3 J. Its rotational moment of inertia is $0.50\,\mathrm{kg\,m^2}$. Its small-oscillation period about the stable orientation, in s, is

(A) $\frac{\sqrt{3}\pi}{3}$ (B) $\frac{\sqrt{6}\pi}{6}$

(C) $\frac{2\sqrt{3}\pi}{3}$ (D) $\frac{\pi}{3}$

19. A small dipole at 60° to the x axis is in field $\mathbf{E}=(E_0+\alpha x)\hat{i}$. At its centre $E_0=4$ N/C, torque magnitude is $\sqrt{3}$ N m and net force magnitude is 2 N. The positive gradient α in $\mathrm{N\,C^{-1}m^{-1}}$ is

(A) 16 (B) 12

(C) 4 (D) 8

20. A dipole in $\mathbf{E}=E_0(1+x/L)\hat{i}$ requires 6 J to rotate from parallel to perpendicular at $x=0$. Starting at $x=0$ and angle 60°, it is moved slowly to $x=L$ and ends at 30°. Net external work in J is

(A) $3+6\sqrt{3}$ (B) $-3+6\sqrt{3}$

(C) $3-3\sqrt{3}$ (D) $3-6\sqrt{3}$

21. A uniformly charged solid nonconducting sphere has radius R and total $Q>0$. An outside point has the same field magnitude as the point $r=R/2$. External work to move $q>0$ from $R/2$ to that outside point divided by k_eQq/R is

(A) $-1+\frac{\sqrt{2}}{2}$ (B) $-\frac{11}{8}+\frac{\sqrt{2}}{2}$

(C) $\frac{11}{8}-\frac{\sqrt{2}}{2}$ (D) $\frac{1}{2}-\frac{\sqrt{2}}{2}$

22. A spherical cavity of radius $R/3$ is removed from a uniformly charged solid sphere of radius R. Cavity centre is displaced by $R/2$ from the sphere centre. Uniform field in the cavity has magnitude E_0. The field of the original uncut sphere at radius $2R$, divided by E_0, is

(A) 2 (B) 0.25

(C) 1 (D) 0.5

23. A point charge $Q > 0$ is at a cube vertex. Uniform field $E\hat{i}$ also acts, with cube side a along $+x, +y, +z$ from the charge and $Ea^2 = Q/(16\varepsilon_0)$. Flux through the three faces not meeting the charge, divided by Q/ε_0, is

(A) 0.1875 (B) 0.0625

(C) 0.125 (D) 0.25

24. A charge sits at the centre of a cube in uniform field along $+x$. Outward fluxes on $+x$ and $-x$ faces are $5\Phi_0$ and Φ_0. The cube side is doubled, keeping charge, centre and field fixed. New outward flux on the $+x$ face divided by Φ_0 is

(A) 11 (B) 20

(C) 7 (D) 12

25. A 1 kg particle of charge +1 C moves along x in potential $V = ax^2 + bx + c$. Its stable equilibrium is at $x = 1$ m and electric force at $x = 0$ is +4 N. Starting at $x = 0$ with velocity $+2\,\mathrm{m\,s^{-1}}$, its first right turning point in m is

(A) $1 + \sqrt{5}$ (B) $1 + \sqrt{3}$

(C) $1 + \sqrt{2}$ (D) 2

26. A positive particle starts at $x = 0$ moving right with kinetic energy $qV_0/2$ in potential $V(x) = V_0[2(x/L)^2 - (x/L)^4]$, with $V_0 > 0$. The first positive turning point divided by L is

(A) $\frac{\sqrt{2\sqrt{2}+4}}{2}$ (B) $\frac{\sqrt{4-2\sqrt{2}}}{2}$

(C) 1 (D) $\frac{\sqrt{2}}{2}$

27. A positively charged oil drop is held stationary by an upward electric field. That field is then reversed without changing its magnitude. Starting from rest, the drop falls through height H. Its impact speed divided by that for an uncharged drop falling from rest through H is

(A) 2 (B) $\sqrt{3}$

(C) 1.5 (D) $\sqrt{2}$

28. A charged pendulum has period twice its uncharged period T_0 in a uniform upward electric field. The same field is turned horizontal without changing magnitude. The small-oscillation period about its new equilibrium divided by T_0 is

(A) $\frac{2\sqrt{5}}{5}$ (B) $\frac{\sqrt{5}}{2}$

(C) 0.8 (D) $\sqrt{2}$

29. An electron accelerated from rest through voltage V enters a uniform transverse field of magnitude 1000 N/C for horizontal length 0.10 m, then drifts 0.20 m. Its total transverse deflection is 5.0 mm. Ignore fringing and relativistic effects. V in volts is

(A) 500 (B) 5000

(C) 2500 (D) 1000

30. An electron accelerated through voltage V crosses two adjacent transverse-field regions of equal horizontal length 0.10 m. Their fields are 1000 N/C upward and 1000 N/C downward. It exits parallel to its incident direction with offset 5.0 mm. Ignoring fringing and relativistic effects, V in volts is

(A) 500 (B) 2500

(C) 1000 (D) 2000

31. A negative bead oscillates with small-amplitude angular frequency ω along the axis of a positively charged ring of radius R. Released from rest at axial distance $\sqrt{3}R$, its speed at the centre divided by ωR is

(A) 1 (B) 0.5

(C) $\sqrt{2}$ (D) $\sqrt{3}$

32. A positive particle of mass m and charge q approaches a fixed positively charged ring of radius R and charge Q along its axis from infinity. Its initial speed satisfies $u^2 = 3k_eQq/(2mR)$. Its positive turning distance from the ring plane divided by R is

(A) $\frac{1}{3}$ (B) $\frac{\sqrt{7}}{2}$

(C) $\sqrt{3}$ (D) $\frac{\sqrt{7}}{3}$

33. Two particles of masses m and $3m$, each with charge $+q$, approach from very large separation along one line with velocities $+u$ and $-u$. Closest separation divided by $k_eq^2/(mu^2)$ is

(A) 1 (B) $\frac{4}{3}$

(C) 0.5 (D) $\frac{2}{3}$

34. Two initially resting particles of masses m and $2m$, charges q and $2q$, are released at separation R. When separation becomes $2R$, lighter speed squared divided by $k_eq^2/(mR)$ is

(A) $\frac{4}{3}$ (B) $\frac{2}{3}$

(C) $\frac{8}{3}$ (D) 2

35. Two concentric thin conducting shells have radii R and $2R$ and charges $+Q$, $-Q$. External quasistatic work to bring a charge q from infinity to radius $3R/2$, divided by k_eQq/R, is

(A) $\frac{1}{6}$ (B) 0.5

(C) $\frac{1}{3}$ (D) 0

36. Concentric thin conducting shells of radii R and $3R$ have potentials $5V$ and V. The field magnitude at $r = 2R$, divided by V/R, is

(A) 0.5 (B) 3

(C) 1.5 (D) 1

37. An infinite straight positive line charge produces field E_0 at distance $2a$. A negative particle of mass m, charge magnitude q, starts at radius a moving outward at speed $2\sqrt{qE_0a/m}$. Its turning radius divided by a is

(A) e (B) $e^{\frac{1}{2}}$

(C) e^2 (D) 2

38. A positive bead moves along $z > 0$ under a fixed charge $Q > 0$ at $z = 0$ and a uniform field E_0 toward negative z. It has stable equilibrium z_0 and small-oscillation frequency ω. Released from rest at $2z_0$, its speed at z_0 divided by ωz_0 is

(A) 1 (B) 0.5

(C) $\sqrt{2}$ (D) $\frac{\sqrt{2}}{2}$

39. A finite dipole has charges $+q$ and $-q$ at $x = +a, -a$. At axial coordinate $r > a$, its exact field magnitude is 81/64 times the far-dipole approximation at the same point. The ratio r/a is

(A) $\sqrt{3}$ (B) 3

(C) 4 (D) 2

40. For a dipole of charges $\pm q$ at $x = \pm a$, the exact equatorial field at distance r from its centre is $8/(5\sqrt{5})$ of its far-dipole approximation there. At the same distance r, exact axial field divided by exact equatorial field is

(A) $\frac{10\sqrt{5}}{9}$ (B) $\frac{5\sqrt{5}}{3}$

(C) $\frac{20\sqrt{5}}{9}$ (D) 2

41–50 / Preserved hard practice

41. A positive particle of charge q and mass m is projected at speed u from level ground in a uniform horizontal electric field pointing along the launch direction. Gravity is g, and $qE = 3mg/4$. Over all launch angles between 0 and 90°, maximum range back to ground divided by u^2/g is

(A) 2 (B) 1.75

(C) 1.25 (D) 1.5

42. A positively charged pendulum bob is at rest in a uniform horizontal electric field, with its string 30° from the downward vertical. Its charge is suddenly reversed without mechanical impulse, while field and string length stay fixed. Neglect resistance. Maximum subsequent string tension divided by mg is

(A) $\frac{2\sqrt{3}}{3}$ (B) $\frac{4\sqrt{3}}{3}$

(C) $2\sqrt{3}$ (D) $\sqrt{3}$

43. Fixed charges $Q > 0$ and $4Q$ are at $x = 0$ and $x = 3a$. A bead of charge $q > 0$, mass m, is constrained to their joining line between the charges. At its stable equilibrium the potential is V_e, and small oscillations have angular frequency ω. Find $qV_e/(ma^2\omega^2)$.

(A) 3 (B) 1

(C) 2 (D) 0.5

44. Two point particles of masses $m, 3m$ and charges $q, 2q$ start at separation a. Their velocities point toward each other and have magnitudes $u, u/3$, respectively, with $u^2 = kq^2/(ma)$. No other forces act. Their minimum separation divided by a is

(A) $\frac{2}{3}$ (B) 0.6

(C) 0.75 (D) 0.5

45. Two fixed positive charges q are at $x = \pm a$. A bead of charge $Q > 0$ and mass m slides without friction along the line joining them, close to $x = 0$. Neglect gravity. Its small-oscillation angular frequency squared, divided by $kqQ/(ma^3)$, is

(A) 4 (B) 8

(C) 1 (D) 2

46. A charged pendulum has zero-field period T_0. A uniform upward field E_1 makes its period $2T_0$, while its equilibrium remains below the pivot. The same particle is then released from rest at height H in a horizontal field $E_2 = 2E_1$, with the electric force horizontal. Neglect drag. Its horizontal displacement magnitude on reaching ground, divided by H, is

(A) 0.75 (B) 2

(C) 1.5 (D) 3

47. A negative bead of charge q, mass m, slides without friction on the axis of a uniformly charged ring of radius R and charge $Q > 0$. It starts from rest at axial distance $\sqrt{3}R$. At the first point where the electric-field magnitude is maximum, its kinetic energy divided by $kQ|q|/R$ is

(A) $\frac{\sqrt{6}}{3}$ (B) $1 - \frac{\sqrt{6}}{3}$

(C) 0.5 (D) $-\frac{1}{2} + \frac{\sqrt{6}}{3}$

48. An electron enters three successive field regions, each of horizontal length L, with horizontal speed v and zero vertical velocity. Their uniform electric fields are vertically upward E, downward $2E$, and upward E, respectively. Gravity and edge fields are neglected. Maximum distance from the original horizontal axis divided by $eEL^2/(mv^2)$ is

(A) 1.5 (B) 1

(C) 0.75 (D) 0.5

49. Two identical conducting spheres of radius R initially carry $3Q$ and $-Q$, at separation $r \gg R$. They are connected until charge redistributes, then disconnected and moved quasistatically to separation $2r$. Use self-capacitance $4\pi\varepsilon_0 R$ and point-charge mutual energies, neglecting induction. Heat released in the connection divided by the magnitude of external work during the later separation is

(A) $\frac{8r}{R}$ (B) $-8 + \frac{8r}{R}$

(C) $\frac{4r}{R}$ (D) $-4 + \frac{4r}{R}$

50. A particle of charge $q > 0$ and mass m moves under attraction to a fixed charge $-Q$. At distance $2R$ it has purely transverse speed u, and its closest approach is R. Find u^2 divided by $kQq/(mR)$.

(A) 0.5 (B) $\frac{1}{3}$

(C) $\frac{2}{3}$ (D) 1

Answer key

Electrostatics

P-ELEC / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**B**	21	**B**	31	**A**	41	**A**
02	**C**	12	**B**	22	**D**	32	**D**	42	**B**
03	**C**	13	**B**	23	**A**	33	**D**	43	**B**
04	**C**	14	**B**	24	**A**	34	**A**	44	**C**
05	**D**	15	**A**	25	**C**	35	**A**	45	**A**
06	**B**	16	**A**	26	**B**	36	**C**	46	**C**
07	**D**	17	**A**	27	**D**	37	**A**	47	**D**
08	**D**	18	**A**	28	**A**	38	**D**	48	**C**
09	**B**	19	**D**	29	**C**	39	**B**	49	**B**
10	**D**	20	**D**	30	**C**	40	**C**	50	**B**

Chapter 17

Capacitance

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A $2\,\mu\text{F}$ and $4\,\mu\text{F}$ series pair is across $12\,\text{V}$. Voltage across $2\,\mu\text{F}$ in volts is

(A) 16 (B) 8

(C) 7 (D) 9

02. A $2\,\mu\text{F}$ and $4\,\mu\text{F}$ series pair is across $12\,\text{V}$. Charge magnitude on either in microcoulombs is

(A) 32 (B) 16

(C) 17 (D) 15

03. A vacuum capacitor C_0 has half its plate area filled with dielectric $K = 3$ across full gap. Equivalent capacitance divided by C_0 is

(A) 2 (B) 3

(C) 4 (D) 1

04. A capacitor gap is filled by two layers of equal thickness with relative permittivities 1 and 3. Capacitance divided by vacuum capacitance is

(A) 1.5 (B) 3

(C) 0.5 (D) 2.5

05. A parallel-plate capacitor is isolated and plate separation doubles. Stored energy is multiplied by

(A) 2 (B) 4

(C) 3 (D) 1

06. A $1\,\mu\text{F}$ capacitor at $6\,\text{V}$ is connected in parallel with identical uncharged capacitor, with like plate orientation. Final common voltage in volts is

(A) 2 (B) 3

(C) 6 (D) 4

07. Two equal capacitors, one charged to V and one uncharged, are connected in parallel. Final stored energy divided by initial stored energy is

(A) 1 (B) -0.5

(C) 1.5 (D) 0.5

08. Capacitors $C, 2C$ are in series with equal charge Q. Their energy ratio U_C/U_{2C} is

(A) 1 (B) 2

(C) 3 (D) 4

09. Four equal capacitors C form two series pairs connected in parallel. Equivalent capacitance divided by C is

(A) 2 (B) 0.5

(C) 1 (D) 0

10. Four equal capacitors C form two parallel pairs connected in series. Equivalent capacitance divided by C is

(A) 0.5 (B) 0

(C) 2 (D) 1

11–40 / New middle-to-hard practice

11. Nodes A and B are held at V and zero. Capacitors AX, XB, AY, YB, XY have values $2C, C, C, 2C, C$. Floating nodes X and Y have zero net free charge. Charge magnitude on capacitor XY divided by CV is

(A) $\frac{1}{3}$ (B) $\frac{1}{15}$

(C) 0.4 (D) 0.2

12. Capacitors AX, XB, AY, YB have values $2C, C, C, 2C$. A and B stay connected to a voltage V. Floating X, Y are neutral. An initially uncharged capacitor C is now connected between X, Y. Additional positive charge supplied at A divided by CV is

(A) $\frac{1}{3}$ (B) $\frac{2}{15}$

(C) 0.2 (D) $\frac{1}{15}$

13. A floating node X lies between capacitors C from A to X and $2C$ from X to B. With A at V and B at zero, X is at $V/2$. Keeping its net free charge fixed, set A to zero and B to V. New X potential divided by V is

(A) $\frac{1}{3}$ (B) $\frac{5}{6}$

(C) 0.5 (D) $\frac{2}{3}$

14. A floating X connects to A through C and to B through $3C$. With A at V and B at zero, X potential is $V/2$. Preserve X net charge, then set A to $2V$ and B to $-V$. Final stored energy divided by CV^2 is

(A) 4.5 (B) 3.5

(C) 3 (D) 2

15. A vacuum parallel-plate capacitor C_0 is charged and isolated. Half its area is then filled across the full gap by dielectric K=2. The other half has layers of equal thickness, one vacuum and one K=2. Final stored energy divided by initial energy is

(A) $\frac{5}{3}$ (B) $\frac{2}{3}$

(C) 0.5 (D) 0.6

16. An isolated capacitor initially vacuum has half its area filled across its gap by dielectric K. Its energy becomes half the original. The same dielectric volume is then rearranged as a full-area slab filling half the gap. Charge stays fixed. Final energy divided by the original vacuum energy is

(A) $\frac{2}{3}$ (B) 0.5

(C) $\frac{1}{3}$ (D) 0.75

17. A charged isolated vacuum capacitor has initial energy U_0 and gap d. An uncharged metal slab of thickness $d/3$ fills its plate area. The remaining gaps are filled with dielectric K=2. Neglect fringing. Total quasistatic external work divided by U_0 is

(A) -2 (B) $-\frac{1}{3}$

(C) $-\frac{2}{3}$ (D) $\frac{2}{3}$

18. A vacuum capacitor C_0 remains at battery voltage V. A floating uncharged metal slab of thickness half the gap is inserted and the remaining gap filled with dielectric K=3. External quasistatic mechanical work divided by C_0V^2 is

(A) -1.5 (B) -2.5

(C) -5 (D) 2.5

19. Capacitor C at voltage V and capacitor $2C$ at voltage $2V$ are connected in parallel with opposite polarities. No source remains. The fraction of initial electric energy dissipated is

(A) $\frac{5}{6}$ (B) $\frac{2}{3}$

(C) $\frac{1}{3}$ (D) 0.5

20. Two identical capacitors C have positive voltages V_1, V_2 with $V_1 + V_2 = 4V$ and total initial energy $5CV^2$. They are joined in parallel with opposite polarities. Heat dissipated divided by CV^2 is

(A) 1 (B) 5

(C) 4 (D) 3

21. Capacitors C and $2C$ are charged in series across V. Battery is removed and the capacitors are reconnected in parallel with like-charged plates together. The fraction of initial stored energy dissipated is

(A) $\frac{2}{9}$ (B) $\frac{1}{6}$

(C) $\frac{1}{3}$ (D) $\frac{1}{9}$

22. Capacitors C and $3C$ are separately charged to V, then connected in series aiding as C between A, X and $3C$ between X, B. An initially uncharged capacitor C is connected across A, B; no battery remains. Its final charge divided by CV is

(A) 0.75 (B) 1

(C) $\frac{6}{7}$ (D) $\frac{4}{7}$

23. A vacuum capacitor C_0 is charged to V and disconnected. Its gap is doubled and then filled completely with dielectric K=3, always isolated and moved quasistatically. Net external mechanical work divided by C_0V^2 is

(A) $\frac{1}{6}$ (B) $\frac{1}{12}$

(C) $-\frac{1}{3}$ (D) $-\frac{1}{6}$

24. A vacuum capacitor C_0 stays connected at V. Its gap is doubled and then filled completely with K=3, quasistatically. Net external mechanical work divided by C_0V^2 is

(A) -0.25 (B) $-\frac{1}{6}$

(C) $\frac{1}{12}$ (D) 0.25

25. An isolated parallel-plate capacitor pulls a movable plate against a spring. Its zero-charge gap is d. At fixed charge Q its equilibrium closure is $d/4$. The charge is suddenly increased to $\sqrt{2}Q$ without a mechanical impulse, and the plate is released from its old rest position. Minimum gap during motion divided by d is

(A) 0.125 (B) 0.5

(C) 0.25 (D) 0

26. A parallel-plate capacitor has one plate attached to a spring of constant k; the zero-voltage gap is d. At fixed battery voltage its stable equilibrium closure is $d/4$. The moving plate has mass m. For small oscillations, $\omega^2/(k/m)$ is

(A) 0.5 (B) $\frac{2}{3}$

(C) $\frac{1}{3}$ (D) 0.75

27. A capacitor is discharged through two resistors in series, heating insulated bodies of heat capacities H and $2H$. Their temperature rises are equal. Reset the capacitor to the same initial charge and discharge it through the same resistors in parallel. The first body's new temperature rise divided by its old rise is

(A) 0.5 (B) 4

(C) 1.5 (D) 2

28. A capacitor discharged through parallel resistors heats two insulated bodies of heat capacities H and $3H$. The first temperature rise is twice the second. After resetting the capacitor, discharge through the same resistors in series. The second body's new temperature rise divided by its old rise is

(A) 1.5 (B) 2

(C) $\frac{1}{3}$ (D) $\frac{2}{3}$

29. A dielectric slab fills the gap of a capacitor over an insertion length x, from zero to plate length L; $C(x) = C_0[1 + (K-1)x/L]$. Charge Q is fixed. The slab's pulling force at $x = L/2$ is one-fourth its value at zero. Positive quasistatic work extracted during insertion from zero to $L/2$, divided by Q^2/C_0, is

(A) 0.25 (B) 0.5

(C) 0.75 (D) 0.125

30. A capacitor C_0 at battery voltage V receives a dielectric K completely. The battery supplies four times its original stored energy. Reset to vacuum, charge again to V and isolate. For $C(x) = C_0[1+(K-1)x/L]$, slab pulling force just before full insertion divided by its force at half insertion is

(A) $\frac{2}{3}$ (B) 2.25

(C) $\frac{4}{9}$ (D) $\frac{1}{3}$

31. Capacitors C and $2C$ have safe voltage limits 100 V and 60 V respectively. They are connected in series to an adjustable source. The greatest safe source voltage, without exceeding either rating, is

(A) 150 (B) 160

(C) 180 (D) 120

32. Two parallel-plate capacitors of equal area are in series. The first has vacuum gap d and breakdown field E_0; the second has gap $2d$, dielectric K=3 and breakdown field $E_0/2$. The greatest safe total voltage divided by $E_0 d$ is

(A) 2 (B) $\frac{4}{3}$

(C) $\frac{5}{3}$ (D) 1.5

33. Two conducting spheres of radii R and $2R$ are far apart, initially carrying Q and zero. They are connected by a thin wire. All lost electric energy heats the wire, of heat capacity H. With $k_e = 1/(4\pi\epsilon_0)$, its temperature rise divided by $k_e Q^2/(RH)$ is

(A) 0.5 (B) $\frac{2}{3}$

(C) $\frac{1}{3}$ (D) $\frac{1}{6}$

34. Eight identical conducting liquid drops, each of radius r and charge q, have total self-energy U. Merge them into one drop, then connect it to an initially uncharged conducting sphere of radius $2r$ far away. Heat dissipated during the wire connection alone, divided by U, is

(A) 2 (B) 3

(C) 1 (D) 4

35. An ideal LC circuit starts with capacitor C charged to V_0, and zero current. When capacitor charge first becomes $CV_0/2$, a dielectric K=3 is inserted instantly; charge and current are continuous. The new maximum capacitor voltage divided by V_0 is

(A) $\frac{\sqrt{10}}{6}$ (B) $\frac{\sqrt{10}}{3}$

(C) $\frac{1}{3}$ (D) $\frac{\sqrt{3}}{3}$

36. An ideal LC circuit starts with charge Q_0 on capacitor C. At charge $Q_0/2$, dielectric K is inserted instantly, preserving charge and current. The new maximum current is $\sqrt{7/8}$ times the maximum without insertion. New maximum charge divided by Q_0 is

(A) $\frac{\sqrt{14}}{4}$ (B) $\frac{\sqrt{7}}{4}$

(C) $\frac{\sqrt{7}}{2}$ (D) $\sqrt{2}$

37. An infinite capacitor ladder starts with a series capacitor C. Beyond it, a shunt C connects to the return terminal in parallel with an identical continuation of the entire ladder. Input voltage is V. Charge on the first shunt capacitor divided by CV is

(A) $-2+\sqrt{5}$ (B) $\frac{3}{2}-\frac{\sqrt{5}}{2}$

(C) $\frac{1}{2}+\frac{\sqrt{5}}{2}$ (D) $-\frac{1}{2}+\frac{\sqrt{5}}{2}$

38. An infinite capacitor ladder begins with series C, followed by shunt C in parallel with an identical continuation, all shunts going to the return terminal. With input voltage V, energy in the first series capacitor divided by total stored energy is

(A) $-\frac{1}{2}+\frac{\sqrt{5}}{2}$ (B) $\frac{1}{2}+\frac{\sqrt{5}}{2}$

(C) $\frac{3}{2}-\frac{\sqrt{5}}{2}$ (D) $\frac{5}{2}-\frac{\sqrt{5}}{2}$

39. Two unknown capacitors measure $6\,\mu$F in parallel and $4/3\,\mu$F in series. They are connected in series across 12 V. Stored energy in the larger capacitor, in μJ, is

(A) 32 (B) 24

(C) 16 (D) 64

40. An unknown capacitor C with parallel stray S measures $5\,\mu$F. Putting C in series with a known $6\,\mu$F capacitor, with S still across the entire pair, measures $4\,\mu$F. C alone is then charged to 12 V, isolated, and filled by K=2. Its final energy in μJ is

(A) 72 (B) 216

(C) 144 (D) 108

41–50 / Preserved hard practice

41. Three isolated conducting nodes A,B,D have fixed net charges $Q, 0, -Q$. They are linked only by ideal capacitors $C_{AB}=C$, $C_{BD}=2C$, $C_{DA}=3C$. A dielectric of relative permittivity 2 is fully inserted quasistatically into AB. Work done by the electric field divided by Q^2/C is

(A) 0.125 (B) $\frac{1}{88}$

(C) $\frac{3}{22}$ (D) $\frac{1}{44}$

42. A dielectric slab of mass M slides into an isolated capacitor of fixed charge Q. Insertion $x \in [0, L]$ gives capacitance $C_0(1+2x/L)$. A spring is unstretched at $x=L/2$ and has $kL^2 = Q^2/(3C_0)$. Released from rest at $x=L/2$, the slab reaches $x=L$ without friction. Its speed squared there divided by $Q^2/(MC_0)$ is

(A) $\frac{1}{12}$ (B) $\frac{1}{6}$

(C) $\frac{1}{3}$ (D) $\frac{1}{24}$

43. A capacitor C charged to V is first joined in parallel through a resistor to an uncharged capacitor $2C$. After redistribution, the pair stays parallel and is connected through a resistor to a battery that makes its final common voltage $-V$, opposite to the original polarity. Total Joule heat from both operations divided by CV^2 is

(A) 3 (B) $\frac{8}{3}$

(C) $\frac{1}{3}$ (D) 4

44. Two conducting spheres of radii R and $2R$ are very far apart and initially carry charges $3Q$ and $-Q$. A thin wire joins them until their potentials are equal. Neglect mutual energies and wire capacitance. Heat released divided by kQ^2/R is

(A) $\frac{2}{3}$ (B) $\frac{49}{12}$

(C) $\frac{41}{12}$ (D) 4.75

45. Capacitors C and $2C$ initially have voltages V and $2V$. Their positive plates are joined, as are their negative plates, through resistive connections. After charge redistribution, a dielectric of relative permittivity 3 is fully inserted quasistatically into the capacitor originally of capacitance C, while the pair stays connected and isolated. Work done by the electric field during insertion, divided by CV^2, is

(A) $\frac{25}{6}$ (B) $\frac{2}{3}$

(C) $\frac{5}{6}$ (D) $\frac{5}{3}$

46. A vacuum parallel-plate capacitor of initial capacitance C, gap d, remains at battery voltage V while its gap is doubled quasistatically. It is then disconnected, and a dielectric of relative permittivity 3 is inserted fully. Net external mechanical work in the two operations divided by CV^2 is

(A) $\frac{5}{12}$ (B) 0.25

(C) $\frac{1}{12}$ (D) $-\frac{1}{6}$

47. A capacitor C at voltage V is connected to an uncharged $2C$ capacitor: negative plates by ideal wire, positive plates through R in series with a parallel pair R and $3R$. Only the $3R$ resistor heats an insulated body of heat capacity C_h. Its temperature rise divided by CV^2/C_h is

(A) $\frac{1}{3}$ (B) $\frac{1}{28}$

(C) $\frac{1}{7}$ (D) $\frac{3}{28}$

48. An ideal LC circuit starts with capacitor voltage V_0 and zero current. When its capacitor charge first falls to half its initial value, a dielectric of relative permittivity 4 is inserted very rapidly into the capacitor. Charge and inductor current do not change during insertion. Subsequent maximum current divided by the original maximum current is

(A) 0.5 (B) $\frac{\sqrt{7}}{4}$

(C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{13}}{4}$

49. One capacitor plate of mass m is attached to a horizontal spring k, initially unstretched at plate gap d. Both plates have area A and carry fixed isolated charges $\pm Q$. The other plate is fixed. The movable plate is released from rest; ignore fringing and friction. At the critical charge where first contact becomes possible, $Q^2/(\varepsilon_0 Akd)$ is

(A) 0.25 (B) 0.5

(C) 1 (D) 2

50. A movable plate of a parallel-plate capacitor is attached to a spring of stiffness k. The uncharged gap is d, plate area is A, and the fixed plate is rigid. A battery maintains voltage V. Neglect fringing. At a static equilibrium the movable plate has shifted $d/3$ toward the fixed plate. Find V^2 divided by $kd^3/(\varepsilon_0 A)$.

(A) $\frac{16}{27}$

(B) $\frac{8}{27}$

(C) $\frac{4}{27}$

(D) $\frac{2}{9}$

Answer key

Capacitance

P-CAPA / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**D**	21	**D**	31	**A**	41	**B**
02	**B**	12	**D**	22	**C**	32	**C**	42	**A**
03	**A**	13	**B**	23	**D**	33	**C**	43	**A**
04	**A**	14	**B**	24	**A**	34	**A**	44	**B**
05	**A**	15	**D**	25	**C**	35	**A**	45	**D**
06	**B**	16	**A**	26	**C**	36	**C**	46	**C**
07	**D**	17	**C**	27	**D**	37	**B**	47	**B**
08	**B**	18	**B**	28	**D**	38	**A**	48	**D**
09	**C**	19	**B**	29	**A**	39	**A**	49	**C**
10	**D**	20	**C**	30	**C**	40	**D**	50	**B**

Chapter 18

Current Electricity

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A wire is stretched to twice its length at constant volume and resistivity. New resistance divided by original is

(A) 8 (B) 4

(C) 3 (D) 5

02. A wire is cut into three equal lengths and all are connected in parallel. Equivalent resistance divided by original is

(A) $\frac{-8}{9}$ (B) $\frac{1}{9}$

(C) $\frac{10}{9}$ (D) $\frac{2}{9}$

03. Two resistors $2,4\,\Omega$ are in series across $12\,\mathrm{V}$. Voltage across $4\,\Omega$ in volts is

(A) 7 (B) 16

(C) 8 (D) 9

04. Resistors $3,6\,\Omega$ are in parallel carrying total $6\,\mathrm{A}$. Current through $3\,\Omega$ in amperes is

(A) 4 (B) 5

(C) 3 (D) 8

05. A cell supplies current $2\,\mathrm{A}$ to $4\,\Omega$ and $1\,\mathrm{A}$ to $10\,\Omega$ separately. Internal resistance in ohms is

(A) 2 (B) 3

(C) 4 (D) 1

06. The cell in an independent experiment gives current $2\,\mathrm{A}$ through $4\,\Omega$ and $1\,\mathrm{A}$ through $10\,\Omega$. Its emf in volts is

(A) 13 (B) 11

(C) 24 (D) 12

07. In a metre bridge, balance length from left is 40 cm. Left resistance is 2 Ω. Right resistance in ohms is

(A) 3 (B) 4

(C) 6 (D) 2

08. Two identical 100 W, 200 V lamps in series are across 200 V. Treat each resistance as fixed at its rating. Total power in watts is

(A) 49 (B) 50

(C) 100 (D) 51

09. Four equal resistors R form two series branches each of two resistors, connected in parallel. Equivalent divided by R is

(A) 0 (B) 0.5

(C) 2 (D) 1

10. A cell charging at current I has emf E and internal resistance r. Terminal voltage is

(A) $E + Ir$ (B) E

(C) $Ir - E$ (D) $E - Ir$

11–40 / New middle-to-hard practice

11. A 10 m potentiometer wire of resistance 40 Ω is driven by an unknown ideal source through a box resistance R. The gradient is 0.4 mV/cm; adding 240 Ω to R changes it to 0.25 mV/cm. The source emf, in volts, is

(A) 6.4 (B) 4

(C) 2.5 (D) 1.6

12. A 4 V ideal driver feeds a 40 Ω, 10 m potentiometer wire through an unknown box R. A test cell balances at 4 m. Adding 240 Ω to R makes the same cell balance at 6.4 m. Its emf, in volts, is

(A) 0.1 (B) 0.4

(C) 0.25 (D) 0.16

13. A cell gives 2 A through 2 Ω and 1 A through 5 Ω in separate trials. Its greatest power delivery to an adjustable external resistor, in watts, is

(A) 6 (B) 4.5

(C) 9 (D) 18

14. Two identical cells in series give 3 A through 2 Ω; in parallel they give 2 A through the same load. The initial ideal short-circuit current of one cell, in amperes, is

(A) 4.5 (B) 9

(C) 3 (D) 18

15. A 6 V cell with $1\,\Omega$ internal resistance and a 3 V cell with $2\,\Omega$ internal resistance are connected in parallel with like terminals together, across $2\,\Omega$. The charging current into the 3 V cell, in amperes, is

(A) 0.375 (B) 0.75

(C) 1.5 (D) 1.125

16. A 6 V cell with $1\,\Omega$ internal resistance is in parallel with a second cell of unknown emf and $2\,\Omega$ internal resistance. They supply 1 A to a $2\,\Omega$ load. Taking the load terminal at +2 V, the second cell is found to have its positive pole at the other terminal. Its emf magnitude, in volts, is

(A) 8 (B) 4

(C) 2 (D) 6

17. Across a 12 V ideal supply, an unknown R is in series with $6\,\Omega$. A $6\,\Omega$ voltmeter across the latter reads 4 V. Replace it by a $12\,\Omega$ voltmeter across the same resistor. Its reading, in volts, is

(A) 4.8 (B) 4

(C) 6 (D) 7.2

18. A linear two-terminal source reads 6 V when loaded by $3\,\Omega$ and 8 V when loaded by $6\,\Omega$. Connecting a $9\,\Omega$ voltmeter alone across it gives a reading, in volts, of

(A) 9 (B) 6.75

(C) 12 (D) 10

19. A and B are held at 12 V and zero. Resistances AX, XB, AY, YB, XY are respectively $2, 4, 4, 2, 2\,\Omega$. The current from X to Y, in amperes, is

(A) $\frac{6}{7}$ (B) $\frac{12}{7}$

(C) 0 (D) $\frac{3}{7}$

20. A Wheatstone bridge has arms AX=$3\,\Omega$, XB=$6\,\Omega$, AY=$4\,\Omega$ and YB unknown. Its central galvanometer shows zero. If 18 V is applied between A and B, total source current, in amperes, is

(A) 4.5 (B) 3

(C) 3.5 (D) $\frac{14}{3}$

21. A wire has resistance $4\,\Omega$ at its reference temperature and coefficient $0.002\,\mathrm{K}^{-1}$. After stretching at constant volume, its resistance at 100 K above reference is $12\,\Omega$. Its new area divided by its original area is

(A) $\frac{\sqrt{10}}{2}$ (B) $\frac{\sqrt{10}}{5}$

(C) 0.4 (D) 0.6

22. A uniform wire of resistance R is cut into equal halves. One half is stretched to twice its length, at constant volume and resistivity; the other stays unchanged. The halves are placed in parallel across the original fixed voltage. Total power divided by the original uncut-wire power is

(A) 5 (B) 2.5

(C) 2 (D) 1.25

23. A resistor obeys $R = R_0(1 + \alpha\Delta T)$ and loses heat at rate $h\Delta T$. Steady temperature rises at currents I and $\sqrt{2}I$ are 10 K and 30 K. The critical current above which this model has no finite stable thermal equilibrium, divided by I, is

(A) 2 (B) $\sqrt{2}$

(C) 3 (D) $\sqrt{3}$

24. A resistor has $R = 10(1 + 0.025\Delta T)\,\Omega$ and loses heat at rate ΔT watts. It is maintained across 20 V. Its steady current, in amperes, is

(A) 1.25 (B) 2

(C) $-1 + \sqrt{5}$ (D) $1 + \sqrt{5}$

25. For $0 \le t \le 4$ s, current through $1\,\Omega$ is $i = a - bt$, with positive a, b. Net charge is 4 C and heat is 16 J. The current reverses during this interval. Total charge counted without regard to direction, in coulombs, is

(A) $\frac{16}{3}$ (B) 8

(C) $\frac{20}{3}$ (D) 4

26. Current through $1\,\Omega$ is $i = at$ for 0 to 2 s and $i = -b$ for 2 to 4 s, where a, b are positive constants in SI units. Net charge is zero and heat generated is 42 J. Total unsigned charge, in coulombs, is

(A) 9 (B) 6

(C) 12 (D) 18

27. A galvanometer becomes a 101 mA ammeter with a $1\,\Omega$ shunt, and a 201 mA ammeter with a $0.5\,\Omega$ shunt. To turn the original galvanometer into a 10 V voltmeter, the required series resistance, in ohms, is

(A) 10000 (B) 4900

(C) 19900 (D) 9900

28. A galvanometer in series with $900\,\Omega$ gives full-scale at 10 V; with $1900\,\Omega$ it gives full-scale at 20 V. The shunt needed to make the original meter read 1 A full-scale, in ohms, is

(A) $\frac{100}{101}$ (B) 1

(C) $\frac{10}{9}$ (D) $\frac{100}{99}$

29. A cell delivers 2 W to either a $2\,\Omega$ or an $8\,\Omega$ load. Its greatest power delivery to an external resistor, in watts, is

(A) 2 (B) 4.5

(C) 2.25 (D) 3

30. A 12 V cell with internal resistance $3\,\Omega$ has a fixed $6\,\Omega$ resistor across its terminals. A variable load is connected across the same terminals. Maximum power in that variable load, in watts, is

(A) 6 (B) 4

(C) 12 (D) 8

31. A 12 V cell of internal resistance $1\,\Omega$ feeds an ideal regulated load that consumes 27 W. Its operating rule permits source current no greater than 4 A. The permitted load voltage, in volts, is

(A) 3 (B) 9

(C) 6 (D) 12

32. A 12 V cell of internal resistance $1\,\Omega$ feeds a 27 W constant-power load and a shunt resistance R in parallel. For any positive steady terminal voltage to exist, the smallest R, in ohms, is

(A) 1 (B) 6

(C) 3 (D) 2

33. A uniform circular wire has total resistance $12\,\Omega$. Two contacts give equivalent resistance $8/3\,\Omega$. Move one contact halfway along the shorter arc toward the other contact. The new equivalent resistance, in ohms, is

(A) $\frac{10}{3}$ (B) $\frac{5}{3}$

(C) $\frac{4}{3}$ (D) 2

34. A uniform circular wire of total resistance $16\,\Omega$ reads $3\,\Omega$ between two contacts. Cut the two arcs apart, stretch the shorter one to twice its length at constant volume, and reconnect the arcs in parallel. Their new equivalent resistance, in ohms, is

(A) 6 (B) 12

(C) $\frac{48}{7}$ (D) $\frac{24}{7}$

35. Two resistors have linear temperature coefficients $+0.004$ and $-0.002\,\mathrm{K}^{-1}$. Their series combination has zero temperature coefficient. At a rise of 100 K, their parallel equivalent resistance divided by its reference-temperature value is

(A) 1.4 (B) 1

(C) 1.12 (D) 0.96

36. Two resistors each have reference resistance R and coefficients $+a$ and $-a$. At a positive temperature rise T their parallel equivalent is 0.96 of its reference value. At the same T, put the positive-coefficient resistor in series with the parallel combination of the other and a temperature-independent R. Total resistance divided by R is

(A) $\frac{74}{45}$ (B) 1.6

(C) $\frac{14}{9}$ (D) $\frac{16}{9}$

37. Two resistors in series across 12 V dissipate 9 W and 3 W. They are reconnected in parallel across the same supply. Total power, in watts, is

(A) 36 (B) 48

(C) 96 (D) 64

38. Two positive resistors consume 12 W in series and 64 W in parallel across the same 12 V supply. The smaller resistance, in ohms, is

(A) 4 (B) 6

(C) 2 (D) 3

39. Two wire segments of equal length and areas A and $2A$ are in series. Electron transit time through the second is twice that through the first. Its mobility is twice the first's; use $\sigma = ne\mu$. The fraction of total Joule heat generated in the first segment is

(A) $\frac{2}{3}$ (B) 0.8

(C) 0.5 (D) 0.75

40. Two equal-length metal segments in series have equal carrier density and areas A and $2A$. Their electric fields obey $E_1 = 4E_2$. In the Drude model $\rho = m/(ne^2\tau)$, with the same carrier mass. Mean numbers of collisions during an electron's transit obey $N_2/N_1 =$

(A) 4 (B) 1

(C) 0.5 (D) 2

41–50 / Preserved hard practice

41. A DC voltage V is applied between A and D of a bridge. Resistances are $AB = R$, $BD = 2R$, $AC = 2R$, $CD = R$, and $BC = R$. What fraction of the total Joule power is generated in BC?

(A) $\frac{1}{14}$ (B) $\frac{1}{21}$

(C) $\frac{1}{7}$ (D) $\frac{1}{35}$

42. A resistor has $R = R_0(1 + \alpha\Delta T)$ and loses heat at rate $G\Delta T$. A constant voltage V produces equilibrium rise $1/(2\alpha)$. It is then driven instead by constant current V/R_0. Assuming the same linear laws remain valid, the new stable temperature rise divided by the old rise is

(A) 4 (B) 1.5

(C) 3 (D) 6

43. A potentiometer wire of total resistance R is driven by a cell of emf E and internal resistance r. A fixed standard cell balances at 40 cm. A resistor R is then connected across the entire potentiometer wire, and balance moves to 60 cm. Driving-cell emf and wire temperature stay fixed. Total external power after adding the shunt, divided by E^2/r, is

(A) $\frac{1}{9}$ (B) $\frac{2}{9}$

(C) $\frac{4}{9}$ (D) 0.25

44. A galvanometer of resistance 100 Ω reaches full scale at 1.00 mA. A series resistor converts it to a 10.0 V voltmeter. This voltmeter is connected across an unknown resistor, the parallel pair being supplied by a 12.0 V cell of internal resistance 1.00 kΩ. The meter reads 9.00 V. Unknown resistance in kΩ is

(A) $\frac{30}{7}$ (B) 3

(C) $\frac{90}{7}$ (D) $\frac{10}{3}$

45. A charged capacitor discharges completely through a resistor R in series with a parallel pair R and $2R$. What fraction of its initial stored energy is dissipated in the $2R$ resistor?

(A) 0.2 (B) 0.4

(C) $\frac{1}{3}$ (D) $\frac{2}{15}$

46. A uniform material forms a wire whose radius increases linearly from a to $3a$ along its length. A brief current pulse passes through it. Neglect heat exchange and temperature dependence of resistivity. The two equal-length halves are then separately allowed to reach uniform temperatures without losing heat. Their temperature-rise ratio, thin half to thick half, is

(A) 3 (B) $\frac{19}{3}$

(C) $\frac{57}{7}$ (D) $\frac{19}{7}$

47. A cell of emf E and internal resistance r feeds a wire. The wire is stretched to twice its original length at constant volume, with unchanged resistivity. Its steady Joule power is identical before and after stretching. What fraction of the cell's total Joule heating is generated inside the cell after stretching?

(A) $\frac{2}{3}$ (B) 0.2

(C) $\frac{1}{3}$ (D) 0.5

48. A thermistor of resistance $R_0/(1+\alpha\Delta T)$ is in series with a temperature-independent resistor R_0 across constant voltage V. Only the thermistor heats the body, whose loss rate is $G\Delta T$. If $V^2\alpha/(GR_0) = 9/2$, its positive equilibrium value of $\alpha\Delta T$ is

(A) 2 (B) 1.5

(C) 0.5 (D) 1

49. A cell of emf E and internal resistance r feeds a variable resistor $R < r$. Its external power is one quarter of the maximum possible external power. The terminal voltage accelerates a particle of mass m and positive charge q from rest. It then enters a perpendicular uniform field B with orbit radius a. Find $qB^2a^2/(mE)$.

(A) $1 - \frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{3}}{2} + 1$

(C) 0.5 (D) $2 - \sqrt{3}$

50. A cell of emf E and internal resistance r feeds a parallel combination of a variable resistor x and a fixed resistor $2r$. Choose x to maximise total external power. Resistor x alone heats an insulated body of heat capacity C_h for time t; neglect resistance changes. Its temperature rise divided by $E^2t/(rC_h)$ is

(A) 0.125 (B) 0.25

(C) 0.0625 (D) $\frac{2}{9}$

Answer key

Current Electricity

P-CUEL / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**B**	21	**B**	31	**B**	41	**D**
02	**B**	12	**D**	22	**B**	32	**C**	42	**D**
03	**C**	13	**C**	23	**A**	33	**B**	43	**B**
04	**A**	14	**B**	24	**C**	34	**C**	44	**A**
05	**A**	15	**A**	25	**C**	35	**C**	45	**D**
06	**D**	16	**B**	26	**C**	36	**A**	46	**C**
07	**A**	17	**A**	27	**D**	37	**D**	47	**C**
08	**B**	18	**A**	28	**D**	38	**D**	48	**D**
09	**D**	19	**A**	29	**C**	39	**B**	49	**A**
10	**A**	20	**C**	30	**D**	40	**B**	50	**A**

Chapter 19

Magnetic Effects of Current

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A galvanometer resistance $G = 100\,\Omega$ has full-scale current $1\,\text{mA}$. A shunt makes range $1\,\text{A}$. Shunt resistance in ohms is

(A) $\frac{100}{999}$ (B) $\frac{1099}{999}$

(C) $\frac{200}{999}$ (D) $\frac{-899}{999}$

02. A $100\,\Omega$ galvanometer with $1\,\text{mA}$ full scale is converted to $10\,\text{V}$ voltmeter. Series resistance in ohms is

(A) 19800 (B) 9899

(C) 9900 (D) 9901

03. A long straight wire carries $10\,\text{A}$. At $0.1\,\text{m}$, with $\mu_0 = 4\pi \times 10^{-7}$, field in microtesla is

(A) 20 (B) 19

(C) 40 (D) 21

04. A particle velocity has parallel component $3\,\text{m/s}$ and cyclotron period $2\,\text{s}$. Helical pitch in metres is

(A) 5 (B) 7

(C) 6 (D) 12

05. A particle has perpendicular speed 4 and parallel speed $3\,\text{m/s}$ in a magnetic field. Its speed after one full turn in m/s is

(A) 5 (B) 4

(C) 6 (D) 10

06. For circular motion in uniform magnetic field, orbit radius doubles at fixed particle and field. Kinetic energy multiplier is

(A) 3 (B) 5

(C) 4 (D) 8

07. A particle kinetic energy quadruples at fixed mass, charge and field. Orbit radius multiplier is

(A) 3 (B) 4

(C) 1 (D) 2

08. A magnetic field along $+z$ acts on a positive charge moving along $+x$. Force direction is

(A) Positive z (B) Negative x

(C) Negative y (D) Positive y

09. A magnetic field along $+z$ acts on an electron moving along $+x$. Force direction is

(A) Negative x (B) Positive y

(C) Negative y (D) Positive z

10. Two parallel wires separated by d have force per length f. Currents both double while separation doubles. New force per length divided by f is

(A) 3 (B) 4

(C) 2 (D) 1

11–40 / New middle-to-hard practice

11. A particle of mass m and charge magnitude q follows a helix of radius r and pitch $2\pi r$ in uniform B. Its kinetic energy divided by $q^2B^2r^2/m$ is

(A) 1.5 (B) 1

(C) 2 (D) 0.5

12. Two particles of equal mass and charge move in the same uniform B. Their helix radii are r and $2r$, and pitches are $2\pi r$ and πr respectively. The second kinetic energy divided by the first is

(A) 2.5 (B) 4

(C) 2 (D) 2.125

13. A particle initially has speed v_0 and magnetic radius r_0. It loses 36% of its kinetic energy, then enters perpendicular to the near boundary of a uniform-field strip of width $0.6r_0$, with velocity perpendicular to B and the same field magnitude. Its velocity component normal to the far boundary on first exit, divided by v_0, is

(A) 0.8 (B) $\frac{\sqrt{7}}{4}$

(C) $\frac{\sqrt{7}}{5}$ (D) 0.6

14. A particle enters along +x into a magnetic strip $0 < x < \sqrt{3}r/2$, perpendicular to B, with orbit radius r and bending toward negative y. Beyond the strip it moves freely to a detector at $x = \sqrt{3}r$. Its displacement magnitude in y there, divided by r, is

(A) 2 (B) 1

(C) 1.5 (D) $\sqrt{3}$

15. A velocity selector has perpendicular fields $E = 6000$ V/m and $B_s = 0.02$ T. Selected positive ions are then accelerated through 1600 V and enter an analysing field 0.1 T normally, tracing radius 0.1 m. Their charge-to-mass ratio, in C/kg, is

(A) 80000000 (B) 50000000

(C) 18000000 (D) 30000000

16. A selector with $E = 6000$ V/m and $B_s = 0.02$ T passes positive ions. They are retarded through 2500 V, then enter a 0.1 T analysing field normally and trace radius 0.2 m. Their charge-to-mass ratio, in C/kg, is

(A) 10000000 (B) 5000000

(C) 22500000 (D) 20000000

17. For a unit positive charge, unknown uniform in-plane E and perpendicular B give force $(4, -2)$ N at velocity $(3, 0)$ m/s and $(8, 4)$ N at velocity $(0, 2)$ m/s. The force-free in-plane speed, in m/s, is

(A) $2\sqrt{2}$ (B) $\sqrt{2}$

(C) 4 (D) 2

18. A positive charge q of mass m is released from rest in uniform $\mathbf{B} = B\hat{z}$ under gravity $-g\hat{y}$. Ignore all other forces. At the first positive time its vertical velocity again becomes zero, horizontal displacement magnitude divided by vertical drop is

(A) $\frac{2}{\pi}$ (B) π

(C) $\frac{\pi}{2}$ (D) 1

19. Two long parallel wires at x=0 and x=d carry I and $4I$ in the same direction. Locate their field null, then reverse only the larger current. The distance between the old and new finite null positions, divided by d, is

(A) $\frac{8}{15}$ (B) 0.8

(C) $\frac{1}{3}$ (D) $\frac{2}{15}$

20. Three long wires at x=0, d,3d carry currents I, J,$4I$. The outer two point along +z. J is chosen so net magnetic force on the wire at zero vanishes. Force per length magnitude on the middle wire divided by $\mu_0 I^2/(2\pi d)$ is

(A) $\frac{8}{3}$ (B) $\frac{4}{3}$

(C) 4 (D) $\frac{2}{3}$

21. A long solid cylinder of radius R carries axial current density $J(r) = J_0 r/R$. Its surface field is B_s. Two radii have field magnitude $B_s/4$. Their separation divided by R is

(A) 3.75 (B) 3

(C) 3.5 (D) 1.5

22. A long hollow wire has inner radius a, outer radius $2a$, and uniform axial current density. At an interior point of radius r the field is half its value at the outer surface. The allowed r/a is

(A) $-\frac{3}{8}+\frac{\sqrt{73}}{8}$ (B) $\frac{3}{8}+\frac{\sqrt{73}}{8}$

(C) $\sqrt{2}$ (D) 1.5

23. A circular wire of radius r is fed by total current I at contacts separated by 60°. The shorter arc has twice the cross-sectional area of the longer, with the same resistivity. Leads contribute no centre field. Centre field magnitude divided by $\mu_0 I/r$ is

(A) $\frac{5}{66}$ (B) 0

(C) $\frac{1}{12}$ (D) $\frac{5}{132}$

24. A circular wire has two semicircular branches of resistance R each. An additional R is placed in one branch far from the centre. Total input current is I. A charge q crosses the centre at speed v making 30° with the ring axis. Its magnetic-force magnitude divided by $\mu_0 Iqv/r$ is

(A) $\frac{1}{12}$ (B) 0

(C) $\frac{1}{6}$ (D) $\frac{1}{24}$

25. Two concentric coplanar loops have radii R and $2R$. Their currents oppose and cancel the field at the centre; the smaller carries I. At axial distance R, field magnitude divided by $\mu_0 I/R$ is

(A) 0.1 (B) $-\frac{\sqrt{2}}{8}+\frac{4\sqrt{5}}{25}$

(C) $\frac{\sqrt{2}}{8}$ (D) $\frac{\sqrt{2}}{8}+\frac{4\sqrt{5}}{25}$

26. Two coaxial loops of equal radius R have centres separated by R and opposite currents. The field at the first loop centre vanishes; its current magnitude is I. At the midpoint, field magnitude divided by $\mu_0 I/R$ is

(A) $-\frac{1}{2}+\sqrt{2}$ (B) $-\frac{4\sqrt{5}}{25}+\frac{8\sqrt{10}}{25}$

(C) $\frac{4\sqrt{5}}{25}$ (D) $\frac{4\sqrt{5}}{25}+\frac{8\sqrt{10}}{25}$

27. A moving-coil galvanometer has current sensitivity $NAB/k = 2\,\mathrm{rad\,A^{-1}}$, where k is torsional constant. Its full-scale angle is 0.02 rad and coil resistance $60\,\Omega$. Series resistance required for a 20 V full-scale voltmeter, in ohms, is

(A) 2000 (B) 940

(C) 2060 (D) 1940

28. A moving-coil galvanometer of resistance G becomes a voltmeter using series resistance $9G$. Rewind with twice as many turns of the same-gauge wire at unchanged mean area; spring constant and full-scale angle stay fixed. Keeping that same series resistor, new voltage range divided by old is

(A) 0.5 (B) 0.45

(C) 0.55 (D) 1.1

29. A current loop in a uniform field has torque magnitude $\sqrt{3}$ N m at an unknown angle between zero and 180°. Quasistatic work to turn it from that angle to anti-alignment is 3 J. Work from that initial angle to 90°, in joules, is

(A) $\sqrt{3}$ (B) 3

(C) 2 (D) 1

30. A current loop on a torsion spring has the same equilibrium axis when a uniform field is parallel or anti-parallel to its moment. Both equilibria are stable. Their small-oscillation periods are T and $2T$ respectively. Magnetic stiffness mB divided by spring stiffness k is

(A) 0.8 (B) $\frac{1}{3}$

(C) 0.6 (D) 0.5

31. A particle of mass m and charge q moves normally to uniform B on radius r. It sticks to a stationary particle of mass $2m$ and charge $-q/2$. Immediately afterward its orbit diameter divided by r is

(A) 4 (B) 6

(C) $\frac{4}{3}$ (D) 2

32. Two particles each of mass m and charge q collide and stick in uniform B. Just before impact both speeds are v, in mutually perpendicular directions normal to B. Each initially has orbit radius r. New orbit radius divided by r is

(A) $\frac{\sqrt{2}}{2}$ (B) $\sqrt{2}$

(C) 1 (D) 0.5

33. A particle circles in fixed B, losing the same kinetic energy in each passage through a thin foil. For a reference length R, its radii after two and five passages are $3R$ and $\sqrt{3}R$. The greatest number of complete passages after which it still has positive kinetic energy is

(A) 6 (B) 8

(C) 5 (D) 7

34. A particle repeatedly passes a foil that retains a fixed fraction a of kinetic energy, then an accelerator that adds a fixed energy b. Its magnetic radii at three successive observations in the same B are $r_0, \sqrt{2}r_0, \sqrt{5/2}r_0$. Its limiting radius divided by r_0 is

(A) $\sqrt{5}$ (B) $\sqrt{3}$

(C) $\frac{\sqrt{10}}{2}$ (D) 2

35. A wire carries current I along the negative x axis toward the origin, then along the positive y axis. At point $(a, a, 0)$, the magnetic-field magnitude divided by $\mu_0 I/(4\pi a)$ is

(A) $\sqrt{2}$ (B) 2

(C) $\sqrt{2}+2$ (D) $2-\sqrt{2}$

36. A wire approaches $(R, 0)$ vertically along x=R from negative y, follows the quarter-circle of radius R counterclockwise to $(0, R)$, then leaves horizontally toward negative x along y=R. Centre field divided by $\mu_0 I/(4\pi R)$ is

(A) $1 + \frac{\pi}{2}$ (B) $\frac{\pi}{2} + 2$

(C) $2 - \frac{\pi}{2}$ (D) $\frac{\pi}{2}$

37. A wire segment carries current I from $(0, 0)$ to $(a, 0)$ to $(a, 2a)$ in field $\mathbf{B} = (B, 2B, 0)$. The net magnetic torque about the origin on just these two straight parts, divided by IBa^2, has magnitude

(A) 2 (B) 3

(C) 0 (D) $\sqrt{5}$

38. A rigid current loop is hinged along a horizontal edge. Its plane is vertical without current. In a vertical magnetic field, current I gives equilibrium tilt 30° from vertical. Double the current, keeping gravity and geometry fixed. New magnetic flux magnitude divided by the old is

(A) 2 (B) $\frac{2\sqrt{7}}{7}$

(C) $\frac{4\sqrt{7}}{7}$ (D) $\frac{\sqrt{7}}{2}$

39. A closely wound flat spiral has turns uniformly distributed in radius from R to $2R$. Its wire length is ℓ and current I; connecting leads are negligible. Magnetic moment divided by $I\ell R$ is

(A) 0.75 (B) $\frac{2}{3}$

(C) $\frac{7}{6}$ (D) $\frac{7}{9}$

40. A flat spiral occupies radii R to $2R$. The number of turns in a thin annulus is proportional to its area, so $dN \propto r\,dr$. Wire length is ℓ and current I. Magnetic moment divided by $I\ell R$ is

(A) $\frac{45}{56}$ (B) $\frac{5}{6}$

(C) 0.9375 (D) $\frac{7}{9}$

41–50 / Preserved hard practice

41. A positive charge enters normally into a magnetic slab of width a, with field perpendicular to the motion plane and orbit radius $2a$. Immediately afterward it crosses a second slab of the same width with equal and opposite field. It leaves parallel to its original velocity. Magnitude of its final transverse displacement divided by a is

(A) $2 - \sqrt{3}$ (B) 1

(C) $-2 + 2\sqrt{3}$ (D) $4 - 2\sqrt{3}$

42. A block of mass m and positive charge q slides on a horizontal rough insulating floor. A uniform horizontal magnetic field perpendicular to its velocity makes the magnetic force vertically upward. Kinetic friction coefficient is μ. Initial speed is $mg/(2qB)$; it never loses contact. Stopping distance divided by $m^2 g/(\mu q^2 B^2)$ is

(A) $\log(2)$ (B) $-\frac{1}{2} + \log(2)$

(C) $1 - \log(2)$ (D) 0.5

43. A positive particle enters at $x = 0$ with velocity $v\hat{x}$ into a field $B_z(x) = B_0(1 + x/L)$ for $x \geq 0$. No electric field acts, and $mv = qB_0L$. Neglect radiation. Its first maximum x coordinate divided by L is

(A) $\sqrt{3}$ (B) 1

(C) $-1 + \sqrt{2}$ (D) $-1 + \sqrt{3}$

44. A rectangular current loop of mass m has both its width and height equal to a. It lies in the plane of a long fixed straight wire, with nearer side parallel to that wire at distance x. Currents I_1, I_2 are held constant, and their directions make the net force repulsive. The loop is constrained to translate without rotation, and an additional restoring force is $-kx$. If $\mu_0 I_1 I_2/(2\pi) = 2ka$, small oscillations about equilibrium have $\omega^2/(k/m)$ equal to

(A) 3 (B) 2.5

(C) 1.5 (D) 2

45. A positive particle first passes undeflected, with velocity perpendicular to both fields, through crossed fields $E = 1000\,\mathrm{V/m}$ and $B = 0.10\,\mathrm{T}$. It is then accelerated through 1.50 V and enters a second perpendicular magnetic field of 0.10 T, where its orbit radius is 2.00 mm. Neglect relativistic effects. Its charge-to-mass ratio in units of $10^8\,\mathrm{C/kg}$ is

(A) 2 (B) 1

(C) 0.5 (D) 4

46. A particle of mass m, positive charge q, is confined to a horizontal plane by an isotropic restoring force $-k\mathbf{r}$ and a perpendicular uniform field B. Gravity is balanced. Uniform circular orbits in the two opposite senses have periods in ratio 2 : 1. Find $qB/\sqrt{mk}$.

(A) $\sqrt{2}$ (B) 1

(C) 0.5 (D) $\frac{\sqrt{2}}{2}$

47. A charged vehicle of mass m moves without friction around a banked circular track of radius R and banking angle 45°. A vertical magnetic field gives a horizontal Lorentz force toward the centre. If $qB/m = \sqrt{g/R}$, its allowed speed squared divided by gR is

(A) $\frac{3}{2} - \frac{\sqrt{5}}{2}$ (B) 2

(C) $\frac{\sqrt{5}}{2} + \frac{3}{2}$ (D) $\frac{1}{2} + \frac{\sqrt{5}}{2}$

48. A particle of charge $q > 0$ and mass m moves at speed v through a semicircle of radius a in a perpendicular field B. It next enters a region with no magnetic field and a uniform electric field directly opposite its velocity; the electric field reverses it, and it returns to the entry boundary. Its farthest electric-region penetration is a. Electric-region round-trip time divided by the magnetic semicircle time is

(A) $\frac{4}{\pi}$ (B) $\frac{\pi}{2}$

(C) 2 (D) $\frac{2}{\pi}$

49. A current loop of N turns, area A, moment of inertia I, is suspended on a torsion fibre of stiffness κ in a horizontal field B. An ideal regulator maintains constant loop current i. Small deflection is assumed, so magnetic torque magnitude is $NABi$, independent of angle. After static equilibrium, the magnetic field is reversed instantly. Maximum ensuing angular speed divided by $NABi/\sqrt{I\kappa}$ is

(A) 0.5 (B) $\sqrt{2}$

(C) 1 (D) 2

50. A particle of charge $q > 0$ and mass m initially has equal velocity components u parallel and perpendicular to a uniform field $B\hat{z}$. A uniform electric field opposite $\hat{z}$ is switched on so that the parallel velocity becomes $-u$ for the first time after exactly half a cyclotron revolution. Find E divided by uB.

(A) $\frac{1}{\pi}$ (B) $\frac{2}{\pi}$

(C) $\frac{4}{\pi}$ (D) $\frac{\pi}{2}$

Answer key

Magnetic Effects of Current

P-MEC / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**B**	21	**C**	31	**A**	41	**D**
02	**C**	12	**D**	22	**B**	32	**A**	42	**B**
03	**A**	13	**C**	23	**D**	33	**A**	43	**D**
04	**C**	14	**A**	24	**D**	34	**B**	44	**B**
05	**A**	15	**B**	25	**B**	35	**A**	45	**B**
06	**C**	16	**A**	26	**B**	36	**B**	46	**D**
07	**D**	17	**A**	27	**D**	37	**D**	47	**C**
08	**C**	18	**C**	28	**C**	38	**C**	48	**A**
09	**B**	19	**A**	29	**D**	39	**D**	49	**D**
10	**C**	20	**B**	30	**C**	40	**A**	50	**B**

Chapter 20

Electromagnetic Induction

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Flux through a loop is $\Phi = 3t^2$ webers in SI units. Induced emf magnitude at $t = 2\,\mathrm{s}$ in volts is

(A) 24 (B) 13

(C) 11 (D) 12

02. Flux through a 5-turn coil is $2t$ webers per turn. Coil resistance is $10\,\Omega$. Induced current magnitude in amperes is

(A) 0.5 (B) 0

(C) 1 (D) 2

03. A one-turn loop of area $0.2\,\mathrm{m}^2$ and resistance $2\,\Omega$ lies normal to a field increasing at $3\,\mathrm{T/s}$. Current in amperes is

(A) 0.6 (B) -0.7

(C) 1.3 (D) 0.3

04. A loop resistance is R. Total flux linkage changes by $\Delta\Lambda$. Magnitude of total charge through the loop is

(A) $|\Delta\Lambda|R^2$ (B) $R|\Delta\Lambda|$

(C) $|\Delta\Lambda|/R^2$ (D) $|\Delta\Lambda|/R$

05. A one-turn loop resistance $2\,\Omega$ has flux changed from 3 to 7 Wb. Total charge magnitude in coulombs is

(A) 2 (B) 1

(C) 3 (D) 4

06. A series RL circuit starts with zero current and source E. Initial current increase rate is

(A) E/R (B) E/L

(C) ER/L (D) L/E

07. A coil has self-inductance 1 H. Current falls from 4 to 2 A. Magnetic energy released in joules is

(A) 6 (B) 5

(C) 12 (D) 7

08. An inductor carries energy 18 J at current 3 A. Inductance in henry is

(A) 4 (B) 3

(C) 5 (D) 8

09. A magnet approaches a loop with its north pole facing the loop. Viewed from the magnet, induced current is

(A) Alternately clockwise and counterclockwise at every instant (B) Counterclockwise

(C) Zero because magnet is uncharged (D) Clockwise

10. A magnet recedes from a loop with its north pole facing the loop. Viewed from the magnet, induced current is

(A) Clockwise (B) Counterclockwise

(C) Zero because magnet is uncharged (D) Always independent of motion

11–40 / New middle-to-hard practice

11. An ideal inductor carries $i = 1 + bt + ct^2$ A, with b, c positive. Its induced-emf magnitudes at t=0 and 2 s are 4 V and 12 V; current increases by 8 A between these times. Magnetic-energy increase, in joules, is

(A) 160 (B) 64

(C) 80 (D) 40

12. A coil with unknown resistance R and inductance L carries prescribed current $i = 2+3t$ A. Terminal voltage in the current direction is 10 V at t=0 and 22 V at t=2 s. Initial magnetic energy, in joules, is

(A) 4 (B) 2

(C) 6 (D) 8

13. Primary current is $i = \sin \omega t + \sin 2\omega t$ A. The mutually induced emf at t=0 has magnitude 6 V. Greatest secondary emf magnitude of polarity opposite to its t=0 polarity, in volts, is

(A) 4.5 (B) 4.125

(C) 3.75 (D) 6

14. A primary current decays exponentially. In a secondary loop of resistance $3\,\Omega$ and negligible self-inductance, induced emf begins at 6 V and halves after $\ln 2$ seconds. Total heat generated in the secondary, in joules, is

(A) 6 (B) 3

(C) 18 (D) 12

15. In the changing field of a long solenoid, initially stationary identical charges at radii 0.2 m and 0.8 m have equal acceleration magnitudes. The solenoid radius lies between them. Greatest induced-electric-field magnitude divided by its value at either charge is

(A) 4 (B) 0.5

(C) 2 (D) $\sqrt{2}$

16. A long solenoid has radius 0.3 m and nonzero uniform field rate. A stationary charge at radius 0.1 m has acceleration magnitude a_0. Two radii have acceleration $1.5a_0$. Their separation, in metres, is

(A) 0.15 (B) 0.3

(C) 0.6 (D) 0.45

17. A 1 kg rod of length 1 m slides on rails in B=2 T, closing resistance $4\,\Omega$. It starts at 3 m/s and then experiences only magnetic drag; circuit inductance is negligible. Distance travelled by the time its kinetic energy halves, in metres, is

(A) $3 - \frac{3\sqrt{2}}{2}$ (B) $3 - \sqrt{3}$

(C) $\frac{3\sqrt{2}}{2}$ (D) 1.5

18. A 1 kg rod of length 1 m slides in B=2 T on rails with resistance $4\,\Omega$. A constant 1 N forward force acts; initial speed is 3 m/s. Inductance and friction are negligible. Heat generated until speed falls to 2 m/s, in joules, is

(A) $\log(4) + \frac{3}{2}$ (B) $\log(2) + \frac{5}{2}$

(C) 3.5 (D) $\log(2) + \frac{7}{2}$

19. A uniform conducting rod of length l is hinged at one end and falls from horizontal to vertical without friction. Field is normal to its plane and varies with pivot distance as $B(r) = B_0(1 + r/l)$. At the lowest point, end-to-end emf divided by $B_0 l\sqrt{gl}$ is

(A) $\frac{5\sqrt{3}}{6}$ (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{\sqrt{6}}{3}$ (D) $\frac{5\sqrt{3}}{3}$

20. A uniform rod of mass m and length l rotates in a plane normal to uniform B about a point one-third of its length from an end. Rotational kinetic energy is K. Voltage magnitude between its two ends divided by $Bl\sqrt{K/m}$ is

(A) $\frac{1}{3}$ (B) $\frac{\sqrt{2}}{2}$

(C) $\sqrt{2}$ (D) $\frac{\sqrt{6}}{2}$

21. A series RL circuit connected to 12 V starts with zero current. Its currents at $t = \ln 2$ s and $2\ln 2$ s are 2 A and 3 A. Magnetic energy at the second time, in joules, is

(A) 9 (B) 4.5

(C) 13.5 (D) 18

22. An inductor discharges through R_1, halving current in $\ln 2$ s; adding R_2 in series halves that halving time. A 12 V source drives 3 A steadily through both resistors and the inductor. Magnetic energy if that source instead drives R_1 alone steadily, in joules, is

(A) 36 (B) 9

(C) 18 (D) 72

23. A 12 V ideal source feeds a $2\,\Omega$ series resistor, then parallel branches: $4\,\Omega$, and $2\,\Omega$ in series with a 2 H inductor. Initially currents vanish. Inductor current at $t = (3/5)\ln 2$ s after connection, in amperes, is

(A) 2.4 (B) 1.2

(C) 0.6 (D) 2

24. A 1 H inductor carries 3 A steadily. Its source is removed, leaving the inductor in series with $2\,\Omega$ and a parallel combination of $3\,\Omega$ and $6\,\Omega$. Total heat ultimately produced in the $3\,\Omega$ resistor, in joules, is

(A) 1.5 (B) 0.75

(C) 3 (D) 2.25

25. Two coupled coils have series aiding and opposing inductances 10 H and 2 H. One coil alone has inductance 2 H. Their coupling coefficient is

(A) $\frac{\sqrt{3}}{2}$ (B) 1

(C) $\frac{\sqrt{2}}{2}$ (D) 0.5

26. Two coupled coils have self-inductances 1 H and 4 H. The secondary is a closed ideal zero-resistance loop, with zero initial flux linkage. When primary current is 2 A, secondary current is -0.5 A in the chosen aiding convention. Total magnetic energy, in joules, is

(A) 2.5 (B) 2

(C) 1 (D) 1.5

27. A tightly coupled 100-turn solenoid has inductance L. Reverse the connections of n individual turns, preserving geometry and total wire length, with $0 < n < 50$. Its inductance becomes $L/4$. The value of n is

(A) 75 (B) 50

(C) 20 (D) 25

28. Two coincident solenoid windings have N turns each and resistances R and $4R$. At a fixed DC voltage, compare parallel aiding connection with series aiding connection. The ratio of their steady core fields, parallel divided by series, is

(A) 6.25 (B) 3.125

(C) 1.25 (D) 2.5

29. An ideal LC circuit initially has positive charge Q_0 and current charging it further. Initial capacitor energy is three times inductor energy. The first maximum charge occurs after what fraction of one LC period?

(A) 0.25 (B) 0.125

(C) $\frac{1}{6}$ (D) $\frac{1}{12}$

30. Two identical capacitors C at voltages V and $3V$ are joined in parallel with opposite polarities, allowed to settle, then connected to an ideal inductor L with zero initial current. Maximum inductor current divided by $V\sqrt{C/L}$ is

(A) 1 (B) $\sqrt{2}$

(C) $\sqrt{5}$ (D) 2

31. A one-turn loop has $A(t) = A_0(1 + t/\tau)$ in normal field $B(t) = B_0(1 - t/\tau)$, from t=0 to τ. Resistance R is constant and self-inductance negligible. Heat generated divided by $A_0^2B_0^2/(R\tau)$ is

(A) $\frac{1}{3}$ (B) 4

(C) $\frac{4}{3}$ (D) 2

32. A loop has area $A = A_0(1 + t/\tau)$ and rotates so its normal makes angle $\pi t/\tau$ with uniform B. At $t = \tau/3$, induced-emf magnitude divided by A_0B/τ is

(A) $\frac{\sqrt{3}\pi}{2}$ (B) $\frac{1}{2} + \frac{2\sqrt{3}\pi}{3}$

(C) $-\frac{1}{2} + \frac{2\sqrt{3}\pi}{3}$ (D) 0.5

33. A rotating coil of negligible inductance and resistance $6\,\Omega$ has peak emf 12 V. At constant speed its average resisting magnetic torque is 3 N m. The peak flux linkage, in webers, is

(A) 1.5 (B) 4

(C) 6 (D) 3

34. A rotating generator's required average external torque is 3 N m at 1 rad/s and 5 N m at 2 rad/s. Mechanical friction torque is constant; electrical drag torque is proportional to speed. Peak emf at 1 rad/s is $\sqrt{8}$ V. Generator resistance, in ohms, is

(A) 4 (B) $\frac{4}{3}$

(C) 2 (D) 1

35. A conducting rod of length 1 m slides on frictionless rails closing only an initially uncharged capacitor C. Ignore resistance and inductance. Under a constant 2 N force its acceleration is 2 m/s^2 without field and 0.5 m/s^2 in B=1 T. Capacitor energy at rod speed 2 m/s, in joules, is

(A) 6 (B) 8

(C) 2 (D) 3

36. A uniform conducting rod of mass m and length l rotates about one end in normal B. Its endpoints connect only to a capacitor C through ideal contacts. Constant applied torque produces half the angular acceleration obtained without B. Neglect resistance, inductance and contact inertia. Required CB^2l^2/m is

(A) $\frac{2}{3}$ (B) $\frac{4}{3}$

(C) 4 (D) $\frac{1}{3}$

37. A superconducting loop of area A and self-inductance L initially has positive current I_0 and positive external flux AB_0. Increase the external field to $3B_0$ without changing geometry. Final current reverses and its stored energy is nine times the initial energy. Current when external field first equals $2B_0$, divided by I_0, is

(A) -2 (B) -1

(C) 0 (D) 1

38. Two superconducting loops each have self-inductance L and mutual inductance $L/2$ in the aiding convention. Initially both currents and external fluxes are zero. Increase external flux through only the first loop by Φ. Total final magnetic energy divided by Φ^2/L is

(A) 0.5 (B) $\frac{2}{3}$

(C) $\frac{4}{3}$ (D) $\frac{1}{3}$

39. A rectangular loop of width a along its motion and height l enters a field region x>0 at speed v. The normal field there is $B(x) = kx$. Resistance is R and self-inductance negligible. Heat generated during entry, divided by $k^2l^2va^3/R$, is

(A) 0.5 (B) $\frac{1}{3}$

(C) $\frac{2}{3}$ (D) 1

40. An isosceles triangular loop has horizontal altitude a and vertical base l. Moving with its vertex first at speed v, it completely crosses a normal-field strip of width $a/2$ and field B. Resistance is R and self-inductance negligible. Total unsigned charge through the wire divided by Bla/R is

(A) 0.5 (B) 0.75

(C) 1 (D) 0.375

41–50 / Preserved hard practice

41. A rectangular loop of resistance R, width a perpendicular to a long wire and length b parallel to it, moves away from the wire. The wire carries steady current I. Neglect loop self-inductance. Initially the near side is at distance a. When half of the total charge that would pass during removal to infinity has flowed, the near-side distance divided by a is

(A) $\sqrt{2}+2$ (B) 2

(C) $\sqrt{2}$ (D) $1+\sqrt{2}$

42. A superconducting loop of area A and self-inductance L has its normal initially along a uniform external field B. Its initial current produces self-flux $+BA$, in the same signed sense as the external flux. It is rotated quasistatically through $90°$, keeping B fixed and conserving total flux linkage. External mechanical work divided by B^2A^2/L is

(A) 0.5 (B) 1

(C) 1.5 (D) 2

43. A rod of mass m and length l falls vertically on smooth conducting rails in perpendicular field B. Total circuit resistance is R, and inductance is negligible. It is released from rest. Write $v_t = mgR/(B^2l^2)$ and $\tau = mR/(B^2l^2)$. Distance fallen when its speed first becomes $3v_t/4$, divided by $v_t\tau$, is

(A) $\log(4)$ (B) $-\frac{1}{2} + \log(2)$

(C) $-\frac{3}{4} + \log(4)$ (D) 0.75

44. Two coils have equal self-inductance L and mutual inductance $M > 0$. Their series-aiding inductance is nine times their series-opposing inductance. At an instant their signed currents are I and $-2I$, where positive signs correspond to aiding flux. Stored magnetic energy divided by the series-aiding inductance times I^2 is

(A) 1.25 (B) 0.9

(C) 0.25 (D) 0.5

45. A conducting rod of length l and mass m slides on smooth horizontal rails in a perpendicular uniform field B. A battery of emf $\mathcal{E}$ and total circuit resistance R drive the rod from rest. Neglect circuit self-inductance. When its speed first reaches $\mathcal{E}/(2Bl)$, total Joule heat divided by $m\mathcal{E}^2/(B^2l^2)$ is

(A) 0.375 (B) 0.125

(C) 0.25 (D) 0.5

46. A closed conducting loop of area A, resistance R, mass m, falls without rotation through a horizontal field whose magnitude is $B(y) = B_0 + \beta y$, with y measured downward. The field is perpendicular to the loop, and its gradient is uniform across it. Neglect inductance. With $v_t = mgR/(\beta A)^2$ and $\tau = mR/(\beta A)^2$, distance from rest when speed is $v_t/2$, divided by $v_t\tau$, is

(A) $-\frac{1}{2} + \log(2)$ (B) 0.5

(C) $\log(2)$ (D) $1 - \log(2)$

47. A square wire loop of initial area A_0 and resistance R_0 expands isotropically in a fixed perpendicular field B. During heating its side is multiplied by $1 + x$, with x increasing from 0 to 1/10. Resistivity stays constant, and all three linear dimensions of the wire scale by $1 + x$, so its resistance is $R_0/(1 + x)$. Neglect inductance. Magnitude of total induced charge divided by BA_0/R_0 is

(A) 0.21 (B) 0.22

(C) $\frac{1331}{1500}$ (D) $\frac{331}{1500}$

48. An ideal inductor L and resistor R carry steady current $\mathcal{E}/R$ from a battery. The battery polarity is suddenly reversed, keeping L, R in series. Joule heat generated until the current first reaches zero, divided by $L\mathcal{E}^2/R^2$, is

(A) $\log(2)$ (B) $1 - \log(2)$

(C) $-\frac{1}{2} + \log(2)$ (D) 0.5

49. An ideal inductor L first carries steady current $\mathcal{E}/R$ from a DC source and series resistor R. It is then isolated from source and resistor and connected to an initially uncharged capacitor C to form an ideal LC circuit. At the first capacitor-voltage maximum, that voltage is used to accelerate a test particle of charge q and mass m from rest; its subsequent orbit in perpendicular field B has radius a. The test particle does not appreciably discharge C. Find $CR^2q^2B^4a^4/(Lm^2\mathcal{E}^2)$.

(A) 1 (B) 2

(C) 0.25 (D) 4

50. A conducting disc of radius R and moment of inertia I spins at ω_0 in a perpendicular field B. Ideal sliding contacts connect centre and rim through resistance r. With negligible circuit inductance, no driving torque and no other loss, time for angular speed to fall to $\omega_0/2$, divided by $Ir/(B^2R^4)$, is

(A) $8\log(2)$ (B) $4\log(2)$

(C) $2\log(2)$ (D) $\log(2)$

Answer key

Electromagnetic Induction

P-ELIN / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**C**	21	**C**	31	**C**	41	**D**
02	**C**	12	**A**	22	**A**	32	**C**	42	**C**
03	**D**	13	**B**	23	**B**	33	**D**	43	**C**
04	**D**	14	**A**	24	**A**	34	**C**	44	**C**
05	**A**	15	**C**	25	**C**	35	**A**	45	**A**
06	**B**	16	**D**	26	**D**	36	**B**	46	**A**
07	**A**	17	**A**	27	**D**	37	**B**	47	**D**
08	**A**	18	**D**	28	**B**	38	**B**	48	**C**
09	**B**	19	**A**	29	**D**	39	**B**	49	**D**
10	**A**	20	**B**	30	**B**	40	**B**	50	**B**

Chapter 21

Alternating Current

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A series LCR circuit has $R = 6\,\Omega$, $X_L = 12\,\Omega$, $X_C = 4\,\Omega$. Impedance in ohms is

(A) 9 (B) 20

(C) 11 (D) 10

02. A series LCR circuit has $R = 6\,\Omega$, $X_L = 12\,\Omega$, $X_C = 4\,\Omega$ and RMS voltage $100\,\text{V}$. Average power in watts is

(A) 601 (B) 1200

(C) 600 (D) 599

03. A series RC circuit has $R = 3\,\Omega$, $X_C = 4\,\Omega$, RMS voltage $10\,\text{V}$. RMS current in amperes is

(A) 1 (B) 3

(C) 4 (D) 2

04. A series RL circuit has $R = X_L$. Power factor is

(A) 0 (B) $1/2$

(C) 1 (D) $1/\sqrt{2}$

05. An AC circuit has $V_{rms} = 100\,\text{V}$, $I_{rms} = 2\,\text{A}$ and power factor 0.6. Real power in watts is

(A) 119 (B) 120

(C) 240 (D) 121

06. In a resonant LCR circuit, L quadruples and C quarters. Resonant frequency multiplier is

(A) 1 (B) 0.5

(C) 2 (D) 0

07. An ideal transformer primary draws 2 A at 200 V. Secondary voltage is 50 V. Secondary current in amperes is

(A) 7 (B) 16

(C) 8 (D) 9

08. A sinusoid $v = 200\cos\omega t$ volts is across $100\,\Omega$. Average power in watts is

(A) 201 (B) 400

(C) 199 (D) 200

09. A sinusoid $i = 4\sin\omega t$ A passes through $2\,\Omega$. Average power in watts is

(A) 32 (B) 16

(C) 15 (D) 17

10. A series circuit has $R = 4\,\Omega$ and impedance $5\,\Omega$. Magnitude of net reactance in ohms is

(A) 4 (B) 6

(C) 3 (D) 2

11–40 / New middle-to-hard practice

11. Current $i = a + b\sin\omega t$ has mean 2 A and rms 4 A. Add the current $b\cos\omega t$ coherently. New rms current, in amperes, is

(A) 6 (B) $4\sqrt{2}$

(C) $2\sqrt{7}$ (D) $2\sqrt{5}$

12. Current $i = a + b\sin\omega t + c\cos\omega t$, with b positive, has mean 3 A, rms $\sqrt{19}$ A, and i(0)=7 A. Add another $2\sin\omega t$ A. New rms current, in amperes, is

(A) $\sqrt{23}$ (B) 5

(C) $\sqrt{21}$ (D) $3\sqrt{3}$

13. A series RLC circuit takes 5 A rms from 100 V rms and consumes 400 W. It is inductive, and $X_L = 3X_C$. Inductor rms voltage, in volts, is

(A) 90 (B) 180

(C) 60 (D) 120

14. A series circuit contains a resistive coil and ideal capacitor. Rms source, coil and capacitor voltages are 100 V, 100 V and 80 V, with current 2 A. Coil resistance, in ohms, is

(A) $10\sqrt{21}$ (B) 50

(C) 30 (D) $20\sqrt{21}$

15. A series RC circuit at fixed source rms voltage has currents I and $\sqrt{2}I$ at frequencies ω and 2ω. Its power factor at 3ω is

(A) $\frac{3\sqrt{13}}{13}$ (B) $\frac{\sqrt{6}}{3}$

(C) $\frac{\sqrt{3}}{3}$ (D) $\frac{3\sqrt{11}}{11}$

16. A series RL circuit has average power P at frequency ω and $2P/5$ at 2ω, with the same source rms voltage. Current at 3ω divided by current at ω is

(A) $\frac{\sqrt{5}}{5}$ (B) $\frac{\sqrt{10}}{5}$

(C) $\frac{1}{3}$ (D) 0.2

17. A series RLC circuit has signed reactances $+3\,\Omega$ at ω and $+9\,\Omega$ at 2ω. Its resonance angular frequency divided by ω is

(A) $\frac{\sqrt{10}}{2}$ (B) $\frac{\sqrt{10}}{5}$

(C) 0.6 (D) 0.4

18. A series RLC circuit of resistance $3\,\Omega$ draws equal current from the same voltage at ω_1 and $4\omega_1$. Power factor there is 0.6. Inductive reactance at resonance, in ohms, is

(A) 4 (B) 8

(C) $\frac{4}{3}$ (D) $\frac{8}{3}$

19. A series RLC circuit has half-power angular frequencies 100 and 400 rad/s. Its inductive reactance at resonance divided by R is

(A) 2 (B) 1.5

(C) $\frac{2}{3}$ (D) 0.5

20. A series RLC circuit with $R = 6\,\Omega$ has half-power angular frequencies 2 and 8 rad/s. Add another $6\,\Omega$ in series. The new lower half-power frequency, in rad/s, is

(A) $6 + 2\sqrt{13}$ (B) 2

(C) $-6 + 2\sqrt{13}$ (D) 4

21. At one frequency, a series circuit has $R = 5\,\Omega$, ideal coils L_1, L_2 and a capacitor. Phase angles of voltage relative to current are 45° with both coils, -45° with L_2 shorted, and $-\arctan 3$ with both shorted. The value of $X_{L_1} + 2X_{L_2}$, in ohms, is

(A) 20 (B) 30

(C) 25 (D) 40

22. A series circuit takes 3 A rms from 60 V rms and consumes 90 W. Add a pure resistance until its power factor becomes $\sqrt{3}/2$. The added resistance, in ohms, is

(A) 10 (B) 30

(C) $10\sqrt{3}$ (D) 20

23. A transformer gives 100 V rms to $20\,\Omega$ from a 200 V rms primary supply. Core loss is 20 W and copper loss is $0.4I_s^2$ W. Primary power factor is 0.8. Primary rms current, in amperes, is

(A) 3.25 (B) 3.125

(C) 3.3125 (D) 2.65

24. At fixed secondary voltage 100 V, a transformer has efficiency 80% at output powers 100 W and 400 W. Losses are constant core loss plus a term proportional to secondary-current squared. Its greatest efficiency is

(A) $\frac{5}{6}$ (B) 0.8

(C) $\frac{8}{9}$ (D) $\frac{10}{11}$

25. A series RL load has $R = 3\,\Omega$, $X_L = 4\,\Omega$ at a given frequency and is across 100 V rms. Add a parallel capacitor so total power factor is 0.8, still lagging. Required capacitive reactance magnitude, in ohms, is

(A) $\frac{100}{7}$ (B) $\frac{100}{9}$

(C) 6.25 (D) 25

26. A series RL load has $\omega L = R$. A parallel capacitor makes its input power factor unity at that frequency. Keeping all elements fixed, the source frequency is doubled. New power-factor magnitude is

(A) $\frac{\sqrt{5}}{5}$ (B) $\frac{\sqrt{10}}{10}$

(C) $\frac{3\sqrt{10}}{10}$ (D) 0.5

27. At resonance, a series RLC circuit across 10 V rms carries 2 A and has inductor voltage 40 V rms. With the same source voltage at twice the resonance frequency, current in amperes is

(A) $\frac{1}{3}$ (B) $\frac{2\sqrt{17}}{17}$

(C) $\frac{2\sqrt{5}}{5}$ (D) $\frac{2\sqrt{37}}{37}$

28. At resonance the capacitor rms voltage in a series RLC circuit is four times source rms voltage. Keeping source amplitude fixed, use half the resonance frequency. Capacitor rms voltage divided by source rms voltage is

(A) $\frac{8\sqrt{17}}{17}$ (B) $\frac{8\sqrt{37}}{37}$

(C) 4 (D) $\frac{4\sqrt{37}}{37}$

29. A black box contains a resistance in series with either one ideal inductor or one ideal capacitor. A 2 A rms current requires 10 V at ω and $2\sqrt{13}$ V at 2ω. Add an inductor of reactance $8\,\Omega$ at the original frequency. New power factor at ω is

(A) 0.8 (B) 0.5

(C) $\frac{3\sqrt{73}}{73}$ (D) 0.6

30. A parallel RL black box across 10 V rms draws 5 A and consumes 30 W. Put this box in series with 1.8 Ω across the same 10 V source. New rms current, in amperes, is

(A) $\frac{50}{19}$ (B) $\frac{50}{17}$

(C) 3.125 (D) $\frac{10}{3}$

31. A periodic current rises linearly from zero to 4 A in half a period and falls linearly to zero in the other half. Replace it by a rectangular pulse current, with the same mean charge per period and Joule heat in the same resistor. Required pulse duty fraction is

(A) $\frac{2}{3}$ (B) 0.5

(C) 0.75 (D) 0.25

32. Current is a for the first third of each period and $-b$ for the remaining two-thirds, with a, b positive. Its net charge per period is zero and rms current $\sqrt{6}$ A. An ideal diode blocks the negative part without changing the positive part. New rms current, in amperes, is

(A) 2 (B) $\sqrt{3}$

(C) $\sqrt{2}$ (D) $\sqrt{6}$

33. Two coils separately resonate with the same capacitor at 2ω and ω. Connected in series aiding, they resonate with that capacitor at $\omega/\sqrt{2}$. Their mutual coupling coefficient is

(A) 0.75 (B) 0.5

(C) 1 (D) 0.25

34. An inductor resonates with a capacitor in parallel with a fixed stray capacitance. External capacitances 1 μF and 4 μF give resonance angular frequencies 1000 and $1000/\sqrt{2}$ rad/s. The inductance, in henrys, is

(A) $\frac{1}{3}$ (B) 1

(C) 0.5 (D) $\frac{1}{6}$

35. An AC bridge has arms AX=10 Ω, XB a capacitor of reactance 10 Ω, AY an inductor of reactance 20 Ω, and YB unknown pure R. The central detector reads zero. Source power-factor magnitude between A and B is

(A) $\frac{3\sqrt{10}}{10}$ (B) $\frac{\sqrt{10}}{10}$

(C) 0.6 (D) 0.8

36. An AC null bridge has AX as 10 Ω in series with C_1, XB as R_2 in series with $C_2 = 4\,\mu$F, and pure resistive arms AY, YB with ratio 2. The product R_2C_1, in microseconds, is

(A) 10 (B) 20

(C) 40 (D) 5

37. Instantaneous input power of a sinusoidal AC load has maximum 9 W and minimum −1 W. Its reactive-power magnitude, in var, is

(A) 4 (B) 5

(C) $\sqrt{5}$ (D) 3

38. Sinusoidal current lags sinusoidal voltage by 60°. Over one period, energy returned to the source divided by positive energy delivered from it is

(A) $\frac{-\pi+6\sqrt{3}}{2(\pi+3\sqrt{3})}$ (B) $\frac{1}{3}$

(C) $\frac{-\pi+3\sqrt{3}}{3\sqrt{3}+2\pi}$ (D) 0.5

39. Two coherent sinusoidal sources of rms voltages 3 V and 4 V in series dissipate 37 W in 1 Ω. Double only the first voltage and reverse the second source. New resistor power, in watts, is

(A) 52 (B) 28

(C) 13 (D) 40

40. Two equal-amplitude coherent sinusoidal sources have rms sum 10 V and rms difference 6 V. Their relative phase is acute. Increase that phase by 90°. The new rms sum, in volts, is

(A) $2\sqrt{2}$ (B) 6

(C) $2\sqrt{5}$ (D) 4

41–50 / Preserved hard practice

41. A series RLC circuit has the same current amplitude, equal to its resonant amplitude divided by $\sqrt{2}$, at angular frequencies ω and 3ω. Applied voltage amplitude is fixed. Its quality factor $Q = \omega_0 L/R$ is

(A) 0.5 (B) $\frac{\sqrt{3}}{2}$

(C) 1.5 (D) $\sqrt{3}$

42. A real coil is in series with a 20 Ω resistor. RMS voltages across resistor, coil, and the complete combination are 60, 100, and 140 V, respectively. Average power dissipated inside the coil, in watts, is

(A) 420 (B) 330

(C) 150 (D) 300

43. A series RLC circuit driven by fixed RMS voltage dissipates power P_0 at resonance. At half the resonant angular frequency it dissipates $P_0/10$. At twice the resonant frequency, inductor RMS voltage divided by applied RMS voltage is

(A) $\frac{\sqrt{10}}{4}$ (B) $\frac{2\sqrt{10}}{5}$

(C) $\frac{\sqrt{10}}{5}$ (D) 2

44. A resistor R is in parallel with a capacitor C, and this pair is in series with an ideal inductor L. Suppose $R^2C = 2L$. At the nonzero angular frequency where input voltage and input current are in phase, capacitor RMS voltage divided by input RMS voltage is

(A) 0.5 (B) 2

(C) $\sqrt{2}$ (D) 1

45. A series RLC circuit has $R = 10\,\Omega$. At angular frequency ω, current lags voltage with $\tan\phi = 2$; at 2ω, it lags with $\tan\phi = 5.5$. The circuit is then driven at resonance by 100 V RMS. RMS voltage across its capacitor is

(A) $50\sqrt{3}$ (B) 100

(C) 200 (D) $100\sqrt{3}$

46. A sinusoidal source has $v(t) = V_0 \sin\omega t$. An ideal switch connects a resistor R only when $|v| > V_0/2$, and otherwise disconnects it. Switching losses vanish. Average heater power divided by the power with the resistor permanently connected is

(A) $\frac{\sqrt{3}}{2\pi} + \frac{1}{2}$ (B) $\frac{\sqrt{3}}{2\pi} + \frac{2}{3}$

(C) $\frac{2}{3}$ (D) $-\frac{\sqrt{3}}{2\pi} + \frac{2}{3}$

47. An ideal transformer has $N_p/N_s = 2$. Its primary is supplied through a $4.00\,\Omega$ resistor by an ideal 8.00 V RMS source. The secondary contains a variable resistor R in series with an ideal inductor of reactance $\sqrt{3}\,\Omega$. The value of R, in ohms, that maximises its own average power is

(A) 4 (B) 1

(C) 2 (D) $\sqrt{3}$

48. A linear series RLC circuit is at resonance at angular frequency ω, with $\omega L = 3R/8$. Its applied voltage is $v(t) = V_0\cos\omega t + V_0\cos 3\omega t$. In steady state, total average power divided by total RMS voltage times total RMS current is

(A) 0.75 (B) 1

(C) 0.5 (D) $\frac{\sqrt{3}}{2}$

49. At resonance a series RLC circuit stores constant total electromagnetic energy 3.00 J in its ideal L and C, and dissipates 2π J in R per cycle. Applied RMS voltage is held fixed while frequency is varied. The ratio of its upper to lower half-power angular frequencies is

(A) $\frac{\sqrt{37}}{18} + \frac{19}{18}$ (B) $\frac{1}{6} + \frac{\sqrt{37}}{6}$

(C) 3 (D) $\frac{10}{9}$

50. An ideal transformer has $N_p/N_s = 4$ and primary RMS voltage 120 V. Its secondary feeds a $3\,\Omega$ resistor in series with an ideal inductor of reactance $4\,\Omega$. At angular frequency 1000 rad/s, an ideal capacitor in parallel with the entire secondary load makes the primary power factor unity. This capacitor is isolated at the peak secondary voltage and discharged fully into an insulated body of heat capacity 0.0720 J/K. Temperature rise in kelvin is

(A) 1 (B) 2

(C) 4 (D) 8

Answer key

Alternating Current

P-ALCU / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**C**	21	**B**	31	**C**	41	**B**
02	**C**	12	**B**	22	**D**	32	**A**	42	**C**
03	**D**	13	**A**	23	**C**	33	**A**	43	**B**
04	**D**	14	**A**	24	**A**	34	**A**	44	**C**
05	**B**	15	**D**	25	**A**	35	**A**	45	**D**
06	**A**	16	**A**	26	**B**	36	**A**	46	**B**
07	**C**	17	**B**	27	**D**	37	**D**	47	**C**
08	**D**	18	**D**	28	**B**	38	**C**	48	**D**
09	**B**	19	**C**	29	**D**	39	**B**	49	**A**
10	**C**	20	**C**	30	**B**	40	**A**	50	**B**

Chapter 22

Electromagnetic Waves

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A wave wavelength is $600\,\text{nm}$. With $c = 3 \times 10^8$, frequency in units of $10^{14}\,\text{Hz}$ is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 5 (B) 10

(C) 6 (D) 4

02. An EM wave electric amplitude is $600\,\text{V/m}$. With $c = 3 \times 10^8$, magnetic amplitude in microtesla is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 1 (B) 3

(C) 4 (D) 2

03. A sinusoidal EM wave has peak electric amplitude E_0. RMS electric amplitude is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) E_0 (B) $E_0/\sqrt{2}$

(C) $\sqrt{2}E_0$ (D) $E_0/2$

04. A wave has angular frequency $6 \times 10^8\,\text{rad/s}$. With $c = 3 \times 10^8$, wave number in rad/m is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 1 (B) 3

(C) 4 (D) 2

05. A changing electric flux has derivative $d\Phi_E/dt = 10^6\,\mathrm{V\,m/s}$. Taking $\epsilon_0 = 9 \times 10^{-12}$, displacement current in microamperes is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 8 (B) 10

(C) 18 (D) 9

06. An EM wave electric amplitude halves while frequency doubles. Intensity becomes what fraction?

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 0.5 (B) -0.75

(C) 1.25 (D) 0.25

07. A radio wave frequency is 30 MHz. With $c = 3 \times 10^8$, wavelength in metres is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 11 (B) 10

(C) 9 (D) 20

08. A microwave wavelength is 3 cm. With $c = 3 \times 10^8$, frequency in GHz is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 9 (B) 10

(C) 11 (D) 20

09. A beam carries average power 20 W uniformly over area $0.01\,\mathrm{m}^2$. Intensity in W per square metre is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 2000 (B) 1999

(C) 4000 (D) 2001

10. An isotropic source emits 36π W. At radius 3 m, intensity in W per square metre is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 0.5 (B) 0

(C) 1 (D) 2

11–40 / New middle-to-hard practice

11. A vacuum plane wave travels along unit vector $(2,-1,2)/3$. At an instant its electric field has $E_x = E_0$ and $E_y = E_z$. Its magnetic-field component B_y, divided by E_0/c, is

(A) 1 (B) 2

(C) $\frac{2}{3}$ (D) -2

12. At an instant a vacuum plane wave has $\mathbf{E} = (2,-1,2)E_0$ and $B_z = 2E_0/c$. Its propagation direction has positive z component. The allowed $B_x/(E_0/c)$ is

(A) 1 (B) -1

(C) 0.4 (D) -2.2

13. A charge q moves along and opposite to a vacuum wave at equal speed v. Peak total force magnitudes in these two trials are F and $3F$. Wave intensity divided by $\epsilon_0 c F^2/q^2$ is

(A) 2 (B) 0.5

(C) 1 (D) 4

14. A vacuum wave travels along +x with E along +y. A positive charge has velocity $c(0.6, 0.3, 0)$. At a nonzero positive E phase, its instantaneous power gain divided by c times total-force magnitude is

(A) 0.75 (B) 0.6

(C) 0.5 (D) 0.8

15. A vacuum wave has $B(t) = B_0(\sin\omega t + \sin 2\omega t)$ along one fixed axis at a detector. At phase $\omega t = \pi/3$, the field magnitude is B_m. Its mean intensity divided by cB_m^2/μ_0 is

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$

(C) 0.5 (D) $\frac{2}{3}$

16. A vacuum plane wave at a detector has $E(t) = a\cos\omega t + b\sin\omega t$. Measured values at phases zero and $\pi/4$ are 3 V/m and $5/\sqrt{2}$ V/m. Mean intensity divided by $\epsilon_0 c$, in $(\text{V/m})^2$, is

(A) 6.5 (B) 13

(C) 5 (D) 6.25

17. A circular capacitor of radius R has gap field $E(r,t) = \alpha t(1 + r^2/R^2)$. Fringing is neglected. Displacement current through the concentric area of radius $R/2$, divided by total displacement current, is

(A) 0.125 (B) 0.3125

(C) 0.1875 (D) 0.25

18. In a circular capacitor, field rate is $A + Br^2/R^2$, with A, B positive. Displacement current inside radius $R/2$ is one-eighth the total. Fraction inside radius $R/\sqrt{2}$ is

(A) 0.25 (B) $\frac{1}{3}$

(C) 0.375 (D) 0.5

19. A vacuum capacitor has fixed area, voltage $V_0(1 + t/\tau)$ and plate gap $d_0(1 + 2t/\tau)$. Displacement-current magnitude at $t = \tau$ divided by its initial magnitude is

(A) $\frac{1}{9}$ (B) $\frac{4}{9}$

(C) $\frac{2}{3}$ (D) $\frac{1}{3}$

20. At a fixed voltage-rise rate, a capacitor with half its area filled across the gap by dielectric K carries three times its all-vacuum current. A surface of area one-quarter the full plate lies wholly in the dielectric region. Its displacement current divided by total is

(A) $\frac{5}{6}$ (B) 0.5

(C) 0.25 (D) $\frac{5}{12}$

21. A lossless linear medium has plane-wave speed $c/3$ and impedance $2Z_0/3$, where Z_0 is vacuum impedance. Using $v/c = 1/\sqrt{\epsilon_r \mu_r}$ and $Z/Z_0 = \sqrt{\mu_r/\epsilon_r}$, its relative permittivity is

(A) 4.5 (B) 1.5

(C) 3 (D) 6

22. Two lossless media carry plane waves of equal intensity. The second has twice the electric amplitude and half the wave speed of the first. Its permeability divided by that of the first is

(A) 8 (B) 2

(C) 4 (D) 16

23. A normally incident vacuum wave reflects one-ninth of its intensity at a lossless medium. Its transmitted speed is c/2 and medium impedance is below Z_0. Use $R = [(Z - Z_0)/(Z + Z_0)]^2$. The medium's relative permittivity is

(A) 1 (B) 8

(C) 4 (D) 2

24. A nonmagnetic slab of refractive index 3 lies in vacuum. At each normal interface use reflected fraction $R = [(n - 1)/(n + 1)]^2$. During one traverse the slab absorbs half the energy inside it. Ignoring all later reflected passes, the incident energy fraction emerging from the far side is

(A) 0.5625 (B) 0.25

(C) 0.375 (D) 0.28125

25. A normally incident beam of intensity I meets a plane surface. One-quarter is transmitted forward; reflected and absorbed fractions are unknown. Radiation pressure is $5I/(4c)$. Fraction absorbed is

(A) 0 (B) 0.5

(C) 0.25 (D) 0.75

26. A non-transmitting plane surface with specular reflection is illuminated at 60° to its normal by beam intensity I. Its physical area A feels normal force $3IA/(8c)$. The absorbed heat power divided by IA is

(A) 0.75 (B) 0.25

(C) 0.5 (D) 0.375

27. Two points on a ray from an isotropic source are separated by d. Nearer intensity is nine times farther intensity. Increase source power fourfold and move the farther detector another distance d outward. Its new intensity divided by the original nearer intensity is

(A) 0.16 (B) 0.04

(C) $\frac{1}{9}$ (D) $\frac{4}{9}$

28. Two incoherent isotropic sources at x=0, d produce equal intensities at x=d/3. Move the detector to a point above the midpoint by $\sqrt{3}d/2$. Total intensity there divided by the total at x=d/3 is

(A) $\frac{5}{9}$ (B) $\frac{1}{3}$

(C) $\frac{5}{18}$ (D) 0.5

29. An intensity detector has unknown constant background. For absorber thicknesses zero, d and $2d$, it reads 10,6,4 units. Actual intensity attenuates exponentially. Electric-field amplitude after thickness $3d$ divided by its initial amplitude is

(A) $\frac{\sqrt{2}}{4}$ (B) 0.25

(C) 0.125 (D) 0.5

30. Two absorbers A, B have exponential intensity attenuation. Thickness d of each in series transmits one-eighth; using $2d$ of A and d of B transmits one-thirty-second. Electric amplitude after d/2 of A and $3d$ of B, divided by its incident amplitude, is

(A) $\frac{\sqrt{2}}{4}$ (B) 0.25

(C) 0.5 (D) 0.125

31. A photodetector's current includes constant dark current. At wavelength λ, powers P and $2P$ give 5 nA and 8 nA. At wavelength 2λ and power P its quantum efficiency becomes three-quarters of the old value. New measured current, in nA, is

(A) 6.5 (B) 8

(C) 7.5 (D) 6

32. A beam is fully absorbed by a detector, exerting force 10^{-9} N. The detector collects 0.5 electron per incident photon and gives current 4.8 mA. Use $c = 3 \times 10^8$, $e = 1.6 \times 10^{-19}$ C and $hc = 2 \times 10^{-25}$ J m. Wavelength, in nm, is

(A) 20 (B) 40

(C) 400 (D) 80

33. Two counterpropagating vacuum waves of the same linear polarisation are in phase electrically at x=0. There, total electric amplitude is $6E_0$ and magnetic amplitude $2E_0/c$. Net mean energy flow is along +x. Its intensity divided by $\epsilon_0 c E_0^2$ is

(A) 10 (B) 4

(C) 8 (D) 6

34. Two equal opposite waves form $E = 2E_0 \sin kx \cos \omega t$ and $B = -(2E_0/c) \cos kx \sin \omega t$. At $x = \lambda/12$, the first positive time with equal electric and magnetic energy densities divided by the wave period is

(A) $\frac{1}{12}$ (B) 0.125

(C) 0.25 (D) $\frac{1}{6}$

35. Unpolarised light of intensity I_0 crosses polarisers at $0°, \theta, 90°$, where $0 < \theta < 45°$. Output is $I_0/16$. Increase only the middle angle by 22.5°. New output divided by its previous value is

(A) 1 (B) 0.5

(C) 2 (D) 4

36. A coherent elliptically polarised beam gives transmitted intensities 2,8 and 7 W/m^2 through ideal linear analysers at $0°, 90°, 45°$. Greatest possible transmitted intensity over all analyser angles, in W/m^2, is

(A) $\sqrt{5} + 5$ (B) $\sqrt{13} + 5$

(C) 8 (D) 10

37. A normally incident plane pulse has electric envelope $E(t) = E_0(1 - |t|/\tau)$ for $|t| \leq \tau$, zero otherwise. A perfect absorber collects energy Q and experiences peak force F_0. Full pulse duration divided by $Q/(cF_0)$ is

(A) 6 (B) 1.5

(C) 3 (D) 2

38. A plane pulse has instantaneous energy flux that rises linearly from zero to its peak in time τ, then falls linearly to zero in time 2τ. A perfect absorber of area A receives momentum p. Peak electric-field squared divided by $p/(\epsilon_0 A\tau)$ is

(A) $\frac{1}{3}$ (B) $\frac{4}{3}$

(C) 2 (D) $\frac{2}{3}$

39. A beam contains photons of energies E and $2E$, total power P. A detector collects one electron per E photon and half an electron per $2E$ photon on average. Its current is $3eP/(5E)$. Power fraction in the higher-energy photons is

(A) 0.4 (B) $\frac{7}{15}$

(C) $\frac{8}{15}$ (D) 0.5

40. A beam has wavelengths λ and 3λ. A detector collecting one electron per photon measures current J and total power P satisfying $P/(J/e) = hc/(2\lambda)$. A second surface perfectly reflects only the shorter-wavelength component and absorbs the other. Its force divided by P/c is

(A) $\frac{4}{3}$ (B) $\frac{5}{3}$

(C) 1.5 (D) 1

41–50 / Preserved hard practice

41. A normally incident circular beam has intensity $I(r) = I_0(1 - r^2/R^2)$ for $0 \le r \le R$. A stationary target absorbs perfectly for $r < R/\sqrt{2}$ and reflects normally and perfectly in the remaining annulus. If incident power is P, total radiation force satisfies $Fc/P =$

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 1.75 (B) 1.5

(C) 1.25 (D) $\frac{4}{3}$

42. Two coherent plane electromagnetic waves of the same frequency and polarisation propagate in opposite directions in vacuum. Their peak electric-field amplitudes are $2E_0$ and E_0. Time-averaged net Poynting-flux magnitude divided by c times time-averaged total electromagnetic energy density is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 0.6 (B) $\frac{1}{3}$

(C) 0.5 (D) 0.8

43. Two coherent plane waves of equal peak electric amplitude E_0 cross in vacuum at 60°. Both electric vectors are perpendicular to their propagation plane. At a point where their electric oscillations are in phase, the magnitude of time-averaged Poynting flux divided by c times average total field energy density is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) $\frac{\sqrt{3}}{2}$ (B) 1

(C) $\frac{4\sqrt{3}}{7}$ (D) $\frac{6}{7}$

44. Between circular capacitor plates of radius R, an axially symmetric electric field is $E(r,t) = E_0(t)(1 - r/R)$. It vanishes outside the plate region; fringing is ignored. The largest azimuthal magnetic-field magnitude at that instant, divided by its magnitude at the rim, is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) $\frac{4}{3}$ (B) 1.5

(C) 0.75 (D) 1.125

45. Between circular capacitor plates of radius R, the axial electric field is $E(r,t) = E_0(t)(1 - r^2/R^2)$. Assume axial symmetry and negligible fringing. At $r = R/2$, the azimuthal magnetic field divided by $\mu_0\varepsilon_0 R\, dE_0/dt$ is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 0.25 (B) 0.4375

(C) 0.1875 (D) 0.21875

46. A perfectly reflecting mirror of mass m is attached to a horizontal spring of stiffness k, initially at rest at equilibrium. A constant-power beam strikes it normally for exactly one quarter of its natural oscillation period, then switches off. Treat radiation force as $2P/c$, neglecting Doppler effects. If the subsequent free-oscillation amplitude is A, find $P/(ckA)$.

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) $\frac{\sqrt{2}}{2}$ (B) 0.25

(C) 0.5 (D) $\frac{\sqrt{2}}{4}$

47. A small perfectly absorbing spherical grain of fixed density is initially at rest at distance r from a star. Radiation force exactly balances the star's gravity. Its radius is suddenly halved at unchanged density, without mechanical impulse; thereafter its mass and radius remain fixed. Ignore every force except stellar gravity and radiation, both varying as inverse distance squared. At distance $2r$, its speed squared divided by GM/r is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 1 (B) 2

(C) 1.5 (D) 0.5

48. A leaky capacitor is modelled as ideal capacitance C in parallel with resistance $2R$. Initially uncharged, it is connected to a battery $\mathcal{E}$ through a series resistor R. Both leakage and displacement current densities are uniform over the same plate area. At the first instant when their magnetic-field contributions inside the gap are equal, $t/(RC)$ is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) $\log(2)$ (B) $\frac{4\log(2)}{3}$

(C) $\frac{2\log(2)}{3}$ (D) $2\log(2)$

49. A photocathode of work function ϕ absorbs a two-colour beam whose photon energies are 2ϕ and 3ϕ. One electron is emitted per absorbed 2ϕ photon, but only one per two absorbed 3ϕ photons on average. All emitted electrons are collected. If total beam power P and saturation current I obey $eP/(I\phi) = 4$, the fraction of power carried by the 3ϕ component is

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 0.5 (B) $\frac{2}{3}$

(C) 0.8 (D) 0.75

50. An opaque surface reflects fraction r of normally incident beam power P and absorbs the rest, without transmission. It is stationary. A spring balances the radiation force, and absorbed power leaves through a thermal link. Measurements give $Fc/H = 3$, where F is force and H is steady heating power. Find $P/(Fc)$.

Enrichment practice: Quantitative EM-wave/displacement-current/radiation extension: optional enrichment beyond a strictly qualitative reading of the pinned BITSAT 2026 EM-wave clause; not certified as core scope.

(A) 0.75 (B) 0.5

(C) $\frac{2}{3}$ (D) $\frac{1}{3}$

Answer key

Electromagnetic Waves

P-ELWA / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**B**	21	**A**	31	**A**	41	**C**
02	**D**	12	**B**	22	**A**	32	**B**	42	**A**
03	**B**	13	**A**	23	**C**	33	**D**	43	**C**
04	**D**	14	**B**	24	**D**	34	**A**	44	**D**
05	**D**	15	**B**	25	**C**	35	**C**	45	**D**
06	**D**	16	**A**	26	**B**	36	**B**	46	**D**
07	**B**	17	**C**	27	**A**	37	**C**	47	**A**
08	**B**	18	**B**	28	**C**	38	**D**	48	**B**
09	**A**	19	**A**	29	**A**	39	**C**	49	**D**
10	**C**	20	**D**	30	**B**	40	**C**	50	**C**

Chapter 23

Ray Optics

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A converging lens focal length 10 cm has object 15 cm in front. Real image distance in cm is

(A) 30 (B) 31

(C) 29 (D) 60

02. A converging lens focal length 10 cm has object 15 cm in front. Magnification is

(A) -2 (B) -4

(C) -1 (D) -3

03. A converging lens focal length 20 cm has object 10 cm in front. Virtual image distance magnitude in cm is

(A) 21 (B) 19

(C) 40 (D) 20

04. A diverging lens focal-length magnitude 10 cm has real object 10 cm in front. Virtual image distance magnitude in cm is

(A) 6 (B) 4

(C) 10 (D) 5

05. A concave mirror focal-length magnitude 10 cm has object 30 cm before it. Real image distance magnitude in cm is

(A) 30 (B) 14

(C) 16 (D) 15

06. A convex mirror focal-length magnitude 10 cm has object 30 cm before it. Image distance behind mirror in cm is

(A) 7.5 (B) 6.5

(C) 15 (D) 8.5

07. A glass slab thickness 6 cm and index 1.5 is viewed normally. Apparent shift of its lower face in cm is

(A) 4 (B) 1

(C) 2 (D) 3

08. Thin lenses of focal lengths 20 and -30 cm are in contact. Equivalent focal length in cm is

(A) 61 (B) 59

(C) 120 (D) 60

09. At minimum deviation a prism apex 60° has index $\sqrt{3}$. Minimum deviation in degrees is

(A) 61 (B) 120

(C) 60 (D) 59

10. A compound microscope has tube length 20 cm, objective focal length 1 cm, eyepiece 5 cm. Using standard long-tube approximation and final image at infinity with $D = 25$ cm, magnification magnitude is

(A) 200 (B) 101

(C) 100 (D) 99

11–40 / New middle-to-hard practice

11. A thin lens has power 10 D in air and 2.5 D in water of index 4/3. In a liquid of index 5/3 its signed focal length, in metres, is

(A) 0.5 (B) 0.2

(C) -2 (D) -0.5

12. An equiconvex glass lens of index 1.5 has air focal length 20 cm. In one liquid its focal length is -40 cm. Cut it across its axis through the centre into plano-convex halves. Put one half in a second liquid whose index is 0.8 times the first liquid index. Its focal length, in cm, is

(A) 320 (B) -320

(C) -160 (D) -80

13. Parallel rays meet a lens of focal length -10 cm and then,5 cm later, a lens of focal length $+20$ cm. With first lens at x=0 and incident propagation along +x, final image coordinate, in cm, is

(A) -60 (B) -55

(C) 25 (D) 65

14. Thin lenses of focal lengths $+20$ cm and -10 cm are separated so a parallel incident beam leaves parallel. A real object is then placed 40 cm before the first. Final image coordinate relative to the first lens, positive along incident propagation, in cm, is

(A) -5 (B) -15

(C) 5 (D) 25

15. A thin converging lens makes a real image with magnification magnitude 2 and object-image separation 90 cm. Move only the object 10 cm farther from the lens. New object-image separation, in cm, is

(A) 100 (B) 80

(C) 70 (D) 90

16. For a fixed 100 cm object-screen separation, a glass lens of index 1.5 has two sharp-image positions 60 cm apart in air. In a liquid the two positions are 20 cm apart. The liquid index is

(A) 1.125 (B) $\frac{4}{3}$

(C) 1.2 (D) 1.5

17. A vessel of depth h has upper layer h/3 of index n and lower layer 2h/3 of index 2n. Its bottom appears h/2 below the rim when viewed normally from air. Drain only the upper layer. New apparent bottom depth from the original rim, divided by h, is

(A) 0.25 (B) $\frac{7}{12}$

(C) $\frac{2}{3}$ (D) $\frac{5}{12}$

18. Two liquid layers each have depth h, with upper index n_1 and lower n_2. Bottom apparent depth from the rim is 7h/6. Drain the upper layer; apparent depth from the same rim becomes 5h/3. Critical-angle sine for light from the original upper into the lower liquid is

(A) 0.5 (B) $\frac{4}{3}$

(C) $\frac{2}{3}$ (D) 0.75

19. An achromatic contact doublet has net power 5 D. Dispersive powers of its positive and negative lenses are 0.02 and 0.05. Increase only the positive lens power by 10%, without changing either dispersive power. New chromatic power difference, in dioptres, has magnitude

(A) $\frac{1}{120}$ (B) $\frac{1}{60}$

(C) $\frac{1}{6}$ (D) $\frac{1}{30}$

20. Two oppositely placed thin prisms give net deviation 0.5 degrees without angular dispersion. Their indices are 1.5 and 1.6 and dispersive powers 0.02 and 0.04 respectively. First apex angle divided by the second is

(A) 2 (B) 1.2

(C) 2.4 (D) $\frac{5}{12}$

21. A plano-convex lens has unknown glass index and radius. With plane face silvered, its equivalent mirror focal length in air is 20 cm. Without silver, its focal length in water of index 4/3 is 160 cm. The glass index is

(A) $\frac{5}{3}$ (B) $\frac{4}{3}$

(C) 2 (D) 1.5

22. The same thin plano-convex lens gives equivalent-mirror focal lengths 30 cm when its plane face is silvered, and 10 cm when its curved face is silvered, with incidence through the other face in air. Sine of its glass-to-air critical angle is

(A) 0.75 (B) $\frac{\sqrt{2}}{2}$

(C) $\frac{1}{3}$ (D) $\frac{2}{3}$

23. At minimum deviation a prism has incidence 45 degrees and deviation 30 degrees in air. Sine of its critical angle to water of index 4/3 is

(A) $\frac{2\sqrt{2}}{3}$ (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{\sqrt{2}}{2}$ (D) $\frac{2}{3}$

24. A prism of index $\sqrt{3}$ has minimum deviation equal to its apex angle in air. Immerse it in a liquid of index $\sqrt{3/2}$. New minimum deviation, in degrees, is

(A) 15 (B) 45

(C) 60 (D) 30

25. A 60-degree prism of index $\sqrt{3}$ receives a ray from air at incidence 30 degrees. The ray reaches the second face in the usual prism path. Greatest coating index there allowing total internal reflection, interpreted as the limiting value, is

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{4}+\frac{\sqrt{33}}{4}$

(C) $\frac{\sqrt{6}}{2}$ (D) $-\frac{1}{4}+\frac{\sqrt{33}}{4}$

26. A 60-degree prism of index 2 is coated on its exit face by a medium of index $\sqrt{2}$. A ray entering from air emerges grazing that interface. Sine of its first-face incidence angle is

(A) $\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}$ (B) $\frac{\sqrt{2}}{2}$

(C) $-\frac{\sqrt{2}}{2}+\frac{\sqrt{6}}{2}$ (D) 0.5

27. A telescope in normal adjustment has magnifying power 9 and length 100 cm. Move its eyepiece to put the final virtual image at the observer's near point 25 cm from that eyepiece. Required lens separation, in cm, is

(A) $\frac{650}{7}$ (B) $\frac{680}{7}$

(C) 95 (D) 100

28. A telescope has normal-adjustment magnifying power 5 and length 60 cm. Keep that separation but place a real object 100 cm before its objective. Final image coordinate measured from the objective along incoming propagation, in cm, is

(A) 68 (B) 60

(C) 70 (D) 65

29. A microscope has objective focal length 1 cm and eyepiece focal length 5 cm. Its exact total magnification for final image at infinity is 100, using near-point distance 25 cm. Without a long-tube approximation, lens separation in cm is

(A) 26 (B) 25

(C) 20 (D) 21

30. A microscope has objective focal length 2 cm, eyepiece 5 cm, and lens separation 17 cm. The eye is at the eyepiece and the final image is at its 25 cm near point. Exact magnification magnitude is

(A) 38.5 (B) 36

(C) 32.5 (D) 30

31. A glass sphere of radius 10 cm and index 1.5 is in air. A small axial object is 30 cm in front of its first surface. In the paraxial approximation, final image distance behind its rear surface, in cm, is

(A) 34 (B) 90

(C) 14 (D) 30

32. A spherical air bubble of radius 10 cm is in water of index 4/3. Parallel paraxial rays enter it. Taking first surface coordinate zero and forward propagation positive, final virtual image coordinate, in cm, is

(A) -5 (B) 5

(C) -25 (D) 25

33. A converging lens of focal length 20 cm has an axial object 30 cm before it. Cut the lens into semicircular halves along its axis and translate them oppositely perpendicular to the axis until their axes are 2 mm apart. Separation of their two images, in mm, is

(A) 2 (B) 4

(C) 6 (D) 8

34. An equiconvex lens of focal length 20 cm is cut across its axis into equal plano-convex halves. One half is placed in contact with a diverging lens of focal length -60 cm. An object 90 cm before the combination has virtual-image distance magnitude, in cm, of

(A) 60 (B) 360

(C) 180 (D) 120

35. Two parallel plane mirrors are 1 m apart and extend from x=0 to 6 m. A ray enters at x=0, one-quarter metre above the lower mirror, with upward slope $dy/dx = 1/2$. Number of reflections before leaving at x=6 m is

(A) 3 (B) 5

(C) 4 (D) 2

36. Two parallel mirrors are separated by d and have length 4d. A ray enters halfway between them, initially upward, undergoes exactly three reflections, and exits at height d/4 above the lower mirror. Its initial dy/dx is

(A) 0.8125 (B) 0.9375

(C) 0.1875 (D) 0.875

37. A plane-parallel slab of thickness t gives lateral ray displacement $t/\sqrt{3}$ for incidence 60 degrees in air. Its refractive index is

(A) $\sqrt{3}$ (B) 2

(C) 1.5 (D) $\sqrt{2}$

38. A lens of focal length 10 cm images a real object 30 cm before it. Insert a 6 cm thick plane glass slab of index 1.5 anywhere between object and lens, normal to the axis. Paraxial image displacement away from the lens, in cm, is

(A) $\frac{10}{9}$ (B) 2

(C) $\frac{5}{9}$ (D) $\frac{5}{6}$

39. A thin lens of focal length 20 cm is at x=0, a plane mirror at x=40 cm, and an object at x=−60 cm. Light passes the lens, reflects, and passes back through the lens. Final real image distance to the left of the lens, in cm, is

(A) $\frac{100}{3}$ (B) 30

(C) $\frac{200}{3}$ (D) 50

40. A lens of focal length 30 cm is 10 cm in front of a concave mirror of focal length 15 cm, facing the lens. Parallel rays enter from the lens side. Taking the lens at zero and mirror side positive, final image coordinate after reflection and return through the lens, in cm, is

(A) $\frac{10}{7}$ (B) 1.5

(C) −1.5 (D) $-\frac{60}{7}$

41–50 / Preserved hard practice

41. A thin convex lens forms sharp images of a fixed object on a fixed screen at two positions. Moving uniformly at 2.0 cm/s, the lens takes 10.0 s to move between them. The larger image is nine times the smaller image in linear size. Find the lens focal length, in cm.

(A) 5 (B) 15

(C) 7.5 (D) 10

42. A fixed object is at $x = 0$. A convex lens of focal length 10 cm moves along $+x$ at constant 2 cm/s. At the instant its centre is at $x = 20$ cm, find the real image acceleration along $+x$, in cm/s^2.

(A) 1.6 (B) 0.8

(C) 0 (D) 0.4

43. An object and screen are 60 cm apart. In air, a thin glass lens has two sharp-image positions separated by $20\sqrt{3}$ cm. When the whole setup is immersed in a liquid, the two positions coincide. Glass index is 1.50. Find light speed in the liquid divided by vacuum speed.

(A) 0.75 (B) $\frac{8}{9}$

(C) $\frac{2}{3}$ (D) 0.8

44. A thin convex lens of focal length 10 cm is centred at $(0, 0)$ with its axis parallel to x. An object is instantaneously at $(-15, 2)$ cm, moving at $+1$ cm/s vertically. The lens translates at $(1, 1/2)$ cm/s without tilting. Find the image vertical velocity, in cm/s, positive upward.

(A) -0.5 (B) 0.3

(C) -0.3 (D) 0.5

45. A fish is 30 cm below a plane water surface and rises at 6 cm/s. Water has index 1.5. A fixed thin convex lens of focal length 10 cm is 10 cm above the surface. For near-normal paraxial rays, the final real image's instantaneous upward speed, in cm/s, is

(A) 3 (B) 4

(C) 2 (D) 1

46. A prism of apex angle 60° is in air. A ray incident at 45° on its first face just emerges grazing along its second face. Vacuum wavelength is λ_0 and wavelength in the prism is λ. Find $(\lambda_0/\lambda)^2$.

(A) 1.5 (B) 3

(C) $\frac{2\sqrt{2}}{3} + 2$ (D) 2

47. A straight optical fibre has core wavelength 400 nm for vacuum wavelength 600 nm. Its core-to-cladding critical angle is 60°. Its flat entrance is normal to the fibre axis and lies in a liquid of index 4/3. For meridional rays, find the sine of the largest external acceptance angle to the axis.

(A) 0.5 (B) 0.5625

(C) 0.75 (D) 0.375

48. A uniform parallel beam of power P and radius R enters an ideal lossless thin convex lens of focal length $f \gg R$. At distance $f/4$ beyond it is a centred circular target of radius $R/2$, normal to the axis. The target absorbs for radius below $R/4$ and perfectly reflects in the remaining annulus. Find $P/(Fc)$, where F is its force. Use leading paraxial order and neglect momentum direction corrections.

(A) 2.25 (B) $\frac{9}{7}$

(C) $\frac{7}{9}$ (D) 1.5

49. Two narrow coherent slits 0.50 mm apart are illuminated simultaneously at vacuum wavelengths 500 and 600 nm. A thin glass lens of index 1.50 and air focal length 20 cm forms their far-field pattern. The entire arrangement is immersed in a liquid of index 4/3, with negligible dispersion. Find the smallest positive distance in mm from the focal-plane axis where both wavelengths have bright fringes.

(A) 2.4 (B) 4.8

(C) 1.2 (D) 3.6

50. A thin convex lens of focal length $10\,\text{cm}$ is at $x = 0$ and a thin concave lens of focal length $-10\,\text{cm}$ is at $x = 15\,\text{cm}$. An object is instantaneously at $x = -20\,\text{cm}$ and moves toward increasing x at $2\,\text{cm/s}$. Light travels toward increasing x. The final image velocity in cm/s, positive along $+x$, is

(A) 2

(B) -8

(C) 8

(D) -2

Answer key

Ray Optics

P-RAOP / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**D**	21	**D**	31	**C**	41	**C**
02	**A**	12	**B**	22	**D**	32	**A**	42	**B**
03	**D**	13	**B**	23	**A**	33	**C**	43	**B**
04	**D**	14	**A**	24	**D**	34	**B**	44	**B**
05	**D**	15	**B**	25	**D**	35	**A**	45	**D**
06	**A**	16	**A**	26	**C**	36	**A**	46	**C**
07	**C**	17	**B**	27	**B**	37	**A**	47	**B**
08	**D**	18	**D**	28	**A**	38	**C**	48	**B**
09	**C**	19	**B**	29	**A**	39	**A**	49	**D**
10	**C**	20	**C**	30	**C**	40	**B**	50	**C**

Chapter 24

Wave Optics

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Two coherent waves of equal individual intensity I have phase difference 60°. Resultant intensity divided by I is

(A) 3 (B) 2

(C) 6 (D) 4

02. For two coherent sources with amplitude ratio $3:1$, visibility $(I_{max}-I_{min})/(I_{max}+I_{min})$ is

(A) 1.2 (B) 0.6

(C) 1.6 (D) -0.4

03. An ideal polariser and analyser have axes 45° apart. Incident unpolarised intensity is I_0. Final intensity divided by I_0 is

(A) -0.75 (B) 0.5

(C) 0.25 (D) 1.25

04. Between crossed ideal polarisers, a third polariser is inserted at 45° to each. For initially unpolarised light, final fraction is

(A) 0.125 (B) 1.125

(C) -0.875 (D) 0.25

05. A glass refractive index is $\sqrt{3}$. For air incidence at Brewster angle, refraction angle in degrees is

(A) 31 (B) 30

(C) 29 (D) 60

06. A single slit $a = 0.1\,\text{mm}$ uses $\lambda = 500\,\text{nm}$. First diffraction minimum small angle in radians is

(A) 0.01 (B) 1.005

(C) -0.995 (D) 0.005

07. A single slit $a = 0.2\,\text{mm}$, $D = 2\,\text{m}$, $\lambda = 500\,\text{nm}$ has central maximum width in mm

(A) 10 (B) 20

(C) 9 (D) 11

08. A thin film in air has index n, thickness t, with one reflected ray receiving a π phase reversal. At normal incidence, the smallest nonzero thickness giving reflected constructive interference is

(A) $2\lambda/n$ (B) λ/n

(C) $\lambda/(4n)$ (D) $\lambda/(2n)$

09. A thin film in air has one reflected-ray phase reversal. Its normal-incidence reflected destructive condition is

(A) $2t = m\lambda$ (B) $nt = m\lambda$

(C) $2nt = m\lambda$ (D) $2nt = (m + 1/2)\lambda$

10. A glass plate index 1.5 and thickness $2\,\mu\text{m}$ is introduced in one Young path. Extra optical path in micrometres is

(A) 0.5 (B) 2

(C) 0 (D) 1

11–40 / New middle-to-hard practice

11. Two plates of indices 1.5 and 1.25 cover different Young slits. At vacuum wavelength 0.5 micrometre, their net phase delay is 3 fringe orders in air and 5/3 orders when the entire setup is in water of index 4/3, in the same signed sense. First plate thickness divided by second is

(A) 1.5 (B) 0.5

(C) 3 (D) 2

12. A transparent plate covering one Young slit shifts the centre by 3 mm in air, where fringe width is 1 mm. With the entire setup immersed in water of index 4/3, it shifts 0.75 mm in the same direction. Geometry and vacuum wavelength stay fixed. Plate refractive index is

(A) $\frac{5}{3}$ (B) 2

(C) 1.5 (D) $\frac{4}{3}$

13. In a Young pattern with shifted centre, third dark fringe on the right is at 1.3 mm and fifth bright on the right at 2.3 mm. Screen distance is 1 m and slit separation 1 mm. Wavelength, in nm, is

(A) 600 (B) 200

(C) 500 (D) 400

14. In a Young pattern, the third bright fringe on the right is at x=1.4 mm, and the second dark fringe on the left at x=−0.4 mm. The shifted central maximum coordinate, in mm, is

(A) 0.2 (B) 0.5

(C) 0 (D) −0.2

15. Two coherent slits give maximum-to-minimum intensity ratio 49:9. A filter transmits one-quarter of the stronger slit intensity and leaves the weaker unchanged. New fringe visibility is

(A) $\frac{40}{41}$ (B) 0.96

(C) 0.8 (D) $\frac{20}{29}$

16. Two coherent beams have total individual intensity $5I_0$ and visibility 3/5. At a chosen point their resultant is $13I_0/2$. Reduce only the stronger individual intensity by a factor 9. Resultant at that same phase point, divided by I_0, is

(A) 2.5 (B) 1.5

(C) 2 (D) 1

17. Two coherent intensities I and 4I give 7I at a point with phase in $[0, \pi]$. Increase the second path delay so phase rises by $\pi/3$. New intensity divided by I is

(A) 3 (B) 7

(C) 5 (D) 9

18. Two equal beams have maximum intensity I_m. A point with initial path difference between zero and $\lambda/2$ has intensity $I_m/4$. Add another path delay $\lambda/4$. New intensity divided by I_m is

(A) $\frac{\sqrt{3}}{4} + \frac{1}{2}$ (B) 0.75

(C) 0.25 (D) $\frac{1}{2} - \frac{\sqrt{3}}{4}$

19. In air, the fourth bright Young fringe is 4 mm from centre. After immersion in an unknown liquid and halving screen distance, the third bright is 1 mm from centre. Vacuum wavelength is 0.6 micrometre. A plate of index 2 and thickness 1 micrometre is now placed in one liquid-filled arm. Its fringe-order shift is

(A) $\frac{5}{6}$ (B) $\frac{5}{3}$

(C) 0.5 (D) 1.25

20. The same Young experiment is immersed in a liquid while slit spacing doubles. Fringe width becomes one-sixth its air value. Another medium has wave speed c/2. Critical-angle sine from the liquid into that other medium is

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) 1.5 (D) 0.5

21. A double-slit pattern has its first missing interference order at the fourth bright fringe because of finite slit width. Add a phase delay π in one slit. Number of interference maxima strictly inside the central diffraction envelope is

(A) 6 (B) 9

(C) 7 (D) 8

22. A double slit has d/a=4 and wavelengths 600 nm and 400 nm. On one side, count common interference maxima where neither colour is at a diffraction minimum, before the first common coincidence at which both colours are missing. The count is

(A) 1 (B) 3

(C) 2 (D) 4

23. A tilted single-slit pattern has its first minimum to the left at x=−1 mm and second minimum to the right at x=3.5 mm, measured from a fixed origin. Screen distance is 1.5 m and wavelength 500 nm. Slit width, in mm, is

(A) 0.5 (B) 1

(C) $\frac{2}{3}$ (D) $\frac{1}{3}$

24. A slit of width $a = 2\lambda$ receives light at 30 degrees to its normal. Use exact Fraunhofer minima $\sin\theta - \sin 30^\circ = m\lambda/a$, for nonzero integer m. How many minima lie strictly within the forward angular interval $-90^\circ < \theta < 90^\circ$?

(A) 1 (B) 2

(C) 3 (D) 4

25. At complete polarisation of reflected light from air, the refraction angle in a transparent medium is 30 degrees. Immersing the interface in a liquid of index $\sqrt{3/2}$, the cosine of its new Brewster angle is

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{3}}{3}$

(C) 0.5 (D) $\frac{\sqrt{6}}{3}$

26. From medium 1 into medium 2, Brewster-angle tangent is 2. From medium 2 into air, critical-angle sine is 1/3. A light wave has wavelength 400 nm in medium 1. Its vacuum wavelength, in nm, is

(A) 400 (B) 600

(C) 1200 (D) 800

27. An optically active slab rotates a linearly polarised beam by an acute angle proportional to thickness. An analyser parallel to the initial polarisation transmits one-quarter after thickness L. Halve the slab thickness and set the analyser 45 degrees from the initial direction, in the same rotation sense. Transmitted fraction is

(A) 0.75 (B) 0.5

(C) $\frac{1}{2} - \frac{\sqrt{3}}{4}$ (D) $\frac{\sqrt{3}}{4} + \frac{1}{2}$

28. Unpolarised intensity I_0 crosses two ideal polarisers and emerges as $I_0/8$, with an acute angle between their axes. Insert a third halfway between their original axes, then rotate only the last axis an extra 30 degrees in the same sense. Final fraction of I_0 is

(A) 0.28125 (B) 0.1875

(C) 0.09375 (D) 0.125

29. A coating of index 1.5 on glass of index 2.25 is in air. At normal incidence,600 nm and 360 nm are consecutive reflected-intensity minima. The coating thickness, in nm, is

(A) 100 (B) 300

(C) 600 (D) 200

30. A 100 nm coating of index 1.5 on higher-index glass is illuminated from air at 60 degrees. Neglect dispersion. Longest wavelength for a reflected interference minimum, in nm, is

(A) $300\sqrt{3}$ (B) $200\sqrt{3}$

(C) $200\sqrt{6}$ (D) 600

31. In reflected light from an air wedge, dark orders 3 and 7 occur at x=1.6 and 3.6 mm. Fill the same wedge with liquid of index 1.25. Coordinate of the fifth bright fringe measured from the same origin, in mm, is

(A) 1.8 (B) 2

(C) 2.1 (D) 1.9

32. Newton rings are formed with a small nonzero air gap at the centre. At wavelength 500 nm, squared diameters of absolute dark orders 5 and 9 are 8 and 16 mm^2. Central gap thickness, in nm, is

(A) 250 (B) 500

(C) 0 (D) 125

33. Young slits use wavelengths 500 and 600 nm. A nondispersive extra optical path 750 nm is added in one slit. Smallest positive geometric path difference giving simultaneous bright fringes for both colours, in nm, is

(A) 1500 (B) 2250

(C) 3000 (D) 750

34. Two Young wavelengths are 450 and 600 nm. On one side of centre, count path differences below 6000 nm that are bright for 450 nm and dark for 600 nm. The count is

(A) 4 (B) 3

(C) 0 (D) 2

35. Two phase-stable optical fields of equal amplitude have frequencies differing by 2 Hz. At a screen point the detector averages intensity over a time interval 1/4 s centred on zero; initial phase varies across the screen. Visibility of the recorded fringes is

(A) 0.5 (B) 0

(C) $\frac{1}{\pi}$ (D) $\frac{2}{\pi}$

36. A mirror in one arm of an interferometer moves so signed optical-path difference is $\delta(t) = \delta_0 + 2vt$, with $v = 1$ micrometre/s. Fringes are visible exactly when $|\delta| < L_c$ and appear from t=2 to 6 s. Initial signed optical-path difference, in micrometres, is

(A) 8 (B) -4

(C) -8 (D) -12

37. A microscope objective views through index 1.5 liquid. Its aperture radius is $\sqrt{3}$ times its axial working distance; use exact cone geometry. For vacuum wavelength 500 nm, Rayleigh resolution $0.61\lambda_0/(n\sin\theta)$, in nm, is

(A) $\frac{610}{3}$ (B) $\frac{1220\sqrt{3}}{9}$

(C) $\frac{610\sqrt{3}}{3}$ (D) 305

38. Two point sources 1 m apart have their separation vector at 60 degrees to the line of sight. A telescope of diameter 0.122 m uses wavelength 500 nm. Using Rayleigh angle $1.22\lambda/D$, greatest resolving distance, in km, is

(A) $100\sqrt{3}$ (B) 200

(C) 100 (D) $200\sqrt{3}$

39. A plane-parallel etalon of index 1.5 has consecutive normal-incidence transmission maxima at wavelengths 600 and 500 nm. Its thickness, in nm, is

(A) 1500 (B) 1000

(C) 750 (D) 500

40. A plane-parallel etalon of index 1.5 transmits a fixed wavelength at order 10 at normal incidence. Tilt it in air until the same wavelength is transmitted at order 8. Sine of the external incidence angle is

(A) 0.6 (B) 0.8

(C) 0.5 (D) 0.9

41–50 / Preserved hard practice

41. Two coherent beams of the same frequency have linear polarisations at an acute angle θ. Separately, ideal detectors with quantum yields 1/2 and 1 measure photocurrents in ratio 9 : 2. Their interference maximum-to-minimum intensity ratio is 2 : 1. Find $\cos\theta$.

(A) $\frac{2}{3}$ (B) 0.8

(C) $\frac{1}{3}$ (D) $\frac{5}{9}$

42. A glass gives thin-lens powers 6 D and 5 D at vacuum wavelengths 500 and 600 nm, respectively, for curvature factor $1/R_1 - 1/R_2 = 10\ \mathrm{m}^{-1}$. A plate of the same glass is put in one Young path in air. The old centre becomes bright at 500 nm and dark at 600 nm. Find the least positive plate thickness in micrometres.

(A) 15 (B) 5

(C) 30 (D) 10

43. Unpolarised light of power 64 W passes through three ideal polarisers whose axes are 0, ωt and $3\omega t$, where $\omega = \pi$ rad/s. All transmitted power heats an insulated body of heat capacity 4 J/K during $0 \le t \le 0.50$ s. Use quasistatic optical response. Find its temperature rise in kelvin.

(A) 2 (B) 1

(C) 4 (D) 0.5

44. Two coherent arms interfere at a fixed detector. One arm has a normally reflecting mirror executing $x = A\cos\omega t$, with $A = 5\lambda/8$ and speed negligible relative to light. The other arm is fixed; at $x = 0$ their phases agree. Mirror displacement changes that arm optical path by $2x$. How many bright maxima pass the detector during one full mechanical period?

(A) 8 (B) 5

(C) 6 (D) 4

45. Unpolarised light of power P passes through three ideal polarisers with axes $0, \theta, 90^\circ$, where $0 < \theta < 45^\circ$. The transmitted power is $P/16$. Find θ, in degrees.

(A) 15 (B) 30

(C) 22.5 (D) 7.5

46. In Young's experiment, $\lambda = 500$ nm, $d = 1.0$ mm and instantaneous screen distance $D = 1.0$ m. The screen moves away at 0.10 m/s. A plate of index 1.50 in one path has instantaneous thickness $2.0\,\mu$m increasing at $2.0\,\mu$m/s. With signed fringe condition $dy/D + (n-1)t_{plate} = m\lambda$, which bright order m has zero instantaneous transverse coordinate velocity?

(A) 18 (B) 20

(C) 24 (D) 22

47. Two coherent beams initially have powers $9P$ and P. An absorbing neutral filter attenuates the stronger beam until fringe visibility is 4/5; the attenuated beam still remains stronger. Find the heating power in the filter divided by the original total beam power.

(A) 0.5 (B) $\frac{5}{9}$

(C) 0.6 (D) $\frac{1}{3}$

48. A Young experiment entirely in a medium of index 1.50 has fringe width 0.40 mm, slit spacing 1.0 mm and screen distance 1.0 m. The same light then illuminates a photocathode of work function 1.50 eV in air. With $hc = 1240$ eV nm, find its stopping potential in volts.

(A) $\frac{31}{15}$ (B) 1.6

(C) $\frac{17}{30}$ (D) 0.6

49. Young fringe widths in air and a liquid are 1.00 and 0.80 mm with all geometric dimensions fixed. A thin glass lens of index 1.50 and air focal length 20 cm is then immersed in that liquid. An object is 40 cm to its left. Find the image coordinate in cm relative to the lens, positive to the right.

(A) -200 (B) 100

(C) 200 (D) -100

50. In Young's experiment, vacuum wavelength is 600 nm. A plate of index 1.50 and thickness $3.60\,\mu$m covers one slit. The entire apparatus is then immersed in a liquid of index 4/3, while the plate index stays 1.50. Slit spacing and screen distance stay fixed. The magnitude of the final central-fringe shift divided by the fringe width of the original air experiment is

(A) 2.25 (B) 1

(C) 0.75 (D) 3

Answer key

Wave Optics

P-WAOP / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**D**	21	**D**	31	**D**	41	**D**
02	**B**	12	**C**	22	**C**	32	**A**	42	**A**
03	**C**	13	**D**	23	**A**	33	**B**	43	**B**
04	**A**	14	**A**	24	**B**	34	**B**	44	**C**
05	**B**	15	**A**	25	**B**	35	**D**	45	**C**
06	**D**	16	**B**	26	**B**	36	**C**	46	**D**
07	**A**	17	**A**	27	**D**	37	**B**	47	**A**
08	**C**	18	**D**	28	**C**	38	**A**	48	**C**
09	**C**	19	**A**	29	**B**	39	**B**	49	**A**
10	**D**	20	**B**	30	**C**	40	**D**	50	**C**

Chapter 25

Dual Nature

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Two metals have work functions 2 and 3 eV. The same 5 eV photons illuminate them. Ratio of maximum electron kinetic energies first/second is

(A) 1.5 (B) 2.5

(C) 0.5 (D) 3

02. Incident photon energy increases from 4 to 6 eV at work function 2 eV. Stopping-voltage ratio final/initial is

(A) 1 (B) 4

(C) 3 (D) 2

03. A photoelectric surface has stopping potentials 1 and 3 V for photon energies 3 and 5 eV. Work function in eV is

(A) 4 (B) 1

(C) 2 (D) 3

04. A beam has photon arrival rate $10^{15}\,\mathrm{s}^{-1}$. Each photon releases one collected electron. With $e = 1.6 \times 10^{-19}\,\mathrm{C}$, saturation current in microamperes is

(A) 320 (B) 161

(C) 159 (D) 160

05. At the same nonrelativistic kinetic energy, particles have masses m and $4m$. Wavelength ratio λ_m/λ_{4m} is

(A) 2 (B) 3

(C) 4 (D) 1

06. An electron and proton have the same kinetic energy. Wavelength ratio is

(A) m_p/m_e (B) 1

(C) m_e/m_p (D) $\sqrt{m_p/m_e}$

07. A metal threshold wavelength is 500 nm. With $hc = 1240\,\text{eV nm}$, work function in eV is

(A) 3.48 (B) 1.48

(C) 2.48 (D) 4.96

08. Incident wavelength $\lambda_0/2$ illuminates a surface with threshold wavelength λ_0. Maximum electron kinetic energy divided by work function is

(A) 0.5 (B) 1

(C) 2 (D) 0

09. A particle wavelength doubles with mass fixed. Kinetic energy becomes what fraction?

(A) -0.75 (B) 0.5

(C) 0.25 (D) 1.25

10. Electron beam accelerating voltage changes from 100 to 400 V. Ratio of final to initial wavelength is

(A) 0.5 (B) -0.5

(C) 1.5 (D) 1

11–40 / New middle-to-hard practice

11. Light of wavelengths λ and 2λ gives stopping potentials 3 V and 1 V on the same metal. At wavelength $4\lambda/3$, maximum electron speed divided by its value at λ is

(A) $\frac{\sqrt{6}}{3}$ (B) $\frac{2}{3}$

(C) $\frac{\sqrt{3}}{3}$ (D) $\frac{\sqrt{6}}{2}$

12. Wavelengths λ and 3λ give stopping potentials 5 V and 1 V on a surface. Threshold wavelength divided by λ is

(A) 1.5 (B) 3

(C) 6 (D) 5

13. Photons of energies 3 eV and 9 eV eject electrons with maximum speeds in ratio 1:2. At photon energy 5 eV, maximum speed divided by its value for 3 eV photons is

(A) 2 (B) $\frac{\sqrt{6}}{2}$

(C) $\sqrt{3}$ (D) $\sqrt{2}$

14. A surface gives maximum electron speeds v and $\sqrt{3}v$ at incident frequencies f and 2f. At frequency $5f/4$, maximum speed divided by v is

(A) $\frac{\sqrt{6}}{2}$ (B) 1.5

(C) $\frac{\sqrt{5}}{2}$ (D) $\sqrt{2}$

15. A stopping-voltage meter adds an unknown constant offset b to the true stopping voltage. It reads 3 V at frequency f and 6 V at 2f. Independent threshold frequency is f/2. Offset b, in volts, is

(A) 3 (B) 1.5

(C) 1 (D) 0

16. A photoelectric voltmeter has unknown scale factor g and offset b: displayed voltage is $gV_s + b$. For monochromatic photon energies 4,6 eV it displays 5,9 V. The metal work function is 2 eV. If a different surface at the same 6 eV energy displays 7 V, that surface work function, in eV, is

(A) 1 (B) 2.5

(C) 3 (D) 4

17. Photoelectrons from one bound shell trace magnetic radii 2 mm and 4 mm in the same B for photon energies 10 and 22 keV. Use nonrelativistic curvature energy. At photon energy 15 keV, radius in mm is

(A) 5 (B) $\sqrt{10}$

(C) 2.5 (D) 3

18. At one fixed B, two shells illuminated by 30 keV photons give orbit radii r and $2r$. At40 keV the first shell gives $\sqrt{3}r$. Using nonrelativistic curvature energy, binding energy of the second shell, in keV, is

(A) 10 (B) 5

(C) 25 (D) 15

19. A nonrelativistic electron initially has momentum along x. A constant electric force along y makes its de Broglie wavelength half the initial value at time τ. At 2τ, wavelength divided by the initial is

(A) $\frac{\sqrt{7}}{7}$ (B) $\frac{\sqrt{13}}{13}$

(C) 0.5 (D) 0.25

20. An electron initially moves at speed v_0 against a constant electric force. At time τ its de Broglie wavelength has doubled and it still moves forward. Total distance travelled by 3τ, divided by $v_0\tau$, is

(A) 0.75 (B) 1.25

(C) 1.5 (D) 1

21. Two positive particles A, B start from rest through the same voltage. Their de Broglie wavelengths obey $\lambda_A/\lambda_B = 2$, while in the same perpendicular B their orbit radii obey $r_A/r_B = 3$. Mass ratio m_B/m_A is

(A) 6 (B) $\frac{1}{6}$

(C) $\frac{2}{3}$ (D) 1.5

22. Two positive particles A, B are accelerated from rest through the same voltage and fly the same distance. Their flight-time ratio is 2 and de Broglie wavelength ratio $\lambda_A/\lambda_B = 1/2$. The mass ratio m_A/m_B is

(A) 2 (B) 4

(C) 1 (D) 8

23. A photon and a nonrelativistic particle have the same wavelength. Photon energy is 100 eV while particle kinetic energy is 2 eV. The particle rest-energy scale mc^2, in eV, is

(A) 2500 (B) 5000

(C) 1250 (D) 50

24. An electron has kinetic energy 1 keV and rest energy 511 keV; use nonrelativistic momentum. A photon has that same kinetic-energy value as its own energy. Photon wavelength divided by electron wavelength is

(A) 511 (B) $\sqrt{511}$

(C) $\sqrt{1022}$ (D) $\frac{\sqrt{1022}}{2}$

25. An electron diffraction apparatus has unknown positive contact voltage V_c, so beam energy is $e(V + V_c)$. For external voltages 100 and 700 V, first-order Bragg sines have ratio 2:1. At100 V the Bragg angle is 45 degrees. Bragg angle at external 300 V, in degrees, is

(A) 15 (B) 45

(C) 60 (D) 30

26. A nonrelativistic electron beam initially has first-order Bragg angle 45 degrees. Increasing its accelerating voltage by 300 V reduces that angle to 30 degrees. Initial accelerating voltage, in volts, is

(A) 300 (B) 150

(C) 100 (D) 600

27. Equal numbers per second of photons with energies $\phi/2$ and 2ϕ form a beam of total power P on a surface of work function ϕ. Each above-threshold photon emits one collected electron and subthreshold photons emit none. Saturation current divided by eP/ϕ is

(A) 0.8 (B) 0.4

(C) 0.5 (D) 0.2

28. An 8 W beam of 310 nm photons is fully absorbed by a work-function 2 eV surface. Collected current is 1 A. In an ideal model every escaping electron has maximum kinetic energy and none returns; all other absorbed energy becomes heat. Use hc=1240 eV nm. Heating power, in watts, is

(A) 4 (B) 6

(C) 8 (D) 2

29. Monoenergetic 4 eV photoelectrons have angular emission rate per solid angle proportional to $\cos\theta$ over the outward hemisphere. A planar 1 V retarding barrier transmits only electrons whose normal kinetic energy exceeds eV. Fraction of zero-bias current collected is

(A) 0.25 (B) 0.5

(C) 0.75 (D) $\frac{\sqrt{3}}{2}$

30. Monoenergetic photoelectrons are emitted uniformly per solid angle over a hemisphere. A planar 1 V retarding barrier reduces current to half its zero-bias value. At retarding voltage 9/4 V, the collected fraction is

(A) 0.25 (B) 0.75

(C) 0.5 (D) 0

31. An electron retarding region has potential-energy rise $U(x) = ax + bx^2$, with x in cm and U in eV. Initial kinetic energies 4 and 12 eV turn at x=1 and 2 cm respectively. A 24 eV electron turns at x, in cm, equal to

(A) 2 (B) 4

(C) 3 (D) $2\sqrt{3}$

32. An electron enters a retarding potential-energy profile $U(x) = U_0(1 - e^{-x/l})$. At x=$l \ln 2$, its de Broglie wavelength is twice the initial value, with forward motion. Its eventual turning distance divided by l is

(A) $\log(2)$ (B) 1.5

(C) $\log(3)$ (D) $\log(4)$

33. A monatomic ideal gas is compressed adiabatically to one-eighth its volume, then heated at fixed volume by an energy equal to its original internal energy. Define effective matter wavelength as $h/\sqrt{3mkT}$. Final wavelength divided by initial is

(A) $\frac{\sqrt{3}}{3}$ (B) $\frac{\sqrt{5}}{5}$

(C) 0.2 (D) 0.5

34. A gas of particles m has drift speed u and random thermal motion at T. Define effective wavelength as $h/\sqrt{\langle p^2 \rangle}$. Raising T to 4 T while keeping u fixed decreases that wavelength by $\sqrt{2}$. Drift kinetic energy divided by random thermal kinetic energy in the original state is

(A) 2 (B) 0.5

(C) 4 (D) 1

35. Particles accelerated from rest through voltage V fly a fixed distance; a timer adds an unknown constant delay. Recorded times at 100 and 400 V are 20 and 15 ns. Recorded time at 900 V, in ns, is

(A) $\frac{35}{3}$ (B) 10

(C) $\frac{40}{3}$ (D) $\frac{50}{3}$

36. Monoenergetic photoelectrons take time 2t to cross a field-free flight tube. Giving them an additional 3 eV before the tube reduces time to t. Incident photons had energy 5 eV. Cathode work function, in eV, is

(A) 2 (B) 1

(C) 3 (D) 4

37. A stationary body of mass M absorbs two photons of energies E and 2E arriving at right angles. To leading nonrelativistic recoil order, kinetic energy gained divided by $E^2/(Mc^2)$ is

(A) 0.5 (B) 4.5

(C) 2.5 (D) 1.5

38. A stationary body absorbs opposite photons of energies E and 3E. Its rest-energy scale is $Mc^2 = 100E$. To leading nonrelativistic recoil order, ignoring mass change in the recoil term, internal-energy increase divided by E is

(A) 3.98 (B) 4

(C) 3.96 (D) 3.99

39. In an ideal matter-wave interferometer, arm lengths are L and 2 L. Their electron kinetic energies are eV and 4eV. Their phase difference is 12 full cycles. Raise only the second energy to 9eV. Number of fringe cycles shifted is

(A) 20 (B) 8

(C) 12 (D) 6

40. Equal-length matter-wave arms initially have electron kinetic energy eU, with unknown positive U. Adding 3 eV to one arm shifts N fringes; adding 8 eV shifts 2N. Outside-arm phases are unchanged. Adding 15 eV to that arm shifts how many times N?

(A) 4 (B) 3

(C) 2 (D) 5

41–50 / Preserved hard practice

41. A photocathode of work function 2.0 eV absorbs a two-colour beam of photon energies 4.0 and 6.0 eV. Unit quantum yield is assumed; all emitted electrons travel normally toward the collector at their maximum kinetic energies. A 3.0 V retarding potential reduces collected current to one third of saturation. Find the incident-power fraction in the 6.0 eV component.

(A) 0.5 (B) $\frac{2}{3}$

(C) $\frac{1}{3}$ (D) $\frac{3}{7}$

42. Photoelectrons from the same cathode are emitted by 5.0 eV and 3.0 eV photons. Fastest electrons from the first beam are accelerated through 1.0 V, then enter a perpendicular field B. Their orbit radius is twice that of fastest electrons from the second beam entering the same B directly. With $hc = 1240$ eV nm, find cathode threshold wavelength in nm.

(A) 248 (B) $\frac{1240}{3}$

(C) 1240 (D) 620

43. A positive nonrelativistic particle has de Broglie wavelength 0.10 nm. It passes undeflected, with velocity perpendicular to both crossed fields $E = 1.0$ V/m and $B = 10^{-3}$ T, then makes a 0.020 m radius orbit in that same magnetic-field magnitude. It is subsequently accelerated through 4.0 V. With $h = 6.4 \times 10^{-34}$ J s and $e = 1.6 \times 10^{-19}$ C, find its final kinetic energy in eV.

(A) 8.02 (B) 16.02

(C) 4.02 (D) 8

44. Fastest electrons from a fixed photocathode have de Broglie wavelengths λ_1, λ_2 after additional accelerating potentials 1 and 9 V, with $\lambda_1/\lambda_2 = \sqrt{5}$. Instead, a retarding potential 0.50 V is applied to the same emitted electrons. Find their resulting wavelength divided by λ_1, assuming transmitted fastest electrons remain nonrelativistic.

(A) $\sqrt{5}$ (B) $\sqrt{2}$

(C) 0.5 (D) 2

45. Light of wavelengths 400 and 600 nm separately illuminates the same photocathode. The de Broglie wavelength of its fastest electron is $\sqrt{3}$ times larger for 600 nm than for 400 nm. With $hc = 1240$ eV nm, threshold wavelength in nm is

(A) 1200 (B) 800

(C) 1000 (D) 600

46. A 1.0 W monochromatic beam illuminates a cathode with quantum yield 1/2. All emitted electrons are normal and have maximum energy, with stopping potential 2.0 V. They are accelerated through 3.0 V and collected by an insulated calorimeter of heat capacity 0.10 J/K, which warms at 5.0 K/s. Find the work function in eV; all electrons are collected and all impact energy becomes heat.

(A) 2 (B) 5

(C) 4 (D) 3

47. A nonrelativistic positive particle is accelerated through V, making its de Broglie wavelength half its initial value. It then enters a retarding field of magnitude $E_0(1 + x/L)$ opposite its velocity, with $x = 0$ at entry and $V = E_0 L$. Find its stopping coordinate divided by L.

(A) $-1 + \sqrt{3}$ (B) $-1 + \frac{\sqrt{33}}{3}$

(C) $\frac{4}{3}$ (D) $-1 + \frac{2\sqrt{6}}{3}$

48. An initially stationary massive particle of charge $q > 0$ absorbs two photons of wavelengths λ and 2λ, arriving at right angles in one plane. A uniform magnetic field B is normal to that plane. Its ensuing recoil orbit radius is r. At leading nonrelativistic order, find total incident photon energy divided by $cqBr$.

(A) $\sqrt{5}$ (B) $\frac{3\sqrt{5}}{5}$

(C) $\frac{\sqrt{5}}{3}$ (D) 1.5

49. A proton of de Broglie wavelength λ makes an elastic head-on collision with a stationary nucleus of unknown mass and charge $+2e$. The proton is reflected with wavelength 2λ. Both then enter the same uniform perpendicular magnetic field. Find nucleus orbit radius divided by reflected-proton orbit radius; neglect further interactions.

(A) 3 (B) 0.5

(C) 2 (D) 1.5

50. A light pulse of energy E strikes an opaque mirror of mass m attached to a horizontal spring of stiffness k, initially at rest at equilibrium. A fraction is absorbed, depositing heat H; the rest reflects normally. The pulse is very short, recoil is nonrelativistic, and Doppler and mass changes are negligible. If subsequent maximum spring extension is x, find E.

(A) $\frac{H}{2} + \frac{c\sqrt{k}\sqrt{m}x}{2}$

(B) $H + c\sqrt{k}\sqrt{m}x$

(C) $\frac{H}{2} + c\sqrt{k}\sqrt{m}x$

(D) $\frac{c\sqrt{k}\sqrt{m}x}{2}$

Answer key

Dual Nature

P-DUNA / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**A**	21	**C**	31	**C**	41	**D**
02	**D**	12	**C**	22	**B**	32	**C**	42	**D**
03	**C**	13	**D**	23	**A**	33	**B**	43	**A**
04	**D**	14	**A**	24	**C**	34	**A**	44	**D**
05	**A**	15	**B**	25	**D**	35	**C**	45	**B**
06	**D**	16	**C**	26	**A**	36	**D**	46	**D**
07	**C**	17	**D**	27	**B**	37	**C**	47	**B**
08	**B**	18	**A**	28	**B**	38	**A**	48	**B**
09	**C**	19	**B**	29	**C**	39	**B**	49	**D**
10	**A**	20	**B**	30	**A**	40	**B**	50	**A**

Chapter 26

Atoms

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Hydrogen transition $n = 4$ to $n = 2$ photon energy in eV is

(A) 1.55 (B) 2.55

(C) 3.55 (D) 5.1

02. Hydrogen transition $n = 3$ to $n = 1$ photon energy divided by ground ionisation energy is

(A) $\frac{16}{9}$ (B) $\frac{17}{9}$

(C) $\frac{8}{9}$ (D) $\frac{-1}{9}$

03. For hydrogen, excitation energy from ground to $n = 2$ in eV is

(A) 10.2 (B) 20.4

(C) 9.2 (D) 11.2

04. For hydrogen, energy needed to excite ground to $n = 3$ divided by ground ionisation energy is

(A) $\frac{-1}{9}$ (B) $\frac{17}{9}$

(C) $\frac{8}{9}$ (D) $\frac{16}{9}$

05. A hydrogen atom absorbs exactly 10.2 eV from ground. Its final principal quantum number is

(A) 3 (B) 1

(C) 4 (D) 2

06. For hydrogen, ratio of longest Lyman wavelength to shortest Lyman wavelength is

(A) $\frac{4}{3}$ (B) $\frac{8}{3}$

(C) $\frac{1}{3}$ (D) $\frac{7}{3}$

07. For hydrogen, ratio of shortest Balmer wavelength to shortest Lyman wavelength is

(A) 8 (B) 4

(C) 5 (D) 3

08. For hydrogen, ratio of orbital radii r_4/r_2 is

(A) 8 (B) 5

(C) 3 (D) 4

09. Taking $hc = 1240\,\text{eV nm}$, an X-ray tube at 20 kV has cut-off wavelength in nm

Enrichment practice: X-ray tube cut-off extension retained as optional enrichment; exact inclusion in pinned BITSAT 2026 core scope not asserted.

(A) 0.124 (B) 0.062

(C) 1.062 (D) -0.938

10. An X-ray photon has energy 10 keV. Its energy is what fraction of the maximum photon energy of a 20 kV tube?

Enrichment practice: X-ray tube cut-off extension retained as optional enrichment; exact inclusion in pinned BITSAT 2026 core scope not asserted.

(A) 1 (B) -0.5

(C) 1.5 (D) 0.5

11–40 / New middle-to-hard practice

11. A hydrogen-like ion has orbit radius $9a_0/2$ and binding energy $54.4/9$ eV, where a_0 is the hydrogen ground-state radius. Use $r = a_0 n^2/Z$ and $E_b = 13.6Z^2/n^2$ eV. Principal quantum number is

(A) 4 (B) 6

(C) 2 (D) 3

12. A hydrogen-like orbit has angular momentum $2\hbar$ and radius $2a_0$. Energy of the longest Lyman photon of this ion, in eV, is

(A) 10.2 (B) 27.2

(C) 54.4 (D) 40.8

13. A transition in He^+ has upper-to-lower orbit radius ratio 9:4 and the same wavelength as hydrogen 3 to 2. Its upper quantum number is

(A) 3 (B) 9

(C) 4 (D) 6

14. A He^+ transition changes principal quantum number by 2 and matches the hydrogen 2 to 1 photon wavelength. Its upper quantum number is

(A) 4 (B) 8

(C) 6 (D) 3

15. An ensemble of hydrogen atoms excited to the same highest level can produce 15 distinct downward-transition lines. A filter removes every line ending at n=1 or n=2. Number of remaining distinct lines is

(A) 6 (B) 5

(C) 10 (D) 3

16. A hydrogen ensemble can emit 28 distinct lines from one highest level. Consider its two-photon cascades ending at n=4, with exactly one intermediate level. In how many such cascades does the first photon have greater energy than the second?

(A) 0 (B) 1

(C) 2 (D) 3

17. A heavy three-level atom emits cascade wavelengths 1000 nm and 500 nm at right angles. Neglect recoil shifts of spectral energies. Cascade recoil momentum magnitude divided by recoil momentum for the direct transition between the same endpoints is

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{2}{3}$

(C) $\frac{\sqrt{5}}{3}$ (D) 1

18. A three-level atom has wavelengths 500 nm for A to C and 750 nm for A to B, with A above B above C. Its B-to-C photon illuminates a metal of work function 0.5 eV. Use hc=1240 eV nm. Maximum photoelectron energy, in eV, is

(A) $\frac{62}{75}$ (B) $\frac{49}{150}$

(C) 0.74 (D) $\frac{49}{75}$

19. A 20 eV photon ionises a hydrogen atom in an unknown level, giving electron maximum energy 16.6 eV. A photon from that initial level to the ground subsequently illuminates a work-function 2 eV metal. Maximum photoelectron energy there, in eV, is

(A) 3.4 (B) 10.2

(C) 8.2 (D) 14.6

20. One photon energy just ionises He^+ at a principal level n, while on hydrogen at the same n it ejects a 10.2 eV electron. The value of n is

(A) 2 (B) 4

(C) 3 (D) 1

21. A hydrogen-like orbit has total energy -3.4 eV and angular momentum $2\hbar$. After excitation to the next principal level, electron potential energy, in eV, is

(A) -6.8 (B) $-\frac{136}{45}$

(C) $-\frac{68}{45}$ (D) -27.2

22. A Bohr electron has angular momentum $3\hbar$ and kinetic energy $4E_R/9$, where $E_R = 13.6$ eV. It jumps to n=2. Change in potential energy divided by E_R is

(A) $-\frac{10}{9}$ (B) $\frac{10}{9}$

(C) $-\frac{5}{9}$ (D) $-\frac{4}{9}$

23. A hydrogen-like orbit has radius twice and period twice those of the hydrogen ground orbit. Frequency of its longest Lyman line divided by the hydrogen longest Lyman frequency is

(A) 8 (B) 2

(C) 4 (D) 0.25

24. A hydrogen-like orbit has period 27 times the hydrogen ground period and binding energy one-ninth the hydrogen ground binding. Orbital radius divided by a_0 is

(A) 9 (B) 81

(C) 27 (D) 3

25. An electron in a Bohr orbit has de Broglie wavelength $2\pi a_0$ and circumference $6\pi a_0$. Photon energy on transition to n=2, divided by hydrogen ground binding E_R, is

(A) 0.75 (B) 2.25

(C) 1.25 (D) $\frac{5}{36}$

26. A hydrogen-like orbit has electron wavelength equal to the hydrogen ground circumference and enclosed area $16\pi a_0^2$. Its orbital current divided by hydrogen ground orbital current is

(A) 0.5 (B) 0.0625

(C) 4 (D) 0.25

27. A charged projectile of mass m and kinetic energy K approaches a stationary target of mass 4 m head-on. Closest separation divided by that for an immovable target of the same charge is

(A) 1 (B) 1.25

(C) 0.8 (D) 1.5

28. A projectile of mass m moves right at v and a target of mass 3 m moves left at v/3, head-on. Their charges are fixed. Compare closest separation to an immovable target struck by that same projectile at v. The separation ratio is

(A) 1 (B) 0.75

(C) 0.5 (D) $\frac{4}{3}$

29. For Rutherford scattering let d be the fixed-target head-on closest distance. A 90-degree scatter has impact parameter $b = (d/2)\cot(90^\circ/2)$. Its closest separation divided by d is

(A) $\sqrt{2}$ (B) 1.5

(C) $\frac{1}{2} + \frac{\sqrt{2}}{2}$ (D) $\frac{1}{2} + \frac{\sqrt{5}}{2}$

30. A charged projectile scatters from an immovable nucleus. Its speed at closest approach is half its incident speed. If d is its head-on closest distance at that incident energy, impact parameter divided by d is

(A) $\frac{1}{3}$ (B) $\frac{4}{3}$

(C) $\frac{2}{3}$ (D) 0.75

31. A Rutherford detector has a constant unknown background. Counts at 90 and 180 degrees are 10 and 4 for equal solid angles. Signal scales as $\sin^{-4}(\theta/2)$. Expected total counts at 60 degrees are

(A) 18 (B) 32

(C) 40 (D) 34

32. A second scattering foil has half the mass per area and twice the atomic mass of the first. At twice the projectile energy it gives one-quarter as many Rutherford counts at the same angle. Use signal proportional to areal atom density times Z^2/K^2. Nuclear-charge ratio second/first is

(A) 2 (B) 1

(C) $\sqrt{2}$ (D) 4

33. An X-ray tube has effective accelerating voltage $V - V_c$, with constant positive contact loss. At applied 20 kV its cutoff is λ; at 50 kV the cutoff is $\lambda/3$. The contact loss V_c, in kV, is

(A) 5 (B) 0

(C) 15 (D) 10

34. An X-ray tube voltage is sinusoidal about 20 kV and always positive. Its shortest observed continuum wavelength is 0.04 nm. Use hc=1240 eV nm and ignore contact loss. Minimum instantaneous tube voltage, in kV, is

(A) 11 (B) 9

(C) 31 (D) 20

35. In an idealised characteristic-line series, energy follows $E = A(Z - s)^2$. Model elements Z=10 and 18 have wavelengths 40 and 10 nm. The same model line for Z=26 has wavelength, in nm, equal to

(A) 5 (B) $\frac{40}{3}$

(C) $\frac{40}{9}$ (D) $\frac{10}{3}$

36. An idealised characteristic-line frequency follows $\nu = A(Z - s)^2$. Model elements Z=12 and 20 give 25 and 81 times a reference frequency. At what Z is that line 121 times the reference? Take positive screened charges.

(A) 24 (B) 26

(C) 28 (D) 22

37. An X-ray tube's cutoff photons eject electrons from a fixed shell with maximum 8 keV; photons of twice that cutoff wavelength eject 2 keV electrons. Increase tube voltage by 50%. New cutoff photons give maximum electron energy, in keV, of

(A) 10 (B) 14

(C) 16 (D) 12

38. Electrons enter an X-ray tube with nonzero initial kinetic energy. At accelerating voltage 10 kV, continuum cutoff is 0.062 nm. If voltage is changed to 30 kV, cathode electron wavelength divided by its final anode wavelength is, nonrelativistically ($hc = 1240\,\text{eV nm}$)

(A) $\sqrt{3}$ (B) 1.5

(C) 2 (D) $\sqrt{2}$

39. Include nuclear recoil through Rydberg factor $R_M = R_\infty M/(M + m_e)$. Two same-Z ions have nuclear masses $2000m_e$ and $4000m_e$. For a fixed transition, heavier-ion wavelength divided by lighter-ion wavelength is

(A) $\frac{4002}{4001}$ (B) $\frac{4001}{4002}$

(C) 0.5 (D) $\frac{4000}{4001}$

40. Two same-Z isotopes have nuclear masses M and 2M. Their fixed-transition frequency ratio heavier/lighter is 3674/3673. Use $R_M = R_\infty M/(M + m_e)$. The ratio M/m_e is

(A) 1836 (B) 3672

(C) 1837 (D) 3673

41–50 / Preserved hard practice

41. A hydrogen-like one-electron ion has ground-state orbit radius $a_0/3$, where a_0 is the hydrogen ground-state radius. A photon emitted in its $n = 3 \to 2$ transition illuminates a metal of work function 5.00 eV. Use $E_n = -13.6Z^2/n^2\,\text{eV}$. Neglect recoil. The stopping potential for the fastest photoelectrons, in volts, is

(A) $\frac{109}{9}$ (B) 17

(C) $\frac{40}{3}$ (D) 12

42. A nonrelativistic alpha particle approaches a fixed positive nucleus from infinity with kinetic energy K. Its distance of closest approach in a head-on collision would be r_0. In another collision at the same energy, the closest distance is $2r_0$. With pure Coulomb repulsion and negligible nuclear recoil, the impact parameter divided by r_0 is

(A) $\sqrt{3}$ (B) 2

(C) 1 (D) $\sqrt{2}$

43. A hydrogen-like one-electron ion is in a circular Bohr state whose radius is $8a_0$, where a_0 is the hydrogen ground-state radius. Its ionization energy from that state is 3.40 eV. Use $r_n = a_0 n^2/Z$, $E_n = -13.6Z^2/n^2\,\text{eV}$ and $hc = 1240\,\text{eV nm}$. The wavelength required to excite that ion from its current state to $n = 8$, in nm, is

(A) $\frac{24800}{17}$ (B) $\frac{12400}{51}$

(C) $\frac{6200}{17}$ (D) $\frac{24800}{51}$

44. A large hydrogen sample occupies only $n = 1, 2, 3$. Its mean excitation energy above the ground state is 6.80 eV per atom. On relaxation, every atom initially in $n = 3$ is constrained to follow $3 \to 2 \to 1$, and every atom initially in $n = 2$ follows $2 \to 1$. The observed number of $3 \to 2$ photons is 0.225 per initial atom. Use $E_n = -13.6/n^2$ eV. The total number of Lyman-alpha photons per initial atom is

(A) 0.225 (B) 0.775

(C) 0.4 (D) 0.625

45. A hydrogen-like ion has ground-state energy -54.4 eV. From $n = 2$ it absorbs a 10.2 eV photon. Across many such ions, all possible downward cascades occur. How many distinct cascade frequencies can eject photoelectrons from a metal of work function 4.00 eV?

(A) 6 (B) 5

(C) 3 (D) 4

46. A circular Bohr orbit in a hydrogen-like ion has orbital magnetic-moment magnitude $3\mu_B$, with $\mu_B = e\hbar/(2m_e)$. Its revolution frequency is 4/27 of that of ground-state hydrogen. Use $L_n = n\hbar$, $\mu = eL/(2m_e)$, $f_n/f_{\mathrm{H},1} = Z^2/n^3$, $E_n = -13.6Z^2/n^2$ eV and $hc = 1240$ eV nm. Wavelength emitted on direct return to $n = 1$, in nm, is

(A) $\frac{6975}{544}$ (B) $\frac{6975}{272}$

(C) $\frac{1395}{34}$ (D) $\frac{6975}{68}$

47. An excited atom initially at rest emits one photon and recoils. The internal transition energy is Δ, and the final atom mass is M. In the stated model, the photon has momentum E_γ/c, atomic recoil energy is $p^2/(2M)$, and $\epsilon = \Delta/(Mc^2) \ll 1$. Keeping the recoil term exactly within this nonrelativistic model, E_γ/Δ equals

(A) $\frac{1}{\epsilon+1}$ (B) $\frac{2}{\sqrt{\epsilon+1}+1}$

(C) $\frac{2}{\sqrt{2\epsilon+1}+1}$ (D) 1

48. Two transitions of the same hydrogen-like ion end at $n = 1$ and begin at consecutive levels n and $n + 1$, with $n \geq 2$. The lower-energy photon has 128/135 times the energy of the higher-energy photon. The higher-energy photon has energy 51.0 eV. Use $E_n = -13.6Z^2/n^2$ eV and $hc = 1240$ eV nm. The ground-state photoionization threshold wavelength, in nm, is

(A) $\frac{775}{34}$ (B) $\frac{1550}{17}$

(C) $\frac{1240}{51}$ (D) $\frac{775}{17}$

49. A hydrogen beam occupies only $n = 2$ and $n = 3$. Its mean binding energy is 34/15 eV per atom. Every atom absorbs one 4.00 eV photon and is ionized; neglect recoil. The emitted electrons then meet a retarding potential of 2.00 V along their direction of motion. Use binding energy $13.6/n^2$ eV. The fraction of the original atoms whose electrons pass the retarding region is

(A) 0.2 (B) 1

(C) 0.6 (D) 0.4

50. Ground-state hydrogen atoms are excited by electrons of kinetic energy 12.75 eV. Each atom has at most one exciting collision, and all permitted downward cascades subsequently occur across the sample. Take $E_n = -13.6/n^2$ eV and neglect recoil. How many distinct emitted frequencies can eject electrons from a metal of work function 2.50 eV?

(A) 4 (B) 3

(C) 6 (D) 1

Answer key

Atoms

P-ATOM / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**D**	21	**B**	31	**D**	41	**D**
02	**C**	12	**D**	22	**A**	32	**A**	42	**D**
03	**A**	13	**D**	23	**C**	33	**A**	43	**D**
04	**C**	14	**A**	24	**A**	34	**B**	44	**D**
05	**D**	15	**A**	25	**C**	35	**C**	45	**B**
06	**A**	16	**B**	26	**D**	36	**A**	46	**B**
07	**B**	17	**C**	27	**B**	37	**B**	47	**C**
08	**D**	18	**B**	28	**B**	38	**C**	48	**A**
09	**B**	19	**C**	29	**C**	39	**B**	49	**C**
10	**D**	20	**A**	30	**C**	40	**A**	50	**A**

Chapter 27

Nuclei

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A sample activity decreases from 160 to 20 Bq. Number of half-lives elapsed is

(A) 4 (B) 6

(C) 3 (D) 2

02. A sample half-life is 5 h. Time for 7/8 of initial nuclei to decay in hours is

(A) 14 (B) 15

(C) 16 (D) 30

03. For a radioactive isotope, mean lifetime divided by half-life is

(A) $1/\ln 2$ (B) $1/2$

(C) $\ln 2$ (D) 2

04. A nucleus $A = 64$ has radius R. A nucleus $A = 8$ has radius divided by R equal to

(A) 1.5 (B) -0.5

(C) 0.5 (D) 1

05. For a radioactive sample, remaining fraction after time t is $1/16$. If $t = 20$ min, half-life in minutes is

(A) 4 (B) 6

(C) 5 (D) 10

06. Two isotopes have half-lives T and $2T$, equal initial atom numbers. Initial activity ratio first/second is

(A) 2 (B) 3

(C) 4 (D) 1

07. Two equal initial samples of one isotope are prepared T apart, where T is its half-life. At a common time after both preparations, older-to-newer activity ratio is

(A) -0.5 (B) 1.5

(C) 0.5 (D) 1

08. A mass defect is $0.02\,u$ for a nucleus $A = 4$. With $uc^2 = 931\,\text{MeV}$, binding energy per nucleon in MeV is

(A) 3.655 (B) 5.655

(C) 9.31 (D) 4.655

09. An alpha-decay chain has three alpha decays and two beta-minus decays. Net atomic-number change is

(A) -5 (B) -4

(C) -8 (D) -3

10. An alpha-decay chain has three alpha decays and two beta-minus decays. Net mass-number change is

(A) -24 (B) -12

(C) -13 (D) -11

11–40 / New middle-to-hard practice

11. A counter has a constant unknown background. At equally spaced times 0, T,2 T it records 100,60,40 counts per unit time from one decaying isotope. Expected reading at 3 T is

(A) 20 (B) 30

(C) 35 (D) 25

12. A sample counter reads 500 counts/min initially and 200 counts/min after one hour. An independent blocked-source measurement gives background 100 counts/min. Time after the initial measurement when the total reading reaches 150 counts/min, in hours, is

(A) 2 (B) 1

(C) 3 (D) 1.5

13. Two isotopes have half-lives T and 2 T. Their total activity is 12 Bq initially and 5 Bq after 2 T. Fraction of initial radioactive nuclei belonging to the shorter-lived isotope is

(A) $\frac{1}{3}$ (B) 0.25

(C) 0.2 (D) $\frac{2}{3}$

14. Two isotopes have half-lives T and 2 T. Total activities at zero and 2 T are 12 and 5 Bq. Total activity at T, in Bq, is

(A) $2\sqrt{2}+4$ (B) 6

(C) $3+3\sqrt{2}$ (D) $2+4\sqrt{2}$

15. Initially pure parent A decays to radioactive B, which decays to stable C. Constants are λ and 2λ. When parent and daughter activities first become equal at a positive time, fraction of initial nuclei still present as A or B is

(A) 0.25 (B) 0.5

(C) 0.75 (D) $\frac{2}{3}$

16. Parent A and daughter B have the same decay constant λ in A to B to stable C. Initially their populations are N_0 and $N_0/2$. Time of maximum daughter activity divided by parent half-life is

(A) $\frac{1}{2\log(2)}$ (B) $\frac{\log(2)}{2}$

(C) $\frac{1}{\log(2)}$ (D) 0.5

17. A nucleus has independent alpha and beta decay channels. Eighty percent of decays are alpha, and its actual half-life is 10 h. If the beta channel could be suppressed without changing the alpha rate, new half-life in hours is

(A) 12.5 (B) 8

(C) 50 (D) 20

18. Alpha and beta counters have efficiencies 1/4 and 1/2. They record alpha events at twice the beta rate from a sample with only those two decay channels. Suppress the alpha channel without changing beta decay. New half-life divided by the original is

(A) 2 (B) 3

(C) 4 (D) 5

19. A stable target is activated at constant production rate R into a radioactive isotope. During irradiation its activities at T and 2 T are 3 and 4 Bq. Switch the source off at 2 T. Activity one further interval T later, in Bq, is

(A) 2 (B) 1.5

(C) 1 (D) $\frac{4}{3}$

20. In one experiment an activated isotope's activity falls by a factor 3 in a time T after irradiation ends. In a fresh identical target, constant-rate irradiation from zero gives 6 Bq after T. Its activity after 2 T of uninterrupted irradiation, in Bq, is

(A) 10 (B) 12

(C) 8 (D) 9

21. A carbon sample contains 25% modern contamination by carbon atoms and 75% ancient carbon. Its measured radioactive isotope fraction is 3/8 of modern carbon. Assuming initial ancient fraction was modern, sample age divided by isotope half-life is

(A) $1+\frac{\log(3)}{\log(2)}$ (B) 2

(C) $-\frac{\log(3)}{\log(2)}+3$ (D) 3

22. Two isotopes initially have number ratio 4:1 and decay constants 2λ and λ. Their measured activity ratio later is 2:1. Elapsed time divided by the second isotope half-life is

(A) 3 (B) 0.5

(C) 1 (D) 2

23. An excited nucleus at rest emits a neutron with 10 MeV and daughter recoil 0.4 MeV. Ground-state mass defects per nucleon of parent and daughter are 0.008 u and 0.0076 u. Use nucleon mass ratios for recoil and $uc^2 = 931$ MeV. Initial nuclear excitation energy, in MeV, is

(A) 27.158 (B) 16.758

(C) 17.58 (D) 26.358

24. A 180 nucleus with binding 7.5 MeV per nucleon splits into two fragments of masses A and 180-A, with respective bindings 8 and 8.5 MeV per nucleon. Q is 140 MeV; initially it is at rest. Neglect mass defects in recoil mass ratios. Lighter fragment kinetic energy, in MeV, is

(A) 80 (B) 70

(C) $\frac{700}{9}$ (D) $\frac{560}{9}$

25. An endothermic reaction has Q=−6 MeV and a projectile of mass m hitting a stationary target 3 m. To leading nonrelativistic order, minimum projectile kinetic energy, in MeV, is

(A) 6 (B) 4.5

(C) 8 (D) 18

26. A projectile m of kinetic energy 8 MeV hits a stationary target 3 m. An exothermic reaction of Q=4 MeV produces two equal-mass fragments. Neglect mass defects in kinematic mass ratios. Greatest lab kinetic energy of either fragment, in MeV, is

(A) 12 (B) 10

(C) $2\sqrt{5}+6$ (D) $6-2\sqrt{5}$

27. An isolated alpha decay gives alpha kinetic energy 4.9 MeV and daughter recoil 0.1 MeV. Treat masses as proportional to mass number. Parent mass number is

(A) 49 (B) 200

(C) 196 (D) 204

28. A mass-number 100 parent decays in flight. Alpha particles observed along the forward and backward laboratory directions have kinetic energies 9 and 1 MeV. Treat masses by mass number and use nonrelativistic kinematics. Decay Q in the parent rest frame, in MeV, is

(A) $\frac{25}{6}$ (B) 4

(C) 4.8 (D) 5

29. A reactor delivers 2 MW electric power at 25% efficiency for 24 h. Each fission releases 200 MeV. Fuel contains 10% uranium-235 by mass, of which only 50% fissions. Use $1\,\text{MeV} = 1.6 \times 10^{-13}$ J and $N_A = 6 \times 10^{23}$ mol^{-1}. Initial fuel mass required, in grams, is

(A) 8.46 (B) 169.2

(C) 33.84 (D) 84.6

30. A critical reactor produces 3 neutrons per fission. Leaked neutrons number 3/2 as many as neutrons lost to nonfission absorption. A retained neutron causes fission with probability p. Halve the leakage fraction and multiply p by 2/3. Expected neutron population after two generations, divided by its value just after this change, is

(A) $\frac{64}{81}$ (B) $\frac{4}{9}$

(C) $\frac{8}{9}$ (D) 0.64

31. A fusion mixture has 8 g total deuterium and tritium, with atom-number ratio D:T=3:2. Allow all possible D+T reactions and no other reactions; use masses 2 and 3 u. Unreacted deuterium mass, in grams, is

(A) $\frac{2}{3}$ (B) $\frac{4}{3}$

(C) $\frac{8}{3}$ (D) 2

32. Start with 100 deuterons. Exactly 50 deuterons are consumed in primary D+D reactions, whose branches to T+p and He-3+n release 4 and 3 MeV. Mean primary Q is 3.6 MeV. Then every produced T fuses with available D, releasing 17.6 MeV each. Total released energy, in MeV, is

(A) 354 (B) 264

(C) 90 (D) 442

33. In a liquid-drop estimate keep only positive surface energy proportional to area and Coulomb energy proportional to Z^2/R. Eight identical nuclei merge at unchanged nuclear density. For fusion to lower these two terms together, initial total Coulomb energy divided by initial total surface energy must be below

(A) $\frac{1}{6}$ (B) 0.5

(C) $\frac{1}{12}$ (D) $\frac{1}{3}$

34. A nucleus and its alpha daughter have radii in ratio $(125/121)^{1/3}$. Assume nuclear radius proportional to $A^{1/3}$. After a total of three alpha decays from the original, final-to-original radius ratio is

(A) $\frac{11^{\frac{2}{3}}}{5}$ (B) $\frac{\sqrt[3]{113}}{5}$

(C) $\frac{\sqrt[3]{117}}{5}$ (D) 0.904

35. A detector has background rate $(\ln 2)/2$ per second and efficiency 1/2 for true radioactive events. Probability of recording zero total events in one second is 1/4. Use a Poisson count model and effectively constant activity. True sample activity, in Bq, is

(A) $\log(8)$ (B) $\frac{3\log(2)}{2}$

(C) $\log(16)$ (D) $\log(4)$

36. In a counting window, true sample decays have effectively constant activity 5 Bq for 2 s and detection efficiency 0.2. Background mean is 1 count. Treat detected sample and background as independent Poisson counts. Given at least one total count, probability that at least one came from the sample is

(A) $1 - e^{-2}$ (B) e^{-1}

(C) $\frac{2}{3}$ (D) $\frac{e\left(1-e^2\right)}{1-e^3}$

37. A chain takes a nucleus with A=232, Z=90 to A=208, Z=82 using only alpha and beta-minus decays. Each alpha release is 5 MeV and each beta mean release 0.5 MeV. Mean total energy release per parent, in MeV, is

(A) 30 (B) 34

(C) 31 (D) 32

38. A parent undergoes five alpha decays and some beta-minus decays to reach A=200, Z=80. Nine charged particles are emitted in all, counting each alpha and beta once. Number of neutrons in the original parent is

(A) 140 (B) 130

(C) 138 (D) 134

39. A nucleus has binding 150 MeV. Its neutron-removed and proton-removed daughters have bindings 145 and 144 MeV; its alpha daughter has 120 MeV. Alpha binding is 28 MeV. Smallest separation energy among n, p, alpha, ignoring reaction barriers, in MeV, is

(A) 6 (B) 2

(C) 5 (D) 30

40. A neutron is transferred from X to Y without other particles. Neutron separation energy of X is 8 MeV. Y and its one-neutron-added product have bindings 120 and 131 MeV. Reaction Q, in MeV, is

(A) 19 (B) -3

(C) 11 (D) 3

41–50 / Preserved hard practice

41. Two independent radioactive species initially contain equal numbers of nuclei. Their half-lives are T and $2T$. At a later instant, their combined activity is one-third of its initial value. What fraction of the surviving radioactive nuclei belongs to the species of half-life $2T$?

(A) $\frac{1}{3}$ (B) 0.5

(C) 0.75 (D) $\frac{2}{3}$

42. A decay chain is $A \to B \to S$, with S stable, $\lambda_A = \lambda$ and $\lambda_B = 2\lambda$. Initially there are N_0 A nuclei and no B or S. For this chain use $N_A = N_0 e^{-\lambda t}$ and $N_B = N_0(e^{-\lambda t} - e^{-2\lambda t})$. At the nonzero time when A and B have equal activities, what fraction of the original nuclei has become S?

(A) 0.125 (B) 0.25

(C) 0.75 (D) 0.5

43. A detector registers $r = R/(1 + R\tau)$ counts per second when the true radioactive event rate is R; its fixed dead time is τ. Background is zero and every decay is otherwise detectable. A source of half-life 600 s gives registered rates $300\,\mathrm{s}^{-1}$ initially and $200\,\mathrm{s}^{-1}$ one half-life later. The initial radioactive population, in units of 10^5 nuclei, is

(A) $\frac{36}{5\log(2)}$ (B) $\frac{18}{5\log(2)}$

(C) $\frac{18}{\log(2)}$ (D) $\frac{9}{5\log(2)}$

44. A complete model decay sequence takes $^{238}_{92}X$ to $^{206}_{82}Y$ using only alpha and beta-minus decays. Each alpha contributes 5.00 MeV of absorbed charged-particle kinetic energy and each beta-minus decay contributes an average 1.00 MeV from its electron. Ignore nuclear recoil and neutrino heating. Total heat per complete sequence, in MeV, is

(A) 46 (B) 40

(C) 54 (D) 48

45. Two independent radioactive species initially have equal activities. Their half-lives are T and $2T$, and their energy releases per decay are $2E$ and E, respectively. At a later instant the combined activity is 3/8 of its initial value. What fraction of the initial thermal power remains, assuming all decay energy is absorbed?

(A) 0.25 (B) 0.375

(C) $\frac{1}{3}$ (D) 0.5

46. A proton strikes a stationary target in an endothermic two-body reaction $p + A \to B + \alpha$ with $Q = -2.00$ MeV. For nonrelativistic kinematics, neglect mass defects in inertial mass ratios and use $m_p : m_A : m_B : m_\alpha = 1 : 7 : 4 : 4$. At threshold, the products have zero relative speed in their centre-of-mass frame. The alpha-particle kinetic energy in the laboratory, in MeV, is

(A) $\frac{2}{7}$ (B) $\frac{1}{7}$

(C) $\frac{8}{7}$ (D) $\frac{16}{7}$

47. A stationary parent emits an alpha particle of mass m and charge $2e$, leaving a daughter of mass $59m$. The alpha enters a perpendicular magnetic field B and follows a circle of radius r. All released energy becomes the two products' nonrelativistic kinetic energies. The decay energy Q divided by $e^2B^2r^2/m$ is

(A) $\frac{120}{59}$ (B) 2

(C) $\frac{60}{59}$ (D) 1

48. A stable target is irradiated for one half-life T of the radioactive product, creating that product at constant rate R. Target depletion is negligible. Production then stops and the sample waits another T. Use $\lambda = \ln 2/T$ and the buildup law $N(t) = R(1 - e^{-\lambda t})/\lambda$ during irradiation. Of all product nuclei created during irradiation, the fraction that has decayed by the end of the waiting period is

(A) 0.75 (B) $1 - \frac{1}{4\log(2)}$

(C) $\frac{1}{4\log(2)}$ (D) $1 - \frac{1}{2\log(2)}$

49. Deuterium and tritium fuse at negligible initial kinetic energy into an alpha particle and a neutron. Use neutral-atom masses $2.0141u$, $3.0160u$, $4.0026u$ and neutron mass $1.0087u$, with $uc^2 = 931.5$ MeV. For kinetic-energy partition take $m_\alpha = 4m_n$. The neutron then collides head-on elastically with a stationary carbon nucleus of mass $12m_n$. The carbon recoil energy, in MeV, is

(A) 14.00976 (B) $\frac{525366}{105625}$

(C) $\frac{1050732}{528125}$ (D) $\frac{2101464}{528125}$

50. A stationary alpha-emitting nucleus releases 6.1 MeV, and daughter mass is 60 times alpha mass. The alpha particle then crosses an accelerating potential of 0.5 MV. Its magnetic-orbit radius in a fixed perpendicular field, divided by its radius before acceleration, is

(A) $\sqrt{2}$ (B) $\sqrt{6/7}$

(C) $\sqrt{7/6}$ (D) $7/6$

Answer key

Nuclei

P-NUCL / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**B**	21	**A**	31	**B**	41	**D**
02	**B**	12	**D**	22	**D**	32	**A**	42	**B**
03	**A**	13	**C**	23	**A**	33	**A**	43	**B**
04	**C**	14	**D**	24	**C**	34	**B**	44	**A**
05	**C**	15	**C**	25	**C**	35	**A**	45	**C**
06	**A**	16	**A**	26	**C**	36	**D**	46	**B**
07	**C**	17	**A**	27	**B**	37	**D**	47	**A**
08	**D**	18	**D**	28	**A**	38	**D**	48	**B**
09	**B**	19	**D**	29	**B**	39	**B**	49	**D**
10	**B**	20	**C**	30	**A**	40	**D**	50	**C**

Chapter 28

Semiconductors

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A transistor has $I_C = 9\,\text{mA}$ and $I_B = 0.1\,\text{mA}$. Common-emitter current gain β is

(A) 89 (B) 180

(C) 90 (D) 91

02. A transistor has $I_C = 9\,\text{mA}$ and $I_B = 1\,\text{mA}$. Emitter current in mA is

(A) 9 (B) 11

(C) 10 (D) 20

03. A transistor has common-base gain $\alpha = 0.98$. Common-emitter gain β is

(A) 48 (B) 98

(C) 50 (D) 49

04. A transistor has $\beta = 99$. Common-base gain is

(A) 0.99 (B) 1.98

(C) -0.01 (D) 1.99

05. A two-input NAND gate has one input fixed at 1 and other A. Output is

(A) $\overline{A}$ (B) A

(C) 0 (D) 1

06. A two-input NOR gate has both inputs tied to A. Output is

(A) $\overline{A}$ (B) 1

(C) A (D) 0

07. For binary inputs, $A + AB$ in Boolean algebra equals

(A) $A + B$ (B) AB

(C) B (D) A

08. For all four equally likely binary input pairs to a two-input OR gate, fraction yielding output 1 is

(A) -0.25 (B) 1.5

(C) 1.75 (D) 0.75

09. A junction diode with 0.7 V forward drop carries 10 mA. Electrical power dissipated in mW is

(A) 14 (B) 8

(C) 6 (D) 7

10. A common-emitter transistor amplifier has small-signal voltage-gain magnitude 20. An instantaneous positive input change of 5 mV produces an output voltage change in mV equal to

(A) -101 (B) -200

(C) -100 (D) -99

11–40 / New middle-to-hard practice

11. A node connects to 12 V through 2 kilohms and to ground through 4 kilohms. An ideal diode has anode at this node and cathode held at 5 V. Current through the diode, in mA, is

(A) 0 (B) 2.25

(C) 3.5 (D) 1.25

12. A node connects to -4 V through 1 kilohm and ground through 2 kilohms. A diode with 0.7 V forward drop has anode held at 2 V and cathode at the node. Diode current, in mA, is

(A) 0.65 (B) 5.3

(C) 0 (D) 5.95

13. A high-impedance node is driven by $4\sin\omega t$ V through a resistor. Ideal diode clamps hold its output within -1 V and+2 V. Fraction of each period when both clamps are off is

(A) 0.5 (B) $\frac{\arcsin\left(\frac{1}{4}\right)+\frac{\pi}{6}}{\pi}$

(C) $\frac{-\arcsin\left(\frac{1}{4}\right)+\frac{\pi}{6}}{\pi}$ (D) $\frac{1}{3}$

14. A symmetric triangular input sweeps linearly between -3 V and+3 V. Through a resistor it drives a high-impedance ideal clamp, clipping below -1 V and above+2 V. Time-average output voltage, in volts, is

(A) 0.75 (B) 0.25

(C) 0.5 (D) 0

15. A 6 V ideal Zener is fed from 12 V through 1 kilohm and shunted by an unknown load. Zener current is 2 mA. Increase supply to 15 V, keeping load and resistor unchanged. New Zener current, in mA, is

(A) 3 (B) 9

(C) 4 (D) 5

16. A 5 V regulator sees supply 10 to 15 V and load current 0 to 5 mA. Its Zener requires at least 1 mA and permits at most 15 mA. Which count of candidate series resistors 680,750,820,1000 ohms keeps regulation within all limits?

(A) 1 (B) 2

(C) 3 (D) 4

17. A 6 V Zener feeds a 1 kilohm load from supply 12 to 18 V. It needs at least 2 mA, with maximum allowed power 60 mW. The series resistance satisfying both extremes, in ohms, is

(A) 1000 (B) 600

(C) 750 (D) 500

18. A Zener has voltage 5.2 V at 2 mA and 5.6 V at 6 mA. Use a linear breakdown model. It is fed from 12 V through 1 kilohm and has a 1 kilohm load. Its operating voltage, in volts, is

(A) 5.2 (B) 6

(C) $\frac{31}{6}$ (D) 5

19. An NPN switch has beta 100, $V_{BE} = 0.7$ V and $V_{CE,sat} = 0.2$ V. Emitter is grounded; collector uses 1 kilohm to 5 V, base uses 80 kilohms to 5 V. Collector current, in mA, is

(A) 0.05375 (B) 4.8

(C) 5.375 (D) 5

20. An NPN has beta 99, $V_{BE} = 0.7$ V. Collector uses 1 kilohm to 10 V, emitter 100 ohms to ground, and base 40 kilohms to 4.7 V. Assume active mode and check it against 0.2 V saturation. Its V_{CE}, in volts, is

(A) 0.8 (B) 0.32

(C) 2.08 (D) 1.28

21. A CE transistor initially has emitter current 5 mA and drops 2.4 V across its 500 ohm collector resistor from an 8 V supply. Increase its base current by 50 microamperes at unchanged gain and active mode. New collector voltage, in volts, is

(A) 5 (B) 5.6

(C) 4.6 (D) 5.2

22. A transistor's collector current rises from 4.8 to 7.2 mA when base current rises by 0.1 mA. Neglect leakage and assume constant active gain. At emitter current 10 mA, collector current in mA is

(A) 10 (B) $\frac{50}{13}$

(C) 9.6 (D) $\frac{125}{12}$

23. A CE stage has unloaded voltage gain -60, input 1 kilohm and output 2 kilohms. A source of peak 0.1 V with 1 kilohm internal resistance drives it, and load is 4 kilohms. Average sinusoidal load power, in mW, is

(A) 0.5 (B) 2

(C) 0.25 (D) 1

24. A CE amplifier with fixed input voltage gives voltage-gain magnitudes 40 and 60 when loaded by 4 and 12 kilohms. Model its output by an unknown series resistance and ideal unloaded gain. Unloaded gain magnitude is

(A) 60 (B) 80

(C) 100 (D) 120

25. Two CE stages have unloaded gains $-12, -15$ and output resistances 1 kilohm each. Stage 2 input is 2 kilohms and final load 4 kilohms; source drives stage 1 ideally. Negative voltage feedback fraction 1/24 is applied. Use $A_f = A/(1 + A\beta_f)$. Closed-loop gain is

(A) 96 (B) 19.2

(C) 24 (D) 9.6

26. Three identical CE-like stages form a positive-feedback ring, each with transfer $H = -G/(1 + j\omega\tau)$, G positive. At a nonzero oscillation frequency, least G allowing unit loop magnitude and zero net loop phase modulo 2π is

(A) 8 (B) 1

(C) 2 (D) 3

27. Independent fair bits A, B, C drive $F = \text{NAND}(\text{NAND}(A, B), \text{NAND}(A, C))$. Probability of F=1 is

(A) 0.25 (B) 0.625

(C) 0.5 (D) 0.375

28. $F = \text{NOR}(\text{NOR}(A, B), \text{NOR}(A, C))$ for fair independent bits. Given exactly one of A, B, C is 1, probability F=1 is

(A) 0.5 (B) $\frac{2}{3}$

(C) 0.75 (D) $\frac{1}{3}$

29. An XOR gate on fair independent A, B is stuck at output 1 with probability p independently of inputs; otherwise it works. Its output is 1 with probability 0.6. A second perfect XOR forms C=F XOR A. Probability C equals B is

(A) 0.9 (B) 0.8

(C) 0.6 (D) 0.95

30. An AND gate on fair independent A, B has its correct output flipped with probability p. The observed output 1 probability is 0.4. Given observed 1, probability the true AND output was 1 is

(A) 0.25 (B) 0.7

(C) 0.4375 (D) 0.375

31. A is high for the first 1 ms of every 2 ms period; B is high for the first 1 ms of every 3 ms period. Their NAND output is 5 V when high and 0 otherwise, driving 250 ohms. Energy in the load over a common 6 ms period, in mJ, is

(A) 0.1 (B) 0.6

(C) 0.2 (D) 0.5

32. A is high in the first 1 ms of each 2 ms period. B is high for 1 ms starting 0.5 ms into each 3 ms period. Output $A\,\overline{B}$ at 5 V drives an LED of 2 V drop through 1 kilohm, otherwise off. Mean LED current, in mA, is

(A) 2 (B) 1.5

(C) 1 (D) 0.5

33. An intrinsic semiconductor obeys $n_i = CT^{3/2}e^{-E_g/(2kT)}$ and mobility $\mu = \mu_0/T$. Conductivity at 2 T is $4\sqrt{2}$ times that at T. The ratio $E_g/(kT)$ is

(A) $\log(16)$ (B) $\log(256)$

(C) $\log(4)$ (D) $\log(64)$

34. An intrinsic semiconductor has $n_i = CT^{3/2}e^{-E_g/(2kT)}$ and $n_i(2T)/n_i(T) = 4\sqrt{2}$. Its mobility varies as $T^{-3/2}$. Conductivity at 4 T divided by that at T is

(A) 2 (B) $4\sqrt{2}$

(C) 4 (D) $2\sqrt{2}$

35. At low temperature a fully ionised n-type sample has electron density 3×10^{15} cm^{-3} and negligible holes. At a higher temperature $n_i = 2 \times 10^{15}$ cm^{-3}. Use $n - p$ fixed, $np = n_i^2$, and unchanged mobilities $\mu_n = 2\mu_p$. New conductivity divided by old is

(A) 1.5 (B) $\frac{5}{3}$

(C) 2 (D) $\frac{4}{3}$

36. An n-type semiconductor at fixed T has $n_i = N$ and equal electron and hole mobilities. Its conductivity is $5e\mu N$. Add fully ionised acceptors of density $\sqrt{21}N$ without changing T or mobility. New conductivity divided by old is

(A) 0.2 (B) 0.4

(C) 0.6 (D) 0.5

37. A solar cell follows $I = I_{ph} - I_0(e^{V/V_T} - 1)$ for V nonnegative. It supplies 9 mA at V=0 and 8 mA at $V = V_T \ln 2$. Connect a load resistor. Greatest resistance keeping output current at least 6 mA, divided by $V_T/(1\,\text{mA})$, is

(A) $\frac{\log(2)}{6}$ (B) $\frac{\log(3)}{3}$

(C) $\frac{\log(2)}{3}$ (D) $\frac{\log(2)}{2}$

38. A reverse-biased photodiode has dark current $aV + b$ and adds constant photocurrent under illumination. Dark readings at 2 and 4 V are 2 and 3 microamperes; illuminated current at 4 V is 9 microamperes. Connect it in reverse bias to 16 V through 1 megohm under the same light. Current, in microamperes, is

(A) 8 (B) 10

(C) 12 (D) 9

39. Identical LEDs emit 620 nm photons and each has forward drop equal to photon energy divided by e. They are in series with 300 ohms across 15 V and carry 10 mA. Use hc=1240 eV nm. Number of LEDs is

(A) 7 (B) 6

(C) 3 (D) 5

40. Red and blue LEDs have forward drops and photon energies 2 V,2 eV and 3 V,3 eV. Separate branches across 9 V use 100 and 150 ohms respectively. Quantum efficiencies are 1/2 and 1/4 photon per electron. Red radiant power divided by blue is

(A) 1.5 (B) $\frac{7}{3}$

(C) 3.5 (D) $\frac{7}{6}$

41–50 / Preserved hard practice

41. A full-wave bridge rectifier feeds a resistance R from $v(t) = 2.00\sin\omega t$ volts. Exactly two diodes conduct in each conducting path, each with fixed drop 0.500 V; otherwise the diodes block ideally. There is no smoothing capacitor. The average load current divided by $(1.00\,\text{V})/R$ is

(A) $\frac{2}{3}$ (B) $-\frac{2}{3} + \frac{2\sqrt{3}}{\pi}$

(C) $-\frac{1}{3} + \frac{2\sqrt{3}}{\pi}$ (D) $\frac{4}{\pi}$

42. An NPN transistor has its collector connected to 12.0 V through 2.00 kΩ, emitter to ground through 1.00 kΩ, and base to 12.0 V through 100 kΩ. Use active gain $\beta = 100$, $V_{BE} = 0.700$ V in either conducting state, and $V_{CE} = 0.200$ V in saturation. The emitter voltage, in volts, is

(A) $\frac{6013}{1510}$ (B) $\frac{11413}{2010}$

(C) 11.3 (D) $\frac{59}{15}$

43. A photovoltaic cell obeys the supplied model $I = I_{\text{ph}} - I_s(e^{V/V_T} - 1)$, with positive constants I_s, V_T. Its short-circuit current is 10.0 mA and open-circuit voltage is $V_T \ln 101$. At terminal voltage $V_T \ln 51$, the delivered power divided by $(10.0\,\text{mA})V_T$ is

(A) $\frac{\log(51)}{2}$ (B) $\log(51)$

(C) $\frac{\log(101)}{2}$ (D) $\frac{\log(51)}{10}$

44. A nondegenerate semiconductor has electron and hole densities n and p, intrinsic density n_i, and fully ionized net donor density $3n_i/2$. Use $n - p = 3n_i/2$, $np = n_i^2$ and $\mu_e = 2\mu_h$. Both carrier types contribute to conventional drift current. The fraction of total drift current carried by holes is

(A) 0.25 (B) $\frac{2}{9}$

(C) 0.2 (D) $\frac{1}{9}$

45. An LED has fixed forward drop 2.0 V and photon energy 2.0 eV. It is in series with a resistor $R = 100(1 + 0.010\Delta T)\,\Omega$ across a 10.0 V supply. The resistor alone heats a body losing $0.00320\Delta T$ W to its surroundings; the LED is thermally isolated from it. At equilibrium, 25% of the LED electrical power becomes light. With $e = 1.6 \times 10^{-19}$ C, emitted photon rate in units of $10^{16}\,\mathrm{s}^{-1}$ is

(A) 25 (B) 12.5

(C) 6.25 (D) 3.125

46. An intrinsic semiconductor obeys $n_i(T) = AT^{3/2}e^{-E_g/(2kT)}$, while the sum of its electron and hole mobilities is $BT^{-3/2}$, with A,B and gap E_g constant over the stated interval. Its conductivity at 600 K is sixteen times that at 300 K. The ratio $E_g/[k(300\,\mathrm{K})]$ is

(A) $16\log(2)$ (B) $32\log(2)$

(C) $4\log(2)$ (D) $8\log(2)$

47. A photodiode produces constant current $I_{\mathrm{ph}} = \eta e\Phi$ under steady illumination, where $\eta = 0.500$ and photon energy is 3.20×10^{-19} J. This current feeds an unknown resistor R in parallel with $C = 1.00\,\mu$F. After long illumination the light is switched off, and voltage decays as $V = V_0e^{-t/(RC)}$. It is 2.00 V at 1.00 ms and 0.500 V at 3.00 ms. Use $e = 1.60 \times 10^{-19}$ C. The earlier incident optical power, in mW, was

(A) $8\log(2)$ (B) $16\log(2)$

(C) $32\log(2)$ (D) $\frac{16}{\log(2)}$

48. A 6.00 V ideal Zener regulator has series resistance 200 Ω and a load resistance R_L. Input voltage can take any value from 11.0 to 13.0 V. Regulation is allowed only when Zener current stays between 5.00 and 20.0 mA, inclusive, at every input voltage. The width of the allowed interval of load resistances, in ohms, is

(A) 100 (B) 400

(C) 300 (D) 200

49. An NPN transistor has grounded emitter and collector connected to 12.0 V through 2.00 kΩ. Its base is fed by an adjustable DC voltage V_B through 100 kΩ. Use $\beta = 60$, active-state $V_{BE} = 0.700$ V, and saturation limit $V_{CE} = 0.200$ V. The base-supply voltage that maximizes transistor collector-emitter power I_CV_{CE}, in volts, is

(A) 5.7 (B) 0.7

(C) 6.7 (D) 10.7

50. Two DC sources, 9.0 V and 6.0 V relative to ground, each feed the same output node through a 1.0 kΩ resistor followed by a diode oriented from source to output. A 2.0 kΩ load connects output to ground. Each conducting diode drops 0.70 V and blocks reverse current ideally. Output voltage is

(A) $\frac{83}{15}$

(B) $\frac{68}{15}$

(C) 5.3

(D) 5.44

Answer key

Semiconductors

P-SEMI / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**B**	21	**A**	31	**D**	41	**B**
02	**C**	12	**D**	22	**C**	32	**C**	42	**A**
03	**D**	13	**B**	23	**A**	33	**B**	43	**A**
04	**A**	14	**B**	24	**B**	34	**D**	44	**D**
05	**A**	15	**D**	25	**B**	35	**A**	45	**C**
06	**A**	16	**C**	26	**C**	36	**B**	46	**A**
07	**D**	17	**C**	27	**D**	37	**C**	47	**B**
08	**D**	18	**C**	28	**D**	38	**B**	48	**A**
09	**D**	19	**B**	29	**A**	39	**B**	49	**A**
10	**C**	20	**D**	30	**C**	40	**B**	50	**A**

Chapter 29

Magnetism & Matter

50 questions / Single correct / Four options

Optional enrichment practice. Magnetism and Matter is retained as an observed engine unit for enrichment. Exact current-syllabus inclusion of its extra material is not asserted.

01–10 / Selected middle practice

01. A magnetic dipole moment $0.5\,\mathrm{A\,m^2}$ rotates from parallel to anti-parallel in $0.2\,\mathrm{T}$. Quasistatic external work in joules is

(A) 1.2 (B) -0.8

(C) 0.2 (D) 0.4

02. A magnetic dipole rotates from parallel to $60°$. External work divided by mB is

(A) 1 (B) -0.5

(C) 1.5 (D) 0.5

03. A dipole rotates from $60°$ to $120°$. External work divided by mB is

(A) 1 (B) 0

(C) 0.5 (D) 2

04. A bar magnet is divided into four equal lengths across its magnetic axis. Total of the four moment magnitudes divided by original magnitude is

(A) 0.5 (B) 1

(C) 2 (D) 0

05. A magnetic dipole at far axial distance r gives field B. At equatorial distance r, field magnitude divided by B is

(A) 1.5 (B) 0.5

(C) -0.5 (D) 1

06. A material obeys Curie law. Susceptibility is 0.006 at 300 K. At 450 K it is

(A) -0.996 (B) 0.008

(C) 1.004 (D) 0.004

07. A paramagnetic sample has $\chi = 0.01$ and $H = 2000\,\text{A/m}$. Magnetisation in A/m is

(A) 20 (B) 40

(C) 21 (D) 19

08. Two equal perpendicular uniform fields have magnitude B each. Resultant field divided by B is

(A) 2 (B) $1/\sqrt{2}$

(C) $\sqrt{2}$ (D) 1

09. A magnetic dipole far-field axial field equals B at distance r. Distance where that field is $B/27$ divided by r is

(A) 6 (B) 2

(C) 3 (D) 4

10. A sample obeys Curie law. To reduce susceptibility to 3/4 its original value, absolute temperature multiplier is

(A) $\frac{4}{3}$ (B) $\frac{7}{3}$

(C) $\frac{8}{3}$ (D) $\frac{1}{3}$

11–40 / New middle-to-hard practice

11. A dipole constrained to the xy plane is rotated in an unknown uniform in-plane field. External works from 0° to 90° and from 90° to 180°, measured from +x, are 2 J and 6 J. External work from the initial 0° direction to stable alignment, in joules, is

(A) -4 (B) $4 - 2\sqrt{5}$

(C) $2 - 2\sqrt{5}$ (D) $-4 + 2\sqrt{5}$

12. A bar magnet is given angular speed at its stable orientation in a uniform field. With no friction it would turn as far as 120° and stop. The field is instead switched off when it first reaches 60°. Its subsequent angular speed divided by the initial speed is

(A) $\frac{\sqrt{6}}{2}$ (B) $\frac{2}{3}$

(C) $\frac{\sqrt{3}}{3}$ (D) $\frac{\sqrt{6}}{3}$

13. A compass oscillates with period T in a horizontal field B_0. Adding an unknown perpendicular horizontal field changes its period to $T/\sqrt{2}$. The added field divided by B_0 is

(A) $\sqrt{2}$ (B) $\sqrt{3}$

(C) 2 (D) 1

14. Two rigidly joined magnets have moments m and $2m$. With parallel axes their small-oscillation period in a uniform field is T. Remount their axes at 90° while keeping total rotational inertia unchanged. New period divided by T is

(A) $\frac{3\sqrt{5}}{5}$ (B) $\frac{\sqrt{3}\sqrt[4]{5}}{3}$

(C) $\frac{\sqrt{15}}{5}$ (D) $\frac{\sqrt{3}\cdot 5^{\frac{3}{4}}}{5}$

15. A uniform bar magnet of moment M is cut across its axis into length ratio 1:2. The pieces are fixed with perpendicular moments. At far axial distance $2r$ along their resultant moment, field divided by the original bar's far axial field at r is

(A) $\frac{\sqrt{5}}{3}$ (B) $\frac{\sqrt{5}}{24}$

(C) $\frac{\sqrt{5}}{12}$ (D) 0.125

16. A bar magnet of moment M is cut lengthwise into equal halves. One half is then cut across its axis into equal pieces. Combine the uncut half and one small piece with moment directions at 120°. Work to rotate this assembly from stable to unstable in B, divided by MB, is

(A) $\frac{\sqrt{3}}{4}$ (B) 0.5

(C) 1.5 (D) $\frac{\sqrt{3}}{2}$

17. A bar magnet has far axial fields B_0 and $B_0/8$ at distances r and $r+d$. Its far equatorial field at distance $2d$, divided by B_0, is

(A) 0.25 (B) 0.125

(C) 0.0625 (D) 0.03125

18. At a point at the same far distance as an axial observation, a dipole field is perpendicular to the dipole moment. Use $\mathbf{B} \propto 3(\hat{m}\cdot\hat{r})\hat{r} - \hat{m}$. Its magnitude divided by the axial field at that distance is

(A) $\sqrt{2}$ (B) $\frac{\sqrt{3}}{3}$

(C) $\frac{\sqrt{2}}{2}$ (D) 0.5

19. A horizontal field 3×10^{-5} T perpendicular to the magnetic meridian deflects a compass by 45°. The local dip angle is 60°. Magnitude of the total terrestrial field, in tesla, is

(A) $\frac{\sqrt{3}}{50000}$ (B) 0.00003

(C) 0.00006 (D) $\frac{3\sqrt{3}}{100000}$

20. The total terrestrial field is equal at two places. A horizontal compass has period T at the first, of dip 30°, and $\sqrt{2}T$ at the second. The second place's vertical field component divided by total field is

(A) $\frac{\sqrt{13}}{4}$ (B) $\frac{\sqrt{7}}{4}$

(C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{3}}{4}$

21. A Curie-law paramagnet has its magnetisation doubled when field H is tripled and its absolute temperature increased by 100 K. Starting from the original state, halve H and lower temperature by 50 K. New magnetisation divided by the original is

(A) $\frac{2}{3}$ (B) 0.5

(C) 1 (D) 0.75

22. A paramagnet follows Curie law in a solenoid whose field is proportional to current. Its steady temperature is $T_0 + \beta I^2$. Equal magnetisations are observed at currents I_0 and $3I_0$. Magnetisation at $\sqrt{3}I_0$ divided by that at I_0 is

(A) $\sqrt{3}$ (B) $\frac{2}{3}$

(C) 1.5 (D) $\frac{2\sqrt{3}}{3}$

23. A paramagnet obeys $\chi = C/(T - \Theta)$. Its susceptibility is 0.004 at 300 K and 0.002 at 500 K. Temperature at which it is 0.001, in kelvin, is

(A) 900 (B) 800

(C) 1000 (D) 700

24. A material obeys $M = CH/(T - \Theta)$. Its magnetisations are equal at $(H, 400\,\text{K})$ and $(2H, 600\,\text{K})$. With H held fixed, susceptibility at 300 K divided by that at 500 K is

(A) 3 (B) 2

(C) $\frac{5}{3}$ (D) 1.5

25. A long core obeys $M = M_r + \chi H$. Measurements give $B/\mu_0 = 6000$ A/m at H=1000 A/m and 10000 A/m at H=2000 A/m. At H=−500 A/m, its pole strength for area $1\,\text{cm}^2$, in A m, is

(A) 0.1 (B) 0.2

(C) 0 (D) 0.05

26. A long solenoid has H=2000 A/m. At temperature T its paramagnetic core has $B = 0.012\pi$ T. The core obeys Curie law. At temperature $2T$ and unchanged current, pole strength for cross-section $2\,\text{cm}^2$, in A m, is ($\mu_0 = 4\pi \times 10^{-7}$)

(A) 5.6 (B) 1.4

(C) 2.8 (D) 3

27. A linear magnetic sample has intrinsic susceptibility 4 and demagnetising factor $N = 1/4$. Use $H_{\text{in}} = H_a - NM$ and $M = \chi H_{\text{in}}$. Its induction B divided by $\mu_0 H_a$ is

(A) 1.25 (B) 2.5

(C) 2 (D) 5

28. A sample has intrinsic Curie susceptibility C/T and $H_{\text{in}} = H_a - NM$. Measured M/H_a is 2 at 300 K and 6/5 at 600 K. Its effective susceptibility at 900 K is

(A) $\frac{6}{7}$ (B) $\frac{2}{3}$

(C) 1 (D) 0.8

29. A magnetic circuit has core length ℓ, relative permeability μ_r and a narrow air gap g, all of equal area. At fixed ampere-turns, its field is B_0 without the gap and $B_0/5$ with it. Then double g and halve μ_r. New field divided by B_0 is

(A) 0.2 (B) $\frac{1}{9}$

(C) 0.125 (D) 0.1

30. Two equal-length, equal-area linear core sections form one series magnetic path, with effective relative permeability 150. One section has relative permeability 100. The same materials are rearranged into equal-area parallel flux paths of the same length. New effective relative permeability is

(A) 250 (B) 300

(C) 200 (D) 150

31. A core's hysteresis power is proportional to frequency f and eddy-current power to f^2. Total losses are 6 W at 50 Hz and 20 W at 100 Hz, under the same field amplitude. Total loss at 75 Hz, in watts, is

(A) 9 (B) 13

(C) 13.5 (D) 12

32. A core of volume $10^{-4}\,\mathrm{m}^3$ has a rhombus B-H loop with H-diagonal 400 A/m and unknown B-diagonal. It heats a calorimeter; loss to ambient is $0.2\Delta T$ W. Initial heating at 50 Hz is 0.1 K/s. At 100 Hz and $\Delta T = 5$ K it is 0.15 K/s. Ignore eddy loss. The B-diagonal, in tesla, is

(A) 0.5 (B) 2

(C) 1 (D) 4

33. Equal volumes of a paramagnet $\chi_p = C/T$ and a diamagnet $\chi_d = -d$ form a non-interacting mixture. Its susceptibility is 0.001 at 300 K and zero at 600 K. Its susceptibility at 900 K is

(A) $\frac{1}{3000}$ (B) $-\frac{1}{3000}$

(C) $-\frac{1}{1500}$ (D) 0

34. A non-interacting mixture contains a Curie paramagnet with susceptibility 0.006 at 300 K and a temperature-independent diamagnet with susceptibility -0.002. At 300 K the mixture susceptibility is 0.002. Heat it to 600 K, then add pure diamagnet of volume equal to the original mixture. Final susceptibility is

(A) -0.00075 (B) -0.0005

(C) 0.00025 (D) -0.001

35. A small magnet of mass M and moment m is constrained to remain anti-parallel to a vertical field $B(z) = k/z^3$ for z>0. Its upward magnetic force balances gravity at height z_0. Its small vertical oscillation period divided by $\sqrt{z_0/g}$ is

(A) $\sqrt{2}\pi$ (B) π

(C) 2π (D) $\frac{\pi}{2}$

36. A small diamagnetic body of mass M has induced-dipole energy $U = -\alpha B^2/2$, with $\alpha < 0$, in field $B = bz$. Gravity acts toward negative z. Its equilibrium coordinate is $z = -h$. Small vertical period divided by $\sqrt{h/g}$ is

(A) π (B) $\sqrt{2}\pi$

(C) $2\sqrt{2}\pi$ (D) 2π

37. A dipole of moment magnitude $1\,\mathrm{A\,m^2}$ has torque $(0, -3, 4)$ N m when aligned with +x, and $(3, 0, -12)$ N m when aligned with +y, in one uniform field. Its minimum potential energy, in joules, is

(A) -19 (B) -13

(C) -5 (D) -12

38. A field $3B$ along +x is combined with an unknown field pointing at 120° to +x in the xy plane. A compass settles at 30°. Work to turn a dipole m from stable to unstable in the resultant field, divided by mB, is

(A) 4.5 (B) 3

(C) $3\sqrt{3}$ (D) $\frac{3\sqrt{3}}{2}$

39. Two identical far-separated dipoles of moment m initially point along their line of centres. Rotate both rigidly by 90° so they remain parallel but become transverse to that line. If the initial field of either at the other is B_a, total external work divided by mB_a is

(A) 3 (B) 1.5

(C) 2 (D) 1

40. Two small magnets lie at $x = -a, +a$, with moments M along +x and an unknown moment along +y. At the midpoint their resultant field is 45° below +x. A tiny test moment m starts along +y and settles in that field. Energy released divided by kMm/a^3, where $k = \mu_0/(4\pi)$, is

(A) $2 + 2\sqrt{2}$ (B) $2\sqrt{2}$

(C) 4 (D) $-2 + 2\sqrt{2}$

41–50 / Preserved hard practice

41. A thin uniform bar magnet oscillates about its centre in a uniform field with period T. It is cut into lengths $L/3$ and $2L/3$. The shorter piece is reversed and rigidly rejoined to the longer one in the original straight shape. Magnetisation magnitude and mass distribution are unchanged. Find its new small-oscillation period divided by T.

(A) $\frac{\sqrt{3}}{3}$ (B) 1.5

(C) $\sqrt{3}$ (D) 3

42. A magnet of moment μ and inertia I is suspended on a torsion fibre of stiffness κ. Its preferred zero-twist direction is parallel to a field B. Reversing B while keeping the magnet near that direction doubles its small-oscillation period; that direction remains stable. Find $\mu B/\kappa$.

(A) $\frac{1}{3}$ (B) 0.8

(C) 0.6 (D) 0.5

43. A magnet rotates freely about a fixed axis in a uniform field, starting from rest at 60° to the field. At its first alignment the field is reversed instantaneously without changing angular velocity. It subsequently makes complete rotations. Find the ratio of greatest to least angular speed squared during those rotations.

(A) 5 (B) 4

(C) 2 (D) 3

44. A weak compass has period T_0 in the horizontal terrestrial field. A much stronger fixed dipole, parallel to that field, makes its period $T_0/\sqrt{3}$ at a distant axial point r away. The compass is moved to an equatorial point $2r$ away; both points are in the dipole far field. Find its period there divided by T_0.

(A) $\sqrt{2}$ (B) $\frac{2\sqrt{14}}{7}$

(C) $\frac{2\sqrt{3}}{3}$ (D) $\frac{\sqrt{14}}{4}$

45. A bar magnet moment m, inertia I has small-oscillation period T. It is cut into two equal lengths perpendicular to its axis, each oscillating about its own centre with transverse dimension negligible. New period divided by T is

(A) -0.5 (B) 0.5

(C) 1.5 (D) 1

46. A thin uniform magnet of mass M and length L has central-axis small-oscillation period T in a uniform field. A nonmagnetic point mass is fixed to one end; about the original centre the period becomes $2T$. The pivot is then moved to the combined centre of mass. Find the new period divided by T, neglecting gravity torque.

(A) $\frac{\sqrt{10}}{2}$ (B) $\frac{\sqrt{6}}{2}$

(C) 2 (D) $\sqrt{2}$

47. A thin nonmagnetic uniform rod of length L and mass M carries a negligibly light localised point dipole μ rigidly at its centre, aligned with the rod. It lies along $+x$ in an axial field $B(x) = B_0 - bx^2$ near $x = 0$, with $b > 0$. Compare small translations of its centre along x with small rotations about its centre. If their periods are equal, find bL^2/B_0.

(A) 3 (B) 1.5

(C) 6 (D) 12

48. A weak paramagnetic sample in a long solenoid obeys Curie law. Initially it is held at temperature T_0 while an ideal voltage V drives the coil, of resistance R_0. Cooling is then relaxed: coil and sample share temperature T, the coil resistance becomes R_0T/T_0, and heat loss is $G(T - T_0)$. If $V^2/(R_0GT_0) = 2$, find final magnetisation divided by its initial value.

(A) 0.75 (B) 0.125

(C) 0.5 (D) 0.25

49. A thin uniform paramagnetic bar has constant mass, total moment K/T, and length $L_0[1 + (T - 300)/3000]$ over the stated temperature range, in kelvin. Its heat capacity is 200 J/K. From 300 K it absorbs 20 W for 3000 s without loss. In the same uniform field it oscillates about its centre. Find final period divided by initial period.

(A) 2.42

(B) 1.1

(C) $\sqrt{2}$

(D) $\frac{11\sqrt{2}}{10}$

50. A small magnet has small-oscillation period T in a horizontal field B_0. A wire adds a perpendicular horizontal field b, making the period $T/\sqrt{2}$. The wire field is then rotated to oppose B_0, without changing b. After settling at the new stable equilibrium, the final small-oscillation period divided by T is

(A) $\frac{1}{\sqrt{-1+\sqrt{3}}}$

(B) $-1 + \sqrt{3}$

(C) $\frac{1}{\sqrt{1+\sqrt{3}}}$

(D) $\frac{\sqrt{2}}{2}$

Answer key

Magnetism & Matter

P-MAMA / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**B**	21	**A**	31	**D**	41	**C**
02	**D**	12	**D**	22	**D**	32	**B**	42	**C**
03	**A**	13	**B**	23	**A**	33	**B**	43	**A**
04	**B**	14	**D**	24	**A**	34	**A**	44	**B**
05	**B**	15	**B**	25	**D**	35	**B**	45	**B**
06	**D**	16	**D**	26	**C**	36	**D**	46	**A**
07	**A**	17	**C**	27	**B**	37	**B**	47	**C**
08	**C**	18	**C**	28	**A**	38	**C**	48	**D**
09	**C**	19	**C**	29	**D**	39	**B**	49	**D**
10	**A**	20	**A**	30	**C**	40	**A**	50	**A**

Chapter 30

Communication Systems

50 questions / Single correct / Four options

Optional enrichment practice. Communication Systems is retained from the engine as extra practice; it is not listed in the pinned official BITSAT-2026 Physics syllabus.

01–10 / Selected middle practice

01. An AM envelope has maximum 9 V and minimum 3 V. Carrier amplitude in volts is

(A) 12 (B) 7

(C) 5 (D) 6

02. An AM envelope has maximum 9 V and minimum 3 V. Message-induced envelope amplitude in volts is

(A) 6 (B) 2

(C) 3 (D) 4

03. A standard single-tone AM signal has carrier power P_c and modulation index 1. Total transmitted power divided by P_c is

(A) 0.5 (B) 3

(C) 2.5 (D) 1.5

04. For standard single-tone AM with carrier power 100 W and index 0.5, total power in watts is

(A) 112.5 (B) 111.5

(C) 225 (D) 113.5

05. At $m = 1$, total sideband power divided by total AM power is

(A) $\frac{-2}{3}$ (B) $\frac{2}{3}$

(C) $\frac{1}{3}$ (D) $\frac{4}{3}$

06. At $m = 0.5$, total sideband power divided by carrier power is

(A) 1.125 (B) -0.875

(C) 0.25 (D) 0.125

07. For standard AM, modulation index doubles from 0.3 to 0.6 at unchanged carrier power. Total sideband power multiplier is

(A) 4 (B) 3

(C) 8 (D) 5

08. A transmitter height quadruples while receiver height is negligible. Approximate line-of-sight range multiplier is

(A) 3 (B) 4

(C) 1 (D) 2

09. An antenna quarter-wave length is 0.75 m. With $c = 3 \times 10^8$, carrier frequency in MHz is

(A) 101 (B) 99

(C) 100 (D) 200

10. A digital system uses 8 bits per sample at 10^4 samples/s. Raw bit rate in kbit/s is

(A) 160 (B) 81

(C) 79 (D) 80

11–40 / New middle-to-hard practice

11. An AM carrier dissipates 25 W in 2 Ω. Its envelope voltmeter has an unknown constant additive offset and reads 15 V maximum and 7 V minimum. Readings represent peak amplitudes before the offset. Modulation index is

(A) 0.5 (B) 0.8

(C) $\frac{4}{11}$ (D) 0.4

12. An AM signal has envelope extrema 12 V and 4 V and total average power 36 W in a fixed resistor. Raise only the carrier peak amplitude to 12 V, retaining the original message amplitude. New total average power, in W, is

(A) 76 (B) 72

(C) 81 (D) 54

13. An AM wave initially has carrier power 100 W and total power 132 W. A selective network halves only its carrier amplitude, leaving both sideband amplitudes unchanged. Fraction of a message period for which the resulting signed envelope is negative is

(A) 0 (B) 0.5

(C) $\frac{\arccos\left(\frac{4}{5}\right)}{\pi}$ (D) $\frac{\arccos\left(\frac{5}{8}\right)}{\pi}$

14. A transmitter has 50 W carrier power and DC input $P_{DC} = P_0 + P_{out}/\eta$. Its input is 80 W at zero modulation and 110 W at unit modulation. At modulation index 0.6, DC input in W is

(A) 100 (B) 99

(C) 90.8 (D) 59

15. A two-tone AM envelope is $A_c[1 + 0.8\cos\omega t + m_2\cos 2\omega t]$, with $m_2 > 0$. Its total average power is $1.5P_c$. Minimum signed envelope divided by A_c is

(A) -0.4 (B) 0.2

(C) $\frac{4}{15}$ (D) 0

16. An AM envelope is $A_c[1+m_1\cos\omega_1 t+m_2\cos\omega_2 t]$, with positive indices and independently attainable phases. Total power is $1.13P_c$ and minimum signed envelope is $0.4A_c$. Larger modulation index is

(A) 0.3 (B) 0.6

(C) 0.5 (D) 0.4

17. An AM signal has sidebands at 1992 and 2008 kHz. Mix it with a 2010 kHz oscillator and retain positive difference frequencies. Greater resulting sideband frequency divided by resulting carrier frequency is

(A) 1.8 (B) 0.2

(C) 0.8 (D) 1.25

18. A spectrum contains carrier 1000 kHz and sidebands $998, 1002, 995, 1005$ kHz. Mix with 1001 kHz and retain only positive difference frequencies. An ideal filter passes strictly $0 < f < 4$ kHz. Number of distinct frequencies passed is

(A) 2 (B) 4

(C) 3 (D) 1

19. An envelope detector is acceptable only when $\tau \geq 10/\omega_c$ and $\tau \leq \sqrt{1-m^2}/(m\omega_m)$. Here $f_c = 1$ MHz, $f_m = 10$ kHz, $m = 0.6$ and $C = 1$ nF. How many resistances $1, 5, 10, 25$ kilohms satisfy both bounds?

(A) 1 (B) 3

(C) 2 (D) 4

20. A detector has carrier-ripple attenuation $H = [1 + (\omega_c\tau)^2]^{-1/2} = 1/\sqrt{101}$ and $f_c/f_m = 100$. It avoids diagonal clipping when $\tau \leq \sqrt{1-m^2}/(m\omega_m)$. Greatest admissible modulation index is

(A) $\frac{\sqrt{101}}{101}$ (B) 1

(C) $\frac{10\sqrt{101}}{101}$ (D) 0.9

21. A superheterodyne uses a fixed IF and an oscillator above the desired carrier. The image frequency also differs from the oscillator by that IF. When tuned to 1000 kHz its image coincides with a station at 1500 kHz. Oscillator frequency when tuned to 1500 kHz, in kHz, is

(A) 1750 (B) 1250

(C) 2000 (D) 1500

22. A receiver uses an oscillator below its desired carrier of 2000 kHz. A lower-frequency image station has AM sidebands at 1085, 1095 kHz. Desired and image carriers lie one IF above and below the oscillator. The desired signal has the same message frequency as the image. Upper desired IF sideband, in kHz, is

(A) 465 (B) 450

(C) 460 (D) 455

23. An FDM allocation uses equal DSB-AM channels, 2 kHz between neighbours, and identical end guards. Four channels occupy 50 kHz; six occupy 74 kHz. Greatest number fitting a 100 kHz allocation with the same rules is

(A) 7 (B) 8

(C) 9 (D) 10

24. An AM channel has edges 996, 1004 kHz. Its message bandwidth is used for both DSB channels and SSB channels in an 86 kHz FDM allocation. Keep four DSB channels, 2 kHz between all neighbours, and 2 kHz at each end. Largest additional number of SSB channels is

(A) 7 (B) 6

(C) 8 (D) 9

25. A sinusoid has frequency strictly between 8 and 12 kHz. Its baseband aliases have frequencies 1 kHz at sampling rate 10 kHz and 3 kHz at rate 12 kHz. Alias frequency is distance to the nearest integer multiple of sampling rate. Original frequency, in kHz, is

(A) 15 (B) 9

(C) 11 (D) 10

26. A sampler has unknown rate strictly between 6 and 13 kHz. Inputs at 14 and 19 kHz produce baseband aliases at 2 and 3 kHz respectively. Define an alias as distance to the nearest integer multiple of the sampling rate. Sampling rate, in kHz, is

(A) 8 (B) 12

(C) 10 (D) 6

27. A PCM link sends a fixed number of bits per sample plus a fixed overhead bit rate. Total rates are 72 kbit/s at 8 ksample/s and 88 kbit/s at 10 ksample/s. Uniform quantisation spans -2 to $+2$ V with rounding to bin centres. Maximum quantisation error magnitude, in V, is

(A) 0.03125 (B) 0.015625

(C) 0.0078125 (D) 0.00390625

28. Three 8 kHz-bandwidth channels are each sampled at 25% above their Nyquist rate. Uniform quantisation requires at least 256 levels. Overhead occupies 25% of the transmitted bits. Least total bit rate, in kbit/s, is

(A) 800 (B) 600

(C) 480 (D) 640

29. Use radio horizon distance proportional to the square root of tower height and add the two horizons. Two towers initially cover 30 km. Quadrupling only the transmitter height increases coverage to 40 km. Instead, retaining its original height and multiplying receiver height by 9 gives coverage, in km, of

(A) 50 (B) 70

(C) 60 (D) 90

30. Two towers with height ratio 4 : 1 have combined radio horizon 60 km. Their total height is fixed but may be redistributed between them. With each horizon proportional to square root of height, greatest possible combined horizon, in km, is

(A) $30\sqrt{5}$ (B) $40\sqrt{2}$

(C) 60 (D) $20\sqrt{10}$

31. A shortened quarter-wave antenna follows $L = v/(4f) - \delta$, with unknown propagation speed v and fixed end correction δ. Resonant lengths are 0.700 m at 100 MHz and 0.325 m at 200 MHz. Length at 125 MHz, in m, is

(A) 0.55 (B) 0.6

(C) 0.52 (D) 0.45

32. Two short receiving dipoles of lengths L and 2 L see the same sinusoidal field. Open-circuit voltage is proportional to length and radiation resistance to length squared. The shorter has radiation resistance $4\,\Omega$; each drives a separate $4\,\Omega$ load. Longer-to-shorter average load-power ratio is

(A) 0.8 (B) 0.25

(C) 4 (D) 0.64

33. A meter reads received radio power plus a constant background. At unchanged transmit power it reads 10 microwatts at distance d and 4 microwatts at 2d. True received power varies as transmit power divided by distance squared. At 2d with transmit power multiplied by 9/4, reading in microwatts is

(A) 4.5 (B) 9

(C) 6.5 (D) 6.25

34. For a free-space link, $P_r = P_t G_t G_r [\lambda/(4\pi r)]^2$. Both antennas have fixed effective physical apertures and $G = 4\pi A_e/\lambda^2$. Double frequency, halve transmitted power and insert a power transmission factor 1/4. To preserve received power, new distance divided by old distance is

(A) $\sqrt{2}$ (B) $\frac{\sqrt{2}}{2}$

(C) 0.5 (D) 2

35. Received signal varies as $1/r^2$; noise power is $kB + N_0$ for bandwidth B. At distance d, SNR is 20 at bandwidth 2 kHz and 10 at 6 kHz. At distance 2d and bandwidth 10 kHz, SNR is

(A) 2.5 (B) $\frac{5}{3}$

(C) $\frac{10}{3}$ (D) 5

36. Input signal power is 1 nW with SNR 10. A power-gain-10 amplifier adds 30 nW output noise. A following filter passes 80% of signal power and 25% of total noise power. Final SNR is

(A) $\frac{8}{31}$ (B) $\frac{32}{31}$

(C) $\frac{10}{31}$ (D) $\frac{80}{31}$

37. A ground-to-ground message passes through one satellite vertically above the midpoint; neglect horizontal distance. Each satellite passage adds a fixed delay τ. Round-trip delay including one acknowledgement is 240 ms at height H and 360 ms at height 2H. One-way delay at height 3H, in ms, is

(A) 180 (B) 300

(C) 360 (D) 240

38. A 1200 km fibre has propagation speed 2×10^8 m/s and attenuation 0.2 dB/km. Each segment permits at most 30 dB loss. Use the least number of internal repeaters, each adding 0.1 ms delay. End equipment adds no delay. Total latency, in ms, is

(A) 6.7 (B) 6.8

(C) 6.1 (D) 6

39. FM bandwidth obeys Carson's rule $B = 2(kA + f_m)$. A meter adds unknown constant offset B_0. It reads 16 kHz for amplitude A=1 and $f_m = 2$ kHz, and 26 kHz for A=2 and $f_m = 3$ kHz. True bandwidth at A=1.5 and $f_m = 4$ kHz, in kHz, is

(A) 24 (B) 18

(C) 20 (D) 22

40. A single-tone FM wave has Carson bandwidth 24 kHz and deviation ratio $\beta = \Delta f / f_m = 3$. Message frequency is then tripled and its amplitude halved; frequency deviation is proportional to message amplitude. New Carson bandwidth, in kHz, is

(A) 24 (B) 27

(C) 36 (D) 18

41–50 / Preserved hard practice

41. An AM signal has carrier amplitude $A_c > 0$ and envelope $A_c[1 + m_1 \cos(\omega t) + m_2 \cos(3\omega t)]$, where $m_1, m_2 > 0$, $m_1 + m_2 < 1$, and the carrier frequency is much greater than $3\omega/(2\pi)$. Its envelope maximum-to-minimum ratio is 9 : 1. Its total power is $1.20P_c$, where P_c is carrier power. Use $P/P_c = 1 + (m_1^2 + m_2^2)/2$ and upper-sideband power $P_c m_j^2/4$ for tone j. The weaker tone's upper-sideband power divided by P_c is

(A) 0.04 (B) 0.01

(C) 0.09 (D) 0.02

42. A radio tuner has fixed inductance L and total capacitance $C + C_s$, where the variable capacitor spans C_0 to $4C_0$ and C_s is fixed stray capacitance. The measured maximum-to-minimum resonant-frequency ratio is $\sqrt{2}$. Starting at $C = C_0$, a dielectric of relative permittivity 2 is fully inserted into the variable capacitor only; stray capacitance stays fixed. The new frequency divided by the frequency just before insertion is

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{2}}{2}$

(C) 0.5 (D) $\frac{\sqrt{6}}{3}$

43. For line-of-sight communication use $D = \sqrt{2R_E h_t} + \sqrt{2R_E h_r}$, with mast heights much smaller than Earth radius. Initially transmitter and receiver heights are 20.0 and 5.00 m. The receiver stays fixed, and the transmitter mast is made taller to double D. The transmitter mast has uniform mass per unit height 2.00 kg/m, with no other equipment mass. With $g = 10.0\,\mathrm{m/s^2}$ and ground as zero potential, the increase in its gravitational potential energy, in kJ, is

(A) 156.25 (B) 152.25

(C) 76.125 (D) 105

44. A flat-layer skywave model has virtual reflection height 300 km and critical frequency 9.00 MHz. Use the secant law $f = f_c \sec i$, where i is the incidence angle from the vertical at reflection. A 15.0 MHz ray makes one symmetric hop. All propagation speeds are $c = 3.00 \times 10^5$ km/s. Its travel time exceeds that of a direct path along the same ground separation by how many milliseconds?

(A) $\frac{4}{3}$ (B) $\frac{2}{3}$

(C) $\frac{8}{3}$ (D) $\frac{10}{3}$

45. An AM transmitter has envelope maximum and minimum amplitudes in ratio 3 : 1. It consumes 90 W DC, with 40% converted to total radiated AM power. Its carrier frequency is 10^8 Hz. With $h = 6.25 \times 10^{-34}$ J s, carrier photon rate in units of $10^{26}\,\mathrm{s^{-1}}$ is

(A) 1.28 (B) 5.12

(C) 2.56 (D) 5.76

46. A ground station transmits $f_0 = 1.00$ GHz toward a satellite moving radially away at speed $u \ll c$. The satellite coherently retransmits at exactly twice the frequency it receives in its own frame. The ground receives a return frequency 400 Hz below $2f_0$. Use the first-order Doppler factor $1 - u/c$ on each outward-separating link and $c = 3.00 \times 10^8$ m/s. The radial speed u in m/s is

(A) 120 (B) 30

(C) 60 (D) 15

47. An amplifier has power gain 10 and adds fixed output noise power N. An ideal attenuator transmits one-quarter of incident signal and noise powers and adds no noise. At the common input, signal power is S and noise power is $N/10$, with S,N positive. Compare order 1 (amplifier then attenuator) with order 2 (attenuator then amplifier). The ratio $\mathrm{SNR}_1/\mathrm{SNR}_2$ is

(A) 2.5 (B) 0.4

(C) 1 (D) 4

48. For this envelope-detector design require $10/f_c \le RC \le 1/(10f_m)$, with $f_c = 1.00$ MHz and $f_m = 5.00$ kHz. Capacitor tolerance allows $0.80C_0 \le C \le 1.20C_0$. Resistance is $R(T) = R_0[1 + \alpha(T - 300\,\mathrm{K})]$, with $\alpha \ge 0$ and $300 \le T \le 350$ K. A nominal R_0C_0 may be chosen freely, but the detector must satisfy both inequalities for every allowed C and T. The largest feasible α, in $\mathrm{K^{-1}}$, is

(A) $\frac{1}{300}$ (B) $\frac{1}{150}$

(C) $\frac{1}{75}$ (D) 0.01

49. A radio link obeys the supplied available-power model $P(d) = Kd^{-2}e^{-\alpha d}$, with positive constant K and fixed absorption coefficient $\alpha \geq 0$. Measurements give $P(r)/P(2r) = 8$. At distance r the delivered power feeds resistance R; at distance $3r$ it feeds $4R$. Ideal matching preserves the available power in each case. The second RMS load voltage divided by the first is

(A) $\frac{1}{3}$ (B) $\frac{1}{6}$

(C) $\frac{2}{3}$ (D) $\frac{1}{9}$

50. A receiver tunes an ideal LC circuit using a variable capacitor $C(V) = C_0/(1 + V/V_0)$ in series with a fixed capacitor C_0. Inductance stays fixed. Starting at $V = 0$, what V/V_0 is required to double the resonant frequency?

(A) 2 (B) 3

(C) 7 (D) 6

Answer key

Communication Systems

P-COMM / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**D**	21	**A**	31	**A**	41	**B**
02	**C**	12	**A**	22	**C**	32	**D**	42	**A**
03	**D**	13	**D**	23	**B**	33	**C**	43	**B**
04	**A**	14	**C**	24	**A**	34	**B**	44	**B**
05	**C**	15	**C**	25	**B**	35	**B**	45	**B**
06	**D**	16	**C**	26	**A**	36	**B**	46	**B**
07	**A**	17	**A**	27	**C**	37	**D**	47	**A**
08	**D**	18	**A**	28	**D**	38	**A**	48	**B**
09	**C**	19	**C**	29	**B**	39	**C**	49	**A**
10	**D**	20	**C**	30	**D**	40	**B**	50	**D**